# CAPABILITY DEVELOPMENT DOCUMENT

**FOR**

**GROUND SOLDIER SYSTEM (GSS)**

**INCREMENT I (Nett Warrior)**

**Acquisition Category (ACAT): Pre-MDAP (ACAT 1D)**

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## Executive Summary

The Ground Soldier System (GSS) integrates multiple Soldier systems and components and leverages emerging technologies to provide operational capabilities to all ground combatant Soldiers, small units and their attachments. The GSS will integrate with current dismounted Soldier situational awareness and battle command capabilities; provide Military Occupational Specialties (MOS) specific capabilities; and when integrated with the Core Soldier System (CSS), meet the needs of all Soldiers who conduct close ground combat in the Future Force (FF). Each Soldier in the Army will receive CSS capabilities and dependent on duty position and MOS will receive additional integrated, modular components making up the GSS, the Mounted Soldier System (MSS) and the Air Soldier System (AirSS). The GSS is a System of Systems (SoS) having Families of Systems (FoS) (attributes) embedded within it. This FoS approach makes the GSS integrated, modular, scalable and tailorable for MOS and mission specific tasks conducted by the Soldier. The intent of this CDD is to approach GSS capabilities incrementally. Central to this incremental strategy is the Ground Soldier System Increment I (GSS Inc I) which connects Soldiers into the Network by providing dismounted leaders and Soldiers Battle Command (BC) and Situational Awareness (SA) capabilities.

GSS Inc I provides leaders with increased BC capabilities, SA and understanding, voice and data communications, position location and network connectivity to upper and lower echelons. The Chief of Staff of the Army led, “Army Capabilities Review General Officer Steering Committee” held 30 May 2008, approved the GSS CDD strategy supporting the production capabilities associated with the GSS Inc I. GSS Inc I capabilities are intended to meet the needs of leaders who conduct ground combat and provide the “dismounted leader component” of the Brigade Combat Team (BCT) and other brigades. GSS Inc I provides a dismounted leader fightable system for modular and BCT Modernized units. BCT Modernization is the Army’s modernization strategy consisting of manned and unmanned systems, connected by a common network that provides our Soldiers and leaders with leading-edge technologies and capabilities allowing them to dominate in complex environments.

GSS capabilities begin with improvements over the Land Warrior (LW) Increment II Capabilities Production Document (CPD) and then build upon the CSS capabilities to meet the needs of all leaders who conduct ground close combat in the FF. This effort includes determining the optimal distribution of operational capabilities across teams and squads to maximize small unit mission performance. System development requires continuous integration of capabilities to increase functionality while reducing weight, cube/bulk, space, power consumption, and logistical footprint. This effort includes systems integration of multiple capabilities within this document and integration of Soldier related operational capabilities required by other proponents, services and agencies. The GSS Inc I provides leaders and small units superior integrated means to accomplish missions with greater efficiency and effectiveness to win decisively against all opponents. GSS Inc I will enable network centric operations at the small unit level.

The complete GSS Increments I and II including Government Furnished Equipment (GFE), will improve protection against threats such as projectiles, blast and Chemical, Biological, Radiological, Nuclear, and high yield Explosives (CBRNE). The Common Operating Picture

(COP) will provide leaders with the actionable information they need to sustain their collaborative initiatives and environment. Interoperability of the BC and SA with Army BCT Modernized Ground Combat Vehicles (GCV) and Army aviation systems, will give leaders access to external assets formerly available only to higher level units. Networking of leaders, weapons, sensors and external assets enables geographically dispersed small units to collaboratively influence larger areas with greater precision, speed and a broader variety of lethal effects. Embedded Training (ET) will enable the leader to meet the explosive growth in demand for future skills and proficiency levels.

GSS Inc I is part of the Army's BCT Modernization beginning in FY11. GSS Inc I will be postured to reach an Initial Operational BCT Capability in FY12, reaching Full Operational Capability (FOC) by FY16 upon fielding of 29 BCTs with 1 additional BCT provided for training. GSS Inc I will support Army BCT Modernization efforts, prioritizing fielding to BCTs aligned with Army Capability Packages in accordance with Army Force Generation (ARFORGEN). This GSS Inc I CDD will be used for a Pre-Planned Product Improvement (P3I) effort addressing the addition of maturing capabilities and technologies into the GSS Inc I, i.e. Joint Tactical Radio System (JTRS) and Soldier Radio Waveform (SRW).

The GSS will provide leaders and small units with superior, integrated means to accomplish missions with greater efficiency and effectiveness and win decisively against all opponents. For Increment I, the GSS manning includes team leaders and above within the BCT. GSS Inc I will provide an interface to a Rifleman Radio (RR) capability which will provide individual Soldiers with voice communications and beaconing of Position Location Information (PLI) to their leaders.

GSS CDD Increment II will build upon the GSS CDD Inc I capabilities and address future Soldier operational needs associated with remaining GSS CDD threshold attributes, Army BCT Modernized Battle Command. GSS Increment III builds upon the GSS Increment II capabilities and addresses future Soldier operational needs associated with GSS CDD objective attributes.

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**Revision History**

| Draft Version | Date | Purpose |
|---|---|---|
| 1.1 | 1 Feb 2006 | The GSS CDD received AROC validation |
| 1.1 | 13 Oct 2006 | The GSS CDD received J2 & J6 certification |
| 1.1 | 6 Nov 2007 | Force Application (FA) Functional Capabilities Board (FCB) |
| 1.1 | May 2008 | Updating the GSS CDD KPPs per new JCIDS guidance and presenting capabilities in an incremental approach |
| 1.0 | 6 May 2008 | Revision Initial Draft |
| 2.0 | 30 Jun 2008 | Initial ARCIC Gatekeeper Staffing |
| 2.0 | 4 Aug 2008 | ARCIC Staffing/Validation |
| 2.1 | 10 Oct 2008 | USAIC (Sponsor) to ARCIC for submission to HQDA |
| 2.2 | 18 Nov 2008 | USAIC Final Draft for ARCIC validation |
| 2.3 | 16 Jan 2009 | Final for submission to HQDA for Army Staffing |
| 2.4 | 2 April 2009 | Revision of 1-Star ARSTAF comments for 3 Star review |
| 2.5a | 13 July 2009 | Revision of 3-Star ARSTAF comments |
| 2.5b | 3 Aug 2009 | Revision w/ Army 3-Star G-2,G-4 & G-6 Comment Resolution |
| 3.0 | 23 Oct 2009 | Revision w/ Army 3-Star ATEC critical comment for $A_m$ and G-3 directed changes |
| 3.1 | 26 Feb 2010 | Revision w/ Phase 1 Joint Staffing Comment Resolution |
| 3.2 | 29 Mar 2010 | Revision w/ addition to KPP 1 Rationale and Appendix A Data Service Exposure Sheets |
| 3.3 | 12 Apr 2010 | ARCIC Validation of revised CDD includes addition to KPP 1 Rationale and Appendix A Data Service Exposure Sheets |
| 3.4 | 24 June 2010 | Revision for J6 Certification |

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1. Capability Discussion.

a. Operating Environment of the System. Ground Soldier System Increment I (GSS Inc I) is a sub-component of the Ground Soldier System (GSS) program which is a member of the Soldier as a System (SaaS) family of Soldier capabilities. The SaaS process modernizes and integrates all Soldier capabilities, either ground, mounted or air, to support current and future Joint operations. The GSS will be employed worldwide during peacetime, conflict and war; in both hostile and non-hostile environments; and in a variety of terrain and climatic conditions. The core mission of GSS equipped leader is to close with the enemy by means of fire and movement to destroy or capture him, or repel his assault by fire, close combat, and counterattack. The GSS enables the individual leader to operate in tactical voice and data networks with other GSS equipped leaders. GSS equipped leaders perform missions requiring connectivity to their supporting platforms, but must have survivability and communications independent of Soldier platforms. In the absence of a support platform, an alternate routing of information will automatically be performed. Ground Soldiers in close combat and in complex terrain bear the greatest vulnerabilities of all combatants. Air and ground platforms possess overmatching capabilities against threat systems; however, greater parity exists with the enemy on the ground. Systems and programs for multiple Soldier system components are not integrated to provide optimal overmatching capabilities. The GSS capability gaps, include small unit battle command (BC), Situational Awareness (SA),embedded training (ET), lethality, mobility, force protection, and sustainability. Overview of capability gaps:

(1) BC and SA Gap. Current BC and SA capabilities do not enable mutual support for geographically dispersed small units (inadequate lightweight squad/team communications). Radio communications do not facilitate BC and SA needs by company commanders and platoon leaders during operations in complex urban terrain. Commanders must move from building to building to sustain communications with subordinates. Networked communications do not accommodate dismounted Soldiers in close combat. Squad members continue to rely on low cost unsecured radios or by yelling from building to building and on hand and arm signals when possible. The lack of effective communications places individual Soldiers at higher risk. Current Soldier network enabled BC capabilities are limited and technological solutions do not yet exist to realize full capabilities through either the Network Centric Information Environment (NCIE) or the Joint Battle Command-Platform (JBC-P) NCIE. Soldiers require enhanced tracking and reporting at all levels of command. An enhanced knowledge of the Soldier's mission, a more complete picture of the battlefield, and the ability for rapid exchange of pertinent information across the full spectrum of military operations is also needed. This includes complete Joint Blue Force Tracking, COP, data fusion, and timely flow of information down to the small unit lower tactical level for both mounted and dismounted forces.

(2) Lethality Gap. Current capabilities limit the small unit's Area of Influence (AI). Geographically dispersed small units lack capabilities for immediate mutual support conduct of cooperative engagement and access to external assets. Well aimed fires require exposure to threat lethal effects. Squad and teams require greater precision, speed and cooperative engagement against immediate targets when required in a timely manner to support missions.

(3) Mobility Gap. Excessive weight of collective capabilities while worn limit the ground Soldiers mobility, especially in complex terrain. Excessive weight limits their mobility and causes exhaustion in well conditioned Soldiers during operations in difficult environments.

(4) Force Protection Gap. Improvised Explosive Devices (IEDs) creatively concealed and un-detected along travel routes continue to inflict casualties on friendly forces. Threat personnel continue to move close to Ground Soldiers and use small arms to target areas not covered by ballistic body protection. Current body armor vulnerabilities include coverage areas, bulk, weight and protection levels.

(5) Sustainability Gap. Current sustainment capabilities limit the small unit area of influence for dismounted Soldiers. Requiring increased mission durations and distances results in overburdening Soldiers with additional supplies such as ammunition, water, rations and batteries. Meeting the BCT mission requirements widens this gap. The gap is defined as the Soldier's inability to carry enough equipment and supplies for sustainment of 24 to 72 hours without re-supply. This gap is based on historical data and current doctrine.

(6) Training Gap. Joint level capabilities of the current force are transitioning to strategic and then tactical levels faster than units can adapt. This decentralization trend requires a greater depth of knowledge and breadth in training for each individual Soldier. Expanding mission sets, sophisticated technologies and decreases in available personnel require broader generalist capabilities of the Soldier.

b. System of System and Family of System discussion.

(1) The GSS, linked to the Battle Command System (BCS), provides enablers facilitating the integration of Ground Soldiers with Army BCT Modernization. The enhanced war fighting capabilities of the GSS provides an increased level of connectivity, lethality, and survivability for the future leader. Variations of the basic suite of equipment will meet the MOS specific needs of Soldiers in the force. The BOIP is in development across the entire BCT formation; however, for GSS Inc I, manning includes team leaders and above within the BCT. GSS Inc I will provide an interface to a Rifleman Radio capability which will provide individual Soldiers with voice communications and beaconing of Position Location Information (PLI) to their leaders.

(2) SoS and FOS Related Documents. LW is the evolutionary base for the GSS capabilities. The GSS improves upon LW capabilities, and when combined with the Core Soldier System (CSS) will meet the needs of all Soldiers who conduct close combat in the FF. The CSS CPD establishes standard capabilities for all Soldiers. Each Soldier in the Army will receive Core Soldier capabilities and dependent on duty position and MOS will receive additional integrated, modular components that make up the GSS. The RR CPD establishes a standalone radio to support unclassified real-time intra-squad communications. The RR allows the Soldier to participate in doctrinal voice networks and transmit position location. The Army BCT Modernization Strategy is currently supported by approved FCS Spin-Out I CPD and draft Ground Combat Vehicle ICD and CDD.

(3) Operational and Organizational Description.

(a) Value added to the Warfighter. GSS equipped units will directly enhance the Army's combat power in two Army mission essential tasks. The GSS will enhance small unit combat power and will enable commanders to more effectively combine the elements of combat power (leadership, information, movement and maneuver, intelligence, fires, sustainment, command and control, and protection) and Situational Understanding (SU) to limit friendly casualties and swiftly end tactical engagements. The GSS program integrates CSS capabilities to enable future Soldiers and small units to dominate conventional and asymmetrical threats in close combat in complex terrain and at tactical standoff through improved BC, SA, ET, lethality, mobility, force protection, and sustainability. The GSS will enable NCIE at the leader and small unit level. The GSS will provide leaders improved SA through the COP. BC and SA will enhance troop leading procedures, tactical problem solving, and operational momentum. GSS will provide capabilities to share communications vertically and horizontally, monitor the movements of small unit combatants, and accurately control organic and supporting fires. GSS will help facilitate networked integration of Soldiers with sensors, weapons and platforms. GSS enables small units and leaders to conduct mutually supporting operations while geographically dispersed. GSS will enable small units to engage in a larger geographic area with greater precision, speed and a broader variety of lethal and non-lethal effects. GSS capabilities integrate Unified Battle Command principles by contributing to the TRADOC approved 10 Battle Command Essential Capabilities of: Provide A Robust Network Capability, Execute Tactical Network Operations, Display/Share Relevant Information, A Standard and Shareable Geospatial Foundation, Enable Collaboration, Create and Disseminate Orders, Battle Command on the Move, Execute a Running Estimate, Joint, Interagency, Intergovernmental, Multinational (JIIM) Interoperability and Rehearsal and Training Support.

(b) Joint Operations Concepts (JOpsC), Army Concept Strategy, and Integrated Architectures. GSS equipped leaders will interact with current, JOpsC, Concept of Operations (CONOPS) and integrated architectures. GSS will also interact with Interagency and Multinational forces as part of theater level task forces. GSS enabled organizations require agility and rapid reconfiguration to meet specific needs for an assortment of changing missions. The Maneuver Center of Excellence (MCOE) and Program Manager - Soldier Warrior (PM-SWAR) will express needs as "capabilities" or "desired effects" to allow for the widest range of solutions. Joint land operations include operational maneuver from operational distances to directly attack centers of gravity in order to achieve the joint force commander's desired objectives. Close combat is a fundamental capability for successful joint land operations. Key components of close combat are maneuver and engagement in order to destroy opposing forces, securing key terrain, controlling vital lines of communications, and establishing local or regional military superiority.

c. Supporting JCIDS Documents. Source ICD: The AROC approved the SaaS ICD on 21 Mar 2005. The JROC validated the SaaS ICD on 21 Oct 2005. The SaaS ICD identifies capability gaps and integrated capabilities required by the Soldier. The SaaS process provides a System of Systems modernization approach to resolving Soldier capability gaps. The SaaS ICD also prescribes a methodology to enhance the capabilities of all Soldiers to perform core tasks, functions and missions. SaaS includes everything a Soldier wears, carries or consumes to perform Warrior Tasks, Battle Drills, and individual & Collective Tasks.

d. Associated Tier 1-3 Joint Capability Areas (JCA). The GSS will support the following Tier 1 through Tier 3 JCAs:

**Table 1-1: Associated Tier 1-3 JCA's**

| Tier 1 JCA | Tier 2 JCA | Tier 3 JCA | GSS Capability(s) | GSS Impact |
|---|---|---|---|---|
| Force Application | Maneuver<br><br>Engagement | Maneuver to Engage<br><br>Kinetic Means | The ability to effectively engage personnel with precision fires<br>The ability to accurately exact the application of force | More precise navigation<br><br>Increase ability to maintain momentum |
| Command & Control | Understand<br><br><br><br>Decide | Employ Shared SA<br><br>Develop knowledge<br><br>Manage Risk | Minimize fratricide and collateral damage<br>Detection and location of concealed combatants<br>Avoid potentially hazardous situations | Operate longer and more effectively<br>Address full range of military operations<br>Increase survivability |
| Battlespace Awareness | Intelligence, Surveillance and Reconnaissance (ISR) | Process & Exploitation<br>Analysis & Production | Avoid potentially hazardous situations<br>The ability to increase operational environment awareness | Enhance force protection<br><br><br>Expedite decisions making |
| Net Centric | Information Transport<br><br>Enterprise Services | Information Sharing<br><br>Information Assurance<br><br>Secure Information Exchange | The system must be able to enter and be managed in the network, and exchange data in a secure manner to enhance mission effectiveness. The system must continuously provide survivable, interoperable, secure, and operationally effective information exchanges | Improve SA and BC through the COP<br><br>Facilitate networked integration of Soldiers with sensors, weapons and platforms |

2. Analysis Summary. The GSS program includes numerous studies shaping capabilities development. Training and Doctrine Command (TRADOC) has completed the LW AoA (June 05), the LW Doctrine, Organization, Training, Materiel, Leadership and Education, Personnel, and Facilities (DOTMLPF) Assessment and In-Country Assessment (May 06 through June 08), and the Ground Soldier System AoA (April 08). An additional GSS AoA is planned to support the MS C/B scheduled for FY 11.

a. LW AoA. The LW Analysis of Alternatives included a functional area analysis (FAA), functional needs assessment (FNA), and a functional solution analysis (FSA) determined materiel solutions are required to fill identified small unit capability gaps. The FNA identified capability gaps in 19 tasks executed by Soldiers, leaders and small units. The FSA found a materiel solution was required for 18 of the 19 capability gaps identified by the FNA. During the detailed assessment of the gaps filled by the non-materiel solutions, the DOTMLPF implications of those non-materiel solutions, and the overall argument for a materiel solution for each of the capability gaps it was determined LW could address 12 of 19 capability gaps. The LW AoA also determined the LW Key Performance Parameter (KPP) threshold capabilities (requirements) were analytically supportable. The one task not requiring a materiel solution is "React to man-to-man contact (combatives). Live experimentation demonstrated leaders could perform critical tasks more effectively with LW equipment.

b. LW DOTMLPF Assessment. As part of the Army Force Generation (ARFORGEN) cycle, the Army fielded LW to a Striker Brigade Combat Team (SBCT) at Ft Lewis during the 3rd and 4th quarters FY 06. The DOTMLPF Assessment (Phase II of the LW AoA) began May 2006 with leaders from 4th Battalion, 9th Infantry Regiment (Manchu) going through New Equipment Training (NET) on the LW system. The training was conducted in four phases: 1) Training of Commanders, Staff, and selected personnel at the Brigade, Battalion, and Company levels; 2) Training of remaining company grade officers and noncommissioned officers; 3) Training of remaining leaders; and 4) Unit collective training (squad, platoon and company level exercises). The battalion conducted a Limited User Test (LUT) and prior to deployment conducted a Mission Rehearsal Exercise (MRE) to further refine their skills. Impacts to the DOTMLPF domains are low in all areas except doctrine, organization, training, and leader education.

c. In-Country Assessment. In Mar 2007, the 4th Battalion, 9th Infantry Regiment (Manchu), a subordinate unit of the 4th Brigade, 2nd Infantry Division (SBCT) deployed to Operation Iraqi Freedom (OIF) with a Basis of Issue Plan (BOIP) of team leader and above equipped with the LW ensemble. PM-SWAR deployed the following teams in support of the unit's deployment; Contractor Logistic Support (CLS) Subject Matter Experts (SMEs), Army Test & Evaluation Cell (ATEC) and representatives from the TRADOC Capabilities Manager Soldier (USAIC) to collect data, surveys and support the unit. Simultaneously with the units deployment TRADOC Army Capabilities Integration Center (ARCIC) directed TRADOC Analysis Center-White Sands Missile Range (TRAC-WSMR) to conduct a GSS AoA with the following scenarios; base case Rapid Fielding Initiative (RFI) equipped dismounted leaders, LW team leader and above, GSS team leader & squad leader and above scenarios using Ft. Lewis terrain imputed into Combat XXI simulations.

d. GSS AoA. On 9 Apr 08, TRAC WSMR submitted the GSS AoA Final report to TRADOC ARCIC to inform the GSS program. The report considered the force effectiveness impact to the DOTMLPF domains, and costs associated with the GSS. Two basis of issue equipping options were considered: GSS to the team leader level and GSS to the squad leader level. Analysis results demonstrated GSS improved small unit force effectiveness at both levels. The Networked GSS capabilities improved the force effectiveness of a Stryker infantry platoon by increasing unit Operating Tempo (OPTEMPO) and enabling greater unit lethality. DOTMLPF impact is low except in the areas of doctrine, training, and leadership and education.

3. Concept of Operations Summary.

a. Actions Before Forces are Joined. Through the use of real-time, immediately available intelligence and mission planning data received via the GSS, friendly force leaders will develop the situation prior to making any visual or physical contact with the enemy. Every tactical formation down to the team leader level will have access to relevant timely information on the terrain, obstacles, and the composition, disposition and intentions of relevant enemy and friendly units. Additionally, information and data pertaining to emergency re-supply and sustainment operations are critical to small unit operations.

b. Actions During Contact. GSS SA enables small units to conduct a higher level of precision maneuver based on intelligence data received prior to military operations thus minimizing engagements characterized by uncertainty such as Movement to Contact. Leaders exploit this readily available BC/SA to better synchronize maneuver and supporting fires while maintaining momentum, not having to slow their pace to confirm friendly and enemy locations; especially between subordinate units operating in complex terrain where visual contact cannot be continuously maintained. GSS enabled BC/SA supports greater dispersion of dismounted maneuver elements and enhances synchronization between dismounted and mounted elements at the lowest levels resulting in quicker, more efficient mission completion rates. GSS minimizes the differences between day and night and limited visibility increasing operational tempo by providing leaders and maneuver units added command and control and better visibility during all military operations.

c. Conduct Tactical Assault. GSS equipped small units exploit digital BC and SA capabilities to dynamically maneuver to positions of advantage to initiate decisive contact at their chosen time and place while integrating fire and maneuver. Integrated capabilities enable leaders and small units in the close fight to employ speed, stealth, and deception to avoid detection, protect movement, retain freedom of action, engage enemy forces while en route, and build and maintain momentum. The GSS equipped unit adapts on the move, adjusting routes and objectives based on changes to the situation, fighting the enemy, not the plan.

d. Conduct Counter Ambush and Security Operations. Through the use of immediately available intelligence and mission planning data received via the GSS, friendly force leaders are able to adapt on the move, adjusting routes and objectives based on changes to the current operational situation based upon information received via GSS. If attacked, support Soldiers will seek immediate cover and concealment, and queued by leaders with digital BC tools, place effective small arms fire on enemy combatants, use indirect weapons viewing and aiming capabilities, making these Soldiers more survivable to perform their primary missions.

4. Threat Summary.

a. Threat to be Countered or Targeted. GSS Inc I is not intended to target a specific threat but support Soldiers while performing their missions and tasks. GSS Inc I applies only to network and communications equipment IAW this CDD.

b. Projected Threat Environment. Adversaries will be difficult to template as they employ adaptive strategies, conventional and unconventional operations to counter US military advantages and technological overmatch. Opponents will attempt to make use of military systemology - the study and exploitation of systems, links and nodes as they apply to military operations - employing multi-dimensional, simultaneous and sequential actions across a wide spectrum of operations to destroy or damage discrete capabilities that will cause the greatest degradation of US forces. They will be aware of the importance of GSS Inc I as a critical asset for US military capability and may take action to degrade or destroy associated elements. They will blur the lines between combatant and non-combatant, often blending in with civilian populations using them as hostages to protect vital facilities and the use of sanctuary structures such as (religious, schools and hospitals) for military purposes. They will employ selective precision strikes and conduct rapid maneuver after setting the conditions for success. Because of the close proximity of forces, tempo for engagements will be important in that adversaries will want to establish a rapid tempo in order to achieve objectives and set conditions for entry denial operations before US forces can establish a foothold in the region. Their operations will be manpower intensive; but possess greater opportunity for surprise. Finally, some adversaries may have the capability to use Chemical, Biological, Radiological, Nuclear, and high yield Explosives (CBRNE) weapons and materiel against US forces. Understanding how US forces operate, adversaries will focus on preservation of their conventional forces and capabilities until the opportunity can be created for their employment.

(1) On the future battlefield, opponents will attempt to counter US strengths by attacking or exploiting perceived weakness, especially the US requirement for points of access into theater and its dependence on C4 and, Intelligence, Surveillance and Reconnaissance (ISR) assets that are vital to the US SoS approach to warfare. Opponents will seek to cloud situational awareness through Information Operations (IO), including Computer Network Operations (CNO), Electronic Warfare (EW), Psychological Operations (PSYOPS), deception, and destruction of US command and control to degrade US strengths. The result is a future battlefield of unprecedented complexity, fluidity, and lethality.

(2) Potential adversaries in this environment will be difficult to template as they will use adaptive responses to counter US conventional military advantages and technological overmatch. They will develop patterns of operation that will change as they achieve success or experience failure in engagements. Their goals and doctrine won't change, but their tactics and methods of operations will. In other words, when facing a regional peer they will operate in a traditional manner but will adopt a more radical approach when fighting US forces. They will conduct decentralized, dispersed, or distributed operations in an attempt to throw US units off balance. Adversaries will field a mixed force comprising of conventional / unconventional forces, paramilitary, and elite special units. It is critical to note that even as they develop and use adaptive means, some potential adversaries will retain and attempt to improve existing force capabilities. Modernization and hybridization of existing conventional weapon systems will increase their effectiveness. When coupled with new adaptive systems and methods, these residual and improved forces will pose a truly significant threat to future US forces.

c. Targets to be Countered or Targeted. The threats to countered will be enemy use of Electronic Warfare (EW), Computer Network Attack (CNO) and physical threats.

d. Threats to the System. The most significant threat to the GSS Inc I will be those threats used to disrupt communications links that can directly degrade the overall operations of the network.

e. References. Detailed threats are described in the System Threat Assessment Report (STAR) for the FCS BCT dated 27 February 2007; the DIA validated SaaS STAR dated 21 January 2010; DIA validated threat reference "Information Operations (IO) Capstone Threat Assessment, 6th edition, DI-1577-37-07, April 2007; and the ONI-TA-076-09, System Threat Assessment Report (STAR) for the Joint Tactical Radio System (JTRS) 15 May 2009.

5. Program Summary.

a. General. The GSS program will initially furnish an integrated wearable GSS Increment I leader solution and will provide an interface to a Rifleman Radio capability which will provide individual Soldiers with voice communications and beaconing of PLI to their leaders without interfering with the highly physical nature of small unit operations. In so doing, GSS will fully leverage GFE (Government Furnished Equipment) communications (including RR capability) and BC software applications. GSS is developed and fielded in increments, each with a basic operational focus further enabling leaders and their Soldiers.

b. GSS Increment I. The objective of GSS Inc I is to establish digital communications connectivity among dismounted team leaders and above within the BCT, using GFE radios and Battle Command software common to other users and able to fit in current and future Army Networks. Additionally, for team, squad and platoon leadership, GSS Inc I will provide these key leaders secure voice communications and an interface to a Rifleman Radio capability which will provide individual Soldiers with voice communications and beaconing of PLI to their leaders.

(1) The primary operational purpose of GSS Inc I is to provide dismounted leaders the ability to:

(a) Send and receive with other GSS Inc I leaders the location of friendlies.

(b) Send and receive with other GSS Inc I leaders reports on enemy and civilian status and locations within the leader's area of influence.

(c) Send and receive with other GSS Inc I leaders Battle Command information, including orders and graphical control measures.

(d) Visualize the leader's area of influence by displaying the above.

(e) Securely communicate in order to collaborate and rehearse the execution of plans and orders.

(2) PM-Ground Soldier (PM-GS) will continue integrating the RR Capability described in KSA 7 until the capability is available and ready. When ready, PM-GS will test the RR capability as part of a separate LUT excursion. RR capability will be fully tested during Initial Operational Test and Evaluation (IOT&E) prior to Full Rate Production (FRP).

(3) During Increment I, selected GSS leaders will carry the RR capability in addition to the GSS GFE radio to conduct voice communication to and from Soldiers. PLI received by the Leader's RR will be directly fed into the GSS by a one way guard to enable the PLI to appear on the leaders GSS display.

(4) As part of a P3I during Increment I, GSS will transition to the JTRS Small Form Factor (SFF) B utilizing the SRW. The decision to incorporate this new radio is dependent upon its technical maturity.

(5) There are MOS specific Soldiers who will receive additional attributes based on mission specific tasks, i.e.; Soldiers performing Explosive Ordinance Disposal (EOD) requiring additional blast protection above what is provided by the Core or GSS. Additional Performance Attributes for GSS Increments are provided by programs that are GFE to GSS. Once the specific attributes are codified, supported by defensible operationally-based evidence and deemed achievable by the responsible acquisition authorities, they will be documented in a CPD for that increment, as authorized in CJCSI 3170.01G, Enclosure A, page A-4, (item g.) dated 1 March 09.

c. GSS Increment II. The primary focus of GSS Increment II is to strengthen digital communications connectivity among dismounted Leaders, while further reducing the burden on both these leaders and the sustainment tail. Increment II will strive to reduce the overall weight of the GSS (system and associated batteries), while maintaining the needed mission profile established for Increment I. GSS Increment II will provide significant improvements in Soldier capabilities by providing increased lethal and non-lethal network fires, embedded training, multi-spectrum images, language translation, hands-free voice control, full color visor display, enhanced power for 48 hrs and Army BCT Modernized Battle Command interoperability as outlined in Appendix D. If Increment II is consistent with the description described in Appendix D, only the changes to the capabilities described in paragraph 5 for Increment II will be provided in an addendum to the initial approved Increment I CDD and provided for validation and approval, as appropriate. If Increment II contains significant revisions to the overall strategy, the capabilities provided by the next or future increments, or changes to the KPPs, will be appropriately revised in an Increment II CDD in accordance with CJCSI 3170.01G. In addition GSS Increment II will provide for MOS specific Soldier Tier I attributes and threshold capabilities also as outlined in Appendix D - GSS Increment II.

d. GSS Increment III. The GSS Increment III builds upon the GSS Increment II capabilities and addresses future operational needs associated with GSS CDD objective attributes with dismounted Soldiers within the BCT. This GSS increment addresses additional capabilities required to achieve full GSS CDD compliance (threshold and objective) and will address gaps provided by further analyses, operational assessments and determined threat.

6. System Capabilities Required for the Current Increment

a. Statutory and Compliance KPPs not appropriate for GSS.

(1) Survivability. The Survivability KPP does not apply because GSS is not a manned platform (as in tank, plane, ship, HMMWV, etc). The GSS is a system of capabilities the Soldier will "wear", just as body armor is not a "manned system." The GSS does not include attributes such as speed, maneuverability, detectability, and countermeasures that reduce a system's likelihood of being engaged by hostile fire, nor attributes such as armor and redundancy or

critical components that reduce the system's vulnerability if it is hit by hostile fire. This does not eliminate the requirement to live fire test the GSS.

(2) Force Protection. Force Protection is not applicable for the GSS Inc I. The GSS Inc I does not contribute to the protection of personnel by preventing or mitigating hostile actions against friendly personnel, military and civilian. The GSS Inc I is provided force protection through the CSS at no less than the current level of protection fielded at the time of GSS Inc I production. The GSS Inc I and RR must be physically compatible with the CSS force protection materiel and not require any modifications affecting the integrity of the system.

(3) Energy Efficiency. Energy Efficiency is not applicable for the GSS Inc I. The GSS is a not a platform system or platform sub-system that obtains power either directly or indirectly from hydrocarbon energy sources. GSS Inc I is powered by a rechargeable battery verses a variety of energy consuming disposable batteries.

(4) System Training. System Training is not applicable for the GSS Inc I. GSS Inc I Training and Program of instruction are provided by the materiel developer. Embedded training capabilities are addressed in the GSS Increment II CDD. Training is not a significant part of the total Life Cycle Costs. GSS Inc I does not require a "stand alone" training device as part of the program.

b. GSS Inc I KPPs**.** Table 6-1 contains the GSS Inc I KPPs with Threshold (T) and Objective (O) values. All leaders receiving the GSS Inc I will have the following KPPs inherent in their individual system as described in Table 6-1 below. Additionally, there are MOS specific Soldiers who will receive additional "attributes" based on mission specific tasks, i.e.; Engineers performing Explosive Ordinance Disposal (EOD) must have additional blast protection above what is provided by the Core or GSS.

**Table 6-1 GSS KPPs Increment I**

| JCA Tier 1/2 | Key Performance Parameter | Development Threshold | Development Objective |
|---|---|---|---|
| Net-Centric (Tier 1)<br>Enterprise Services (Tier 2)<br>Information Assurance (Tier 2)<br>Information Transport (Tier 2)<br>NET Management (Tier 2) | KPP 1: Net Ready: The system must support Net Centric military operations. The system must be able to enter and be managed in the network, and exchange data in a secure manner to enhance mission effectiveness. The system must continuously provide survivable, interoperable, secure, and operationally effective | The capability, system, and/or service must fully support execution of joint critical operational activities and information exchanges identified in the DOD Enterprise Architecture and solution architectures based on integrated DODAF content, and must satisfy the technical requirements for transition to Net-Centric military operations to include:<br>1) Solution architecture products compliant with DOD Enterprise Architecture based on integrated DODAF content, including specified operationally effective information exchanges<br>2) Compliant with Net -Centric Data Strategy and Net-Centric Services Strategy, and the principles and rules | The capability, system, and/or service must fully support execution of all operational activities and information exchanges identified in DOD Enterprise Architecture and solution architectures based on integrated DODAF content, and must satisfy the technical requirements for transition to Net-Centric military operations to include:<br>1) Solution architecture products compliant with DOD Enterprise Architecture based on integrated DODAF content, including specified operationally effective information exchanges<br>2) Compliant with Net –Centric Data Strategy and Net-Centric Services Strategy, and the principles and rules identified in the DOD IEA, excepting |

| JCA Tier 1/2 | Key Performance Parameter | Development Threshold | Development Objective |
|---|---|---|---|
| | information exchanges to enable a Net-Centric military capability. | identified in the DOD Information Enterprise Architecture (DOD IEA), excepting tactical and non-IP communications<br>3) Compliant with GIG Technical Guidance to include IT Standards identified in the TV-1 and implementation guidance of GIG Enterprise Service Profiles (GESPs) necessary to meet all operational requirements specified in the DOD Enterprise Architecture and solution architecture views<br>4) Information assurance requirements including availability, integrity, authentication, confidentiality, and non-repudiation, and issuance of an Interim Authorization to Operate (IATO) or Authorization To Operate (ATO) by the Designated Accrediting Authority (DAA), and<br>5) Supportability requirements to include SAASM, Spectrum and JTRS requirements. | tactical and non-IP communications<br>3) Compliant with GIG Technical Guidance to include IT Standards identified in the TV-1 and implementation guidance of GESPs, necessary to meet all operational requirements specified in the DOD Enterprise Architecture and solution architecture views<br>4) Information assurance requirements including availability, integrity, authentication, confidentiality, and non-repudiation, and issuance of an ATO by the DAA, and<br>5) Supportability requirements to include SAASM, Spectrum and JTRS requirements. |
| Command & Control (Tier 1)<br>Understand (Tier 2)<br>Decide (Tier 2) | KPP 2. BC | Provide voice and data exchange among networked GSS Inc I Leaders, tactically distributed within the leader's Area of Operation (AO) (T) using standard communications devices and battle command applications. | and Area of Interest (AI) (O) |
| Battlespace Awareness (Tier 1)<br>ISR (Tier2) | KPP 3. COP: Shared Friendly Situational Awareness | Must display 90% (T) of network populated PLI, tactically distributed within the leader's AO and AI within 5 seconds (T) of the entity being reported to the network and received by the GSS Inc I, as defined in KPP 2. | and 100%<br>and 3 seconds |
| Battlespace Awareness (Tier 1)<br>ISR (Tier2) | KPP 4. COP: Shared Survivability Information (Enemy, Neutral, Non-Combatant, Unknown). | The GSS Inc I must display 90% (T) of shared survivability information reported to the network within the small unit leaders danger zone (AO and AI); within 5 Seconds (T) of the entity being reported to the network and received by the GSS Inc I, as defined in KPP 2. | and 100%<br>and 3 Seconds |
| Logistics (Tier 1)<br>Maintain (Tier 2) | KPP 5. Sustainment (Availability) | The GSS Inc I shall have an Operational Availability > 90% (T) and a Materiel Availability of .> 72% | Operational Availability > 95% and Materiel Availability of (T=O). |

(1) KPP 1 Net-Ready. The capability, system, and/or service must fully support execution of joint (T) and all (O)critical operational activities and information exchanges identified in the DOD Enterprise Architecture and solution architectures based on integrated

DODAF content, and must satisfy the technical requirements for transition to Net-Centric military operations to include: 1) Solution architecture products compliant with DOD Enterprise Architecture based on integrated DODAF content, including specified operationally effective information exchanges 2) Compliant with Net -Centric Data Strategy and Net-Centric Services Strategy, and the principles and rules identified in the DOD Information Enterprise Architecture (DOD IEA), excepting tactical and non-IP communications 3) Compliant with GIG Technical Guidance to include IT Standards identified in the TV-1 and implementation guidance of GIG Enterprise Service Profiles (GESPs) necessary to meet all operational requirements specified in the DOD Enterprise Architecture and solution architecture views 4) Information assurance requirements including availability, integrity, authentication, confidentiality, and non-repudiation, and issuance of an Interim Authorization to Operate (IATO) (T) or Authorization To Operate (ATO) (T=O) by the Designated Accrediting Authority (DAA), and 5) Supportability requirements to include SAASM, Spectrum and JTRS requirements.

***Rationale:*** *The GSS Inc I will operate as part of a Joint team across the full spectrum of military operations and, supporting operations vertically and horizontally, from alert through deployment and employment. Currently, the IBCT lacks a digital transport infrastructure to support GSS Inc I external interoperability with other IBCT digital network assets. This lack of infrastructure will prevent digital and voice GSS Inc I transmission over extended ranges and between GSS Inc I dismounted leaders and FBCB2/BFT equipped platforms. This capability gap will persist until the IBCT's digital transport infrastructure is sufficiently robust. GSS Inc I provides small unit SA/C2 functionality that fills a critical capability gap, so it must be delivered as soon as possible. The MCoE has documented a synchronized, time phased requirement for achieving GSS Inc I external interoperability with other IBCT digital network assets.*

*Phase 1: Required at GSS Inc I IOT&E test event and through fielding of the first four BCTs: GSS Inc I will exchange friendly and shared survivability situational awareness information to the level defined in KPPs 3 and 4 with other GSS Inc I systems in its area of operations/area of interest as defined in KPP2. GSS Inc I will also be capable of exchanging C2 messages (orders, overlays, SPOT reports, etc) with other GSS Inc I systems as defined in this CDD. Additionally GSS Inc I will be capable of interoperating with the RR capability as defined in the KSA 7 when it becomes available. The MCoE is willing to accept GSS Inc I fielding without the RR capability if it is not available.*

*Phase 2: Required by BCTs 5 through 30: Building upon Phase 1, GSS Inc I will exchange friendly and shared survivability situational awareness information to the level defined in KPP 3 and 4 with existing fielded Army Battle Command Systems defined in Capability Sets 13/14 and 15/16 in its area of operations/area of interest as defined in KPP2. GSS Inc I will also be capable of exchanging C2 messages (orders, overlays, SPOT reports, etc) with fielded Army Battle Command Systems as defined in this CDD.*

(2) KPP 2 BC. The GSS Inc I must provide voice and data exchange among networked GSS Inc I Leaders, tactically distributed within the leader's Area of Operation (AO) (T) and Area of Interest (AI) (O) using standard communications devices and battle command applications. This GSS requirement contributes to the TRADOC approved Battle Command Essential Capability to Execute Tactical Network Operations and Enable Collaboration.

***Rationale:*** *GSS Inc I is the leader worn system that provides situational awareness in the small unit tactical fight. GSS Inc I requires a communications system to enable small units the ability* *to quickly and accurately communicate by both voice and data with leaders and subordinates in rapidly changing operational environments. GSS Inc I will enable appropriate and timely tactical actions, focus of organic fires, and request of joint supporting fires while minimizing the potential for fratricide. The distance covered by the maximum effective range of the squad's direct fire weapons is 800 meters. The AO and AI will contract from maximum depending upon the leader's echelon and is defined by a radius as follows: for the squad the AO is 1 km and the AI is 2.5 km; Platoon AO radius is 3 km and AI is 7.5 km; Company AO radius is 9 km and AI is 22.5 km. The squad AO is determined by adding to the range an approximate 20% overmatching effective range (as determined by the Small Arms CBA) from expected enemy direct fire weapons. Platoon (3 km) and Company (9 km) AO's are based on the maximum separation of 3 squad or 3 platoon AOs (3 on-line worse case) that allow for organic direct fire, organic indirect fire, and supporting fires. AI is the leader's area of concern that may extend into enemy held territory and to the objectives of current or planned operations. The squad's AI (2.5 km) is determined by the maximum range of organic anti-tank weapons and mortars. Platoon (7.5 km) and Company (22.5 km) AI's are based on the maximum separation of 3 squad or 3 platoon AIs (3 on-line worse case) that allow for organic direct fire, organic indirect fire and supporting fires. The AO and AI will expand or contract depending on the echelon and operational environment (urban, jungle) the unit is in. The small unit will normally occupy less terrain in urban areas. The density of buildings and rubble and street patterns will dictate the unit frontage. This assumes that the squad/platoon/company are kept within tactical contact and METT-TC is the over-riding determination of AO and AI distances. The network utilized by GSS Inc I can be comprised of advantaged nodes (non-Soldier worn platform equipment) and disadvantaged nodes (Soldier worn equipment). GSS Inc I unique data exchange messages include Free Text, Call for Medic and Tactical Chem Light. All three unique messages are internal to the GSS Network, however; a Call for Medic with User intervention could generate a Call for MEDEVAC message.*

(3) <u>KPP 3 COP: Shared Friendly Situational Awareness:</u> The GSS Inc I must display 90% (T) and 100% (O) of network populated PLI, tactically distributed within the leader's AO and AI within 5 seconds (T) and 3 seconds (O) of the entity being reported to the network and received by the GSS Inc I; as defined in KPP 2, using standard communications devices and battle command applications. This GSS Inc I requirement contributes to the TRADOC approved Battle Command Essential Capability to Display/Share Relevant Information.

***Rationale:*** *GSS Inc I is a leader worn system that provides friendly situational awareness in the small unit tactical fight. This knowledge enables commanders and leaders to execute their assigned mission, meet their higher commander's intent, determine if subordinates are executing that intent, and make real time adjustments to the battle plan, if required. For example, at company and lower level, the performance parameter of displaying 90% of the GSS Inc I equipped leaders AO is based on the maneuver unit's requirement to fight two levels down and is the minimum acceptable standard. This parameter comes from the validated Land Warrior ORD/CPD and it permits the company commander to track his elements by squad or platoon. For most missions, the GSS Inc I display of 90% of operational systems (supplemented by voice radio nets and visual observation of the battlefield) enables the company commander to track his platoons to control their maneuver and distribution of fires. As the company maneuvers to attack, the company commander will control the movement of his platoons to maintain their* *mutual support. As the company goes into the assault, the commander may synchronize some platoons' assault with other platoons' establishment of support by fire positions. At the platoon level, 90% of friendly SA provides sufficient information for the platoon leader to enable successful execution of their primary mission in the close fight by tracking to the fire team level – maneuvering to a position of advantage (even during limited visibility), killing the enemy and protecting their force. At the squad level, the squad leader is able to track the positions of his two team leaders during the fight with a high level of confidence. Information displayed within 5 seconds (or near real time) improves combat effectiveness by preventing delays in command and control, enabling leaders to rapidly shape the fight, and assists in the prevention of fratricide by disseminating timely and accurate Friendly SA of all GSS Inc I equipped forces operating in the AO.*

(4) KPP 4 COP: Shared Survivability Information (Enemy, Neutral, Non-Combatant, Unknown). The GSS Inc I must display 90% (T), 100% (O) of shared survivability information reported to the network within the small unit leaders danger zone (AO and AI); within 5 Seconds (T) and 3 Seconds (O) of the entity being reported to the network and received by the GSS Inc I, as defined in KPP 2. This GSS requirement contributes to the TRADOC approved Battle Command Essential Capability to Display/Share Relevant Information and Execute a Running Estimate.

***Rationale:*** *The GSS Inc I must enable the visualization and dissemination of essential information for display on the COP. SA is the immediate knowledge of the conditions of the operation, constrained geographically and in time. COP is a single display of relevant information within a commander's AI tailored to the user's requirements and based on common data and information shared by more than one command. Survivability Information is all relevant enemy and neutral information reported to the network, this includes enemy Soldiers, ground vehicles, mortars, artillery, aircraft, antiaircraft, bridges, minefields and CBRN hazards. A minimum of 90% of the threat displayed on the GSS Inc I should allow sufficient local awareness so that those who are aware of the threat can notify other Soldiers in the vicinity. Timely and accurate knowledge of enemy locations and observed survivability information at operationally relevant distances is critical at all echelons of command for executing any form of maneuver in conjunction with direct and indirect fires. This knowledge enables commanders and leaders to execute their assigned mission and make real time adjustments to the concept of operations, if required.*

(5) KPP 5 Sustainment (Availability). Availability will consist of two components: Operational Availability ($A_o$) and Materiel Availability ($A_m$). The components provide availability percentages from an operational unit level, and a corporate, fleet-wide perspective respectively.

(a) The $A_o$ of the GSS Inc I in the field/operational environment, is no less than 90 % (T) 95 % (O) during the 8-day wartime usage period (consisting of two aggregate 96-hour wartime scenarios) depicted in the Operational Mode Summary/Mission Profile (OMS/MP). The assessment of $A_o$ includes both GSS Inc I Contractor Furnished Equipment (CFE) + GSS GFE combined and shall be consistent with equipment usage factors described in the OMS/MP, the repair cycle Administrative and Logistics Delay Time (ALDT) of 18.5 hours per failure projected for the field level support concept of fix-forward repair through replacement of GSS Inc I Line Replaceable Units (LRUs) at operator level, where Essential Function Failures (EFFs) as described by the GSS Reliability Failure Definition and Scoring Criteria (FDSC) are the primary indicator of equipment not ready/available conditions.

***Rationale:*** *The threshold $A_o$ requirement provides GSS Inc I equipped leaders with the requisite high level of effectiveness when assessed on the basis of 148.7 GSS Operating Hours (OHs) for the 8-day wartime usage cycle, 4,462 OHs hours per year (wartime), the stated ALDT, and the GSS Inc I attribute requirement for the maximum time to repair. Objective requirement is consistent with (reflects) the $A_o$ required) for Army BCT Modernized units. $A_o$ requirement for the high intensity pulse operating period (8-days) involving execution of the two aggregate 96-hour combat scenarios conducted back-to-back provides assurance that 90 % of the GSS Inc I are in operational status (i.e., mission capable) throughout, wherein maintenance actions for all failures not resulting in an EFF are deferred until the operating pulse has been completed.*

(b) The $A_m$ of the GSS Inc I in the field/operational environment shall be no less than 72% (T = O). The assessment of A(m) consists of the % of the total inventory of a system operationally capable (ready for tasking) of performing an assigned mission at a given time, based on materiel condition. This can be expressed mathematically as the number of operational end items divided by the total population and is measured over the total lifecycle of the entire fleet, including both peacetime and wartime operations. The A(m) addresses the total population of end items planned for operational use, including those temporarily in a non-operational status once placed into service (such as for depot-level maintenance). The total life-cycle timeframe, from placement into operational service through the planned end of service life, must be included.

***Rationale:*** *As a metric, Materiel Availability provides a measure of system performance, taking into account that all of the major sustainment elements are considered, planned for, and measured. The A(m) supports the management of the acquisition and materiel readiness processes supporting the required Operational Availability. For A(m) the entire population of the system must be accounted for, as does all time during the planned service life (using the ARFORGEN model this includes the Reset & Training Pool, Ready Pool, and Available Pool). Nonoperational units and time are included to provide a complete picture of the total investment and sustainment required across the entire program life cycle. For GSS Inc I the A(m) calculation, time starts when the system has been accepted by the unit (after NET and property transfer). A materially down system is not immediately available for tasking caused by failures, combat damage, maintenance delays, Reset, and packed for transportation time. There are no known improvements in any of these areas that will permit an increase of A(m) beyond the threshold value. Failure to achieve the A(m) value will decrease the number of operationally available systems causing undue risk for the combatant commander as he propagates the war fight. This increased risk will directly translate to an unacceptable increase in loss of life or equipment and may result in the inability of the Soldiers to achieve their assigned tasks or missions.*

c. <u>Key System Attributes (KSA)</u>. The following table is a summary of the GSS Inc I KSAs.

**Table 6-2 GSS KSAs Increment I**

| JCA Tier 1/2 | Key System Attribute | Development Threshold | Development Objective |
|---|---|---|---|
| Logistics (Tier 1) Maintain (Tier 2) | KSA 1. Sustainability: (Materiel Reliability) | The GSS Inc I (GSS CFE + GSS GFE combined) must provide operational reliability of not less than a 85% (T) of operating for 24 hours without incurring an EFF. The GSS Inc I CFE will not degrade reliability of GFE . | 92% (O) (T=O). |
| Corporate Management and Support (Tier1) Acquisition (Tier 2) | KSA 2. Ownership Cost | Ownership Costs for the GSS Inc I is $2,161 (BY2009, $M) (T). GSS Inc I maintenance procedures will minimize the need for special tools within the supporting unit organization and require no special tools for operator level troubleshooting and corrective action procedures commensurate with the envisioned fix-forward maintenance support concept (T). | $1,964 (BY2009, $M) (O). GSS Inc I will not require special tools for maintenance through the field Maintenance level (O) |
| Logistics (Tier 1) Maintain (Tier 2) | KSA 3. Sustainability: Power | Rechargeable, detachable power source and integrated power management solution to achieve an operational system runtime of 24 Mission Hours (T). The system will have recharging compatible with tactical vehicles and a stand-alone recharging capability. | 48 mission hours (O) (T=O) |
| NET-Centric (Tier 1) Enterprise Services (Tier 2) | KSA 4. Geospatial Data Exchange (Geospatial Intelligence (GEOINT)) | The GSS Inc I will provide standardized mapping data and geospatial tools and will receive (load) and use standard NGA and commercial geospatial data. The GSS Inc I will build to the approved geospatial information architecture so as to enable the net-centric, interoperable, geospatial enterprise supporting BC with a unified COP (T=O). The GSS Inc I will load via a direct connection (T=O). Using standard NGA and commercial geospatial data, Mapping data, initial Operation Orders, overlays and imagery shall be loaded within 30 minutes per system; Fragmentary Orders within 10 minutes. | (T=O) |
| Force Application (Tier 1) Maneuver (Tier 2) | KSA 5. Mobility | GSS Inc I leader solution will not weigh more than 14.0 lbs. (T) | 10 lbs (O) |
| Battlespace Awareness (Tier 1) ISR (Tier 2) | KSA 6. BC: Transmission Range | The GSS Inc I equipped leader must communicate via voice and data to ranges of 1.8 km in complex urban conditions (NLOS) and 1 km in dense foliage (T). | 5 km in all environments (O). |
| Command & Control (Tier 1) Understand (Tier 2) Decide (Tier 2) Battlespace Awareness (Tier 1) ISR (Tier2) | KSA 7. Rifleman Radio Interoperability | The GSS Inc I must be interoperable with the Rifleman Radio capability defined in its CPD. | T=O |

(1) <u>KSA 1: Sustainability (Materiel Reliability)</u>. The GSS Inc I (GSS CFE + GSS GFE combined) must provide operational reliability of not less than a 85% (T) and 92% (O) of

operating for 24 hours without incurring an EFF, as defined in the Reliability FDSC. The GSS Inc I CFE will not degrade reliability of GFE (T=O).

***Rationale:*** *RAM analysis based on the OMS/MP, consideration of the Land Warrior – Stryker Interoperable (LW-SI) System support concept implemented in OIF and associated ALDT and repair time incurred per failure (factoring in GSS Inc I logistical considerations/implications relative to units other than the SBCT), as well as input from PM – Soldier Warrior representatives regarding the support concept currently envisioned for GSS Inc I (field level fix-forward repair through replacement of GSS LRUs at operator level, LRU repair at sustainment level (depot/vendor)) indicates the stated reliability requirements will enable the GSS Inc I to achieve both steady state and 8-day pulse operational availability of not less than the 90 % threshold and 95 % objective (when assessed in conjunction with factors identified in the Materiel Availability KPP requirement and rationale paragraph). Requirements are defined operationally (as percentage values) in accordance with current TRADOC guidance and have been expressed in relation to 24 hours of operation to maintain consistency (a common basis) with KSA 3 – Sustainability (Power); GSS Inc I threshold requirement correlates to 146 OHs Mean Time Between Essential Function Failure (MTBEFF) and the objective requirement correlates to 291 OHs MTBEFF, based on expectation of the GSS Inc I having a rate of failure which does not increase at the system level over its normal useful service life). Analysis of $A_m$ based on simulation of fleet-wide GSS Inc I operations over a 15 year life cycle consistent with the Army Force Generation Model (encompassing peacetime training, wartime usage, and periods of equipment reset) indicates the 146 OH MTBEFF provided by the threshold reliability requirement adequately supports achievement of the required $A_m$. Sensitivity analyses conducted over a broad range of reliability (MTBEFF) values and replacement/spare parts delay time factors indicates GSS Inc I $A_m$ will vary from 81% to 83% with reliability equal to the threshold requirement (depending on the extent of replacement parts immediately available for the repair of failures), thereby permitting optimization of $A_m$ through tradeoffs in acquisition and support costs.*

(2) KSA 2: Ownership Cost. Ownership Cost provides balance to the sustainment solution by ensuring that the operations and support (O&S) costs associated with materiel readiness are considered in making decisions. For consistency and to capitalize on existing efforts in this area, the Cost Analysis Improvement Group O&S Cost Estimating Structure (now OSD Cost Assessment and Program Evaluation (CAPE)) will be used in support of this KSA. Only the following cost elements are required:

- 2.0 Unit Operations (2.1.1 (only) Energy (Fuel, Petroleum, Oil and Lubricants [POL], Electricity)): GSS Inc I does not have any costs associated with this cost element.
- 3.0 Maintenance (All) : The GSS Inc I costs associated with labor (outside of the scope of unit-level) and materials at all levels of maintenance in support of the primary system, simulators, training devices and associated support equipment are $1,857 (BY2009, $M) (T), $1,688 (BY2009, $M) (O).
- 4.0 Sustaining Support: The GSS Inc I costs under Sustaining Support are directly allocated to cost element 4.4 Sustaining Engineering and Program Management. These costs are those incurred in providing continued systems engineering and program management oversight to manage the program. They are $287 (BY2009, $M)(T), $261 (BY2009, $M)(O).
- 5.0 Continuing System Improvements (All): The GSS Inc I costs under Continuing System Improvements are directly allocated to cost element 5.2 Software Maintenance and

Modifications, which consist of the labor incurred after deployment in supporting the update, maintenance, integration and configuration management of software. They are $17 (BY2009, $M) (T), $15 (BY2009, $M) (O).

**Table 6-2.2 GSS Inc I Ownership Costs (BY2009, $M)**

| Objective | Threshold |
|---|---|
| $1,964 | $2,161 |

All GSS Inc I ownership costs reflect the planned 15 year operational lifecycle timeframe which is also consistent with the timeframe used in the Materiel Availability KPP. GSS Inc I maintenance procedures will minimize the need for special tools within the supporting unit organization and require no special tools for operator level troubleshooting and corrective action procedures commensurate with the envisioned fix-forward maintenance support concept (T). GSS Inc I will not require special tools for maintenance through the field Maintenance level (O).

***Rationale**: GSS Inc I uses the Automated Cost Estimating Integrated Tools (ACE-IT) for all of its cost analysis efforts in accordance with Army guidance. The GSS Inc I cost model that supports the ownership costs above is the same that was part of the GSS Inc I Milestone A Certification and was deemed reasonable by Deputy Assistant Secretary of the Army for Cost Estimation and reviewed ("Quick Look") by OSD CAPE during that process. All GSS Inc I cost analysis efforts follow the Army's Cost Analysis Manual which includes the cost analysis process, cost estimating techniques and methods. The cost model continues to be refined as new data becomes available.*

(3) KSA 3: Sustainability (Power). Provide a rechargeable, detachable power source and integrated power management solution to achieve an operational system runtime of 24 Mission Hours (T) and 48 Mission Hours (O). The system will have recharging, compatible with tactical vehicles and a stand-alone recharging capability (T=O).

***Rationale:** Sustainable and quickly restorable power supports the required momentum of small unit operations. "Dismounted forces must be self-sustaining during continuous operations for at least 24 Hours." BCT O&O, 6.12.10. Units properly trained and ready for combat operations can sustain themselves until a unit re-supply is conducted. An enhanced power system will reduce the logistical burden of the GSS Inc I by reducing the weight, variety, and number of power sources carried by the Soldier and the logistical structure. Dismounted Soldiers, light infantry, airborne, air assault, Special Operations Soldiers require additional capabilities above CSS items to increase their ability to sustain combat operations and survive on today's modern battlefield.*

(4) KSA 4: Geospatial Data Exchange (Geospatial Intelligence (GEOINT)). The GSS Inc I will provide standardized mapping data and geospatial tools and will receive (load) and use standard NGA and commercial geospatial data. The GSS Inc I will build to the approved geospatial information architecture, to include the logical data models, formats and standards; so as to enable and extend the net-centric, interoperable, geospatial enterprise supporting BC with a unified COP (T=O). The characteristics of a net-centric geospatial enterprise system shall include: use of National Geospatial Intelligence Agency (NGA) and leading commercial sources of geospatial data and products; establishment and maintenance of distributed databases and

architectures based on a common core of selected geospatial software, standards, formats, management protocols, and algorithms; operational execution of geospatial information by using units, or forces via collection, generation, storage, management, fusion, and dissemination vertically and horizontally, peer to peer, and National to Soldier level and back (T=O). The GSS Inc I will load all mission support data via a direct connection (T=O). Using standard NGA and commercial geospatial data, Mapping data for the entire area of operations, initial Operation Orders, overlays and imagery must be loaded during the initial mission planning phase within 30 minutes per system; Fragmentary Orders containing new overlays and imagery must be loaded within 10 minutes. (T=O). This GSS Inc I requirement contributes to the TRADOC approved Battle Command Essential Capability to provide A Standard and Shareable Geospatial Foundation.

***Rationale:*** *The Army Geospatial Enterprise is directed by HQDA to ensure interoperability of current and emerging geospatial solutions. This applies to all systems that use, store, and/or disseminate geospatial information. GSS Inc I will use geospatial information from the NGA and appropriate commercial sources. The GSS Inc I receives data exchange (SA, orders, and overlay) information from unit Tactical Operations Centers and platforms. The length of time required to load data will not hinder or delay the mission planning process. This data exchange facilitates intelligence supportability for all military operations, i.e. for use at check points and within cities, villages and towns.*

(5) KSA 5: Mobility (Weight). The GSS Inc I Leader solution will not weigh more than 14.0 lbs (T) and 10 lbs (O).

***Rationale:*** *Excluding the RR capability and ancillary voice radios the current estimate for GSS Inc I Leader solution is 14 lbs to achieve an operational system runtime of 24 mission hours. The current acceptable technology solutions to digital communications, computing, navigation providing 24 mission hours weighs 14 lbs. The leader's load, both weight and cube, must not degrade the mobility of the dismounted leader or slow the momentum of small unit operations in close combat in complex terrain. The dismounted leader will maneuver through upper floor windows, underground sewers, holes in walls, over walls, and over rubble in all environments. He will have to roll left or right and then sprint three to five seconds, through varied terrain, to the next covered and concealed position, while under hostile enemy fire. The leader and small unit need the mobility to pursue and defeat a fleeting enemy in complex terrain. Reducing the collective weight of items carried by the leader improves his mobility and survivability in difficult terrain.*

(6) KSA 6: BC: Transmission Range. The GSS Inc I equipped leader must communicate via voice and data to ranges of 1.8 km in complex urban conditions (NLOS) and 1 km in dense foliage (T), 5 km in all environments (O). Transmission range requirements will vary upwards dependent on the echelon of the GSS Inc I leader. This GSS requirement contributes to the TRADOC approved Battle Command Essential Capability to Execute Tactical Network Operations and Enable Collaboration.

***Rationale:*** *The flow of information must support geographically dispersed small units. NLOS capability in complex terrain allows the leader to communicate while behind cover to avoid threat* *weapons effects.*

(7). KSA 7: Rifleman Radio Interoperability. The GSS Inc I must be interoperable with the Rifleman Radio capability defined in its CPD (T=O).

***Rationale:*** *Soldiers require the ability to provide position location information to the GSS Inc I network for use by higher echelons. Additionally, team leaders require voice communications with their Soldiers. At the squad level, the squad leader is able to track the positions of his two team leaders and six Soldiers during the fight with a high level of confidence. The RR will be added to the team leader when it is defined, interface standards are established, and a clear method of the required IA certification of PLI crossing from the CUI domain to the secret domain is approved.*

d. GSS Attributes for Inc I**.** The following are additional attributes for the GSS Inc I with threshold and objective values.

**Table 6-3 GSS Attributes Increment I**

| Attribute | Development Threshold | Development Objective |
|---|---|---|
| COP: Geospatial Data Services (Mapping)<br>(Geospatial Intelligence (GEOINT)) | Store display color digital map and terrain representations using NGA format .(T) 1m image maps and 6-inch color image maps in complex and urban terrain (T). | 3-D Terrain at all elevation data resolutions. Building Floor Plans Adaptive to portray rubble & craters. Use Commercial Joint Mapping Toolkit (CJMTK). Display terrain reports generated by engineer marking. (O) |
| COP: Friendly Location Tracking | Provide location and tracking of GSS Inc I leaders, Soldiers and digital platforms (T). The system must track individuals within a vehicle and continue tracking them as they transition between mounted and dismounted operations (O). | and GSS Inc I leader's digital platforms (O) |
| COP: Refresh Rate | ≤ 5 seconds | < 3 seconds |
| COP: Identification of Friendly, Enemy and Non-combatants | Via display, distinguish between friendly forces and unknown personnel. | Identify friendly, enemy forces and noncombatants in all environments |
| COP: Marking Items of Interest | Leaders and sensors ability to mark items of interest. | T=O |
| Networked Communications: User Roles, Rights & Privileges | Assign user rights and privileges to roles. Assign roles to leaders. Alter roles, rights and privileges. Allow leaders to assume > 1 role. | Automatically and seamlessly re-assign leader roles based on the leadership-determined succession of command. |
| Networked Communications: Leader Interactions | GSS Inc I equipped leaders must communicate with, interact with, and interface with other GSS Inc I systems, including but not limited to Rifleman Radio capabilities, via encrypted (up to Secret) voice and/or data to manned platforms, and Soldier borne equipment (T) | via a two channel radio with embedded Global Positioning Satellite and certified for simultaneous operation in two security domains, classified (up to Secret) and protected unclassified (Controlled Unclassified Information) operation (O) |
| Networked | The system must fully support all Critical | All IERs (O). The system must be |

| Attribute | Development Threshold | Development Objective |
|---|---|---|
| Communications: SA | IERs and at least 70% of Non-Critical IERs listed in Appendix A (T). | interoperable with and independent of the Army BCT Modernization BCS (O). |
| Networked Communications: Security Levels | Speed of full SA/SU not degraded | T=O |
| Leader (User) Interface: Hands Free Efficiency. | Using a common interface, the system must streamline leader input processes for greater efficiency while allowing him control of all GSS Inc I Battle Command related functions (T). | Leader data entry processes must keep dismounted leader's hands free to execute critical tasks while in combat (O); all tasks during operations under all environmental conditions (O). Each hands free capability must have a corresponding manual back up (O). |
| Leader Mission Planning and Rehearsal | The GSS Inc I must include mission planning and rehearsal tools for leaders in small units and special team operations. This capability must accommodate participation by geographically dispersed leaders whether moving (mounted) or stationary. This capability must facilitate all mission planning functions such as mission analysis, course of action development, route planning, wargaming, orders production, SOP integration, orders dissemination, brief-backs and orders updates Fragmentary Orders (FRAGOs). This capability must incorporate visual tools such as the mapping capability, concept sketches, color graphics and icons, images and video in addition to simple text. | T=O |
| Information Management: General | Interoperable with the network. | One integrated system based on end-to-end view of all Soldiers and small unit processes. |
| Information Management: Customization by Leader | Capability to customize system to accommodate SOPs and TTPs (T). Choose or subscribe to information based on mission requirements and individual capacity (T). Capability to select information for subordinates (T). | T=O |
| Information Assurance (IA) | Incorporate all applicable IA protocols IAW DODI 8500.2 (T). The system must ensure identification, access control and accountability of system users (T). Using network infrastructure, must be Joint and Interagency interoperable (T) | Defend against 95% of known attacks. (O). Multinational interoperable (Cross Domain Solutions not required) (O). Enable doctrinal communications by the small unit leader to adjacent unit leaders (O). Have SA of adjacent units in COP (O). |
| Navigate | Fully support GB GRAM-SAASM standard IAW CJCSI 6140.01A, dated 31 March 04. Determine time within one minute (T), and its position within 3 meters (T) with the ability to | position within 1meter (O). |

| Attribute | Development Threshold | Development Objective |
|---|---|---|
| | convert to Longitude and Latitude as provided in the Military Grid Reference System (MGRS), to NGA Datums and directional orientation to the nearest degree (T=O). The GSS Inc I must facilitate planning and tracking of Soldiers' movement (T=O). | |
| Sustainability: Enhanced Power | Reduce number of power sources. Interact with power outlets of current transportation systems. Replace ("hot-swap") power source without shutting down and/or re-starting the system. Power consuming capabilities must be scalable/adjustable. Leaders must be able to share power sources between systems (T=O). | Only one power source. Smaller and lighter than LW Block II and with greater capability. Interact with power outlets of all transportation systems |
| Preventive Maintenance Checks and Services (PMCS) | Automate to simplify checks and to reduce workload. Non-automated portions of the PMCS must require no more than 5 minutes to complete | T=O |
| Operating Environment | –25 deg F (-32 deg C) to +130 deg F (+54 deg C). The system (excluding headsets) will remain operational while submerged for 2 hours at 1 meter (T) depth. The system must be transported and will operate in induced environments such as vibration and shock (T=O) | to + 150 deg F (+66 deg C)<br><br>4 meters (O) depth |
| Sustainability: Environment, Safety, and Occupational Health | Train, operate, maintain, and dispose of in full compliance with applicable US, foreign, and international regulations (T). Ensure that hardware, software, data, and facility resources are archived, sanitized, or disposed of in a manner consistent with system security plans and DOD IA requirements (T). Eliminate or minimize adverse environmental quality impacts (T). Not require ozone depleting chemicals during any life cycle. Comply with safety and occupational health laws and regulations (T). | T=O |
| GSS Inc I Mobility: Ergonomic Load Placement | A modular and integrated load management system must ergonomically place components on the Leader to maximize overall capability. | T=O |
| Maximum Time to Repair (Max TTR) | Max TTR for 95 % of all GSS Inc I malfunctions correctable at field level of support in $\leq$ 20 minutes (dedicated maintenance personnel and operator repair actions assessed separately). | Max TTR for 95 % of all GSS Inc I malfunctions correctable at field level of support in $\leq$ 15 minutes (dedicated maintenance personnel and operator repair actions assessed separately). |
| Maintenance-Related Reliability | Reliability of the GSS (GSS CFE + GSS GFE combined) operating for 24 hours without incurring an Unscheduled Maintenance Demand (UMD) will be no less than 82%. The GSS CFE will not degrade | T=O |

| Attribute | Development Threshold | Development Objective |
| --- | --- | --- |
| | maintainability of GFE. | |

(1) GSS Inc I BC/SA:

(a) COP.

1. COP: Geospatial Data Services (Mapping) (Geospatial Intelligence (GEOINT)). The GSS Inc I must store and display color digital maps and terrain representations as part of the COP using the NGA format (T). Geospatial products must be based on METT-TC; 1m image maps will meet the need of small unit operations in all terrain types, and 6-inch color image maps in complex and urban terrain (T). Terrain representations must provide sufficient detail to allow the small unit leader to visualize how individual enemy and friendly personnel will use the terrain (T). GSS Inc I will have the ability to display all data in 3-D, at all elevation data resolutions (O). Terrain representations include building and floor diagrams for the AO (O). Terrain representations also include adaptive capability to portray current changes in terrain caused by the effects of war such as rubble and large craters (O). GSS Inc I shall use Commercial Joint Mapping Toolkit (CJMTK) compliant tools to show data updates and battle damage assessments; this makes GSS Inc I compatible with higher echelons of battle command systems (O). The display must provide a high level of fidelity on terrain reports generated by Engineer marking and reporting of hazards (O). This GSS requirement contributes to the TRADOC approved Battle Command Essential Capability to Display/Share Relevant Information and provide A Standard and Sharable Geospatial Foundation.

***Rationale**: Leaders and small unit leaders require maps to visualize their AO and AI. 3-D visualization tools using color will help the leader process large amounts of information quickly. The Commercial Joint Mapping Tool Kit will enable this capability. The dismounted leader's fight is three-dimensional in combat in close, complex terrain. The leader's map requires a level of fidelity allowing navigation, position tracking of each Soldier, and enemy situation monitoring. The leader will achieve SA by using his mapping capability to plan, rehearse and execute his future actions and template enemy actions. He will update the same visual aid based on reality updates to the COP.*

2. COP: Friendly Location Tracking. The GSS Inc I must provide location and tracking of GSS Inc I leaders, Soldiers and digital platforms (T) and GSS Inc I leader's digital platforms (O). The system must track individuals within a vehicle and continue tracking them as they transition between mounted and dismounted operations (O). This GSS requirement contributes to the TRADOC approved Battle Command Essential Capability to Execute Tactical Network Operations, Display/Share Relevant Information and execute Battle Command on the Move.

***Rationale:** Position location and tracking provides the basis for maintaining friendly SA. Leaders must account for all of their Soldiers and equipment during all phases of operations. The close, dismounted fight is a three-dimensional fight taking place above and below ground. Close combat in complex terrain includes clearing rooms and buildings where a 3-D location tracking error greater than 3 meters presents an inaccurate COP to the dismounted small unit* *leader. The dismounted, urban fight may proceed with friendly and enemy forces inter-mixed between different rooms and floors of the same building. While conducting precision room clearing, fire team members eliminate or neutralize threat combatants while moving along the walls to points of domination. A location tracking error greater than 1 meter can depict team members outside the specific room, on the wrong floor, and outside the specific building.*

3. COP: Refresh Rate. The GSS Inc I must provide the leader and a COP refresh rate of 5 seconds or less (T), less than 3 seconds (O). COP refresh need only occur when the operational situation has changed (T). The refresh rate must be adjustable (T). The time standard for any operational message from the sending leader system to the receiving leader system is network dependant, but no more than 5 seconds (T). The time standard for any operational message to get from the sending leader system to the receiving platform system is network dependant, but no more than 5 seconds (T). This GSS requirement contributes to the TRADOC approved Battle Command Essential Capability to Execute Tactical Network Operations and Battle Command on the Move.

***Rationale:*** *The COP refresh rate must not impede the speed and momentum of small unit operations in complex terrain. The small unit leader must know the precise location of his Soldiers during combat operations. The disposition of small unit members can change significantly in a matter of seconds. Survivability in combat requires the Soldier to rush between firing positions in three to five seconds or less. Close combat in complex terrain includes clearing rooms and buildings where a refresh rate greater than five seconds can present an inaccurate COP to the dismounted small unit leaders.*

4. COP: Identification of Friendly, Enemy and Non-combatants. The GSS Inc I must enable the leader, via display, to distinguish between friendly forces and unknown personnel (T), enemy forces and noncombatants in all environments (O). This GSS requirement contributes to the TRADOC approved Battle Command Essential Capability to Display/Share Relevant Information and Execute a Running Estimate.

***Rationale:*** *Identifying friendly and enemy forces and noncombatants assists in populating the COP and enhances collaborative SA. This non-cooperative capability enables synchronization of operations and reduces the likelihood of fratricide and injury to noncombatants. This capability is enabled by inputs from shared situational awareness systems and the unit's organic sensors.*

5. COP: Marking Items of Interest. The GSS Inc I must provide leaders and sensors a capability to mark items of interest (T=O). This GSS requirement contributes to the TRADOC approved Battle Command Essential Capability to Execute Tactical Network Operations and Display/Share Relevant Information.

***Rationale:*** *Coalition and Current Force dismounted leaders in adjacent or follow-on units may not have immediate access to the COP and may require a visual means of recognizing items of interest. This capability ensures interoperability with these forces. Marking items of interest will contribute to the leader's total SA and assist him in warning other Soldiers of hazards and passing on information needed for operations. The ability to mark rally points, clear rooms and* *buildings, collection points, target reference points, obstacles, hazardous areas, sectors of fire and designate the front line trace for lead and flank elements is critical for the leader to discriminate targets, engage targets, and reduce fratricide in his engagement area. The ability to mark areas cleared by friendly forces and the ability to mark sectors of fire further enhances his SA and increase his OPTEMPO.*

(b) Networked Communications (NC).

1. NC: User Roles, Rights and Privileges. The Network system employed by GSS Inc I must allow the leader to assign user rights and privileges to roles and then assign roles to leaders (T). The leader must be able to alter these roles, rights and privileges after assigning them (T). The system must be re-configurable during operations to allow the leader to assume a new role (T). The system must automatically and seamlessly re-assign leader roles based on the leadership-determined succession of command (O). This GSS requirement contributes to the TRADOC approved Battle Command Essential Capability to Execute Tactical Network Operations, Display/Share Relevant Information, Enable Collaboration and Create and Disseminate Orders.

***Rationale:*** *The system will uniquely and efficiently identify each leader and their current role (duty position). Automatic role shift allows mission agility without inhibiting the momentum of small unit operations. If a leader becomes disabled, the next leader in the succession of command must immediately assume that leadership role and acquire all necessary user rights and privileges to BC/SA. Small unit leaders usually communicate succession of command information through OPORDs or unit SOPs. The system must inform the leader immediately when his role has changed ("battlefield promotion") so the small unit does not lose operational momentum when taking casualties. If the leader's GSS fails, he may take another system and must re-assume his role.*

*Assignment of roles, rights and privileges must be a leadership decision. Personnel shortages may require leaders to assume additional duties. New personnel replacements should trigger leadership action to cross-level duties of subordinates. Leaders from different small units may consolidate and reorganize into one unit during operations and the system will need to seamlessly continue tracking their identities and roles. Although all leader systems may possess Networked Fires access capability, the higher echelon leader may decide which subordinate leaders will designate targets, and he may reserve the right to release targeting data into Networked Fires.*

2. NC: Leader Interaction. GSS Inc I equipped Leaders must communicate with, interact with, and interface with other GSS Inc I systems via encrypted (up to Secret) voice and/or data to manned platforms, and Soldier borne equipment (T); via a two channel radio with embedded Global Positioning Satellite (GPS) and certified for simultaneous operation in two security domains, classified (up to Secret) and protected unclassified (Controlled Unclassified Information) operation (O). GSS Inc I will provide an interface to a Rifleman Radio capability which will provide individual Soldiers with voice communications and beaconing of PLI to their leaders (T=O). This GSS requirement contributes to the TRADOC approved Battle Command Essential Capability to Execute Tactical Network Operations, Display/Share Relevant Information and Enable Collaboration.

***Rationale:*** *Improved communications capabilities will enable synchronization of all elements of combat power and empower commanders no matter where they are physically located with real time combat information. The small unit leader will conduct cooperative, enroute planning and rehearsals with subordinates who are geographically separated between different vehicles. Robust communications enable leaders to designate targets to the COP to facilitate collaborative target engagement using organic and inorganic systems.*

3. NC: SA. The GSS Inc I must support SA/SU at the leader and small unit level in all environments. The system must fully support all Critical IERs and at least 70% of Non-Critical IERs listed in Appendix A (T), all IERs (O). The system must be interoperable with and independent of the Army BCT Modernized BCS (O). This GSS requirement contributes to the TRADOC approved Battle Command Essential Capability to provide A Robust Network Capability, Execute Tactical Network Operations, Display/Share Relevant Information and Enable Collaboration.

***Rationale:*** *Shared SA through the COP provides leaders the information to sustain collaborative initiative – self synchronization. Networked communications enable the geographically dispersed small teams to quickly generate and synchronize effects through collaborative engagement. Networked access to the BCT infrastructure provides the leader and small unit with support from organic and external assets at a much more decentralized level than the Current and Stryker Force capabilities allow. The GSS INC I IERs support Army BCT Modernization interoperability by satisfying Army BCT Modernization Soldier related IERs.*

4. NC: Security Levels. The speed of information exchange must be maintained regardless of security level policies (T). The speed and momentum of small unit operations are enabled through rapid exchange of information (T=O). This GSS requirement contributes to the TRADOC approved Battle Command Essential Capability to Execute Tactical Network Operations, Display/Share Relevant Information and execute Battle Command on the Move.

***Rationale:*** *Policies must not inhibit – but enable combat efficiency and effectiveness of the Leader and small unit engaged in close combat.*

(c) Leader (User) Interface.

1. Leader (User) Interface: Hands Free Efficiency. Using a common interface, the system must streamline leader input processes for greater efficiency while allowing him control of all GSS Inc I Battle Command related functions (T). Leader data entry processes must keep dismounted leader's hands free to execute critical tasks while in combat (O); all tasks during operations under all environmental conditions (O). Each hands free capability must have a corresponding manual back up (O).

***Rationale:*** *The leader must not lose focus on the close fight in complex terrain. User interfaces must not encumber - but enable. Greater efficiency is defined as reducing to a minimum the number of user inputs to accomplish a task, providing short cuts to menus and tasks, menu pull downs, and user tools that facilitate normally complex calculations. i.e. computing route* *azimuths and distances. The leader must maneuver tactically, operate his weapon, designate targets, engage targets with external assets and lead the small unit in combat without distractions from laborious, manual data entry tasks. The dismounted leader using GSS Inc I must engage targets with his individual weapon and external assets simultaneously with the same level of operational efficiency as he would today from using only his individual weapon. The small unit must efficiently employ resources to maximize combat power during close combat operations.*

2. Leader Mission Planning and Rehearsal. The GSS Inc I must include mission planning and rehearsal tools for GSS Inc I Leaders in small units and special team operations (T). This capability must accommodate participation by geographically dispersed leaders whether moving (mounted) or stationary (T). This capability must facilitate all mission planning functions such as mission analysis, course of action development, route planning, wargaming, orders production, SOP integration, orders dissemination, brief-backs and orders updates Fragmentary Orders (FRAGOs) (T). This capability must incorporate visual tools such as the mapping capability, concept sketches, color graphics and icons, images and video in addition to simple text (T=O). This GSS requirement contributes to the TRADOC approved Battle Command Essential Capability to execute Rehearsal and Training Support.

***Rationale:*** *Leaders must be able to conduct mission planning and rehearsals while being geographically dispersed. This reduces the overall time needed to do those tasks since the Leaders will not require close physical proximity to each other. One version of a GSS Inc I leader system is being developed, therefore one size must fit all users. Mission planning and rehearsal systems must reduce the small unit decision cycle and enable sustainment of initiative – self synchronization and allow small combat units the ability to quickly develop multiple Courses of Action (COA) for a given mission. Development must consider integration of mission planning and execution processes.*

(d) Information Management

1. Information Management: General. The GSS Inc I must be interoperable with the network to ensure all relevant information reaches the right person at the right time, in a usable and digestible form, in order to enable sound and timely decisions (T). The information system must be one integrated system based on an end-to-end view of all Soldier and small unit processes (T=O). This GSS requirement contributes to the TRADOC approved Battle Command Essential Capability to provide A Robust Network Capability, Display/Share Relevant Information, Battle Command on the Move and Execute a Running Estimate.

***Rationale:*** *Information is an element of combat power. An efficient information management system is essential for maintaining an accurate COP. Since GSS Inc I will share data elements used across the DOD enterprise, using DOD standard data elements and information infrastructure for application development will ensure seamless information sharing across all supporting joint capabilities. The user needs to access all necessary capabilities using only one system.*

2. Information Management: Customization by Leader. The leader must have the capability to customize their system to accommodate their unit's current SOPs and TTPs (T). Individual users must be able to choose or subscribe to information most relevant to them based on requirements for each mission and their individual capacity to absorb information (T). Leaders must have the capability to select information for their subordinates (T=O). This GSS requirement contributes to the TRADOC approved Battle Command Essential Capability to Execute Tactical Network Operations, Display/Share Relevant Information and execute Battle Command on the Move.

***Rationale:*** *TTPs and SOPs often differ between units and change often based on changes in leadership and missions. Such changes should update "business rules" of the information system. Units often produce SOPs to standardize procedures and simplify mission planning for operation order content that would otherwise be repetitive, and the system should accommodate these SOPs. For example, if the unit SOP contains the succession of command, this rule should drive the reassignment of roles to individuals when leaders become casualties (See leader roles under the communications requirement). Individual capacity to absorb information can differ between users. Each user's capacity to absorb information can improve with practice.*

(e) Information Assurance (IA). The standard communications network employed by GSS Inc I must be jam resistant, protected from network exploitation, be encrypted, be tamperproof, ensure process and data integrity, be electromagnetic compatible, and be frequency managed, by the organization's S6, using standard Network tools. The GSS Inc I integrated C4ISR function must incorporate all applicable IA protocols IAW DODI 8500.2 to ensure timely, efficient, and reliable exchanges of information (T). This capability must support and not hinder information and network reliability, availability, integrity, authentication, confidentiality and non-repudiation. The system must ensure identification, access control and accountability of system users (T). The system must provide the ability to protect and defend information and information systems against known external threats for 95% of all known attacks (O). The GSS Inc I, using the network infrastructure, must be joint and Interagency interoperable (T), Multinational interoperable (Cross Domain Solutions not required) (O). The system must enable doctrinal communications by the small unit leader to adjacent unit leaders (O). Each GSS Inc I equipped Leader must have SA of adjacent units in his COP (O). This GSS requirement contributes to the TRADOC approved Battle Command Essential Capability to provide A Robust Network Capability.

***Rationale:*** *IA ensures the continuous availability of the COP enabling force protection through the defense of information and information systems. Leaders and small unit operations must proceed without disruption from information threats such as Denial of Service (DOS) attacks to the individual or the network. TRAC-WSMR determined the requirement for 95% as operationally significant and relevant to achieve information superiority in the FF. (The network's ability to update and restore operations as a result of an attack will provide the a self healing network protected to a reasonable level of risk).*

(f) Navigate. The GSS Inc I will enable Leaders to navigate in their operational environment. Fully support GB GRAM-SAASM standard IAW CJCSI 6140.01A, dated 31 March 04. The system must be capable of determining time within one minute (T), and its position within 3 meters (T) and 1 meters (O) with the ability to convert to Longitude and Latitude as provided in the Military Grid Reference System (MGRS), to NGA Datums and directional orientation to the nearest degree (T=O). The GSS Inc I must facilitate planning and tracking of Soldiers' movement (T). This GSS requirement contributes to the TRADOC approved Battle Command Essential Capability to Display/Share Relevant Information and execute Battle Command on the Move.

***Rationale:*** *Navigation capability enhances Soldier and small unit mobility by giving the Leader the SA to know where he is and where he is going. He will plan routes and track movements along those routes. He will make route adjustments when encountering obstacles and danger areas. He will report movement progress including estimated time of arrival. An enhanced navigation capability should help him accomplish these tasks more efficiently. Increased navigation accuracy is necessary for some Leaders in order to plan and emplace targets, obstacles, sensors, to mark and report explosives and hazards, and assist in detailed reconnaissance. In addition, robotics navigation will require better accuracy in order to perform intricate tasks assigned to some robots.*

(g) GSS Inc I Sustainability.

1. Sustainability: Enhanced Power. The integration effort must combine power sources to allow the Soldier to carry the fewest number of them (T). If only one power source is not possible, they must be interoperable (T). The GSS Inc I must be able to operate using one common power source (O). A power source smaller and lighter than that of LW and with greater capability is desired (O). The GSS Inc I must interact with current and future BCT vehicles that transport and support Soldiers and wall outlets, US (110v), World (220v), in order to replenish power (T), all transportation systems such as Air Force aircraft, rotary wing aircraft, watercraft, tactical trucks and rail and their power outlets (O). The Leader must be able to quickly and conveniently replace ("hot-swap") his power source without shutting down and/or re-starting the system. Any GSS Inc I capability consuming power must be scalable and/or adjustable by the user. Leaders must be able to share power sources between systems (T=O).

***Rationale:*** *Power is a critical component of sustainability and weight reduction. An enhanced power system will reduce the logistical burden by reducing the weight and number of power sources carried by the Soldier and the logistical structure. The Leader will use the GSS Inc I to maintain networked communications and collaborative SA while using all modes of transportation. He must leverage all external power sources sustain system capability. Restoration of power must not degrade the speed and momentum of small unit operations. The leader must quickly and conveniently access and exchange his power source while sustaining system functionality. Power sharing can be accomplished by hot swapping spare batteries. Smart power management enables leaders to operate for longer durations.*

2. PMCS. Operator PMCS capabilities must be automated to the maximum extent possible to simplify checks and reduce workload (T). Non-automated portions of PMCS must take 5 minutes or less to complete (T).

***Rationale:*** *The operator of GSS Inc I is a Warfighter first and foremost. PMCS requires automation to ensure this primary job is not adversely impacted by extensive maintenance related time and tasks. Non-mission capable status contributes to an inaccurate COP to inform leaders of mission capability.*

3. Operating Environment. The GSS Inc I must operate in ambient temperatures from –25 degrees Fahrenheit (-32C) to +130 degrees Fahrenheit (+54C), inherent to the system or achieved with the addition of a kit (T), +150 degrees Fahrenheit (+66C) (O). The GSS Inc I must operate in all environments to include snow, ice, rain, sand, dust, salt-water spray and when exposed to battlefield contaminants. The system (excluding headsets) will remain operational while submerged for 2 hours at 1 meter (T) 4 meters (O) depth. The system must be transported and will operate in induced environments such as vibration and shock (T=O).

***Rationale:*** *This capability ensures GSS Inc I operability in anticipated environmental conditions. The system will comply with MIL STD-810G environmental testing.*

4. Sustainability: Environment, Safety and Occupational Health. The user of the GSS Inc I shall have the ability to train, operate, maintain, and dispose of the systems in full compliance with existing and near-term forthcoming applicable US, foreign, and international environmental quality policies, statutes, laws and regulations (T). Ensure that hardware, software, data, and facility resources are archived, sanitized, or disposed of in a manner consistent with system security plans and DOD IA requirements (T). The design, production, operation, maintenance, and disposal of the GSS Inc I shall eliminate or minimize to the greatest extent possible adverse environmental quality impacts and the use of hazardous materials that would preclude use and disposal in operating environments. (T). The GSS Inc I shall not require the use of ozone depleting chemicals during any phase of their life cycles. The GSS Inc I will comply with safety and occupational health laws and regulations (T=O).

(h) GSS Inc I Mobility: Ergonomic Load Placement. A modular and integrated load management system must ergonomically place components on the Leader to maximize overall capability (T=O).

***Rationale:*** *Given reductions in weight and cube of capabilities, maximizing Leader mobility requires an optimal combination and balance between integration and modularity to minimize fatigue and the space claim of components on the Leader and maximize his capabilities.*

(i) Maximum Time to Repair (Max TTR). The Max TTR for 95 % of all malfunctions correctable at the field level of support will not exceed 20 minutes (T) 15 minutes (O) for dedicated maintenance personnel and for operators (each assessed separately).

***Rationale:*** *Threshold requirement constrains the burden placed on dedicated field maintenance personnel and operators to the level achieved with the baseline LW-SI System in OIF. However, based on RAM and OMS/MP analysis, it will not preclude the need for one or more additional dedicated Table of Organization and Equipment (TOE) field level maintainers per BCT (dependent on the level of leaders to which the GSS Inc I is fielded) if the GSS Inc I support concept adopted were to involve traditional Army maintenance without the fix-forward* *benefits of the concept currently envisioned by PM-Soldier Warrior (field level Contractor Logistic Support (CLS) with fix-forward repair through replacement of GSS Inc I LRUs at operator level and back-up maintenance capability provided by dedicated Field Service Representatives (current concept employed for LW-SI System)). The fix-forward aspect of the GSS Inc I support concept currently envisioned by PM-Soldier Warrior, involving a designated Master Warrior operator with additional maintenance training who performs the operator level troubleshooting and repair within his respective platoon, results in a maintenance time burden of under one hour per day for that individual. Objective requirement reduces the burden placed on dedicated maintainers who provide back-up support to the platoon Master Warriors (reducing sustainment maintenance costs accordingly), as well as the Master Warrior's time spent performing corrective maintenance (which is an additional duty having the potential to adversely impact on his role as a leader); the need for one additional dedicated TOE maintainer (as applicable) would not be eliminated even at objective requirement level. The requirement for additional maintainer force structure within the BCTs will be verified through the Level of Repair Analysis (LORA) and Logistics Demonstration.*

(j) Maintenance-Related Reliability. Reliability of the GSS (GSS CFE + GSS GFE combined) operating for 24 hours without incurring an Unscheduled Maintenance Demand (UMD) will be no less than 82% (T=O). A UMD is defined as a demand for corrective maintenance resulting from a failure event that causes the loss of either essential or non-essential GSS functionality. The GSS CFE will not degrade maintainability of GFE (T=O).

*Rationale: To ease the maintenance burden in support of achieving the overarching A(m) and* $A_o$ *threshold performance by limiting the UMD rate which GSS generates. This required capability, together with the Max TTR threshold for UMDs, establishes the allowable maintenance burden the system can place on repair personnel. Requirement is defined operationally (as a percentage value) in accordance with current TRADOC guidance and has been expressed in relation to 24 hours of operation to maintain consistency (a common basis) with KSAs 1 and 3; threshold requirement correlates to 122 OHs Mean Time Between Unscheduled Maintenance Demand (MTBUMD). Analysis conducted in support of the required A(m) capability based on simulation of fleet-wide GSS operations over a 15 year life cycle consistent with the Army Force Generation Model (encompassing peacetime training, wartime usage, and periods of equipment reset) indicates the threshold requirement for maintenance-related reliability (together with the reliability KSA) supports achievement of the required A(m). The maintenance-related reliability requirement has been determined from the operational reliability KSA threshold, based on the ratio of EFFs to the collective sum of essential and non-essential unscheduled maintenance demands yielded by analysis of baseline system data).*

e. GSS Attributes. The intent of this paragraph is to provide a base reference document for ground Soldier requirements. There are MOS specific Soldiers who will receive additional attributes based on mission specific tasks, i.e.; Soldiers performing Explosive Ordnance Disposal (EOD) must have additional blast protection above what is provided by the Core or GSS Inc I. These Additional Performance Attributes for GSS are provided by other programs that are GFE to GSS. It is not intended that these GSS attributes be fielded or tested as part of this increment. Each of these GSS attributes will require a standalone CPD which will require its own programming and testing not provided under this document.

**Table 6-4 GSS GFE Attributes Increment I**

| Attribute | Development Threshold | Development Objective |
|---|---|---|
| NC: Subterranean Communications | Communicate with each other, and to interact with and control devices and equipment without relays or wires in LOS and NLOS complex natural terrain, urban terrain, in subterranean environments, and through water (T=O). Signals must have a nominal range of 100 feet (T) through fresh and salt water, all foliages, sand, earth, rock, man-made objects, and natural structures regardless of the physical barrier to direct linkage. | Nominal range of 8 kilometers regardless of the physical barrier to direct linkage (O) |
| Countermine: Foot Protection | Protection from AP mine of 50 grams explosive filler | Protect from AP mine of 75 grams explosive filler additional layer up to 200 grams |
| Countermine: Facial Protection | Protect V-50 penetrations of projectiles 30 grains or less at 1000 feet per second | Protect V-10 penetrations of projectiles 40 grains or less at 1500 feet per second |
| Countermine: Body Armor | Improved protection to blood vessels, neck, and joints | Protect from mine threats both current and foreseen |
| CB Protection: Uniform Pocket Access | Protective garment must have closing pockets that won't interfere with load carry equipment | T=O |
| CB Protection: Modularity | Modular design to facilitate varying levels of CBRNE Threat for 24 hrs | Decontaminable for reuse, Micro-climate coolers, tubes, and pass-through connectors will not interfere with Soldier access or weapon employment |
| Liquid Protection | Supplemental liquid protection against decontaminants, CBRN, and selected Toxic Industrial Chemicals/Toxic Industrial Materials (TIC/TIM) | Compatible with SaaS Architecture and the Microclimate Cooling System |
| Level "A" Protection | Scalable levels of protection | Embedded as one protective system |
| CB Protection: Above the Neck | Above neck protection for 24 hrs | Above neck protection for 36 hrs continuous exposure |
| CB Protection: Mask Communication | Permit receiving and transmitting communication, integrated microphone | T=O |
| CB Mask and Target Engagement | Increased compatibility with sights of individual and crew-served weapon systems and combat vehicle systems | T=O |
| GSS Mobility: Soldier's Load | Manage integration of GSS components to keep the total weight of the leader's approach march load not greater than 72 lbs (T) and the fighting load not greater than 65 lbs (T) Continue SaaS volume and weight reduction initiatives for capabilities currently carried by the Soldier | Manage integration of GSS components to keep the total weight of the leader's approach march load not greater than 65 lbs (O) and the fighting load not greater than 50 lbs (O) |

(1) NC: Subterranean Communications**.** Soldiers must be able to communicate with each other, and to interact with and control devices and equipment without relays or wires in LOS and NLOS complex natural terrain, urban terrain, in subterranean environments, and through water (T=O). Communications and control signals must have a nominal range of 100 feet through fresh and salt water, all foliages, sand, earth, rock, man-made objects, and natural structures regardless of the physical barrier to direct linkage (T), a nominal range of 8 kilometers regardless of the physical barrier to direct linkage (O). This GSS requirement contributes to the TRADOC approved Battle Command Essential Capability to provide A Robust Network Capability, Execute Tactical Network Operations, Display/Share Relevant Information and Enable Collaboration.

***Rationale:*** *Leaders using GSS need voice communications, access to and control of sensors, interface with robotic devices, and positive control over demolitions and other explosive devices through heavy forests, concrete structures, urban infrastructure, earth and stone materials in tunnels, caves, basements, building floors above ground level, and fresh and salt water without wires or relays.*

*All Soldiers, whether they be combatants, humanitarian workers, law enforcement, or firefighters, who may find themselves clearing and securing complexes such as caves, subways, basements, buildings, and sewer tunnels need to communicate between the street level security, the upper level sweep teams, and the underground teams, as well as to interface un-tethered with remote scouting devices and access-denial devices. Engineer divers performing surf zone and littoral clearing missions, port opening, waterway clearing, and port construction missions must be able to communicate through ship hulls and through underwater terrain features diver to diver and diver to supervisor, without wires or relays. Current communications use technology defeated by the presence of noisy water and by objects intervening between the divers.*

*The (T) nominal range of 100 feet (33 yards) is the minimal range required for safe operation of robotics while underground or in upper stories of buildings where various hazards are expected. It also is a reasonable minimum working distance that fire team members / rescuers / firefighters can use between those above ground and those in basements-bunkers-vaults-tunnels underneath the surface / on upper structure floors above ground level / upper decks and interiors of vessels.*

(2) Countermine.

(a) Countermine: Foot Protection. The GSS must provide protection to feet and lower extremities for troops in mine-infested theaters suitable for full-time daily wear (boots) which can prevent amputations due to blast Anti-Personnel (AP) mine damage when the wearer steps on a blast AP mine of 50 grams of explosive filler (T) 75 grams of explosive filler (O). GSS counter-mine foot protection must provide an additional level of layered protection during counter-mine operations capable of reducing severity of amputations to below the knee surgeries due to blast mines of up to 200 grams of explosive filler (O).

***Rationale:*** *Soldiers performing countermine functions are exposed to increased levels of blast devices during counter-mine, counter-booby-trap, and breaching operations. Protection must be scalable to the level of expected risk in the mission. This attribute is special for MOS specific* *duties such as Engineers clearing unexploded ordinance and is above the capabilities/attributes of the GSS Inc I.*

(b) Countermine: Mine Facial Protection. Soldiers performing counter-mine and counter explosive hazard missions must have enhanced facial blast and fragment protection from explosive hazards beyond that provided to the standard issue Soldier (T). Facial protection must be V-50 for penetrations of projectiles 30 grains and less at velocities of 1,000 feet per second and less at ambient temperatures (T), V-10 penetrations of projectiles 40 grains and less at velocities of 1,500 feet per second and less at ambient temperatures (O), both right cylinder and round ball (T=O).

***Rationale:*** *Soldiers performing countermine functions are exposed to increased levels of blast and fragmentation devices during counter-mine, counter-booby-trap, and breaching operations. Protection must be scalable to the level of expected risk encountered in the mission. Wall breaching operations, sub-terrain explosions, booby-traps will expose troops to higher levels of fragmentation, to blast-wave, and to over-pressure effects. This necessitates improved protection for the face, throat, ears, and tissue internal of the cranium from the fragments and concussion wave, and is critical to performance of the mission as well as individual countermine protection. This attribute is special for MOS specific duties such as Engineers clearing unexploded ordinance and is above the capabilities/attributes of the GSS Inc I.*

(c) Countermine: Body Armor. Soldiers performing counter-mine and counter explosive hazard missions must have enhanced blast and fragment protection from explosive hazards beyond that provided by the CSS (T). Fragment protection for the body must provide a light system weight, hard and or soft armor components with minimal impact on user range of motion (T), no reduction in ranges of motion versus wear of the GSS (O), no exposure to fragment projectiles at vulnerable body sections where major blood vessels may be struck (O); must not accumulate solar heat gain (O); must integrate the GSS Micro-Climate system (T=O) to maintain user comfort in temperatures from –50 degrees F to +150 degrees F (O); must minimally protect to the established ballistic standards of Body Armor Suit, Individual, Countermine (BASIC) for body protection (T) while improving protection to blood vessels, neck, and joints (T), and must protect from mine threats identified both for current and foreseen future mines (O).

***Rationale:*** *Soldiers performing countermine functions are exposed to increased levels of blast and fragmentation devices during counter-mine, counter-booby-trap, and breaching operations. Protection must be scalable to the level of expected risk encountered in the mission. Fragmentation devices require additional modularity for protection while engaging, explosions and booby-traps. Wall breaching operations, sub-terrain explosions, booby-traps will expose troops to higher levels of blast-wave / over-pressure effects, and protection of ears and internal organs from the concussion wave are critical to performance of the mission as well as individual protection.. This attribute is special for MOS specific duties such as Engineers clearing unexploded ordinance and is above the capabilities/attributes of the GSS Inc I.*

(3) CBRNE Protection.

(a) <u>Chemical Biological (CB) Protection: Uniform Pocket Access</u>. The CB Protective overgarment must have closing pockets, which do not interfere with the wear of load carrying equipment (T=O).

***Rationale:*** *Closing pockets allow Soldiers to easily access essential items that are required to perform mission tasks and reduce the risk of losing essential items stored in those pockets. Load carrying equipment includes current and developmental modular body armor and chassis.*

(b) <u>CB Protection: Modularity</u>. Items must be CBRN-resistant for 24-hrs (T), be removable without self-contamination (T) and decontaminable for reuse (O). Micro-climate coolers, tubes, and pass-through connectors will not interfere with Soldier access or weapon employment (O).

***Rationale:*** *Soldiers will have to perform their mission in all environments including CB contamination. In order to protect Soldiers, maximize their endurance, and ensure successful completion of the mission, GSS will incorporate a modular design permitting the Soldiers to vary the level of protection IAW with the mission and/or threat requirements.*

(c) <u>Liquid Protection</u>. Chemical Corps technicians will wear the liquid protection addition to the standard protective over-garment (T). Saturation with decontaminant can severely degrade the standard garment and render it unserviceable. The sealed, impermeable liquid protection system will provide the chemical Soldier with supplemental, liquid protection against decontaminants, CBRN, and selected TIC/TIM. The system must be modular, lightweight, and comfort fitting (T=O). It must be compatible with the SaaS architecture and the micro-climate system (O).

***Rationale:*** *The liquid protection system is worn in addition to the standard protective over-garment. The protective over-garment will become severely degraded and unserviceable if saturated with decontaminants. The impermeable liquid protection system worn over the protective over-garment will provide the chemical Soldier with enhanced protection to conduct decontamination operations. The system must be lightweight and comfort fitting to enhance the chemical Soldier's performance and minimize fatigue.*

(d) <u>Level "A" Protection</u>. Soldiers performing as Chemical Corps technicians must have the capability to conduct Weapons of Mass Destruction (WMD) sensitive site exploitation (T). The Level "A" Protection will provide percutaneous and respiratory protection and will provide chemical Soldiers the ability to select scalable levels of specialized protection equipment to adequately protect against specified CBRN threats and TIC/TIMs (T). The scalable degrees of protection are based upon the duration of the mission and the probability of encountering hazardous liquid splash or direct contact with hazardous chemicals (T). The scalable levels of protection will include hands-free communication systems and, mission duration dependent, self-contained breathing apparatuses and powered cooling systems (T). The scalable levels of protection (T), embedded as one protection system (O), are worn over the CSS and will operate under all environmental and extreme weather conditions. The equipment must collect and contain susceptible WMD samples, conduct presumptive analysis (O), and facilitate packaging of samples in preparation for transportation to laboratories for definitive analysis (T).

***Rationale:*** *The TAA 11, FF, organizational structure includes chemical units requiring the capability to conduct WMD sensitive site exploitation, presumptive analysis, and facilitate in packaging susceptible WMD samples in preparation for transportation to laboratories for definitive analysis.*

(e) CB Protection: Above the Neck. The CB protective mask will provide the wearer above-the-neck protection from CB agents, radioactive fallout particles, and TIMs as listed in appropriate threat documents for 24 hours (T), 36 hours (O) of continuous exposure.

***Rationale:*** *Protection requirements are consistent with the 2 June 1997 Joint Service General Purpose Mask (JSGPM) STAR and RESPO 21 Hazard Assessment, Phase 3 Summary Report, dated June 1997. Challenge levels and duration of protection required equal to Joint Services Lightweight Integrated Suit Technology (JSLIST) suit protection.*

(f) CB Protection: Mask Communication. The CB protective mask must permit the use of receiving and transmitting communication devices (T=O).

***Rationale:*** *Soldiers rely on the ability to leverage the communications devices resident in their system and in the host platform to transmit and receive tactical information. They require this capability to maintain the tactical edge and must retain the ability to effectively transmit and receive voice communications while wearing the protective mask.*

(g) CB Protection: CB Mask and Target Engagement. The CB protective mask must provide increased compatibility, beyond current compatibility, with the sights of individual and crew-served weapon systems as well as those proposed for all other combat vehicle systems (T=O).

***Rationale:*** *Identified requirement reflects mission profile and employment scenario as well as the results from: Concept Development Studies For Respiratory Protection System 21 (ERDEC-TR-285) dated Sep 95. A Health Hazard Assessment (HHA) and Safety System Assessment (SSA) will be conducted to identify potential safety or health problems not eliminated during design and engineering.*

(4) GSS Mobility: Soldiers Load. Manage integration of GSS components to keep the total weight of the leader's approach march load not greater than 72 lbs (T), 65 lbs (O) and the fighting load not greater than 65 lbs (T), 50 lbs (O). Continue SaaS volume and weight reduction initiatives for capabilities currently carried by the Soldier.

***Rationale:*** *The combat load consists of mission essential equipment determined by the commander and required for the Soldier to fight and survive in immediate combat operations. The combat load consists of two levels: approach march load and fighting load. Analytical studies and doctrine (FM 21-18 "Foot Marches", 1990) consistently show fighting loads of 30% of Soldier body weight or less and approach march loads of 45% or less best support Soldier operational performance. The 65 lb fighting load (T) represented by 30% of the average Soldiers body weight, provides a technically achievable improvement over the current LW II system* *although is exceeds the actual Analytical studies provided in FM 21-18 Foot Marches, which is the goal of this process. Analytical studies, doctrine and command guidance consistently point toward a 40-50 lb fighting load limit. A 65 lb fighting load, by the 30% standard, and a 72 lb approach march load, by the 45% standard accommodate the 50th percentile Soldier. The objective standards for the approach march load of 65 lbs and the fighting load of 50 lbs accommodate over 90% of all Soldiers. The 1998 Study of Soldier Loads at the Joint Readiness Training Center (JRTC) further validates this conclusion that the fighting load should not exceed 30% of a Soldier's body weight. Conclusions of the Physical Performance Benefits of Offloading the Soldier support a "30-33% of bodyweight recommend maximal load" from the US Army Institute of Environmental Medicine (USARIEM) Soldier Load Study, 2003.*

*The representative fighting load is common for all operating environments. Specific Soldier roles, specialized tasks and distribution of capabilities within the small unit can add to this load. Different extremes in operational environments can also modify this load. CFE components for GSS are focused specifically within the fighting load, and are therefore additive to the fighting load of 57.950 lbs. excluding the Rifleman Radio Capability and ancillary voice radios the current estimate for GSS is 14 lbs to achieve an operational system runtime of 24 mission hours. The table below illustrates the resultant fighting load when GFE and CFE are integrated or combined.*

**Table 6-4.1 Fighting load when GFE and CFE are integrated or combined**

| **Fighting Load** | **Weight (lbs)** |
|---|---|
| Fighting Load | 57.950 |
| GSS Inc I (w/power supply) | 14.00 |
| **Total** | **71.95** |

7. Family of Systems and System of Systems Synchronization. The intent of the GSS strategy is alignment with Army BCT Modernization in order to meet the needs of all Ground Soldiers. Mature Advanced Technology Demonstration (ATD) technologies will transition into the Ground Soldier System program, postured to support FCS Spin-Out 1 with a BCT capability beginning in FY 12. Subsequent increments will incorporate feedback from System Development and Demonstration (SDD) activities of this current increment. See paragraph 13, Schedule and IOC / Full Operational Capability (FOC) Definitions.

a. Relationship of this System to other systems contributing to this capability. GSS equipped leaders must communicate with, interact with, and control other Soldiers, manned platforms, networked fires, smart munitions, unmanned air and ground vehicles, sensors, ET, Soldier borne equipment and other future capabilities that become available. Other systems include Joint Command and Control – Army (JC2-A), Joint Effects Targeting System (JETS), GIG, DCGS-A, JTRS HMS, MERS and Warriors Information Network – Tactical (WIN-T). Soldier-to-Soldier interaction includes Army, joint, interagency and multinational exchanges. The OVs in the operational architecture section, Appendix A depict these interfaces and interactions with explanations. OVs also depict unit types and support dependencies.

b. Additional Systems supporting the SaaS ICD. This GSS Inc I CDD builds upon the CSS CPD to address capabilities for ground Soldiers.

**Table 7-1 Supported ICDs and Related CDDs/CPDs**

| Capability | CDD Contribution | Related CDDs | Related CPDs | Tier 1&2 JCAs |
|---|---|---|---|---|
| The GSS gives dismounted leaders additional capability above the CSS, specifically oriented to network integration of the individual leaders, protection, mobility, sustainability, reliability and ET. (SaaS ICD) | An integrated leader system focused on small units performing ground missions. | Mounted Warrior Soldier System CDD | LW CPD<br><br>Core Soldier System CPD<br><br>Air Warrior Soldier System | **Force Application (Tier 1)**<br>• Maneuver (Tier 2)<br>- Maneuver to Engage (Tier 3)<br>• Engagement (Tier 2)<br>- Kinetic Means (Tier 3)<br>**Command & Control (Tier 1)**<br>• Understand (Tier 2)<br>- Develop Knowledge & Situational Awareness (Tier 3)<br>• Decide (Tier 2)<br>- Manage Risk Tier 3<br>**Battlespace Awareness (Tier 1)**<br>• ISR (Tier 2)<br>- Process & Exploitation (Tier 3)<br>- Analysis & Production (Tier 3) |
| The GSS gives dismounted leaders additional capability above the CSS, specifically oriented to network integration of the individual leaders, protection, mobility, sustainability, reliability and ET. (SaaS ICD) | Builds upon LW CPD and CSS capabilities to meet the needs of all ground leaders in the FF | Leverages Soldier ATD's to rapidly bring emerging technology to the ground Soldier. | Core Soldier System<br><br>Draft:<br>Mounted Soldier System<br>Air Soldier System | **Force Application (Tier 1)**<br>• Maneuver (Tier 2)<br>- Maneuver to Engage (Tier 3)<br>• Engagement (Tier 2)<br>- Kinetic Means (Tier 3)<br>**Command & Control (Tier 1)**<br>• Understand (Tier 2)<br>- Develop Knowledge & Situational Awareness (Tier 3)<br>• Decide (Tier 2)<br>- Manage Risk Tier 3<br>**Battlespace Awareness (Tier 1)**<br>• ISR (Tier 2)<br>- Process & Exploitation (Tier 3)<br>- Analysis & Production (Tier 3) |
| The GSS gives dismounted leaders additional capability above the CSS, specifically oriented to network integration of the individual leaders, protection, mobility, sustainability, reliability and ET. (SaaS ICD) | Support the formalization of the DOTMLPF process to fully integrate leaders through a SoS approach. | Continues the development of an integrated Soldier System begun in LW CPD. | Core Soldier System<br><br>Draft:<br>Mounted Soldier System<br>Air Soldier System | **Force Application (Tier 1)**<br>• Maneuver (Tier 2)<br>- Maneuver to Engage (Tier 3)<br>• Engagement (Tier 2)<br>- Kinetic Means (Tier 3)<br>**Command & Control (Tier 1)**<br>• Understand (Tier 2)<br>- Develop Knowledge & Situational Awareness (Tier 3)<br>• Decide (Tier 2)<br>- Manage Risk Tier 3<br>**Battlespace Awareness (Tier 1)**<br>• ISR (Tier 2)<br>- Process & Exploitation (Tier 3)<br>- Analysis & Production (Tier 3) |

8. <u>Information Technology and National Security Systems (IT and NSS)</u>.

a. GSS Inc I is the initial Increment of GSS which will provide leaders with increased BC capabilities, SA and understanding, voice and data communications, position location and network connectivity to upper and lower echelons. GSS Inc I is an integrated system of state-of-the-art navigation, processing and communications technology. GSS Inc I will be interoperable with suitable existing military radio systems and emerging JTRS, e.g. SINCGARS, EPLRS, SRW, SAASM, etc. Criteria for incorporation in GSS Inc I are mission suitability of the waveforms, support of Network Centric operations, SWaP compliance of the hardware, appropriate COMSEC support for classified operation, and spectrum approval/supportability in the form of an approved DD-1494 Application for Equipment Frequency Allocation. GSS Inc I will provide an interface to a Rifleman Radio capability which will provide individual Soldiers with voice communications and beaconing of PLI to their leaders. The initial BOI to achieve the Initial Operational Capability (BCT) will be 641 GSS Inc I and 1658 Rifleman Radio capable devices. This enables each Soldier with voice and PLI and each leader with GSS Inc I.

b. Operationally GSS Inc I will enable Network Centric Information Environment at the leader and small unit level. The GSS Inc I will provide leaders improved SA through the COP. BC and SA will enhance troop leading procedures, tactical problem solving, and operational momentum. GSS Inc I will provide capabilities to share communications vertically and horizontally, monitor the movements of small unit combatants, and accurately control organic and supporting fires. GSS Inc I will help facilitate networked integration of leaders with sensors, weapons and platforms. GSS Inc I enables small units and leaders to conduct mutually supporting operations while geographically dispersed. GSS Inc I will enable small units to engage in a larger geographic area with greater precision, speed and a broader variety of lethal and non-lethal effects.

c. GSS Inc I provides voice and data communications between Leaders and an interface to a Rifleman Radio capability which will provide individual Soldiers with voice communications and beaconing of PLI to their leaders. Subsequent Increments will incorporate increased capabilities video, imagery and network lethality. GSS Inc I will be operated as a classified system at the secret level, and will interface with an unclassified Rifleman radio capability as defined by its CPD.

d. The Operational Mode Summary/Mission Profile identified nine combat operations used to develop the Data Throughput (Table 8-1). These missions are considered tactically relevant for the contemporary operating environment. While the total duration of these nine operations is calculated at 96 hours, no one mission is projected to last 24 hours and units may or may not conduct each type of operation over the 96 hours. However, the unit is forward deployed for the duration of the 96 hours and must establish and maintain security during the planning and preparation phases prior to execution, requiring near continuous operation of most GSS Inc I. To determine the Transmission Frequency, the total number of transmissions during the 96 hours of missions was divided by 4 to calculate the 24 hour average of voice and data transmissions.

**Table 8-1 GSS Inc I Data Throughput Table**

| File Type | | Size | Transmission Frequency |
|---|---|---|---|
| Voice | Yes | Small | Medium |
| SA Data File | Yes | Small | High |

| | | | |
|---|---|---|---|
| C2 Data File | Yes | Small | Medium |
| Stream Video | No | N/A | N/A |

Small = 1-Byte-100 KB
Medium = 101-999KB
Large = 1 MB or larger

Low = 1 – 50
Medium = 51-100
High = 101 or greater

| **Table 8-2 RR Data Throughput Table** | | | |
|---|---|---|---|
| **File Type** | | **Size** | **Transmission Frequency** |
| Voice | Yes | Small | Medium |
| SA Data File | Yes | Small | High |
| Stream Video | No | N/A | N/A |

Small = 1-Byte-100 KB
Medium = 101-999KB
Large = 1 MB or larger

Low = 1 – 50
Medium = 51-100
High = 101 or greater

e. The GSS Inc I must be compliant with the Army Software Blocking Policy, Blocks I–III baseline upon fielding. The IT and NSS capability of the GSS Inc I will meet the requirements essential to deploying in support of the National Strategy and any specific Combatant Command requirements. All Ports, Protocols, and Services (PPS) visible to DOD managed network components shall undergo a vulnerability assessment; be assigned to an assurance category; be appropriately registered; be regulated based on their potential to cause damage to DOD operations and interest if used maliciously; and be limited to only the PPS required to conduct official business or required to address quality of life issues authorized by competent authority. GSS Inc I requires assignment of a DOD 8500.2 Mission Assurance Category of 'I' and a Confidentiality Level of 'Classified'.

f. IT/NSS Supportability. The capability, system, and/or service must fully support execution of joint critical operational activities and information exchanges identified in the DOD Enterprise Architecture and solution architectures based on integrated DOD Architecture Framework (DODAF) content, and must satisfy the technical requirements for transition to Net-Centric military operations to include:

(1) Solution architecture products compliant with DOD Enterprise Architecture based on integrated DODAF content, including specified operationally effective information exchanges.

(2) Compliant with Net -Centric Data Strategy and Net-Centric Services Strategy, and the principles and rules identified in the DOD Information Enterprise Architecture (DOD IEA), excepting tactical and non-IP communications.

(3) Compliant with GIG Technical Guidance to include IT Standards identified in the TV-1 and implementation guidance of GIG Enterprise Service Profiles (GESPs) necessary to meet all operational requirements specified in the DOD Enterprise Architecture and solution architecture views.

9. Intelligence Supportability. Networking of dismounted leaders, platforms and sensors down to the individual and small unit with access to network assets will facilitate real time or near real

time information updates to GSS Inc I. The GSS Inc I receives data exchange (situational awareness, orders, and overlay) information from unit Tactical Operations Centers and platforms.

- To support commanders' development of objectives, guidance and intent.
- Maps from FBCB2 are used with Raster feature data (from Compressed ARC Digitized Raster Graphics of 1:250,000 scale and larger), and elevation data (Digital Terrain Elevation Data Level-2 (30m resolution), and High Resolution Terrain data at 10m and 5m resolutions) (T) and imagery data (from, Controlled Image Base at 5m and 1m resolutions) (O) to support the Military Decision Making Process (MDMP).
- Have the ability to download standard map files from a commercial source, geo-references, specific spot locations for demonstrations at various training sites; Fort Lewis/Yakima, Washington; Yuma Proving Grounds, Arizona; National Training Center (NTC), California; Aberdeen Proving Ground (APG), Maryland; Fort Huachuca, Arizona; JRTC, Fort Polk, Louisiana.
- For target development (to include derivation of coordinates), validation, nomination and prioritization.
- Adjacent unit boundaries, phase lines and contaminated areas.
- To support planners at national, strategic and tactical/operational levels.
- To support capabilities analysis and force assignment.
- To support mission planning and execution (e.g., time-sensitive targeting support such as weaponeering, target imagery notation, and coordinate verification at the unit level.
- To support operational execution (e.g., time-sensitive targeting such as target identification, coordinate derivation and weaponeering.
- To support the combat assessment process (to include BDA, MEA and supporting re-attack recommendations).

a. Intelligence Support to Development and Testing. No unique intelligence support requirements to support the testing of GSS Inc I systems have been identified. Threat analysis was provided by the SaaS STAR and is covered in paragraph 4 'Threat Summary' of this document. GSS Inc I systems are fielded and operate within the framework of current force BCT intelligence processes and capabilities.

b. Intelligence Training. The NET program for a BCT requires unit intelligence personnel attend two courses; the Operator Course, and the Mission Data Support Equipment (MDSE) Course.

c. Intelligence Support to Training. New Equipment Training (NET)/Doctrine and Tactics Training (DTT). Once established, the NET and DTT team cadre will play an integral role during operational testing and learning how the systems and organization functions. Functional proponents and integrating centers will participate in the development and review of NET/DTT packages and will be included in the NET/DTT teams. The NET program for a BCT requires the conduct of three courses; the Operator Course, the Mission Data Support Equipment (MDSE) Course, and the Unit Level Maintenance Course. The Materiel Developer (MATDEV) will develop and provide NET to all units being fielded GSS Inc I. Prior to fielding GSS Inc I, the MATDEV will ensure all units receive a New Materiel Information Briefing (NMIB) that will describe not only NET and the impact upon the unit, but the conduct of Doctrine and Tactics Training (DTT). The MATDEV will develop all training products and will resource NETT that will train units during NET. Training products will be delivered as a Training Support Package (TSP), prepared in accordance with (IAW) TR 350-70 and will be verified by USAIS and the TRADOC Capability Manager - Soldier (TCM-S). Existing organizations and billets will provide sufficient support and the GSS Inc I will not require additional, dedicated intelligence personnel.

d. Intelligence Support to Operations. Intelligence personnel will access organic information, intelligence reports, geospatial and imagery data from the NGA, NSA's signals intelligence databases, signature data from the National Signature Program (NSP), and near real-time broadcast from various theater and national intelligence centers to produce a networked fused and tailored threat picture for delivery to the GSS Inc I systems. No new collective intelligence tasks have been identified. There are no anticipated changes to BCT intelligence manpower requirements or processes. Products and services provided by existing BCT systems, processes and personnel (i.e. mapping and geodesy, military targeting assets, etc.) will support GSS Inc I systems operations.

e. Intelligence Security Requirements. The GSS Inc I will use the Battle Command network to process, send, receive, display, and store information classified at multiple levels up to Secret. The GSS Inc I will not support Top Secret (TS)/Sensitive Compartmented Information (SCI). The system must comply with information assurance requirements as stated in applicable policies, directives, and regulations. Information assurance supportability includes availability, integrity, authentication, confidentiality, and non-repudiation, and issuance of an Interim Authorization to Operate (IATO) or Authorization To Operate (ATO) by the Designated Accrediting Authority (DAA). The system hardware must include the provision of safeguards applicable to appropriate classification levels, releasability, and physical facility implications. Paragraphs 6c(4) KSA 4: Geospatial Intelligence, 6d(1)(a)1: Cop Geospatial Data Services (mapping) and 6d(1)(e) Information Assurance sections address IA capabilities for authentication, jam resistant networks, network exploitation, system capture, and encryption. The Soldier requires information with various security classifications. The embedded spreadsheet 'SV-6 System Data Exchange' in Appendix A, Page A-76 depicts security classifications for each IER. GSS Inc I will incorporate all applicable IA protocols and security requirements in DODI 8500.2 that identify the requirements for processing classified information. GSS Inc I systems will not process or transmit SCI. GSS Inc I systems will not interface directly with foreign or coalition information systems; interface remains through current force BCT battle command systems.

f. Potential Intelligence Support Shortfalls. There are no known or projected shortfalls in required intelligence support capabilities, to include manpower, training, processes or systems that would degrade the operational effectiveness of the GSS Inc I or impede system development, testing or Soldier and unit training.

g. Proposed Solutions. No known shortfalls have been identified.

h. Signature Support. GSS Inc I will have sensors that need signatures for algorithm design, and testing. The GSS Inc I program will coordinate with the Signatures Support Program (SSP) to document their signatures requirements in a Life-cycle Signature Support Plan (LSSP) per DODD 5250.01. Signatures (including "blue" signatures) will be provided to the SSP to become part of the distributed national signatures pool, these signatures provided by GSS Inc I program shall adhere to established signature standards to ensure interoperability.

10. Electromagnetic Environmental Effects (E3) and Spectrum Supportability.

a. E3. GSS Inc I electronic components and subsystems shall be self-compatible and mutually compatible in their intended operational Electro-Magnetic Environment (EME). GSS Inc I will operate in the presence of electronic warfare and electromagnetic pulse (T).

(1) E3 shall not degrade the operational performance of GSS Inc I. GSS Inc I shall not be operationally degraded or fail due to exposure to electromagnetic environmental effects, including High Intensity Radio Frequency (HIRF) transmissions or High-altitude Electromagnetic Pulse (HEMP). EMC performance requirements are specified in MIL-STD-464A (platform level) and MIL-STD-461E (equipment and subsystem/system level) for all electromagnetic disciplines. (T)

(2) Potential E3 Compatibility Issues. Potential E3 compatibility issues for the GSS Inc I operating within the BCT are unknown. As the program matures, the materiel developer will identify and resolve these issues and address risk mitigation strategies if needed.

b . DOD Supportability. GSS Inc I will comply with the following: Electromagnetic Environmental Effects (E3) control and Spectrum Supportability. GSS Inc I equipment will comply with all applicable DOD, national, and international spectrum management statutes, policies and regulations, to include obtaining spectrum supportability in all host nations where deployment of the system or equipment is planned. Only transport systems with Military Communications Electronics Board (MCEB) Joint Frequency Panel approved DD Form 1494 documentation will be incorporated. (T).

(1) Permission is expected to be obtained from designated authorities of sovereign ("host") nations (including the United States) to use GSS Inc I within their respective borders. GSS Inc I must operate compatibly with other spectrum-dependent equipment already in the intended operational electromagnetic environment (EME). This information and the requirements of 2 and 3 below will be included in the Safety System Assessment (SSA), which shall be submitted prior to Milestone and Readiness Reviews. (T).

(2) GSS Inc I must be mutually compatible with other systems in their electromagnetic environment and not be degraded below operational performance requirements due to electromagnetic environmental effects. (T)

(3) Spectrum Supportability Procurement or acquisition of this wireless, spectrum dependent device will be conducted IAW DOD guidance (e.g., DODD 3222.3, DODI 4650.01, DODI 4630.8, DODD 5000.1 and DODI 5000.2) as well as applicable Military Department (MILDEP) publications. In accordance with DODI 4650.01, identify and mitigate regulatory, technical and operational spectrum risks using a Spectrum Supportability Risk Assessment. GSS

Inc I documentation will reference the approved DD-1494 of the various candidate radios. The DD Form 1494, Application for Equipment Frequency Allocation, will be releasable for coordination purposes to those foreign countries (host nations) in which permanent deployment or lengthy temporary use is contemplated. The PM acknowledges that, before assuming contractual obligations for deployment, testing, production, or procurement of this spectrum dependent system, the required spectrum support is or will be available in those host nations determined by the PM or procurer for the equipment's intended use. The PM has (will develop) a plan to obtain appropriate equipment allocation guidance/status prior to MS B or MS C as outlined in DODD 4650.1 in order to progress to the next phase.

(4) Spectrum Supportability.

(a) As GSS Inc I evolves to JTRS, system must comply with DOD, national and international spectrum management policies." All future requirements for radio-based communications that fall within the JTRS spectrum range (2MHz-2GHz) will be satisfied by the current JTRS requirements document unless ASD(NII)/DOD CIO grants authorization for a specific procurement, in accordance with DOD policies (references pp and qq).

(b) Selective Availability Anti-Spoofing Module (SAASM). GSS Inc I must comply with CJCSI 6130.01D, which directs specific measures to protect GPS. GSS will use Ground-Based GPS Receiver Application Module (GB-GRAM) SAASM as a component part for installation into the command, control, communications and computer sub-system. GSS will incorporate SAASM standard IAW CJCSI 6140.01A, dated 31 March 04. security, providing extremely accurate and robust Position, Navigation and Timing information to the host system. GSS will use keyed Precise Positioning Service (PPS) rated GPS User Equipment. GSS leaders will carry the RR capability in addition to the GSS GFE radio to conduct voice communication to and from Soldiers. GSS is a Type I Encrypted system and PLI received by the Leader's RR will be directly fed into the GSS by a one way guard to enable the PLI to appear on the leaders GSS display.

(c) Tactical Data Link (TDL). GSS will Implement DOD Joint family of TDL message standards, which includes Link 16 to Air Force Air Craft, GSS Unique and Variable Message Format (VMF) communications to other networked GSS leaders. Interfaces for TDL dissemination across joint interfaces will be implemented through the use of Gateways to combat vehicle platforms. Joint Mission Area interoperability assessment of TDL participants requires identification of platform TDL implementation details.

(d) Bandwidth Analysis. Data Through-Put table 8-2 address Bandwidth requirements the information sharing requirements between GSS, Rifleman's Radio and its potential impact to the Global Information Grid; to include Data Type, Size and Frequency requirements in order for DOD Networks/resources to accommodate the program's future bandwidth considerations that must be made for Terrestrial Transport, Satellite Transport, and Internet Protocol (IP), to include Voice and Video.

(5) GSS Inc I must comply with DODD 4650.1, "Policy for Management and Use of the Electromagnetic Spectrum" Spectrum supportability and E3 control requirements must be addressed IAW 3170.01, as briefly outlined below. (T)

(a) By Technology Development – a Stage 2 (Experimental) spectrum supportability determination must be obtained for GSS Inc I prior to the Technology Development Stage. If a spectrum supportability determination has not been obtained, DODD 4650.1 requires that specific authority be received from the Milestone Decision Authority (MDA) for the program to proceed into the System Development and Demonstration phase and to provide to the USD(AT&L), the ASD(NII), the DOT&E, and Chair, MCEB, a justification and plan to obtain spectrum supportability. Program Manager (PM) will work with the J-6 spectrum office in order to determine justification and plan format. (T)

(b) By System Development and Demonstration – a Stage 3 (Developmental) spectrum supportability determination must be obtained for GSS Inc I prior to Milestone B decision. If a spectrum supportability determination has not been obtained, DODD 4650.1 requires that specific authority be received from the MDA for the program to proceed into the Production and Deployment phase and to provide to the USD(AT&L), the ASD(NII), the DOT&E, and Chair, MCEB, a justification and plan to obtain spectrum supportability. PM will need to work with the J-6 spectrum office in order to determine justification and plan format. (T)

(c) By System Production and Deployment – a Stage 4 (Operational) spectrum supportability determination must be obtained prior to GSS Inc I Milestone C decision. If a spectrum supportability determination has not been obtained, DODD 4650.1 requires that specific authority be received from the MDA for the program to proceed into the Production and Deployment phase and to provide to the USD(AT&L), the ASD(NII), the DOT&E, and Chair, MCEB, a justification and plan to obtain spectrum supportability." PM will need to work with the J-6 spectrum office in order to determine justification and plan format. (T)

(6) GSS Inc I must be compliant with other regulatory documents for spectrum supportability and certification such as the "Manual of Regulations and Procedures for Federal Radio Frequency Management" (a.k.a. "NTIA Redbook"), Office of Management and Budget (OMB) Circular A-11 (Part 2, section 33.4). Additionally, "Electromagnetic Spectrum," of the Defense Acquisition Guidebook (reference aa) where additional information including detailed Milestone requirements, additional mandatory policies, timelines, sample wordings for documents, and other specific requirements for gaining spectrum supportability are covered. (T)

(7) GSS Inc I will document spectrum certification as required. Commercial and non-developmental items must also comply with DOD policy on E3 and Spectrum Supportability. (T)

(8) Host-Nation Approval (HNA). To ensure compatibility as well as interoperability, GSS Inc I, equipment intended for operation in host nations will require HNA approval coordinated by the MCEB and the appropriate combatant commanders prior to use. Spectrum supportability shall be determined for the life of the system through the MCEB. (T)

(9) Need for Frequency Spectrum. The frequency range of GSS Inc I emitters must be

flexible enough to adapt to changes in DOD and non-DOD Government frequency spectrum allocations, as well as civilian bands, both in the US and overseas. Each GSS Inc I emitter requires mutual compatibility with other spectrum dependent devices within its intended environment(s). RF dependent GSS Inc I components must comply with applicable national and international spectrum management statutes, policies and regulations, to include obtaining spectrum supportability in all host nations where deployment is expected. (T)

(10) System Design. The GSS Inc I must ensure the use of the electromagnetic spectrum complies with US national electromagnetic regulations. Electromagnetic emitters must meet radio spectrum guidance requirements specified in DOD Directive 4650.1. Spectrum allocation guidance and authorization will be obtained prior to contractually obligating the government for the modification development, and/or purchase of any spectrum dependent systems. (T)

***Rationale:*** *These steps ensure mitigation of risk that could arise due to incompatibility between GSS Inc I and other RF spectrum dependent systems operating in the same environment. Radio frequency (RF) spectrum support is assessed in an Spectrum Supportability Risk Assessment which is cited in DODI 4650.01. Spectrum regulatory issues are addressed in spectrum certification, frequency assignments, and host nation coordination. RF spectrum allocation is an area requiring careful evaluation for compliance with existing rules and effects of proposed reallocations. Failure to comply with these rules violates DOD and other regulations and can delay programs and negatively affect training and equipment deployability.*

11. Technology Readiness.

a. Critical Technology Elements.

(1) The components comprising GSS Inc I have been derived from previous LW technology and Soldier Advanced Technology Demonstration (ATD) equipment tested and proven in operational use during OIF. These previous efforts indicated the GSS Inc I critical technologies are the computer, the GPS navigation augmentation and the radio antenna. Further analysis will take place to determine if requirements such as network management, radio frequency spectrum availability, etc will result in additional GSS Inc I critical technology elements. An independent assessment by the Army S&T Executive will be performed as part of the TRA preparation for Milestone C.

(2) The technology maturation schedule, Table 11-1 indicates the Technology Readiness Level (TRL) that each component is expected to achieve throughout the TD Phase, which starts at Milestone (MS) A. The current TRL-5 (Component or Breadboard Validation in Relevant Environment) is based on the work done by the Sep 2007 Future Force Warrior Program ATD, and the lessons learned from the LW-SI system in Operation Iraqi Freedom. The technologies identified in the table are expected to mature throughout this phase to reach a minimum of TRL-7 (System Prototype Demonstration in an Operational Environment). Developmental Testing (DT) will address the technical performance and all of the environmental requirements. Survivability is addressed during the LUT. Reliability is examined beginning at field evaluation and continuing during DT and LUT. Logistics considerations (reliability, maintainability, material availability, MTBF, etc.) is addressed during LUT and Log Demo. A draft Technical

Readiness Assessment must be developed after CDR to assist in the evaluation of technology maturity during the LUT.

(3) A LUT is scheduled in 4QFY 10 to support the LRIP decision at the MS C. If the system has not matured to the point to allow the PM to go directly to a MS C, a second LUT will be conducted to support the LRIP decision after additional development. In both cases, an IOT&E must be conducted to support a FRP decision. The expected scope of the LUT is a company with two GSS Inc I equipped platoons. The expected scope of the IOT is a battalion with two GSS Inc I equipped companies. Further details are included in the TEMP.

(4). If a TRL-7 (System Prototype Demonstration in an Operational Environment) is not achieved, the program will proceed to MS B. A separate TRA is planned for each GSS increment.

**Table 11-1 Technology Maturation Schedule**

| Component | MS A 4QFY08 | IPR (SFR) 2QFY09 | IPR (PDR) 3QFY09 | IPR (CDR) 1QFY10 | TEST (LUT) 4QFY10 |
|---|---|---|---|---|---|
| Computer | 5 | 5 | 5 | 6 | 7 |
| Navigation Augmentation | 5 | 5 | 5 | 5/6 | 7 |
| Radio Antenna | 5 | 5 | 5 | 5/6 | 7 |

b. Manufacturing Readiness Challenges. Based upon the fact GSS Inc I is a COTS/NDI integration effort, and the components of this effort lie within industry core competencies associated with mobile computing, there are not any forecasted manufacturing readiness challenges associated with GSS Inc I at this time.

12. Assets Required to Achieve Initial Operational Capability (IOC). The total number of GSS Inc I required to achieve IOC will consist of the sum total of GSS Inc I and maintenance spares required to equip a BCT beginning in FY12. For planning purposes, the number of assets required to achieve IOC is 2299 systems, which is 641 GSS Inc I and 1658 RR for one BCT.

13. Schedule and IOC / Full Operational Capability (FOC) Definitions.

a. Considerations Driving the Incremental Plan. The prototyping phase will occur in FY 09 thru FY 10. Once the prototyping phase is complete, two schedule outcomes are possible.

(1) Outcome #1: If the prototyping phase produces a system of sufficient technical maturity, the system will progress directly from Milestone A in FY 09 to a Milestone C decision in FY 11.

(2) Outcome #2: If the prototyping phase does not produce a system of sufficient technical maturity, the system will follow a typical development approach with the Milestone A followed by a MS B in FY 11 and an MS C in FY 12.

b. Milestone A to Milestone C.

(1) Increment I First Unit Equipped (FUE ). FUE means complete issue of GSS Inc I and RR, and New Equipment Training (NET) to a Battalion size organization in FY 12.

(2) IOC. IOC means complete issue of GSS Inc I and RR, NET, and Training Aids Devices Simulators and Simulations (TADSS) are available, and the maintenance and support structure is in place to support the first BCT. GSS Inc I IOC is FY 12.

(3) FOC. FOC means complete issue of GSS Inc I, RR, NET and TADSS available to support 30 BCTs (29 BCTs plus 1 additional BCT for unit training) to include maintenance and support structure. Priority of issue is to the BCTs and then other units according to Army priorities. GSS Inc I will be fielded in support of Army Capability Packages providing 2 BCTs in FY12, 4 in FY13, and 8 BCTs per year in FYs 14 to 16. GSS Inc I will be integrated into schools for familiarization training, in professional development courses and operator and maintainer training courses beginning in FY13. Additional 93 GSS Inc I sets required to support TRADOC (Maneuver Battle Lab and Army Evaluation Task Force) will be provided beginning in FY13. GSS Inc I FOC is FY 16.

c. Milestone A to Milestone B.

(1) Increment I FUE. FUE means complete issue of GSS Inc I, RR capability, and NET to a BN size organization. GSS Inc I FUE is FY 13.

(2) IOC. IOC means complete issue of GSS Inc I, RR capability, NET, and TADSS are available, and the maintenance and support structure is in place to support the first BCT. GSS Inc I IOC is FY 14.

(3) FOC. FOC means complete issue of GSS Inc I, RR capability, NET and TADSS are available to support 30 BCTs (29 BCTs plus 1 BCT for unit training); to include maintenance and support structure. Priority of issue is to the BCTs and then other units according to Army priorities. GSS Inc I will be fielded in support of Army Capability Packages providing 2 BCTs in FY14; 4 in FY15; and 8 BCTs per year in FYs 16 to 18. GSS Inc I will be integrated into schools for familiarization training, in professional development courses and operator and maintainer training courses beginning in FY14. Additional 93 GSS Inc I sets required to support TRADOC (Maneuver Battle Lab and Army Evaluation Task Force) will be provided beginning in FY14. GSS Inc I FOC is projected in the FY 18 timeframe.

14. Other DOTMLPF and Policy Considerations.

a. Doctrine. Multiple doctrinal changes are necessary in training, maintenance, and tactics manuals. In order to standardize the system’s TTPs across the Army as well as specify tactical employment, it will be necessary to modify ARTEPs, MTPs, CTT manuals, and squad, platoon, and company level field manual’s and SOP’s. These changes would become the responsibility of the MCoE.

b. Organization. No organizational changes are planned or required for fielding of the GSS.

c. Training.

(1) System Training Plan (STRAP) Summary. GSS training will produce leaders who are proficient in operating and maintaining the GSS Inc I, leaders who can effectively plan for and employ it in combat operations, and units that can execute and sustain GSS operations and training. NET will provide initial proficiency for individual operator and maintenance skills, as well as leader's employment skills. Institution training on employment of the GSS Inc I will be integrated into Professional Military Education courses. Unit or operational training is responsible for sustaining individual and organizational proficiencies after NET, relying on unit trained NCOs and the training support package(s) delivered during NET. Upon fielding of the GSS Inc I to the force, the US Army Infantry School will commence two new functional courses designed to qualify leaders on the operation, maintenance, and employment of the GSS Inc I for duty in a GSS Inc I equipped unit.

(2) Individual, Unit, and Crew Training. Initial operator training for the GSS Inc I consists of performance oriented, hands-on training. This training program shall include an introduction, which covers the capabilities of the system and identification of components; the assembly, installation of batteries, disassembly, donning, and doffing of the GSS Inc I; operation of the computer system which includes initialization, logging on, interfacing with the graphic displays, use of the various input devices, accessing the various programs (weapons peripherals, maps, embedded features, and reach-back routines), zeroizing or purging the system, and shutting the system down; operating the communications features to perform voice communications; to create, edit, send, manage, and receive messages and free text; use of the maps to include uploading, downloading, accessing, learning the symbology, familiarization on the selection, creation, editing, storing, retrieving, and transmitting of various overlays, and position location information/features. Training will include initial and annual IA training in accordance with DODD 8570.01. Employment and leader training will cover characteristics and capabilities of the GSS Inc I; planning and employment considerations; maintenance and sustainment; and an overview of TTPs. Those selected as unit GSS instructors will receive, in addition to all the training above, training on planning and execution of individual and collective sustainment training and the operation and maintenance of GSS TADSS. The GSS Inc I will be integrated into unit collective training at team through company level.

(3) NET. The GSS Inc I is fielded to units under the Unit Set Fielding (USF) concept at home station. Units are fielded the GSS Inc I, all applicable TADSS, and Training Support Package (TSP) during USF. NET focuses on three functions – operations and maintenance, employment of GSS Inc I, and conducting unit sustainment training. NET is consolidated at Brigade Combat Team level (or higher where the fielding plan and unit schedules permit). The NET team is composed of contractors and will use a train the trainer approach. NET will leverage computer based training, Interactive Multimedia Instruction, and leaders will train on the newly fielded systems, and TADSS. The NET concept requires the materiel developer conduct a GSS Inc I NET of all combat and combat support leaders within a BCT. During the BCT NET, the intent is to train all personnel on the platforms they will use; not just the GSS Inc I. Hence, a high degree of pre-NET coordination by concerned proponents and the materiel developers is required.

(a) Test Training Support Package (TTSP). The PM in conjunction with USAIC Soldier Requirements Division (SRD) and USAIC, Directorate of Training (DOT) will develop an initial TTSP and provide it to the test manager at least 12 months before Operational Testing (OT). The PM will deliver the final TTSP not later than (NLT) 90 days before testing begins. The initial TTSP will also support Instructor and Key Personnel Training (IKPT) as well as user training for OT. The TTSP will meet content requirements established in TRADOC Regulation 370-50, paragraph II-6-4.

(b) NET TSP. The PM in conjunction with USAIC SRD and DOT will develop a TSP to support NET. It will be based on the TTSP, modified by lessons learned during OT. The TSP will meet content requirements established in TRADOC Regulation 370-50.

(c) Interactive Media Instruction (IMI) Products. The PM in conjunction with the USAIC SRD and DOT will develop IMI modules which will support individual training in the institutional, operational, and self-development domains. The three training modules, operation and maintenance, employment, and conduct unit training, are included in the TSP fielded in NET. The modules will provide standalone computer based training as well as web-based training over the Internet.

(4) Reach-back Training. All institutional courses must be available in IMI as either Computer Based Training (CBT) in a standalone digital media format or as web-based training hosted on the Army Learning Management System. Courseware will comply with the Shareable Content Object Reference Model (SCORM).

(5) Unit and Institutional TADSS. Three simulators should be evaluated and considered for development in support GSS Inc I training. These TADSS will primarily support the operational training domain.

(a) Desktop Trainer. This largely software solution would allow training on individual operator tasks on the GSS Inc I on a typical personal computer or laptop. The system would use the actual operator controller unit interfaced with the computer to provide simulated GSS Inc I operations in varied scenarios and missions.

(b) Multiple Integrated laser Engagement System (MILES). N/A, normal blank fire can produce similar result based on the acoustic signature.

(c) Integration into Close Combat Tactical Trainer (CCTT). The capability to host simulations of GSS Inc I operations within the CCTT's Virtual Soldier Module and the Virtual Soldier Multifunction Workstation would allow not only individual operator training but also its integration into collective training within CCTT.

(6) Combat Training Center Instrumentation and Interface Requirements. GSS Inc I will be integrated into Combat Training Center (CTC) instrumentation. Live Force-on-Force (FOF) training at home station, local training areas, maneuver CTC, and deployed training sites are

required to validate the ability of units to employ GSS Inc I within the force, and mission rehearsal needs.

(7) MOS Specific Training in the Institutional Training Base. There is no MOS specific training conducted at the institutional training base validated during Pre-LUT Logistics Demonstration.

(8) Post Fielding Training Effectiveness Analysis (PFTEA). A post-fielding training evaluation ensures GSS Inc I training capabilities trains leaders, and units to standard. The PM will fund a USAIC conducted PFTEA approximately 1-year following FUE.

d. Materiel. GSS Inc I does not require any new materiel in order to enable the total capability of the system.

e. Leadership & Education. Institution training on employment of the GSS Inc I will be integrated into Professional Military Education courses. Unit or operational training is responsible for sustaining individual and organizational proficiencies after NET, relying on unit trained NCOs and the training support package(s) delivered during NET.

f. Personnel. GSS Inc I will not require the identification and management of Soldiers based on GSS Inc I experience and will not require the creation of a GSS specific MOS, Area of Concentration (AOC), Additional Skill Identifier (ASI), Specialty Skill Identifier (SSI), or Project Development Skill Identifier (PDSI).

g. Facilities.

(1) Facility, Shelter, and Supporting Infrastructure. Lessons learned from the execution of OT will assist in determining the range capabilities required to support NET, unit containment training, and institutional training.

(a) Classrooms. Lessons learned from the execution of the GSS Inc I OT may cause a modification to the classroom requirements for NET, unit sustainment training, and institutional training.

(b) Training Areas/Ranges. It is anticipated the current Army's training areas and ranges will support training for the GSS Inc I. Lessons learned from the execution of OT may highlight certain terrain features that enhance or in some way affect GSS Inc I training.

(c) Storage Facilities. The GSS Inc I may require storage facilities beyond the current arms room capabilities of some units. A number of units receiving the GSS Inc I may require a Military Construction Army (MCA) project for GSS Inc I storage facilities in garrison environments. Each unit will require an assessment to determine adequate storage facility requirements prior to fielding the GSS Inc I. Each GSS Inc I system requires an estimated 2 cubic feet square feet of storage space. The home-station training activity will require a secure room for the GSS Inc I. Weapons and sensitive items are stored IAW US Army regulations.

(d) Battery Charging Facilities. PM-SWAR is currently in the process of determining the battery charging facilities capabilities caused by the fielding of GSS Inc I to operational units. The TRADOC Installation and Home-Station Training Activity, however, will require a battery charging facility contiguous with the classroom.

(e) Logistics and Maintenance Facilities. PM-SWAR, with the assistance of the Army Communities of Excellence (ACOE), is currently in the process of determining the logistics and maintenance facilities capabilities caused by the fielding of GSS Inc I.

(2) Transportation and Deployability. The GSS Inc I will use existing and projected transportation assets available upon fielding. GSS Inc I will not increase capabilities for cargo or personnel space on current Air Mobility Command (AMC) aircraft, other Air Force aircraft, Army and normal military, ground transportation modes. The GSS Inc I shall be transportable on standard road, rail, marine, or air transport systems. It shall also be capable of being deployed through air assault and airborne operations. The GSS Inc I shall be packaged to support transportation in a commercially acceptable container that will adequately protect it from damage and the elements.

(3) Maintenance. The PM will perform a Business Case Analysis to determine a supportability strategy to assure system availability and reliability. The strategy will include plans to support the system in peacetime, as well as in the event of a contingency.

(a) Performance Based Logistics (PBL) or Performance Based Agreements (PBA) Requirements. As applicable, Contractor Logistics Support (CLS) or Interim Contractor Support (ICS) contract(s) must include provisions for maintenance data collection to include: maintenance performance data, maintenance task frequency for both field and sustainment level tasks, man-hour data for task performance duration, repair part usage, demand stockage, and all other repair part provisioning information. The contractor will use a Logistics Information Systems (LIS), formerly known as Standard Army Management Information Systems (STAMIS) reporting format or a format that is readily convertible to the LIS. This information will be consolidated and reported to the Government Program PM on a monthly basis for the duration of the CLS contract. All other PBL or PBA requirements and metric analysis will be developed and managed by the PM. The PM will develop a transition plan as part of the system Supportability Strategy (SS). Planning for transition from CLS to organic support is essential to continuous sustainment of the fielded systems. The content of the transition plan must include:

1. Logistics functions included in the CLS.

2. The length of time CLS will be required.

3. Procedures for possible extension of the CLS.

4. Funding requirements.

5. Control structure for CLS.

6. A checklist of actions to be completed before transition can take place.

7. Milestone dates for major actions leading up to transition date.

8. Tracking and reporting procedures for transition.

9. Contract data on maintenance actions, repair parts consumption, and other data beneficial to establishing organic support.

(b) Level of Repair Analysis: A LORA will be conducted as part of the Logistics Management Information (LMI) collection process. As part of the post deployment evaluation, the LORA will be rerun no earlier than 1 year and no later than 3 years from the FUE date using reliability data collected by materiel developers from fielded equipment. The LORA will be rerun every 5 years throughout the system's life cycle. Military Standard (MIL STD) Technical Manual (TM) Maintenance Allocation Charts (MAC) will be updated to reflect any changes in the LORA outcome. The PM shall resource this effort throughout the system life cycle.

(c) Provisioning Plan. Contract performance specifications must include provisions to provide National Stock Number (NSN) data for spares to the Government. The PM will fully provision and resource sufficient spares to ensure each unit fielded the GSS Inc I maintains the reliability requirement outlined in KSA 1.

(d) Supportability Test & Evaluation Program. The GSS Inc I will undergo a logistics demonstration to verify operator and maintenance tasks, capture projected annual maintenance man-hour data, and form the basis for developing the BOIPFD and Manpower Requirements Criteria (MARC) data. Contractor validated TMs are required for the logistics demonstration. The Government verification process of the TMs will be completed after Operational Testing and prior to the TM authentication and Full Materiel Release. These requirements will be identified within the system Test and Evaluation Master Plan (TEMP).

(e) Technical Data:

1. Technical Manuals. TMs for Operators and Maintainers will be produced to MIL STD and undergo a contractor validation and Government verification process to ensure accuracy and completeness. Operator, field and sustainment levels of maintenance will be called out in the Maintenance Allocation Chart (MAC) found in the Field and Sustainment Maintenance TMs. All calibration requirements, procedures, schedules will be identified in operator and maintainer TMs.

2. Technical Data Package. The technical data package for each GSS Inc I component will be procured by the Government to accommodate cost effective material change, configuration control, re-procurement, and parts commonality requirements.

(4) Support Equipment. Also see sustainability requirements in paragraph 6.b.(5) KPP 5 Sustainment (Availability).

(a) Required Test, Measurement, and Diagnostic Equipment (TMDE) is at the field level of support, and provides a cost effective means of unambiguous fault isolation to the failed LRU level with at least 95% accuracy.

(b) Electro-optic LRUs must be evaluated for fault isolation, if required to a Shop Replaceable Unit (SRU) by use of Integrated Family of Test Equipment (IFTE).

(c) Power Source. Equipment for restoring power to power sources, operated by the battalion task force, must be designed to minimize manpower, charge time, transportation, float assets, and tactical electrical power. Battery charging equipment must be deployable with units to remote operating and training locations with capability integrated as needed into the unit standard logistic re-supply concept.

(d) TMDE. TMDE, Built-In-Test (BIT) and Built-In-Test-Equipment (BITE) are required to provide a cost effective means of unambiguous fault isolation to the failed LRU with at least 80% (T), 95% accuracy (O). The BITE/BIT function will isolate and report failures or faults that have occurred in the system and used to the maximum extent possible. Electro-optic LRU's must be evaluated for fault isolation to a SRU by use of IFTE (T). The extent of on-line BIT and off-line TMDE to isolate faults of an LRU must be determined by the LMI process.

(e) Data Support Equipment. Generate, maintain, and transfer mission data packages containing critical information required to enable GSS Inc I equipped leaders to conduct operational missions. Mission data packages shall consist of digital maps, overlays, OPORDs, Unit Task Organization (UTO)-data, images, logistics authorizations data, unit-generated help documents in HyperText Markup Language (HTML) and technical reference materiel’s. Data support equipment should provide the capability to prepare and edit Soldier access devices.

(5) Environmental Compliance Requirements. The user shall have the ability to train, operate, maintain, and dispose of the system in full compliance with applicable US, foreign, and international environmental quality laws and regulations. The design, production, operation, maintenance and disposal of the system shall eliminate, or minimize to the greatest extent possible, adverse environmental quality impacts. Ozone depleting substances will not be utilized in the production of the GSS Inc I.

15. Other System Attributes.

a. Storage Temperature. The GSS Inc I must be not be affected under storage conditions from -28° F (-33° C) to +160°F (+71° C).

b. Embedded Instrumentation. The GSS Inc I will have embedded diagnostics that can identify errors or faults down to the LRU/Line Replaceable Module (LRM) level.

c. Environment, Safety and Occupational Health. The user of GSS Inc I shall have the ability to train, operate, maintain, and dispose of the systems in full compliance with existing and near-term forthcoming applicable US, foreign, and international environmental quality policies, statutes, laws and regulations. The design, production, operation, maintenance, and disposal of

GSS Inc I shall eliminate or minimize to the greatest extent possible adverse environmental quality impacts. GSS Inc I shall not require the use of ozone depleting chemicals during any phase of their life cycles. GSS Inc I will comply with safety and occupational health laws and regulations. See MANPRINT considerations in paragraph 15.h.

***Rationale**: GSS Inc I systems must comply with all applicable environmental, safety, and occupational health requirements to prevent environmental, health and safety hazards to users.*

d. Information Operations and EW. The primary threat to GSS information systems comes from IO. Supported by intelligence exploitation, the IO threat includes the following tactics: Computer Network Attack (CNA), Computer Network Exploitation (CNE), EW, perception management, and physical attack. The GSS Inc I requires protection from these attacks to enable dispersed small units to collaboratively influence larger areas with greater precision, speed and a broader variety of lethal effects. This protection ensures leaders receive the actionable information they need to sustain collaborative initiative.

e. Conventional Weapons Effects. The GSS Inc I is not designed to survive conventional weapons effects.

f. Initial Nuclear Weapons Effects (INWE). The GSS Inc I is not a INWE mission critical system. The equipment shall withstand the INWE of thermal radiation, and initial nuclear radiation to the same level where 50% the operators remain combat effective long enough to execute the mission (T). Equipment shall also be survivable to High-Altitude Electromagnetic Pulse (HEMP) to the levels specified in MIL STD 2169B. GSS Inc I does not have to operate through a HEMP event. System can be rebooted or power cycled to return to operational configuration within 30 minutes after a HEMP event (T).

g. CBRN Survivability. The GSS Inc I is not mission critical, but it will be CBRN contamination and decontamination survivable against the effects of CBRN agents and decontaminates so that it remains operational in all CBRN environments, with the exception of rubber and canvas field replaceable items, and is compatible with personnel operating and maintaining while in MOPP IV.

h. Detonation Prevention/Unplanned stimuli. The GSS Inc I must not produce a stimulus that detonates its own armaments while carried.

i. Human System Integration/MANPRINT. Each MANPRINT Domain assessment plays an integral role throughout the GSS Inc I lifecycle by providing insight into the Soldier system integration aspects of system development. MANPRINT will receive equal consideration with technical and programmatic decisions during system trade-off analyses. The system will not require addition of manpower for operations or support to the force structure. It will require no change in personal characteristics of Army personnel

***Rationale**: These factors, if left out of critical design, will increase the size of the military footprint*

(1) Manpower. Introduction of the GSS Inc I capability shall not increase the overall number of personnel, both, military and civilian, required to operate, maintain, and support the item.

(2) Personnel. The operation, maintenance, and support of the GSS Inc I capability shall not require aptitudes, skills, or capabilities beyond those currently present in the user population.

(3) Training. The instruction and resources required providing the Warfighter and maintainer with knowledge, skills and abilities in proper operation, maintain, and support Army systems shall not significantly increase due to the introduction of the GSS Inc I capability.

(4) Human Factors Engineering. The GSS Inc I capability design shall promote effective Soldier-machine integration for optimal total system performance. Design principles taking into account human capabilities and limitations shall be incorporated into system definition, design, development, and evaluation. This includes concepts of human-computer interface (e.g., ease of perception and comprehension of displays, ease of use of controls) and compatibility of GSS Inc I capability with other mission-essential equipment (including but not limited to use with standard combat gear, CBRN, and environmental clothing). The GSS Inc I capability should not interfere with the performance of common Soldier tasks. Equipment design must consider mission-dependent tasks and demands through consultation with subject matter experts in order to maximize ease of use, minimize workload and enhance mission performance.

j. Safety Issues. MANPRINT addresses Soldier system safety issues.

(1) Safety requirements needed to protect the system. All safety hazards require elimination or reduction to an acceptable level of risk as outlined in appropriate regulations and MIL Standards and as required to enable effective system operation and maintenance. The design process will include a system safety assessment to ensure the system is free from conditions, which can cause death, injury, or illness to target audience leaders. This assessment must be updated prior to each Milestone Decision Review (MDR) and results will be incorporated into the MANPRINT assessment submitted prior to each MDR.

(2) Hazards of Electromagnetic Radiation to Personnel (HERP). Health Hazards: The system shall be designed to eliminate or control all potential health hazards including noise, heat, musculoskeletal trauma, radiation (laser, radiofrequency, and ionizing), and chemical substances. A health hazard assessment is required, completed as early as practical in the program to identify potential health hazards. The health hazard assessment will be updated prior to each MDR and results incorporated into the MANPRINT assessment.

(3) Hazards of Electromagnetic Radiation of Ordnance (HERO). The system shall be designed to eliminate high intensity radio frequency fields produced by modern radio and radar transmitting equipment which can cause sensitive electro explosive devices (EED's) contained in ordnance systems to actuate prematurely.

(4) Documentation of Hazards. The resolution of all hazards must be formally documented through a hazard tracking system and the risk associated with the residual hazard, if any, accepted by the designated approving authority IAW AR 385-10/DA Pam 385-16.

k. Expected Mission Capability. The GSS Inc I is mission capable in all environments. The system must meet basic cold and hot weather conditions and remain operational in adverse weather conditions with no more than 20% degradation of basic capabilities (Conduct GSS Operations; Execute BC; Subscribe, Get and Publish information; Populate COP; Provide Position location) as required by MIL STD-810G environmental testing.

l. Physical and Operational Security Needs. The GSS Inc I is considered physically secured when it is being worn by the Soldier. When not carried and the information stored in the system is classified, this device will be secured (IAW AR 190-11) in an arms room or safe behind locked doors. If the GSS Inc I has been "purged" of all classified information the system is physically secured in the same way as other property book items.

16. Program Affordability. The life cycle cost of the GSS Inc I is a key factor in determining the Army's ability to provide this critically needed capability for the ground Soldier.

a. Life cycle cost. The life cycle cost estimates provided below are for developing, procuring, and sustaining 30 Infantry Brigade Combat Teams (IBCT) of GSS Inc I and its training base. The Basis of Issue (BOI) per IBCT for the GSS Inc I is based on current TRADOC ARCIC planning of 2299 (641 GSS Inc I and 1658 Rifleman Radios).

**Table 16-1 GSS Inc I Life-Cycle Costs (BY2009, $M)**

| Objective | Threshold |
|---|---|
| $3,929 | $4,322 |

b. Affordability.

**Table 16-2 GSS Inc I Program Affordability**

| TY,$M | FY10 | FY11 | FY12 | FY13 | FY14 | FY15 |
|---|---|---|---|---|---|---|
| RDT&E Cost | 57.3 | 36.1 | 25.6 | 12.3 | 3.0 | 1.9 |
| Funding | 57.3 | 36.1 | 25.6 | 12.3 | 3.0 | 1.9 |
| UFR | 0 | 0 | 0 | 0 | 0 | 0 |
| Procurement Cost | 1.8 | 110.5 | 235.4 | 525.4 | 532.2 | 540.4 |
| Funding | 1.8 | 110.5 | 235.4 | 525.4 | 532.2 | 540.4 |
| UFR | 0 | 0 | 0 | 0 | 0 | 0 |
| Sustainment Cost | 0 | 0 | 0.7 | 1.5 | 3.3 | 0 |
| Funding | 0 | 0 | 0 | 0 | 0 | 0 |
| UFR | 0 | 0 | 0.7 | 1.5 | 3.3 | 0 |
| Total UFR | 0 | 0 | 0.7 | 1.5 | 3.3 | 0 |

(1) Research, Development, Test, & Evaluation Appropriation (RDT&E). The RDT&E costs shown above are for the development and testing of GSS Inc I (including the Pre-planned Product Improvement)

(2) Other Procurement Army (OPA) Appropriation. The GSS Inc I procurement estimate is based on procuring 30 IBCTs beginning in FY 11 with two IBCTs, four IBCTs in FY12, and eight IBCTs per year from FY13-15. Estimates include NET and Fielding one year after procuring systems.

(3) Operation and Maintenance (O&M) Costs. The GSS Inc I O&M costs fall outside the FY10-15 window that is part of the Program Affordability Table 16-2. GSS Inc I sustainment costs for a 15 year operational life cycle are part of the GSS Inc I Life Cycle Costs within Table 16-1.

(4) Other Costs. Other costs such as Military Personnel are currently not broken out separately. No additional military personnel are currently envisioned to operate or maintain the GSS Inc I.

(5) Total System Production Quantities. TRADOC ARCIC and the SaaS Integrated Concept Team (ICT) representing every Army MOS, developed a process to calculate a quantity requirement of GSS Inc I in Army Brigade types. TRADOC ARCIC will continue to conduct analysis to refine the BOIP. The modular Army is made up of an approximate total of 73 Brigades and the GSS Inc I Army Acquisition Objective may increase beyond the current 30 IBCTs to include SBCTs, Heavy Brigade Combat Teams (HBCT) and other brigades IAW ARFOGEN.

c. Source of funding. The source of funding for this program is a new start program in FY 09.

## Appendix A NR-KPP CAPABILITY DESCRIPTION DOCUMENT (CDD)
## For
## Ground Soldier System GSS Increment I
## TAB A Views

### 1. Net Ready Key Performance Parameter NR-KPP Statement

| KPP | Threshold (T) | Objective (O) |
|---|---|---|
| Net-Ready: The capability, system, and/or service must support Net-Centric military operations. The capability, system, and/or service must be able to enter and be managed in the network, and exchange data in a secure manner to enhance mission effectiveness. The capability, system, and/or service must continuously provide survivable, interoperable, secure, and operationally effective information exchanges to enable a Net-Centric military capability. | The capability, system, and/or service must fully support execution of joint critical operational activities and information exchanges identified in the DOD Enterprise Architecture and solution architectures based on integrated DODAF content, and must satisfy the technical requirements for transition to Net-Centric military operations to include:<br>1) Solution architecture products compliant with DOD Enterprise Architecture based on integrated DODAF content, including specified operationally effective information exchanges<br>2) Compliant with Net -Centric Data Strategy and Net-Centric Services Strategy, and the principles and rules identified in the DOD Information Enterprise Architecture (DOD IEA), excepting tactical and non-IP communications<br>3) Compliant with GIG Technical Guidance to include IT Standards identified in the TV-1 and implementation guidance of GIG Enterprise Service Profiles (GESPs) necessary to meet all operational requirements specified in the DOD Enterprise Architecture and solution architecture views<br>4) Information assurance requirements including availability, integrity, authentication, confidentiality, and non-repudiation, and issuance of an Interim Authorization to Operate (IATO) or Authorization To Operate (ATO) by the Designated Accrediting Authority (DAA), and 5) Supportability requirements to include SAASM, Spectrum and JTRS requirements. | The capability, system, and/or service must fully support execution of all operational activities and information exchanges identified in DOD Enterprise Architecture and solution architectures based on integrated DODAF content, and must satisfy the technical requirements for transition to Net-Centric military operations to include<br>1 Solution architecture products compliant with DOD Enterprise Architecture based on integrated DODAF content, including specified operationally effective information exchanges<br>2) Compliant with Net -Centric Data Strategy and Net-Centric Services Strategy, and the principles and rules identified in the DOD IEA, excepting tactical and non-IP communications<br>3) Compliant with GIG Technical Guidance to include IT Standards identified in the TV-1 and implementation guidance of GESPs, necessary to meet all operational requirements specified in the DOD Enterprise Architecture and solution architecture views<br>4) Information assurance requirements including availability, integrity, authentication, confidentiality, and non-repudiation, and issuance of an ATO by the DAA, and 5) Supportability requirements to include SAASM, Spectrum and JTRS requirements. |

**Table A-1. NR-KPP Compliance Statement**

**2. DOD Information Enterprise Architecture (DIEA)**

a. IAW CJCSI 6212.01E, the GSS program architecture will comply with current version of the DOD Architecture Framework (DODAF) and as guided by the laws, regulations, and policies defined in the rules and constraints of the DOD Information Enterprise Architecture reference [DODD 8000.01], including the DOD Information Enterprise Architecture (DOD IEA) (reference tt).

b. GSS does not directly connect to the GIG. The GSS leverages the Battle Command proxy services and has no inherent contiguous transport connectivity to the GIG. GSS broadcasts Position Location Information (PLI) to multicast groups via VMF messages over the transport layer only. There are no underlying web services utilized. GSS operates exclusively in a "disadvantaged" communications environment . GSS employs only JVMF messages for data exchange. GSS employs "Point to Point" information exchanges. In addition, select information exchanges are broadcast. Specifically the K05.1 "Position Report/Update" and K05.19 "Entity Data" messages are broadcast to subscribers of the Community of Interest (COI) "GSS SA Multicast Group." GSS uses Internet Protocol (IP) as a means of communication.GSS is secret high providing information exchanges with a pre-defined pool of BCT members identified in Data Product for that unit. GSS does not support a NCES Federated Search which results in an active link, that when selected provides access to a website displaying the PLI data.

c. The GSS PLI DDMS compliant metadata would be stored within an Enterprise Catalog. GSS program restrictions on Foreign National access to program data will warrant review. GSS will not post PLI onto the web or web pages. GSS will broadcast PLI to multicast groups via VMF messages over the transport layer only. There are no underlying web services (Defined by WSDL) utilized. GSS exposes data directly to users/operators. DDMS – DoD Discovery Metadata Specification, GSS Position Location Information in the Enterprise Catalog will be based on the JVMF message standard, the Metadata for the GSS PLI cannot be posted within the unclassified NIPRNet instance of the DoD Metadata Registry.

d. Only GSS Enterprise level data or Services consumed by GSS Inc I are map products. As a Secret high system the COI primarily consists of leaders at Company level of command and lower.

(1) Net Centric Data Strategy. The Maneuver Center of Excellence (MCoE) has documented a synchronized, time phased requirement for achieving GSS Inc I external interoperability with other IBCT digital network assets.

(a) Phase 1: Required at GSS Inc I IOT&E test event and through fielding of the first four BCTs: GSS Inc I will exchange friendly and shared survivability situational awareness information to the level defined in KPPs 3 and 4 with other GSS Inc I systems in its area of operations/area of interest as defined in KPP2. GSS Inc I will also be capable of exchanging C2 messages (orders, overlays, SPOT reports, etc) with other GSS Inc I systems as defined in this CDD. Additionally GSS Inc I will be capable of interoperating with the RR capability as defined in the KSA 7 when it becomes available.

(b) Phase 2: Required by BCTs 5 through 30: Building upon Phase 1, GSS Inc I will exchange friendly and shared survivability situational awareness information to the level defined in KPP 3 and 4 with existing fielded Army Battle Command Systems defined in Capability Sets 13/14 and 15/16 in its area of operations/area of interest as defined in KPP2. GSS Inc I will also be capable of exchanging C2 messages (orders, overlays, SPOT reports, etc) with fielded Army Battle Command Systems as defined in this CDD.

**3. Global Information Grid (GIG) Technical Guidance (GTG)**

**3.a DISR Compliance: Refer to the TV-1 page A-76.**

**3.b Key Interface Profiles (KIPs) Compliance Statement**

J6 & DISA have restructured the 17 original KIPs into three KIPs families. These new KIPs are service oriented and designed to help enable Programs of Record (PORs) to be Network Centric. KIPs are stored in DISR online within the DISR Profile Registry Area. This is the authoritative list of the technical standards specifications that compose a KIP.

Table A-2 identifies those KIPs that apply to GSS

| **Key Interface Family/Name** | ***VER*** | ***Applicable*** **(Yes or No)** | DISR Status **(Emerging or Mandated)** | **Implementation Phase (Objective or Threshold)** | **Consumer or Provider** | **Implementation Issues/KIP Options** |
|---|---|---|---|---|---|---|
| **Transport Family** | *1 Jun 07* V 2.0 | | | | | |
| UHF-Band SATCOM | | No | NA | NA | NA | NA |
| C-Band SATCOM | | No | NA | NA | NA | NA |
| X-Band SATCOM | | No | NA | NA | NA | NA |
| Ku-Band SATCOM | | No | NA | NA | NA | NA |
| Ka-Band SATCOM | | No | NA | NA | NA | NA |
| EHF/AEHF Band | | No | NA | NA | NA | NA |

| | | | | | | |
|---|---|---|---|---|---|---|
| SATCOM | | | | | | |
| Global Broadcast System | | No | NA | NA | NA | NA |
| DISN IP Router Network Layer (Table H-6 of DISN Upgrade KIP) | | No | NA | NA | NA | NA |
| DISN IP Router Optional Network Interface Feature (Table H-7 of DISN Upgrade KIP) | | No | NA | NA | NA | NA |
| DISN IP Router Physical and Data Link Layers (Table H-5 of DISN Upgrade KIP) | | No | NA | NA | NA | NA |
| DISN Multi-Service Provisioning Platform (MSPP) (Table H-4 of DISN Upgrade KIP) | | No | NA | NA | NA | NA |
| DISN Optical Digital Cross-Connect (ODXC) (Table H-3 of DISN Upgrade KIP) | | No | NA | NA | NA | NA |
| DISN Optical Transport Services (OTS) (Table H-2 DISN Upgrade KIP) | | No | NA | NA | NA | NA |
| GPS Space Segment to Ground | | Yes | Mandated | Threshold | Consumer | None |

| | | | | | | |
|---|---|---|---|---|---|---|
| Interface (Table I-1 of GPS KIP) | | | | | | |
| GPS Ground Segment to DISN Interface (Table I-2 of GPS KIP) | | No | NA | NA | NA | NA |
| Joint Tactical Information Distribution System | | No | NA | NA | NA | NA |
| CENTRIXS | | No | NA | NA | NA | NA |
| Integrated Broadcast System | | No | NA | NA | NA | NA |
| **Application Enterprise Services Family** | 24 Feb 06 Review Draft V 1.0 | | | | | |
| Service Security | | Yes | NA | NA | NA | NA |
| Service Discovery | | Yes | NA | NA | NA | NA |
| Content Discovery | | Yes | NA | NA | NA | NA |
| Messaging | | Yes | NA | NA | NA | NA |
| Content Staging | | Yes | NA | NA | NA | NA |
| Web Services | | Yes | NA | NA | NA | NA |
| Computing Infrastructure Family[1] | | | | | | |
| NetOps | 1.0 | Yes | Mandated in DISR Baseline | JTRS Network Manager Interface | Consumer & Provider | |
| Applications Staging | TBD | TBD | FY 06 Requirement | TBD | TBD | |
| Computing Infrastructure | TBD | Yes | New Requirement | FBCB2 JCR | TBD | |
| Content Staging | 1.0 | No | New requirement | N/A | N/A | |

**Table A-2: Key Interface Profiles (KIPs) Compliance Statement***

*Per CJCSI 6212.01E, 15, December 2008, format from http://kips.disa.mil [now located on AKO DISA Standards Management] from Key Interface Profile (KIPs) Declaration Table. (dated 29 June 2007, downloaded 28 Mar 08) Please note that only Transport Family of KIPs (TV-1s) are approved by ISOP and posted on the DISR online.

## 4. Information Assurance Accreditation Compliance Statement.

The GSS will be in full compliance with DOD Directive 8500.1 and DOD Instruction 8500.2, DOD Instruction 8500.40, DODI 8510.01., CJCS 6510 series directives, instructions and manuals and will make the required Information Assurance documentation available to the Joint Staff J-6 for review. The GSS will be in compliance with DODD 5205.2 (OPSEC). GSS will be tested and certified IAW Joint Test Command (JITC) procedures per DODD C-5200.5 (paragraph 4G). The PM-SWAR point of contact for GSS Information Assurance Documentation is Mr. Grover, (256) 479-9913.

## 4.a Data Service Exposure Tracking Sheet.

| | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| **Data Exposure Verification Tracking Sheet for Ground Soldier System (GSS)** | | | | | | | | | |
| **Key** | N Not Started | I In-Progress | R Progress at Risk | S Progress Stopped | | A Objective Achieved | X Not Applicable | | |
| **Program Manager** | LTC Roland Gaddy | **Project POC** David Veney | | | | | | | |
| **Telephone #** | (703) 704-3846 | **Telephone #** (571) 437-6527 | | | | | **# of Objectives Achieved Since** | (Initial Submission) | |
| **E-Mail Address** | Roland.GaddyJr@US.Army.Mil | **E-Mail Address** David.Veney@US.Army.Mil | | | | | **Previous Submission** | NA | |
| **Web Page URL** | Not applicable - data not posted to the web | | **Visible** | **Accessible** | | **Understandable** | | | |
| **IT System** | Ground Soldier System (GSS) | | **CD&D (1.a)** | **Policy (2.a)** | **Oper (2.b)** | **User (3.a)** | | | |
| **DITPR Number** | 10098 | | | | | | **Submission Date** | 25-Feb-10 | |
| **Top Level JCA** | **Data Asset** | **Description** | **(Note: Use above "Key" to assign values for columns below)** | | | | **Issues/Comments** | **Exposure Start / Complete Date** | |
| Command & Control | K05.1 Position Report/Update (To provide friendly unit location data) | Position Location Information (PLI) - GSS processes digital position reports from lower echelon Soldiers within range. This is accomplished by providing all required information to transmit/receive a properly formated K05.1 Joint Variable Message Format (JVMF) message. (JVMF standard utilized is MIL-STD-6017, header is 2045-47001c) | I | I | N | A | Currently no GSS related work is being done to support a NCES Federated Search which results in an active link, that when selected provides access to a website displaying the PLI data.<br><br>GSS is a Secret high - U.S. Army soldier borne tactical weapon system used at the Company level of command and lower. The data sheet, as directed by the J68 staff, is being provided, but its unclear why its being requested. | 25-Feb-10 | 31-Dec-30 |
| | K05.19 Entity Data Message (To provide supplemental situational understanding information) | Position Location Information (PLI) - GSS provides relevant information to the Soldier in a usable and digestible form to facilitate timely situational understanding and decision making. This is accomplished by providing all required information to transmit/receive a properly formated K05.19 Joint Variable Message Format (JVMF) message. (JVMF standard utilized is MIL-STD-6017, header is 2045-47001c) | I | I | N | A | Currently no GSS related work is being done to support a NCES Federated Search which results in an active link, that when selected provides access to a website displaying the PLI data.<br><br>GSS is a Secret high - U.S. Army soldier borne tactical weapon system used at the Company level of command and lower. The data sheet, as directed by the J68 staff, is being provided, but its unclear why its being requested. | 25-Feb-10 | 31-Dec-30 |
| | Asset #3 | Description #n | | | | | | | |
| (JCA) Category | Asset #1 | Description #1 | | | | | | | |
| | Asset #2 | Description #2 | | | | | | | |
| | Asset #3 | Description #n | | | | | | | |
| (JCA) Category | Asset #1 | Description #1 | | | | | | | |
| | Asset #2 | Description #2 | | | | | | | |
| | Asset #3 | Description #n | | | | | | | |

2010.03.01 ver
97-03 GSS Data Expo

## 5. (AV-1) Overview and Summary Information Ground Soldier System (GSS)

The AV-1 for this CDD can be viewed on the CADIE website under the AV-1 Worksheet Submission Tool, under the title Ground Soldier System CDD, AV-1 ID #235. The CADIE website is: https://cadie.army.mil/cadie/Portal/Default.aspx

**Name:** Ground Soldier System (GSS) Increment I

**Description:** The GSS architecture describes the operational underpinnings for the development of the system architecture by identifying the mission need lines, operational activities, tasks, and processes.

**Architect:** Soldier Requirements Division, Ft. Benning, GA

**Organization Developing the Architecture:** TRADOC ARCIC, AIMD, PEO Soldier

**Assumptions and Constraints:** There will be P3I additions to incorporate new technologies that will evolve and develop beyond present forecast, and future increments will be necessary. GSS program is constrained by operational needs timelines and within the existing technology currently available.

**Approval Authority:** Joint Staff J6

**Date Completed:** Draft Version 0.5, 15 August 2008

**Level of Effort and Costs:** Determination ongoing.

**Purpose and Viewpoint:** This architecture represents the objective state for the Ground Soldier System GSS from the GSS equipped Soldier viewpoint. This architecture identifies the Interoperability of the BC and SA with the Army BCT Modernized, Stryker, and Light and Heavy ground combat systems. Networking of Leaders, weapons, sensors and external assets enables geographically dispersed small units to collaboratively influence larger areas with greater precision, speed and a broader variety of lethal effects.

**Context:** This architecture illustrates the GSS capability to execute: Troop Leading Procedures and Conduct Actions on the Objective at the squad, platoon, and company echelon. It details the GSS information exchanges directly supporting the infantry squad, platoon, and company.

**Scope:** The GSS architecture demonstrates the net-centric effects to be brought to tactical war fighting capabilities that support the information needs/information exchange requirements of maneuver units. GSS equipped units will primarily utilize the system to close with and destroy the enemy. These products reflect the equipping/fielding of GSS to an IBCT in FY 12.

**Intended Use:** Document the core set of processes, information exchanges, operational and system nodes, and system functions. Support POM and Annual Planning and Programming Guidance (APPG). Support requirements development and early milestone decision reviews.

**Scenario:** The GSS equipped unit will be employed as part of an Infantry Brigade Combat Team. The GSS will provide the means to enhance early entry combat power across the spectrum of tactical actions, missions, and operations. Tasks associated with these mission areas require a system that enables success in close combat.

**Findings:** The opportunity to migrate to the JTRS SFF B radio during increment one will enhance the networked capability of the GSS.

**6. Architecture products:**

**OV-1 High- Level Operational Concept Graphic:** This graphic describes a mission and highlights main operational nodes and interesting or unique aspects of operations. It provides a description of the interactions between the subject architecture and its environment, and between the architecture and external systems.

**OV-2 Operational Node Connectivity:** This is a high level picture of operational nodes that must exchange information to accomplish the mission. The main features are identification of the node and information path such as terrestrial, tactical internet, or satellite, etc.

**OV-3 - Operational Information Exchange Matrix:** Provides information exchanged between nodes and the relevant attributes of that exchange.

**OV-4 Organizational Relationship Chart:** This chart illustrates the relationships among organizations. The relationships indicate the command and control that influences the connectivity needed. GSS is organic to the Stryker Brigade Combat Team (SBCT)/Brigade Combat Team (BCT).

**OV-5 Operational Activity Model:** Activities, relationships among activities, inputs and outputs will be modeled to the Universal Joint Task List (UJTL) and Army Universal Task List (AUTL) level based upon mission thread decomposition and structured task analysis. A high level OV-5 node tree is developed initially to identify tasks associated with the GSS. Inputs, outputs, mechanisms, and controls will be identified at high level, then incrementally decomposing to data item level.

**OV-6c Operational Event-Trace Description:** The Operational Event-Trace Description provides a time-ordered examination of the information exchanges between participating operational nodes as a result of a particular scenario. Each event-trace diagram should have an accompanying description that defines the particular scenario or situation.

**OV-7 Logical Data Model:** The Logical Data Model describes the structure of an architecture domain's system data types and the structural business process rules (defined in the architecture's Operational View) that govern the system data. The Logical Data Model (OV-7) and Physical Schema (SV-11) are applicable when the domain deals with "persistent system data."

**SV-1 System Interface Description:** The SV-1 depicts systems nodes and the systems resident at these nodes to support organizations/human roles represented by operational nodes of the Operational Node Connectivity Description (OV-2).

**SV-2 Systems Communications Description:** SV-2 depicts the implementation of the interfaces from SV-1 as specific communications systems, communications links, communications paths, or communications networks and the details of their configurations through which the system interface.

**SV-4 Systems Functionality Description:** The SV-4 describes the functionality of the GSS with other systems in the conduct of the reconnaissance, security, and fire support threads.

**SV-5 Operational Activity Model:** Operational Activity to Systems Function Traceability Matrix is a specification of the relationships between the set of operational activities applicable to the architecture and the set of system functions applicable to that architecture.

**SV-6 Systems Data Exchange Matrix:** The Systems Data Exchange Matrix specifies the characteristics of the system data exchanged between systems. This product focuses on automated information exchanges (from OV-3) that are implemented in systems.

**SV-8 Systems & Services Evolution Description:** The Systems & Services Evolution Description captures evolution plans that describe how the system, or the architecture in which the system is embedded, will evolve over a lengthy period of time.

**SV-10c System Event Trace Description:** The Event Trace Description lays out sequence of data exchanges for mission scenarios.

**SV-11 Physical Schema:** The Physical Schema defines the structure of the various kinds of system data that are utilized by the systems in the architecture.

**TV-1 Technical Standards Profile:** The Technical Standards View (TV) provides the technical systems-implementation standards upon which engineering specifications are based, common building blocks are established, and product lines are developed. This is generated from DISR online (see http://disronline.disa.mil/). To view the TV-1 after it has been published, go to the SIPRNET https://jcpat.ncr.disa.smil.mil and select DISR online then the desired platform.

**TV-2 Technical Standards Profile:** The Technical Standards Forecast contains pertinent expected changes in technology-related standards and conventions, which are documented in the TV-1 product. They will be implemented when mature and become mandated.

**7. Tools and File Formats.**

The primary tools and associated file formats used in the development of this architecture are:

| Tools | File Formats |
|---|---|
| System Architect/ Telelogic/Popkin ProVision | Popkin Graphical Database |
| Microsoft Office Suite | Various Output Formats |
| MS Word | .doc |
| MS PowerPoint | .ppt |
| MS Visio | .vsd |
| MS Excel | .xls |
| MS Access | .mdb |
| Adobe Acrobat | .pdf |

**Architecture Introduction:** The presentation of this architecture is to first present the "High-Level" views and then to further decompose these views into more detailed focused vignette scoped views. The high level views are constant regardless of the vignette and operational and system views. The two vignettes; Troop Leading Procedures, and Actions on the Objective, include the logical supporting OV-2, OV-5, and the OV-6c. The SV products that require further clarity are defined as GSS increment I and GSS P3I and are found after the vignette views in the system views tabs.

**TAB B High Level Architecture Views**

**High-level Operational Concept Graphic (OV-1) Ground Soldier System (GSS)**

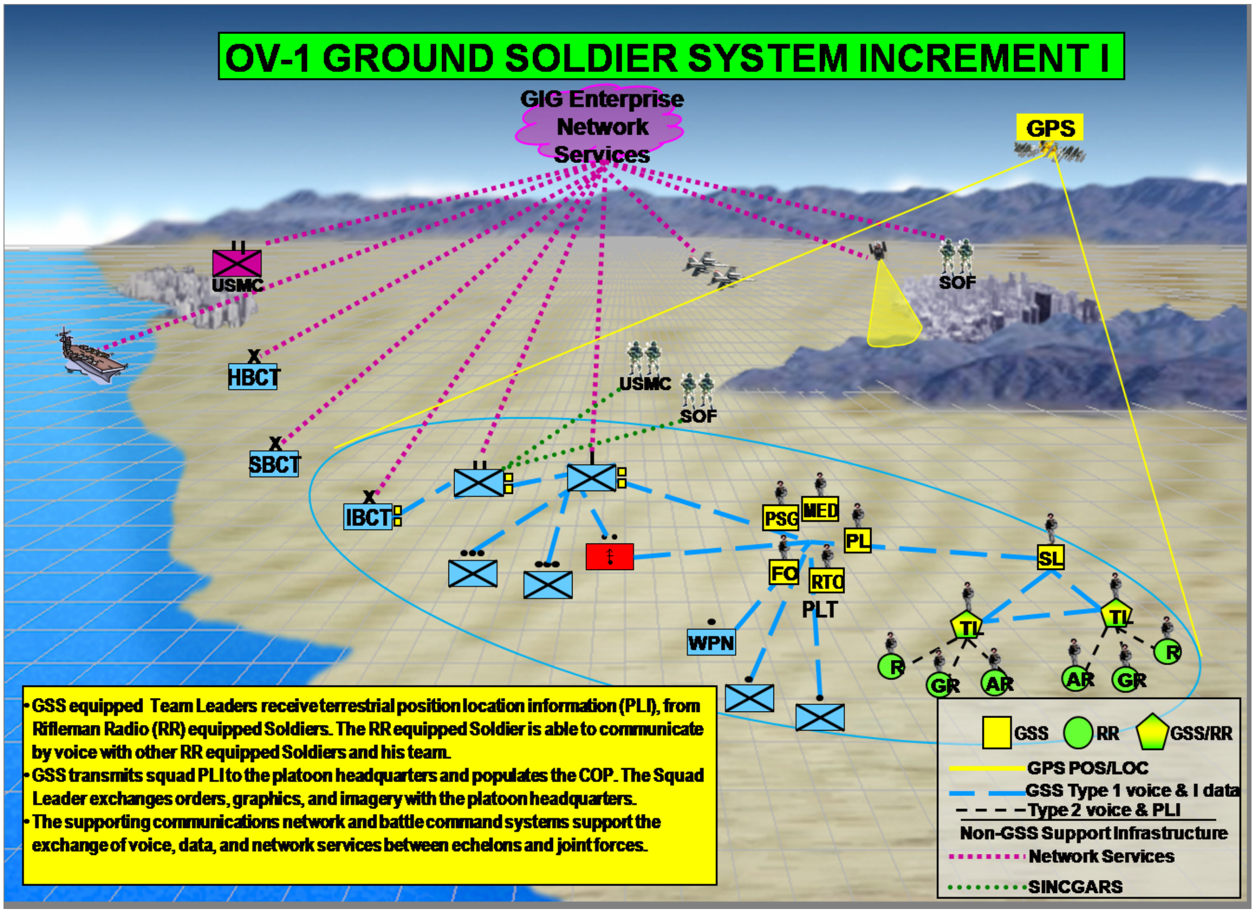



**Figure A-B-1: OV-1 – High Level Operational Concept**

This OV-1 illustrates the operational employment concept for the GSS equipped small combat unit at the squad and platoon echelon. In the squad both team leaders (TL), and the squad leaders (SL), are equipped with the GSS. At the platoon echelon all members of the platoon headquarters are fielded with GSS. Selective Soldiers at echelons above the platoon and other organizations are fielded GSS based on their duty positions. GSS equipped units and Soldiers exchange, publish, and subscribe to the SA Network Services that are resident on numerous manned and un-manned platforms, ground or air vehicles, and other sensors.

## Organizational Relationships Chart (OV-4) Ground Soldier System (GSS)

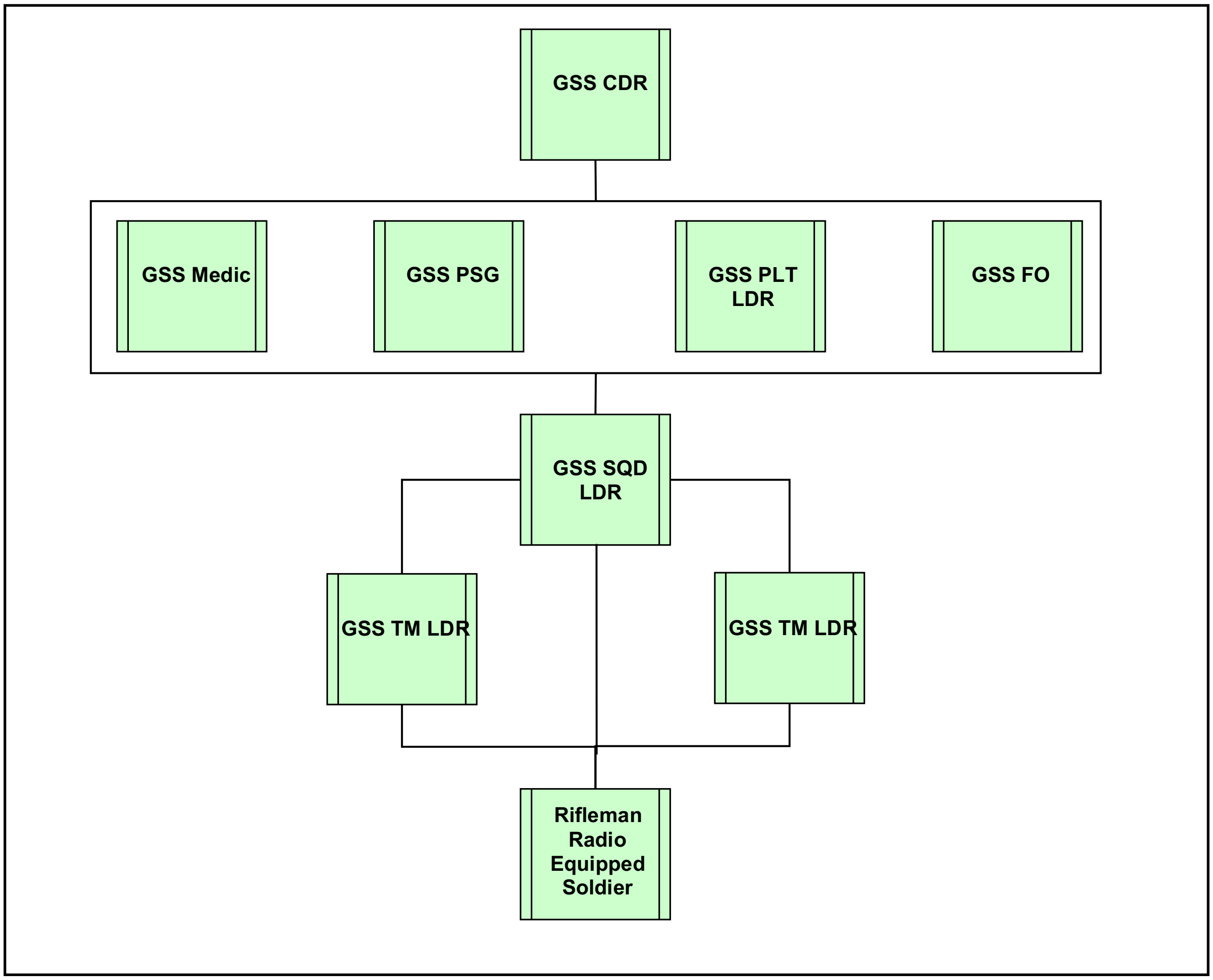



**Figure A-B-2: OV-4 – Organizational Relationships Chart**

The GSS is predominantly assigned to Infantry Platoon formations, but will also be issued to other formations in the BCT.

## High Level Logical Data Model (OV-7) Ground Soldier System (GSS)

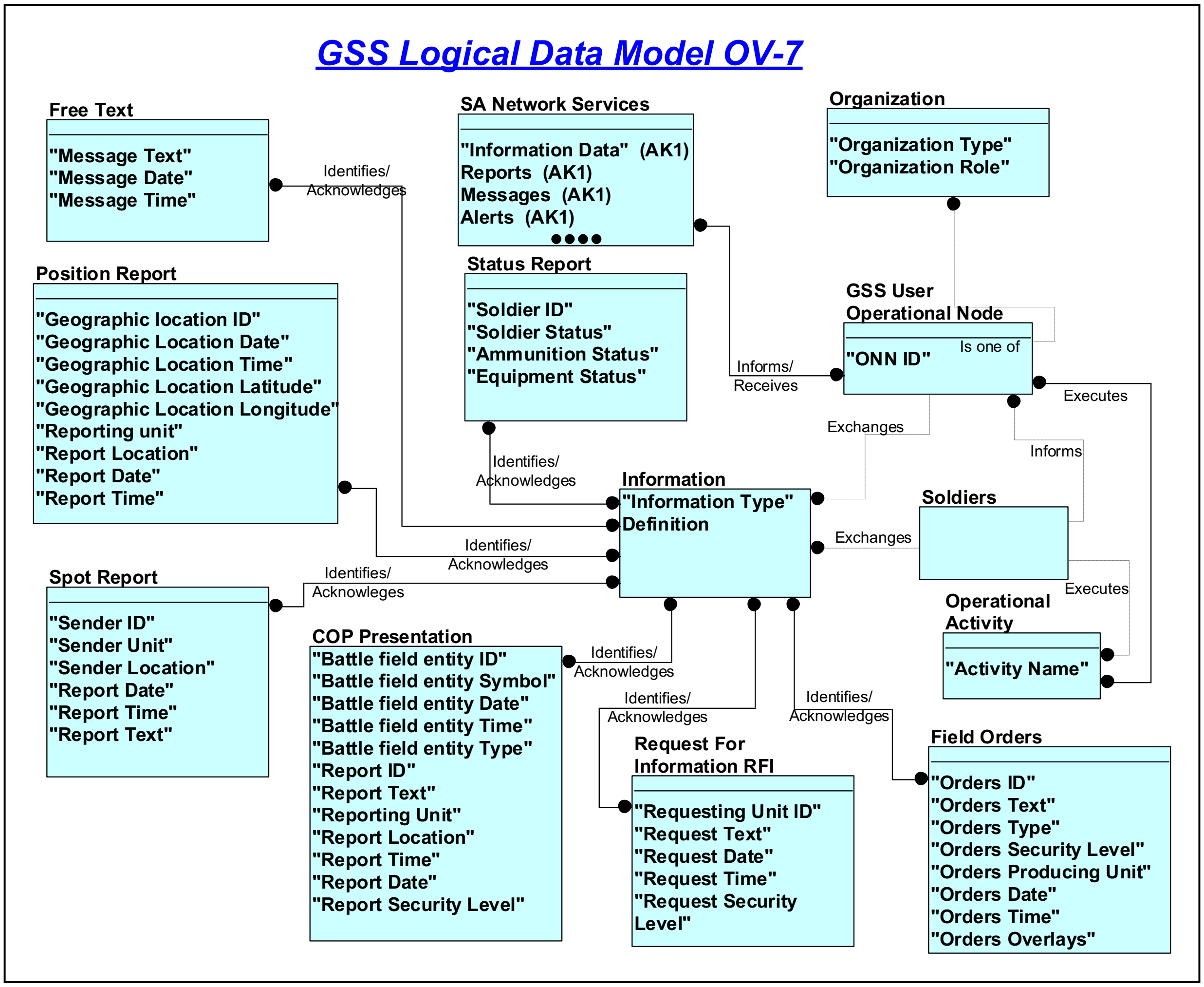



**Figure A-B-4: OV-7 – Logical Data Model**

This diagram depicts the relationship between data elements in the architecture.

**Entity: Soldiers**
**Description:** Identifies Soldiers equipped with the Rifleman Radio which provides position location information and voice communication with the GSS
**Relationship line:** Exchanges
- **Cardinality:** One Soldiers Exchanges Zero, One or Many Information
- **Description:** transfer and receipt of information

**Relationship line:** Executes
- **Cardinality:** One Soldiers Executes Zero, One or Many "Operational Activity"
- **Identifying:** F
- **Description:** Describes the performance of operational tasks and activities by Soldiers.

**Relationship line:** Informs
- **Cardinality:** One Soldiers Informs Zero, One or Many "GSS User Operational Node"

**Description:** Provides position location information to the GSS.

**Entity: GSS User Operational Node**
**Description:** Identifies Leaders equipped with the GSS
ONN ID
**Attribute:** ONN ID
**Column Name:** ONN_ID
**Relationship line:** Exchanges
**Cardinality:** One "GSS User Operational Node" Exchanges Zero, One or Many Information
**Relationship line:** Executes
**Cardinality:** Zero, One or Many "GSS User Operational Node" Executes Zero, One or Many "Operational Activity"
**Description:** Describes the performance of operational tasks and activities by GSS Leaders and other Soldiers.
**Relationship line:** Informs
**Cardinality:** Zero, One or Many "GSS User Operational Node" Informs Zero, One or Many "SA Network Services"
**Description:** Submits information to the receiver.
**Relationship line:** Informs
**Cardinality:** One leader Informs Zero, One or Many "GSS User Operational Node"
**Description:** Provides position location information to the GSS.
**Relationship line:** Is one of
**Cardinality:** One "GSS User Operational Node" Is one of Zero, One or Many Organization
**Description:** Operational node is a member of the organization.

**Entity: Information**
**Description:** Identifies the types of information exchanged by GSS and Soldiers.
**Primary Keys:**
Information Type
Definition
**Attribute:** Information Type
**This attribute is a primary key**
**Column Name:** Information Typed
**Attribute:** Definition
**This attribute is a primary key**
**Column Name:** Definition
**Relationship line:** Exchanges
**Cardinality:** One "GSS User Operational Node" Exchanges Zero, One or Many Information
**Relationship line:** Exchanges
**Cardinality:** One Soldiers Exchanges Zero, One or Many Information
**Relationship line:** Identifies

**Cardinality:** Zero, One or Many Information Identifies Zero, One or Many "Free Text"
**Description:** Categorizes and labels information
**Relationship line:** Identifies
**Cardinality:** Zero, One or Many Information Identifies Zero, One or Many "Position Report"
**Relationship line:** Identifies
**Cardinality:** Zero, One or Many Information Identifies Zero, One or Many "COP Presentation"
**Description:** Categorizes and labels information
**Relationship line:** Identifies
**Cardinality:** Zero, One or Many Information Identifies Zero, One or Many "Spot Report"
**Description:** Categorizes and labels information
**Relationship line:** Identifies
**Cardinality:** Zero, One or Many Information Identifies Zero, One or Many "Status Report"
**Description:** Categorizes and labels information
**Relationship line:** Identifies
**Cardinality:** Zero, One or Many Information Identifies Zero, One or Many "Request For Information RFI"
**Description:** Categorizes and labels information
**Relationship line:** Identifies
**Cardinality:** Zero, One or Many Information Identifies Zero, One or Many "Field Orders"
**Description:** Categorizes and labels information

**Entity: Operational Activity**
**Description:** Describes the operational task and activities executed by Soldiers and GSS equipped leaders.
**Non-Key Attributes:**
Activity Name
**Attribute:** Activity Name
**Column Name:** Operational Activities
**Relationship line:** Executes
**Cardinality:** Zero, One or Many "GSS User Operational Node" Executes Zero, One or Many "Operational Activity"
**Description:** Describes the performance of operational tasks and activities by GSS leaders and other Soldiers.
**Relationship line:** Executes
**Cardinality:** One Soldiers Executes Zero, One or Many "Operational Activity"
**Description:** Describes the performance of operational tasks and activities by Soldiers.

**Entity: Organization**
**Description:** Army organization of which the operational node is a member of.
**Non-Key Attributes:**
Organization Type
Organization Role
**Attribute:** Organization Type
**Column Name:** Data
**Attribute:** Organization Role
**Column Name:** Data
**Relationship line:** Is one of
**Cardinality:** One "GSS User Operational Node" Is one of Zero, One or Many Organization
**Description:** Operational node is a member of the organization.

**Entity: SA Network Services**
**Description:** Identifies the categories of information accessible to and provided by GSS and other entities.
**Comments:**
**Non-Key Attributes:**
Information Data
Reports
Messages
Alerts
Requests
**Attribute:** Information Data
**Column Name:** Information
**Attribute:** Reports
**Column Name:** Reports
**Attribute:** Messages
**Column Name:** Messages
**Attribute:** Alerts
**Column Name:** Alerts
**Attribute:** Requests
**Column Name:** Requests
**Relationship line:** Informs
**Cardinality:** Zero, One or Many "GSS User Operational Node" Informs Zero, One or Many "SA Network Services"
**Description:** Submits information to the receiver.

**Entity: COP Presentation**
**Description:** The data that is provided to the user system either requested or sent by directives and then tailored into the COP presentation
**Primary Keys:**
Battle field entity ID
Reporting Unit
Report Date

Report Location
Battle field entity Symbol
Report ID
Report Security Level
Battle field entity Time
Battle field entity Date
Report Time
Report Text
Battle field entity Type

**Non-Key Attributes:**

**Attribute:** Battle field entity ID
**This attribute is a primary key**
**Column Name:** Battle field entity ID

**Attribute:** Reporting Unit
**This attribute is a primary key**
**Column Name:** Reporting Unit

**Attribute:** Report Date
**This attribute is a primary key**
**Column Name:** Report Date

**Attribute:** Report Location
**This attribute is a primary key**
**Column Name:** Report Location

**Attribute:** Battle field entity Symbol
**This attribute is a primary key**
**Column Name:** Battle field entity Symbol

**Attribute:** Report ID
**This attribute is a primary key**
**Column Name:** Report ID

**Attribute:** Report Security Level
**This attribute is a primary key**
**Column Name:** Report Security Level

**Attribute:** Battle field entity Time
**This attribute is a primary key**
**Column Name:** Battle field entity Time

**Attribute:** Battle field entity Date
**This attribute is a primary key**
**Column Name:** Battle field entity Date

**Attribute:** Report Time
**This attribute is a primary key**
**Column Name:** Report Time

**Attribute:** Report Text
**This attribute is a primary key**
**Column Name:** Report Text

**Attribute:** Battle field entity Type
**This attribute is a primary key**
**Naming Prefix:**

**Column Name:** Battle field entity Type
**Relationship line:** Identifies
**Cardinality:** Zero, One or Many Information Identifies Zero, One or Many "COP Presentation"

**Entity: Position Report**
**Description:** Provides the locations, direction of movement, and speed of friendly elements.
**Non-Key Attributes:**
Report Time
Geographic location ID
Geographic Location Time
Geographic Location Date
Report Date
Report Location
Geographic Location Longitude
Reporting unit
Geographic Location Latitude
**Attribute:** Report Time
**Column Name:** Report Time
**Attribute:** Geographic location ID
**Column Name:** Geographic location ID
**Attribute:** Geographic Location Time
**Column Name:** Geographic Location Time
**Attribute:** Geographic Location Date
**Column Name:** Geographic Location Date
**Attribute:** Report Date
**Column Name:** Report Date
**Attribute:** Report Location
**Column Name:** Report Location
**Attribute:** Geographic Location Longitude
**Column Name:** Geographic Location Longitude
**Attribute:** Reporting unit
**Column Name:** Reporting unit
**Attribute:** Geographic Location Latitude
**Column Name:** Geographic Location Latitude
**Relationship line:** Identifies
**Cardinality:** Zero, One or Many Information Identifies Zero, One or Many "Position Report"

**Entity: Free Text**
**Description:** Permits the creation and publishing of information in a format that allows non standard data type.
**Non-Key Attributes:**
Message Date
Message Time

Message Text
**Attribute:** Message Date
**Column Name:** Message Date
**Attribute:** Message Time
**Column Name:** M
**Attribute:** Message Text
**Column Name:** Message Text
**Relationship line:** Identifies
**Cardinality:** Zero, One or Many Information Identifies Zero, One or Many "Free Text"
**Description:** Categorizes and labels information

**Entity: Spot Report**
**Description:** Report initiated by any activity to report any observed information to include Activity, Location, Time, and Equipment.
**Non-Key Attributes:**
Sender ID
Report Date
Sender Location
Report Text
Report Time
Sender Unit
**Attribute:** Sender ID
**Column Name:** Sender ID
**Attribute:** Report Date
**Column Name:** Report Date
**Attribute:** Sender Location
**Column Name:** Sender Location
**Attribute:** Report Text
**Column Name:** Report Text
**Attribute:** Report Time
**:**
**Column Name:** Report Time
**Attribute:** Sender Unit
**Column Name:** Sender Unit
**Relationship line:** Identifies
**Cardinality:** Zero, One or Many Information Identifies Zero, One or Many "Spot Report"
**Description:** Categorizes and labels information

**Entity: Request For Information RFI**
**Description:** Request made for information pertaining to mission execution
**Non-Key Attributes:**
Requesting Unit ID
Request Text
Request Security Level

Request Time
Request Date

**Attribute:** Requesting Unit ID
**Column Name:** Requesting Unit ID
**Attribute:** Request Text
**Column Name:** Request Text
**Attribute:** Request Security Level
**Column Name:** Request Security Level
**Attribute:** Request Time
**Column Name:** Request Time
**Attribute:** Request Date
**Column Name:** Request Date
**Relationship line:** Identifies
**Cardinality:** Zero, One or Many Information Identifies Zero, One or Many "Request For Information RFI"
**Description:** Categorizes and labels information

**Entity: Status Report**
**Description:** Information indicating the operational and/or fault status of a Soldier or item of equipment. Also includes the status of supplies on a Soldier.
**Non-Key Attributes:**
Equipment Status
Soldier ID
Soldier Status
Ammunition Status
**Attribute:** Equipment Status
**Column Name:** Equipment Status
**Attribute:** Soldier ID
**Column Name:** Soldier ID
**Attribute:** Soldier Status
**Column Name:** Soldier Status
**Attribute:** Ammunition Status
**Column Name:** Ammunition Status
**Relationship line:** Identifies
**Cardinality:** Zero, One or Many Information Identifies Zero, One or Many "Status Report"
**Description:** Categorizes and labels information

**Entity: Field Orders**
**Description:** Orders consist of OPORD, FRAGO and WARNO. Specifically: An OPORD is a directive issued by a commander to a subordinate to coordinate the execution of an operation. An OPORD always specifies an execution time and date. It is in the five-paragraph field order format (situation, mission, execution, service support, and command and signal) and includes detailed annexes and appendixes. A warning order (WO) is provided as a warning for future operations. The WO must include type of operation, general location of the operation, initial

timeline, and movement or reconnaissance to initiate. Immediately after the commander gives his guidance, the staff sends subordinate and supporting units a WO, which must contain the restated mission, the commander.

**Non-Key Attributes:**

Orders Date
Orders ID
Orders Security Level
Orders Producing Unit
Orders Overlays
Orders Type
Orders Text
Orders Time

**Attribute:** Orders Date
**Column Name:** Orders Date
**Attribute:** Orders ID
**Column Name:** Orders ID
**Attribute:** Orders Security Level
**Column Name:** Orders Security Level
**Attribute:** Orders Producing Unit
**Column Name:** Orders Producing Unit
**Attribute:** Orders Overlays
**Column Name:** Orders Overlays
**Attribute:** Orders Type
**Column Name:** Orders Type
**Attribute:** Orders Text
**Column Name:** Orders Text
**Attribute:** Orders Time
**Column Name:** Orders Time
**Relationship line:** Identifies
**Cardinality:** Zero, One or Many Information Identifies Zero, One or Many "Field Orders"
**Description:** Categorizes and labels information

**Systems Interface Description (SV-1) Ground Soldier System (GSS)**

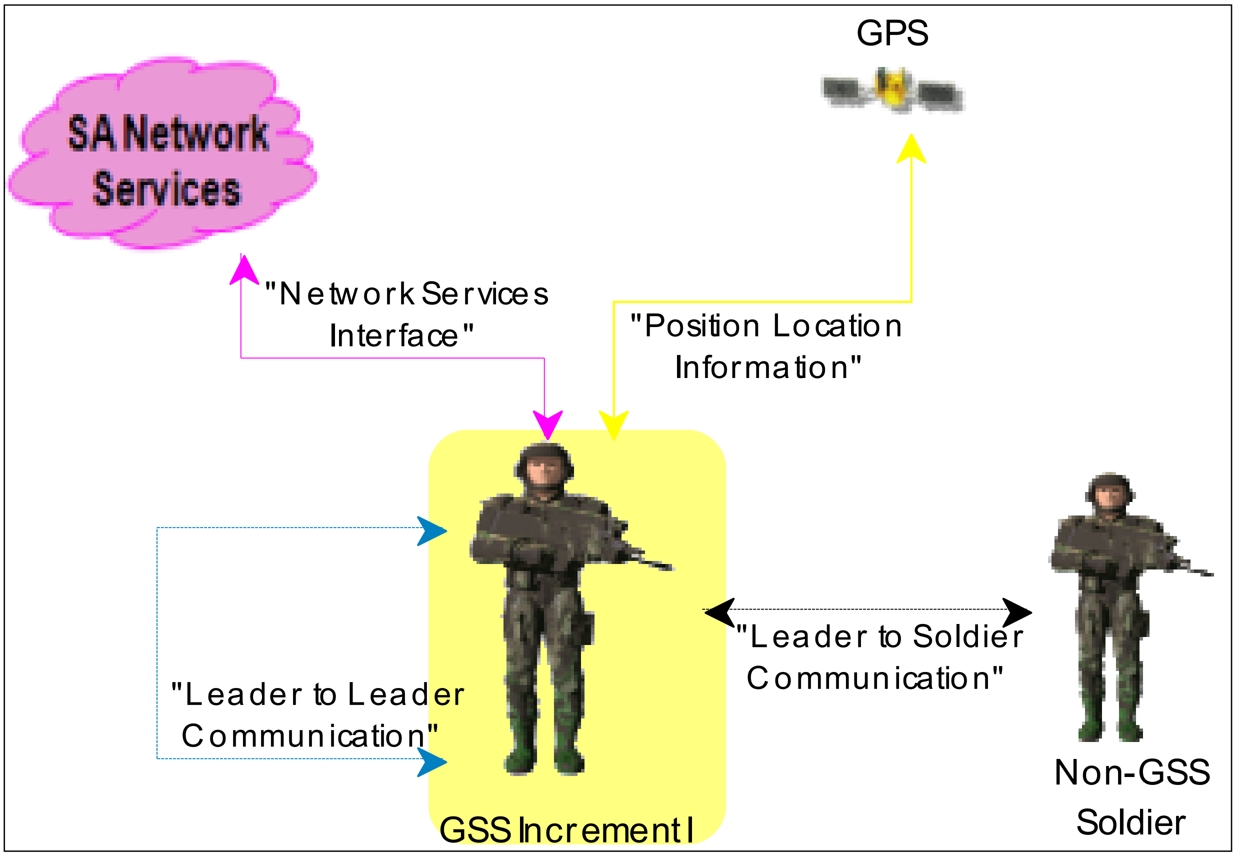



**Figure A-B-5: SV-1 – Systems Interface Description**

The Ground Soldier System (GSS) Increment I Soldier interfaces with other GSS Increment I leaders, Non-GSS Soldiers, the GPS System, and SA Network Services as depicted in the GSS OV-2.

**Physical Schema (SV-11) Ground Soldier System (GSS)**

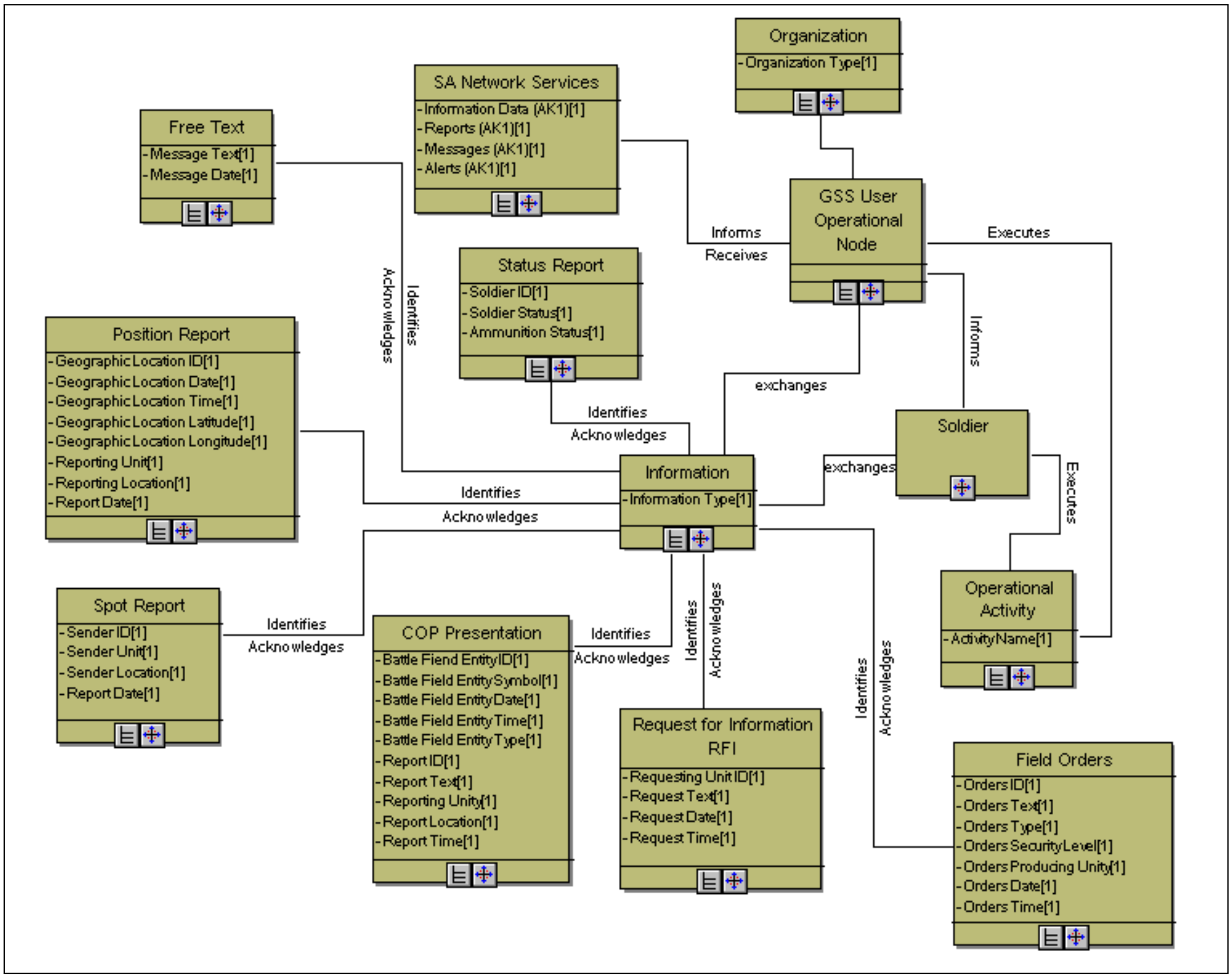



**Figure A-B-6: SV-11 – Physical Schema**

A description of the structure of the data utilized in the Ground Soldier System as shown in the SV-11 above is provided below:

**COP Presentation**
Type: Concrete
Each COP Presentation remembers:
a Battle Fiend Entity ID (Undefined values)
a Battle Field Entity Symbol (Undefined values)
a Battle Field Entity Date (Undefined values)
a Battle Field Entity Time (Undefined values)
a Battle Field Entity Type (Undefined values)
a Report ID (Undefined values)
a Report Text (Undefined values)
a Reporting Unity (Undefined values)

a Report Location (Undefined values)
a Report Time (Undefined values)
a Report Date (Undefined values)
a Report Security Level (Undefined values)
For each COP Presentation:
we will find an Information it Identifies

**Field Orders**
Type: Concrete
Each Field Orders remembers:
an Orders ID (Undefined values)
an Orders Text (Undefined values)
an Orders Type (Undefined values)
an Orders Security Level (Undefined values)
an Orders Producing Unity (Undefined values)
an Orders Date (Undefined values)
an Orders Time (Undefined values)
an Orders Overlays (Undefined values)
For each Field Orders:
we will find an Information it Identifies

**Free Text**
Type: Concrete
Each Free Text remembers:
a Message Text (Undefined values)
a Message Date (Undefined values)
a Message Time (Undefined values)
For each Free Text:
we will find an Information it Identifies

**GSS User Operational Node**
Type: Concrete
Each GSS User Operational Node remembers:
an ONN ID (Undefined values)
For each GSS User Operational Node:
we will find an Information it exchanges
we will find an Operational Activity it Executes
we will find an Organization it is associated with
we will find an SA Network Services it Receives
we will find a Soldier it Informs

**Information**
Type: Concrete
Each Information remembers:
an Information Type (Undefined values)
a Definition (Undefined values)

For each Information:
we will find a COP Presentation it Acknowledges
we will find a Field Orders it Acknowledges
we will find a Free Text it Acknowledges
we will find a GSS User Operational Node it is associated with
we will find a Position Report it Acknowledges
we will find a Request for Information RFI it Acknowledges
we will find a Soldier it is associated with
we will find a Spot Report it Acknowledges
we will find a Status Report it Acknowledges

**Operational Activity**
Type: Concrete
Each Operational Activity remembers:
an Activity Name (Undefined values)
an Activity Description (Undefined values)
For each Operational Activity:
we will find a GSS User Operational Node it is associated with
we will find a Soldier it is associated with

**Organization**
Type: Concrete
Each Organization remembers:
an Organization Type (Undefined values)
an Organization Role (Undefined values)
For each Organization:
we will find a GSS User Operational Node it is associated with

**Position Report**
Type: Concrete
Each Position Report remembers:
a Geographic Location ID (Undefined values)
a Geographic Location Date (Undefined values)
a Geographic Location Time (Undefined values)
a Geographic Location Latitude (Undefined values)
a Geographic Location Longitude (Undefined values)
a Reporting Unit (Undefined values)
a Reporting Location (Undefined values)
a Report Date (Undefined values)
a Report Time (Undefined values)
For each Position Report:
we will find an Information it Identifies

**Request for Information RFI**
Type: Concrete
Each Request for Information RFI remembers:

a Requesting Unit ID (Undefined values)
a Request Text (Undefined values)
a Request Date (Undefined values)
a Request Time (Undefined values)
a Request Security Level (Undefined values)
For each Request for Information RFI:
we will find an Information it Identifies

**SA Network Services**
Type: Concrete
Each SA Network Services remembers:
an Information Data (AK1) (Undefined values)
a Reports (AK1) (Undefined values)
a Messages (AK1) (Undefined values)
an Alerts (AK1) (Undefined values)
For each SA Network Services:
we will find a GSS User Operational Node it Informs
Each SA Network Services can perform:
1 (F) TLP Send Orders
1a (F) Seal the Objective
1b (F) Seal the Objective
1c (F) Seal the Objective
1f (F) TLP
2a
2a (F) TLP Enable Network Management
2b (F) TLP
3d
3d (F) TLP Access & Process C2 Data
4e
4e (F) TLP
4f
4f (F) TLP Provide Position and Navigation Information
A.1
Assault to the LOA
Provide Suppressing Fire

**Soldier**
Type: Concrete
For each Soldier:
we will find a GSS User Operational Node it is associated with
we will find an Information it exchanges
we will find an Operational Activity it Executes

**Spot Report**
Type: Concrete
Each Spot Report remembers:

a Sender ID (Undefined values)
a Sender Unit (Undefined values)
a Sender Location (Undefined values)
a Report Date (Undefined values)
a Report Time (Undefined values)
a Report Text (Undefined values)
For each Spot Report:
we will find an Information it Identifies

**Status Report**
Type: Concrete
Each Status Report remembers:
a Soldier ID (Undefined values)
a Soldier Status (Undefined values)
an Ammunition Status (Undefined values)
an Equipment Status (Undefined values)
For each Status Report:
we will find an Information it Identifies

**TAB C Vignette Architecture Views**
**Operational Node Connectivity Description (OV-2) Ground Soldier System (GSS)**

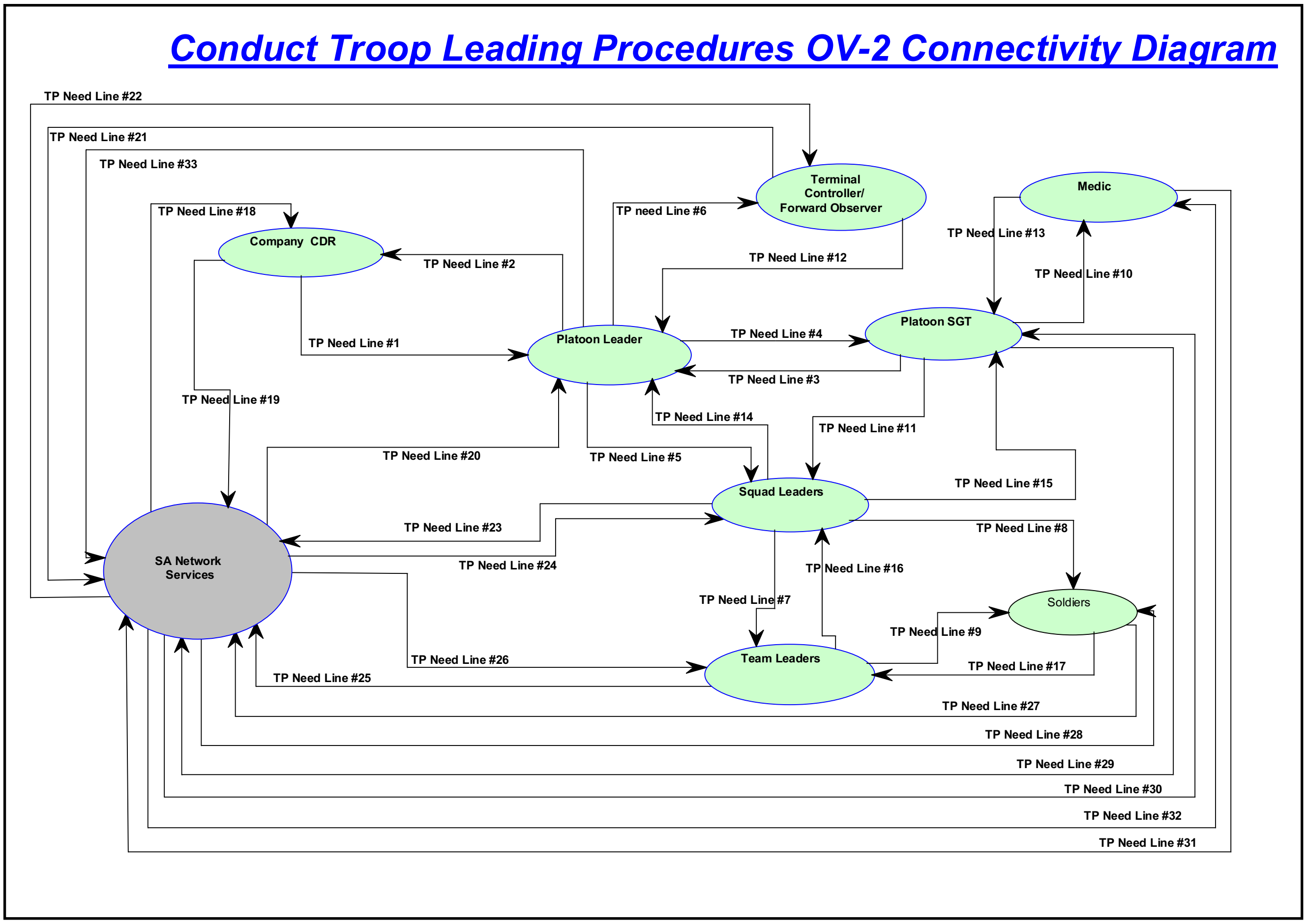



**Figure A-C-1: OV-2 – Operational Node Connectivity Description**

**Scope**: This thread depicts a GSS equipped infantry rifle company and its subordinate platoons and squads conducting troop leading procedures upon receiving a mission from its higher headquarters to execute an assault on an objective.

**TP Need Line #1**(Company CDR to Platoon Leader): Tactical Information: Mission Data
**TP Need Line #2**(Platoon Leader to Company CDR): Tactical Information: Mission Data
**TP Need Line #3** (Platoon SGT to Platoon Leader): Tactical Information: Status Data
**TP Need Line #4** (Platoon Leader to Platoon SGT): Tactical Information: C2 Data
**TP Need Line #5**(Platoon Leader to Squad Leaders): Tactical Information: C2 Data
**TP Need Line #6** (Platoon Leader to Terminal Controller/Forward Observer): Tactical Information: C2 Data
**TP Need Line #7** (Squad leader to Team Leaders): Tactical Information: C2 Data
**TP Need Line #8** (Squad leader to Team Leader): Tactical Information: C2 Data

**TP Need Line #9** (Team Leaders to Soldiers): Tactical Information: C2 Data
**TP Need Line #10** (Platoon SGT to Medic): Tactical Information: C2 Data
**TP Need Line #11** (Platoon SGT to Squad Leader): Tactical Information: C2 Data
**TP Need Line #12** (Terminal Controller/Forward Observer to Platoon Leader): Tactical Information: Fire Support Data
**TP Need Line #13** (Medic to Platoon SGT): Tactical Information: Medical Data
**TP Need Line #14** (Squad leader to Platoon Leader): Tactical Information: Status Data
**TP Need Line #15** (Squad leader to Platoon SGT): Tactical Information: Status Data
**TP Need Line #16** (Team Leaders to Squad Leaders): Tactical Information: Status Data
**TP Need Line #17** (Soldiers to Team Leaders): Tactical Information: Status Data
**TP Need Line #18** (SA Network Services to Company CDR): Tactical Information: COP Data
**TP Need Line #19** (Company CDR to SA Network Services): Tactical Information: COP Data
**TP Need Line #20** (Platoon Leader to SA Network Services): Tactical Information: COP Data
**TP Need Line #21** (Terminal Controller/Forward Observer to SA Network Services): Tactical Information: COP Data
**TP Need Line #22** (SA Network Services to Terminal Controller/Forward Observer): Tactical Information: COP Data
**TP Need Line #23** (Squad Leaders to SA Network Services): Tactical Information: COP Data
**TP Need Line #24** (SA Network Services to Squad Leaders): Tactical Information: COP Data
**TP Need Line #25** Team Leaders to SA Network Services): Tactical Information: COP Data
**TP Need Line #26** (SA Network Services to Team Leaders): Tactical Information: COP Data
**TP Need Line #27** Soldiers to SA Network Services): Tactical Information: COP Data
**TP Need Line #28** (SA Network Services to Soldiers): Tactical Information: COP Data
**TP Need Line #29** (Platoon SGT to SA Network Services): Tactical Information: COP Data
**TP Need Line #30** (SA Network Services to Platoon SGT): Tactical Information: COP Data
**TP Need Line #31** (Medic to SA Network Services): Tactical Information: COP Data
**TP Need Line #32** (SA Network Services to Medic): Tactical Information: COP Data

## Operational Activity Model (OV-5) Ground Soldier System (GSS)

### Conduct Troop Leading Procedures

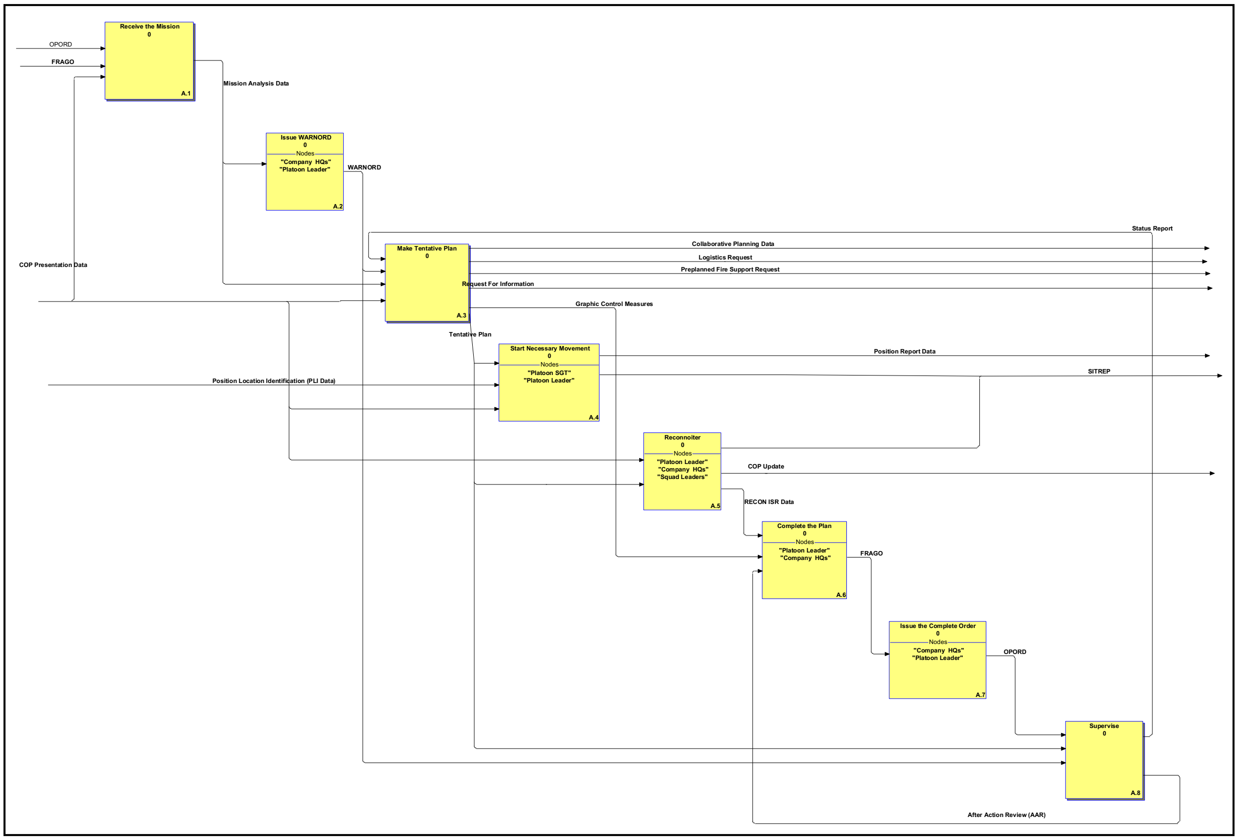
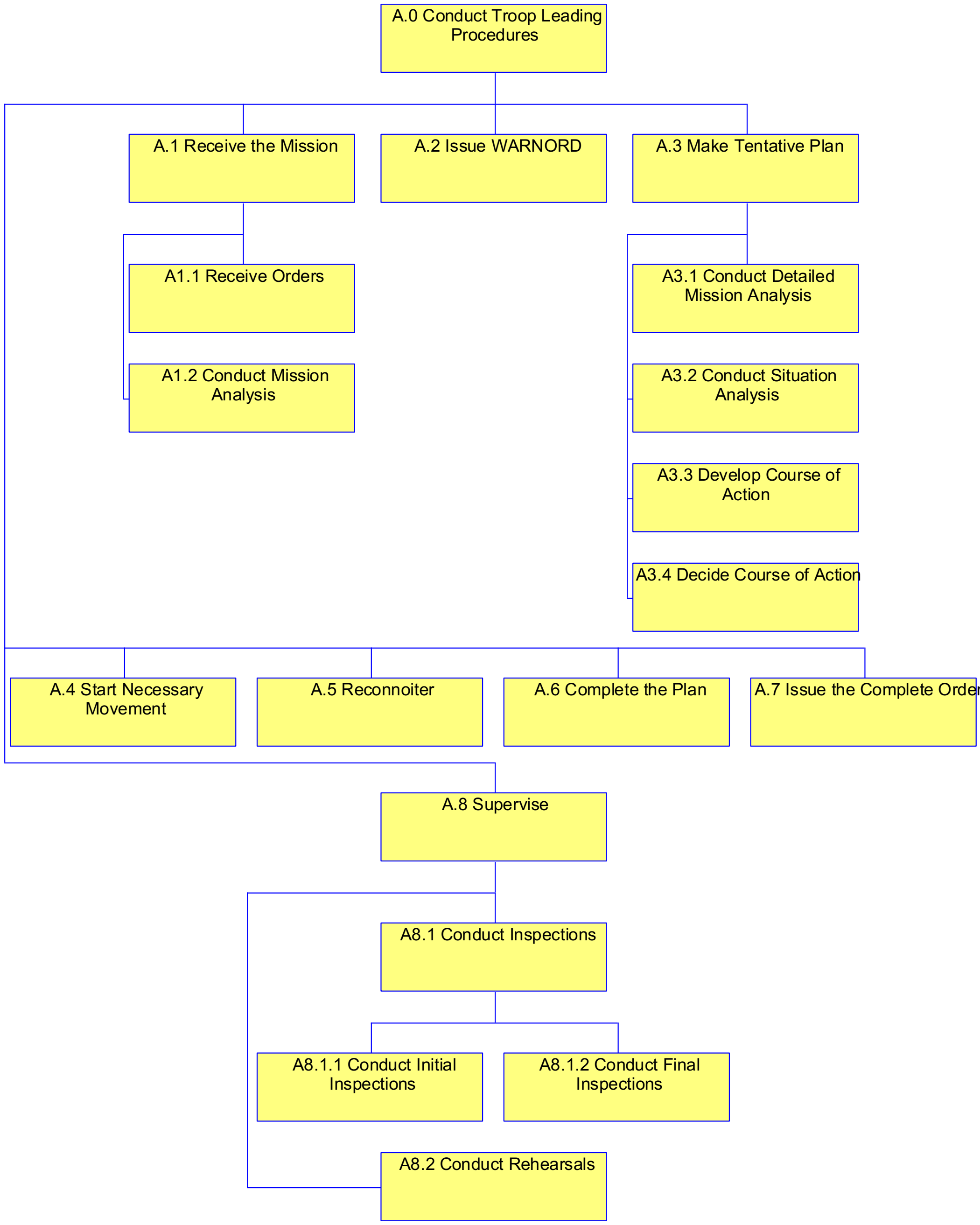



**Figure A-C-2: OV-5 – Operational Activity Model**

### Operational Activity Model Operational View (OV-5)

The following OV-5 activity modeling describes the operational activities executed by a GSS equipped infantry rifle company conducting troop leading procedures, and the activities of its subordinate platoons and squads, upon receiving a mission from its higher headquarters.

## GSS Conduct Troop Leading Procedures OV-5 Node Tree [OV-05 Node Tree]




**GSS Conduct Troop Leading Procedures OV-5 Node Tree [OV-05 Node Tree]**

**Scope**: This model depicts a GSS equipped infantry rifle company and its subordinate platoons and squads conducting troop leading procedures upon receiving a mission from its higher headquarters to execute an assault on an objective.

**Diagram Properties for: Conduct Troop Leading Procedures**
Audit ID: BENNNBT
IDEF0 Model:
Diagram Status: Working
Diagram Text: Troop leading is a process a leader goes through to prepare his unit to accomplish a tactical mission. This task is a subset of ART 7.4 "Plan Tactical Operations Using MDMP/TLP

Update Date: 7/18/2008

**Operational Activity: Receive the Mission**
**Duration:** Time allocated to Mission Planning is dictated by METT-T**:**
Glossary Text: The leader may receive the mission in the form of an OPORD, WARNORD, or FRAGO
ICOM line: COP Presentation Data
Input: coming from <off page>
Glossary Text: User requested information is presented in the tailored view of the user in the format decided by the user and is updated on a user defined schedule. Information determined by the Commander's information policy as essential to the user is also rendered as determined by that policy.
ICOM line: FRAGO
Input: coming from <off page>
Glossary Text: Provides timely changes of existing orders to subordinate leaders
ICOM line: Mission Analysis Data
Output: going to <off page>
Glossary Text: Data that provides insight into the mission requirements and the unit's ability to execute the mission
ICOM line: OPORD
Input: coming from <off page>
Glossary Text:

**Operational Activity: Issue WARNORD**
Duration: 30 Minutes
Glossary Text: The leader provides initial instructions in a warning order. The warning order contains enough information to begin preparation as soon as possible.

ICOM line: Mission Analysis Data
Input: coming from <off page>
Glossary Text: Data that provides insight into the mission requirements and the unit's ability to execute the mission
ICOM line: WARNORD
Output: going to <off page>

Glossary Text: Warning Order; A formatted preliminary notification of an order or action to follow. The WARNORD clearly informs the recipient of what tasks he must do now as well as informs him of possible future tasks

**Operational Activity: Make Tentative Plan**

Duration: Time allocated to Mission Planning is dictated by METT-T

Glossary Text: The leader develops an estimate of the situation to use as the basis for a tentative plan.

ICOM line: COP Presentation Data

Input: coming from <off page>

Glossary Text: User requested information is presented in the tailored view of the user in the format decided by the user and is updated on a user defined schedule. Information determined by the Commander's information policy as essential to the user is also rendered as determined by that policy.

ICOM line: Collaborative Planning Data

Output: going to <off page>

Glossary Text: Collaborative Planning between echelons is a critical enabler in the mission planning and decision making process

ICOM line: Graphic Control Measures

Output: going to complete the Plan as input

Glossary Text: Visual portrayal of controls using products of the cartographic and photogrammetric art, emplaced by leaders for coordination and force protection.

ICOM line: Logistics Request

Output: going to <off page>

Glossary Text: Request for logistical support

ICOM line: Mission Analysis Data

Input: coming from <off page>

Glossary Text: Data that provides insight into the mission requirements and the unit's ability to execute the mission

ICOM line: Preplanned Fire Support Request

Output: going to <off page>

Glossary Text: Request for planned effects that is submitted to higher HQ as a result of the mission planning process.

ICOM line: Request For Information

Output: going to <off page>

Glossary Text: Request made for information pertaining to mission execution

ICOM line: Status Report

Input: coming from Supervise as output

Glossary Text: Report submitted by a subordinate Soldier or leader detailing the battle readiness of personnel and equipment.

ICOM line: Tentative Plan

Output: going to <off page>

Glossary Text: Tentative plans are the basis for the OPORD. The leader uses the commander's estimate of the situation to analyze METT-T information, develop and analyze a COA, compare courses of actions and make a decision that produces a tentative plan and the focus of the reconnaissance effort

ICOM line: WARNORD
Input: coming from <off page>
Glossary Text: Warning Order; A formatted preliminary notification of an order or action to follow. The WARNORD clearly informs the recipient of what tasks he must do now as well as informs him of possible future tasks

**Operational Activity: Start Necessary Movement**
Duration: Movement time allocated depends on METT-T
Glossary Text: The platoon may need to begin movement while the leader is still planning or forward reconnoitering.

ICOM line: COP Presentation Data
Input: coming from <off page>
Glossary Text: User requested information is presented in the tailored view of the user in the format decided by the user and is updated on a user defined schedule. Information determined by the Commander's information policy as essential to the user is also rendered as determined by that policy.
ICOM line: Position Location Identification (PLI Data)
Input: coming from <off page>
Glossary Text: A satellite constellation that provides highly accurate position, velocity, and time navigation information to users. Also called GPS
ICOM line: Position Report Data
Output: going to <off page>
Glossary Text: Provides the locations, direction of movement, and speed of object
ICOM line: SITREP
Output: going to <off page>
Glossary Text: Situation report: Report submitted by a subordinate to a leader describing the current tactical situation
ICOM line: Tentative Plan
Input: coming from <off page>
Glossary Text: Tentative plans are the basis for the OPORD. The leader uses the commander's estimate of the situation to analyze METT-T information, develop and analyze a COA, compare courses of actions and make a decision that produces a tentative plan and the focus of the reconnaissance effort

**Operational Activity: Reconnoiter**
Duration: METT-T Dependent
Glossary Text: : If time allows the leader makes a personal reconnaissance to verify target intelligence, terrain analysis, adjust his plan, confirm the usability of routes, and time critical movements. When time does not allow, the leader must make a map reconnaissance.
ICOM line: COP Presentation Data
Input: coming from <off page>
Glossary Text: User requested information is presented in the tailored view of the user in the format decided by the user and is updated on a user defined schedule.

Information determined by the Commander's information policy as essential to the user is also rendered as determined by that policy.

ICOM line: COP Update

Output: going to <off page>

Glossary Text: Update Common Operational Picture with current information.

ICOM line: RECON ISR Data

Output: going to complete the Plan as input

Glossary Text: Leader Reconnaissance data that confirms terrain analysis, confirms usability of routes and times any critical movements. Data may be as a result of a physical reconnaissance or a map reconnaissance depending on time and risk constraints involved in the mission

ICOM line: SITREP

Output: going to <off page>

Glossary Text: Situation report: Report submitted by a subordinate to a leader describing the current tactical situation

ICOM line: Tentative Plan

Input: coming from <off page>

Glossary Text: Tentative plans are the basis for the OPORD. The leader uses the commander's estimate of the situation to analyze METT-T information, develop and analyze a COA, compare courses of actions and make a decision that produces a tentative plan and the focus of the reconnaissance effort

**Operational Activity: Complete the Plan**

Duration: Time allocated to Mission Planning is dictated by METT-T

Glossary Text: The leader completes his plan based on his reconnaissance and any changes to the situation.

ICOM line: After Action Review (AAR)

Input: coming from Supervise as output

Glossary Text: After action review conducted after rehearsal by leadership to include all Soldiers to evaluate the tentative plan and recommend changes

ICOM line: FRAGO

Output: going to Issue the Complete Order as input

Glossary Text: Provides timely changes of existing orders to subordinate leaders

ICOM line: Graphic Control Measures

Input: coming from Make Tentative Plan as output

Glossary Text: Visual portrayal of controls using products of the cartographic and photogrammetric art, emplaced by leaders for coordination and force protection.

ICOM line: RECON ISR Data

Input: coming from Reconnoiter as output

Glossary Text: Leader Reconnaissance data that confirms terrain analysis, confirms usability of routes and times any critical movements. Data may be as a result of a physical reconnaissance or a map reconnaissance depending on time and risk constraints involved in the mission

**Operational Activity: Issue the Complete Order**

Duration: 15 Minutes

Glossary Text: Company Commander, Platoon and squad leaders issue the complete order to their subordinates
ICOM line: FRAGO
Input: coming from Complete the Plan as output
Glossary Text: Provides timely changes of existing orders to subordinate leaders
ICOM line: OPORD
Output: going to Supervise as input

**Operational Activity: Supervise**
Duration: Supervision is continuous throughout the planning procedure
Glossary Text: The leader supervises the unit's preparation for combat by conducting rehearsals and inspections
ICOM line: After Action Review (AAR)
Output: going to complete the Plan as input
Glossary Text: After action review conducted after rehearsal by leadership to include all Soldiers to evaluate the tentative plan and recommend changes
ICOM line: OPORD
Input: coming from Issue the Complete Order as output
Glossary Text:
ICOM line: Status Report
Output: going to Make Tentative Plan as input
Glossary Text: Report submitted by a subordinate Soldier or leader detailing the battle readiness of personnel and equipment.
ICOM line: Tentative Plan
Input: coming from <off page>
Glossary Text: Tentative plans are the basis for the OPORD. The leader uses the commander's estimate of the situation to analyze METT-T information, develop and analyze a COA, compare courses of actions and make a decision that produces a tentative plan and the focus of the reconnaissance effort
ICOM line: WARNORD
Input: coming from <off page>
Glossary Text: Warning Order; A formatted preliminary notification of an order or action to follow. The WARNORD clearly informs the recipient of what tasks he must do now as well as informs him of possible future tasks

**Receive the Mission**

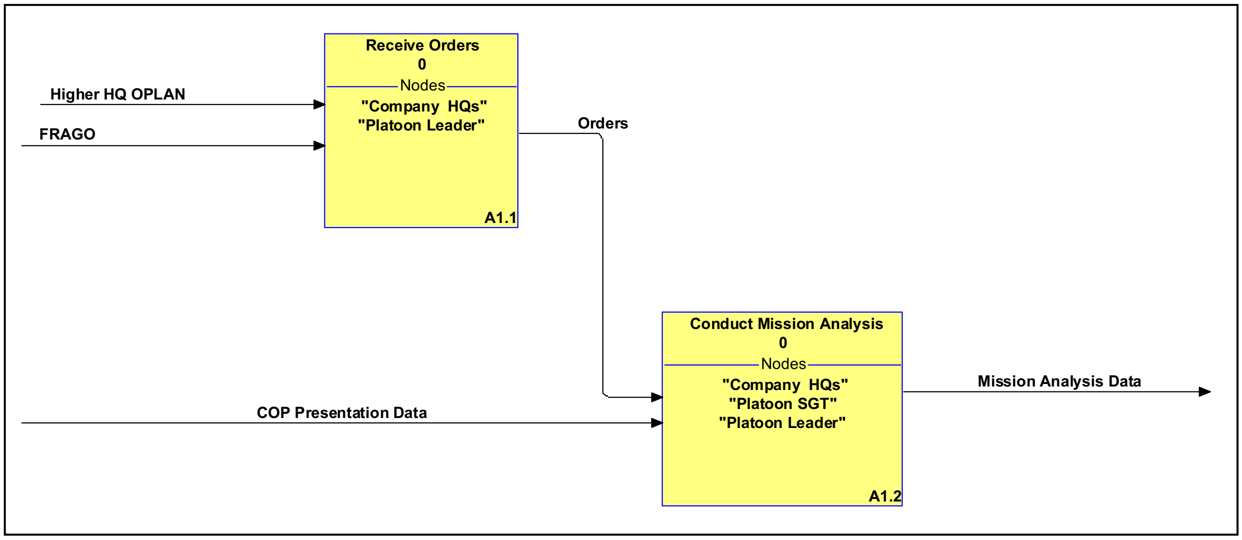



**Figure A-C-3: OV-5 – Operational Activity Model**

**Diagram Properties for: Receive the Mission**
Audit ID: BENNNBT
IDEF0 Model:
Diagram Status: Working
**Diagram Text:** The leader may receive the mission in the form of an OPORD, WARNORD, or FRAGO
Update Date: 7/17/2008
**Operational Activity: Receive Orders**
Duration: 30 minutes
Glossary Text: The leaders receive verbal or written orders in the form of an OPORD, WARNORD, or FRAGO
ICOM line: FRAGO
Input: coming from <off page>
Glossary Text: Provides timely changes of existing orders to subordinate leaders
ICOM line: Higher HQ OPLAN
Input: coming from <off page>
Glossary Text: Plan a commander uses to conduct military operations. Commanders may initiate preparation of possible operations by first issuing an OPLAN
ICOM line: Orders
Output: going to Conduct Mission Analysis as input
Glossary Text: A written or oral communication directing action. Includes OPORDs, FRAGOs, and WARNORDs, This information will include all available annexes with any available graphic control measures and IPB products. Orders could be delivered in data format with additional information in various formats including voice, charts and graphics, video and imagery.

**Operational Activity: Conduct Mission Analysis**

Duration: Planning Time is allocated by the CDR based on METT-T

Glossary Text: Mission analysis may consist of a seventeen step process describes in detail in FM-105-5 on p.5-5. METT-T will determine the time the leader has to allocate to the mission analysis process and the full seventeen step analysis may be abbreviated accordingly.

ICOM line: COP Presentation Data

Input: coming from <off page>

Glossary Text: User requested information is presented in the tailored view of the user in the format decided by the user and is updated on a user defined schedule. Information determined by the Commander's information policy as essential to the user is also rendered as determined by that policy.

ICOM line: Mission Analysis Data

Output: going to <off page>

Glossary Text: Data that provides insight into the mission requirements and the unit's ability to execute the mission

ICOM line: Orders

Input: coming from Receive Orders as output

Glossary Text: A written or oral communication directing action. Includes OPORDs, FRAGOs, and WARNORDs, This information will include all available annexes with any available graphic control measures and IPB products. Orders could be delivered in data format with additional information in various formats including voice, charts and graphics, video and imagery.

**Make Tentative Plan**

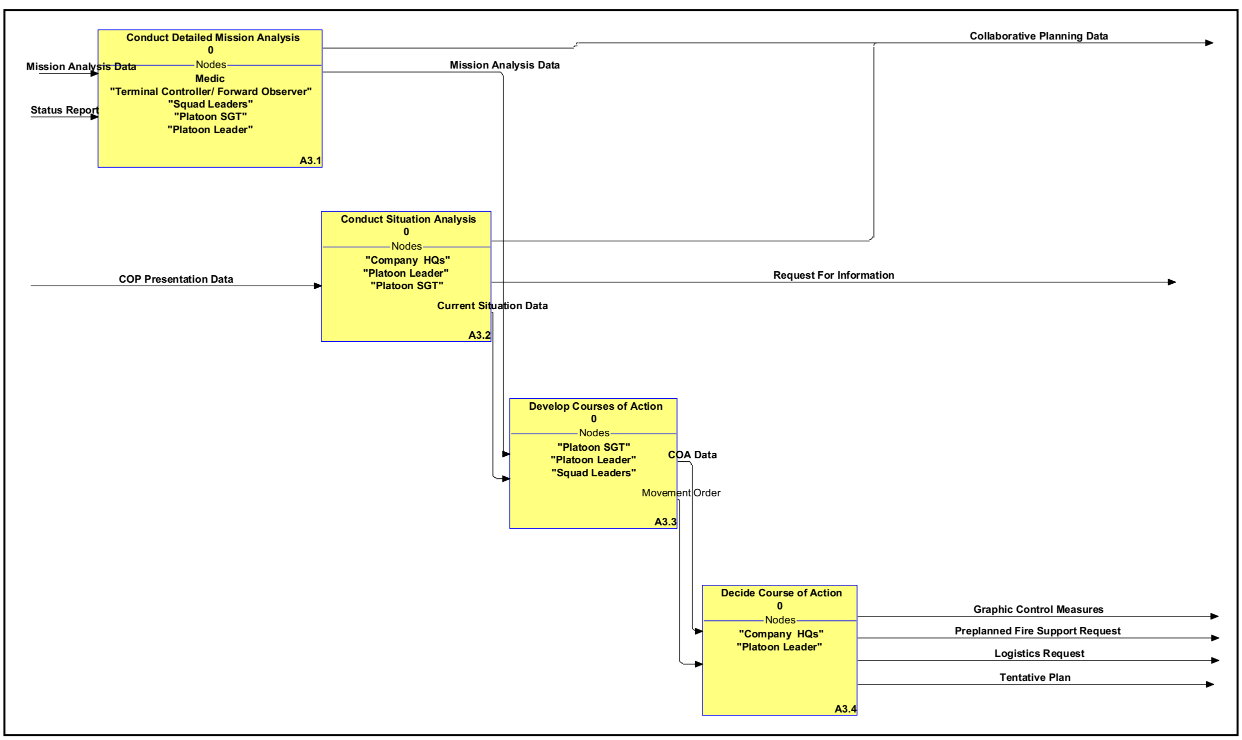



**Figure A-C-4: OV-5 – Operational Activity Model**

**Diagram Properties for: Make Tentative Plan**
Audit ID: BENNNBT
IDEF0 Model:
Diagram Status: Working
Diagram Text: The leader develops an estimate of the situation to use as the basis for a tentative plan.
Update Date: 7/17/2008

**Operational Activity: Conduct Detailed Mission Analysis**
Duration: Planning time is allocated by the commander based on METT-T
Glossary Text: The leader analyzes the mission in light of the commander's intent and derives the essential task his unit must perform in order to accomplish the mission.
ICOM line: Collaborative Planning Data
Output: going to <off page>
Glossary Text: Collaborative Planning between echelons is a critical enabler in the mission planning and decision making process
ICOM line: Mission Analysis Data
Output: going to Develop Courses of Action as input
Glossary Text: Data that provides insight into the mission requirements and the unit's ability to execute the mission
ICOM line: Mission Analysis Data
Input: coming from <off page>
Glossary Text: Data that provides insight into the mission requirements and the unit's ability to execute the mission
ICOM line: Status Report
Input: coming from <off page>
Glossary Text: Report submitted by a subordinate Soldier or leader detailing the battle readiness of personnel and equipment.

**Operational Activity: Conduct Situation Analysis**
Duration: Planning time is allocated by the commander based on METT-T
Glossary Text: Analysis of enemy and friendly situation as well as terrain, weather, and time available.
ICOM line: COP Presentation Data
Input: coming from <off page>
Glossary Text: User requested information is presented in the tailored view of the user in the format decided by the user and is updated on a user defined schedule. Information determined by the Commander's information policy as essential to the user is also rendered as determined by that policy.
ICOM line: Collaborative Planning Data
Output: going to <off page>
Glossary Text: Collaborative Planning between echelons is a critical enabler in the mission planning and decision making process
ICOM line: Current Situation Data

Output: going to Develop Courses of Action as input
Glossary Text: Data related to information on friendly and enemy forces, terrain, weather, and available time
ICOM line: Request For Information
Output: going to <off page>
Glossary Text: Request made for information pertaining to mission execution.

**Operational Activity: Develop Courses of Action**
Duration: Planning time is allocated by the commander based on METT-T
Glossary Text: Represents the various COAs developed for analysis
ICOM line: COA Data
Output: going to Decide Course of Action as input
Glossary Text: Represents the various COAs developed for analysis
ICOM line: Current Situation Data
Input: coming from Conduct Situation Analysis as output
Glossary Text: Data related to information on friendly and enemy forces, terrain, weather, and available time
ICOM line: Mission Analysis Data
Input: coming from Conduct Detailed Mission Analysis as output
Glossary Text: Data that provides insight into the mission requirements and the unit's ability to execute the mission
ICOM line: Movement Order
Output: going to Decide Course of Action as input
Glossary Text: A stand alone order that facilitates a unit’s movement. Movements are typically administrative.

**Operational Activity: Decide Course of Action**
Duration: Planning time is allocated by the commander based on METT-T
Glossary Text: The decision on a course of action represents a tentative plan
ICOM line: COA Data
Input: coming from Develop Courses of Action as output
Glossary Text: Represents the various COAs developed for analysis
ICOM line: Graphic Control Measures
Output: going to <off page>
Glossary Text: Visual portrayal of controls using products of the cartographic and photogrammetric art, emplaced by leaders for coordination and force protection.
ICOM line: Logistics Request
Output: going to <off page>
Glossary Text: Request for logistical support
ICOM line: Movement Order
Input: coming from Develop Courses of Action as output
Glossary Text: A stand alone order that facilitates a unit’s movement. Movements are typically administrative
ICOM line: Preplanned Fire Support Request
Output: going to <off page>

Glossary Text: Request for planned effects that is submitted to higher HQ as a result of the mission planning process.
ICOM line: Tentative Plan
Output: going to <off page>
Glossary Text: Tentative plans are the basis for the OPORD. The leader uses the commander's estimate of the situation to analyze METT-T information, develop and analyze a COA, compare courses of actions and make a decision that produces a tentative plan and the focus of the reconnaissance effort

**Supervise**

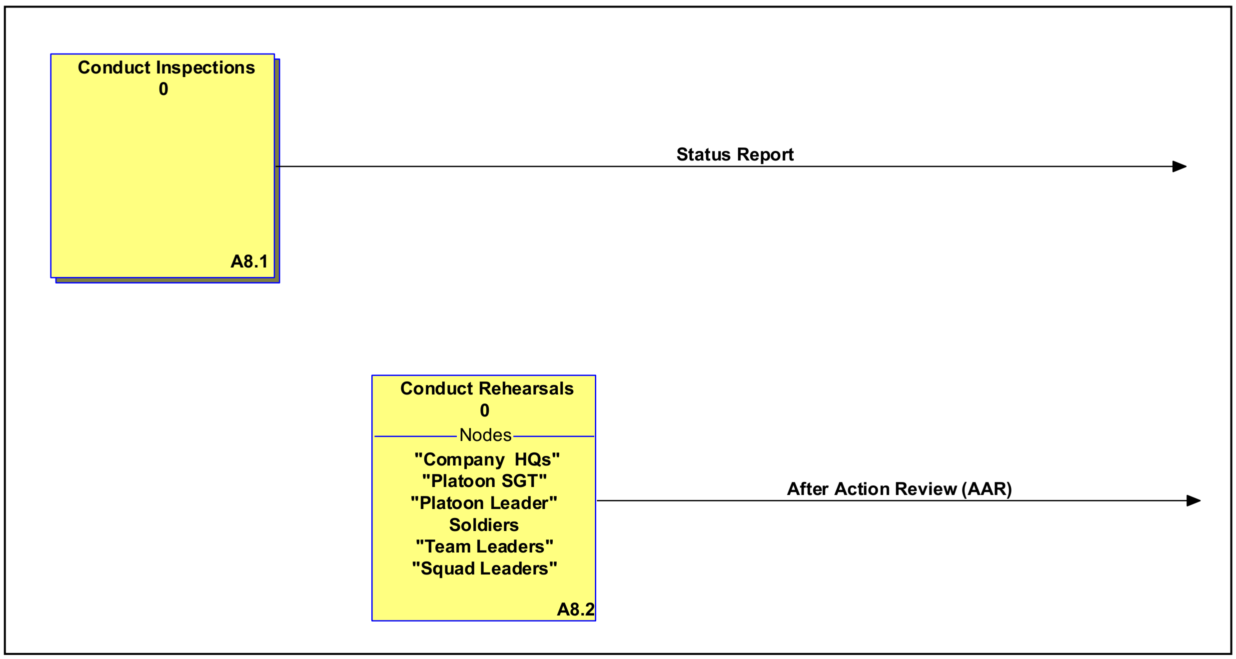



**Figure A-C-5: OV-5 – Operational Activity Model**

**Diagram Properties for: Supervise**
Audit ID: BENNNBT
IDEF0 Model:
Diagram Status: Working
Diagram Text: The leader supervises the unit's preparation for combat by conducting rehearsals and inspections:
Update Date: 7/17/2008

**Operational Activity: Conduct Inspections**
Duration: Supervision is continuous during mission planning Time Measure: day
Glossary Text: Inspections should be conducted to include weapons, ammunition, equipment, Soldier's mission understanding, communications, rations, camouflage
ICOM line: Status Report

Output: going to <off page>
Glossary Text: Report submitted by a subordinate Soldier or leader detailing the battle readiness of personnel and equipment.

**Operational Activity: Conduct Rehearsals**
Duration: Time determined by METT-T
Glossary Text: Rehearsals are essential to improving the Soldiers understanding of the mission and revealing weakness or problems in the plan. Rehearsals will help coordinate the actions of subordinates
ICOM line: After Action Review (AAR)
Output: going to <off page>
Glossary Text: After action review conducted after rehearsal by leadership to include all Soldiers to evaluate the tentative plan and recommend changes.

**Conduct Inspections**

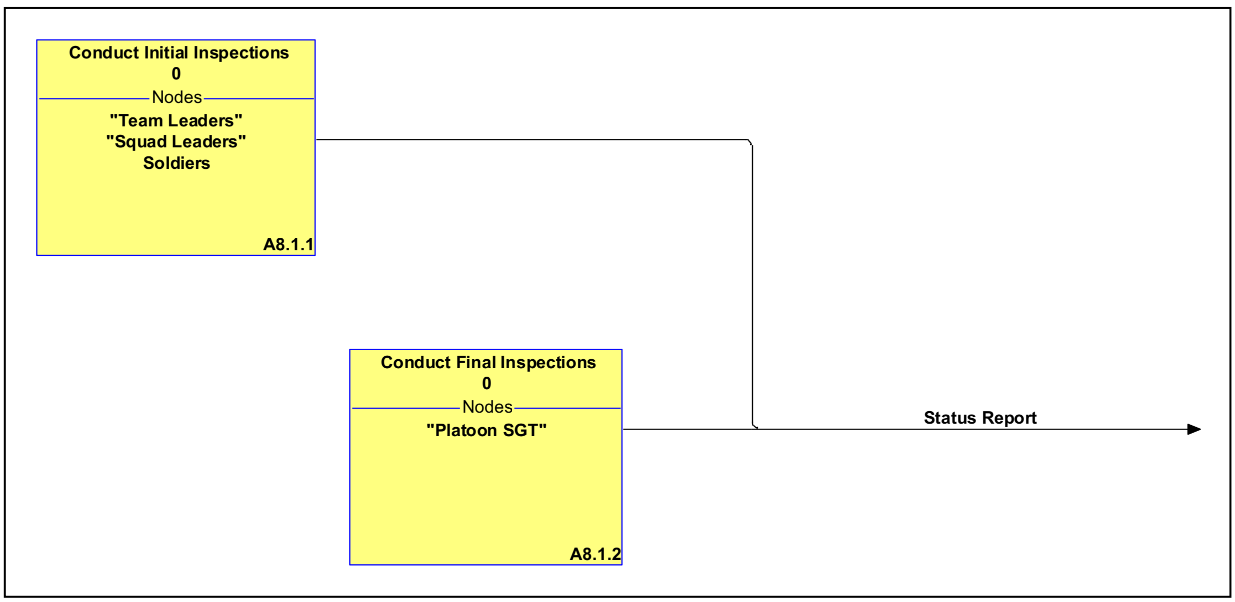



**Figure A-C-6: OV-5 – Operational Activity Model**

**Diagram Properties for: Conduct Inspections**
Audit ID: BENNNBT
IDEF0 Model:
Diagram Status: Working
Diagram Text: Inspections should be conducted to include weapons, ammunition, equipment, Soldier's mission understanding, communications, rations, camouflage
Update Date: 7/17/2008

**Operational Activity: Conduct Initial Inspections**
Duration: 30 Minutes
Glossary Text: Initial Inspections should be conducted by the first line supervisor to determine all deficiencies

ICOM line: Status Report
Output: going to <off page>
Glossary Text: Report submitted by a subordinate Soldier or leader detailing the battle readiness of personnel and equipment.

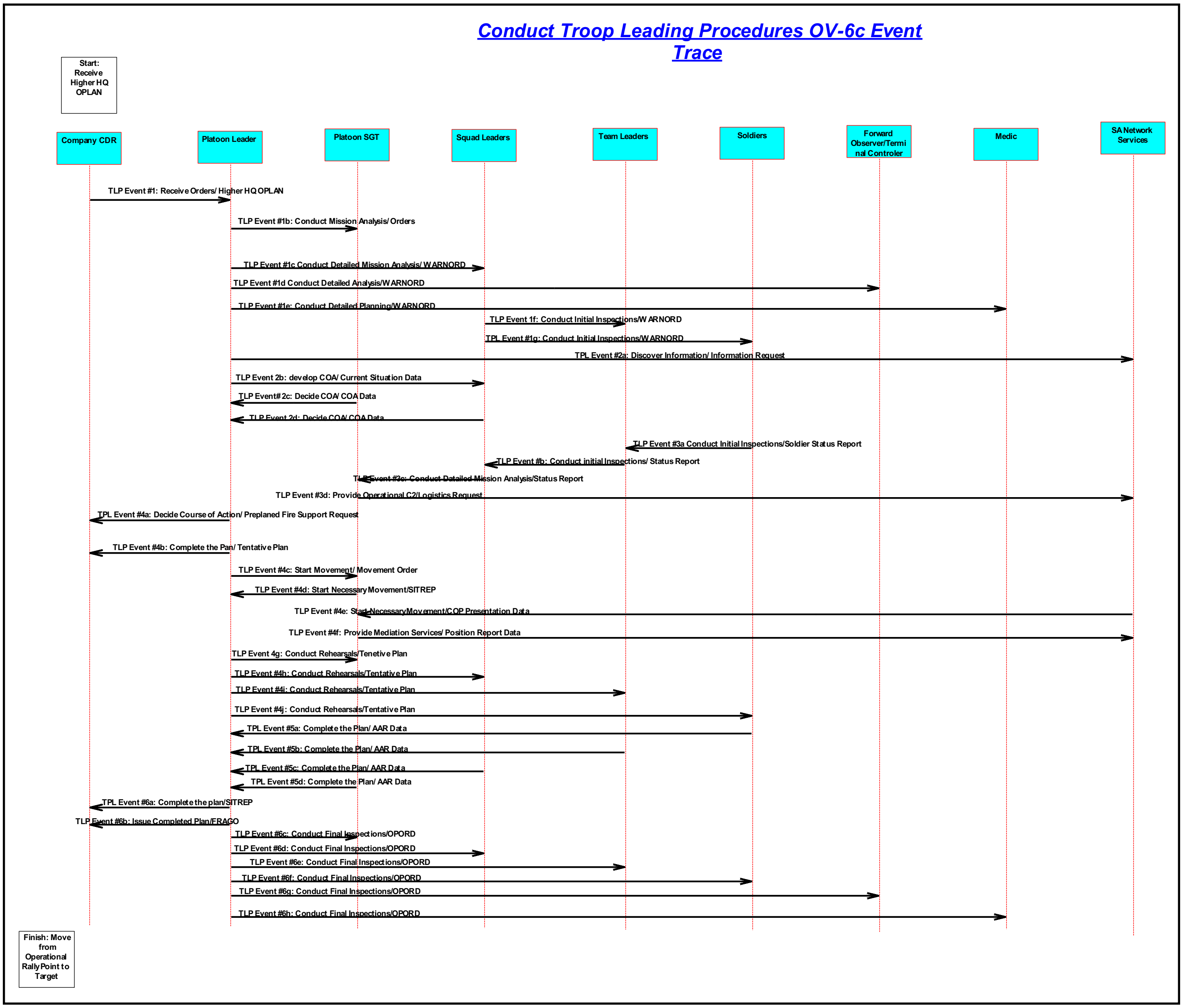
**Operational Activity: Conduct Final Inspections**
Duration: 30 Minutes
Glossary Text: Final Inspections should be conducted by the Mission Commanders and the Unit NCIOC.
ICOM line: Status Report
Output: going to <off page>
Glossary Text: Report submitted by a subordinate Soldier or leader detailing the battle readiness of personnel and equipment.

## Operational Information Exchange Matrix Operational View (OV-3).

The GSS OV-3 Operational Information Exchange Matrix is embedded as an excel spreadsheet below. The OV-3 details exchanges and identifies "who exchanges what information, with whom, why the information is necessary, and how the information exchange must occur.

Troop Leading Procedures OV-3 .xl

## Operational Event-Trace Description (OV-6c) Ground Soldier System (GSS)




Operational Event-Trace Description (OV-6c) Ground Soldier System (GSS)

**Figure A-C-7: OV-6c – Operational Event Trace Description**

### Conduct Troop Leading Procedures

**Scope:** This thread depicts a company conducting troop leading procedures upon receiving a mission from its higher headquarters to execute an assault on an objective. In this vignette multiple tasks and collaborative planning actions would occur.

**Data Element Description**

**TLP Event #1a**: Platoon receives the company OPLAN
**TLP Event #1b**: Platoon leader and the platoon SGT conduct initial analysis of the order.
**TLP Event #1c**: After initial analysis the Platoon Leader Issues a WARNORD to the Platoon Leadership. (Event 1c, 1d, and 1e occur at the same time when the Platoon leader issues the WARNORD to the platoon leadership).
**TPL Event #1d**: FO receives WARNORD (Event 1c, 1d, and 1e occur at the same time when the Platoon leader issues the WARNORD to the platoon leadership)
**TPL Event #1e**: Medic receives WARNORD (Event 1c, 1d, and 1e occur at the same time when the Platoon leader issues the WARNORD to the platoon leadership)
**TPL Event #1f**: Squad Leader Issues WARNORD to Squad (Event 1f and 1g are completed simultaneously)
**TPL Event #1g**: Squad Leader Issues WARNORD to Squad (Event 1f and 1g are completed simultaneously)
**TPL Event #1h**: Squad Leader Issues WARNORD to Squad (Event 1f and 1g are completed simultaneously)
**TPL Event #2a**: Platoon leader queries for information/imagery or other target information.
**TPL Event #2b**: Platoon leader provides COA development teams with latest situational updates
**TPL Event #2c**: Platoon SGT assists the Platoon Leader in developing COAs
**TPL Event #2d**: Squad Leaders assist Platoon Leader in developing COAs.
**TPL Event #3a**: Initial Inspection of equipment is conducted to identify all shortages and insure mission readiness.   Soldiers inspect all personal and team equipment and report status to their team leaders.
**TPL Event #3b**: Team Leaders report the status of team equipment and personnel to their squad leaders
**TPL Event #3c**: Squad leader reports squad status and equipment shortages to the platoon SGT
**TPL Event #3d**: Based on the status reports he has received from the initial inspections and the mission analysis that has been done, the platoon SGT submits logistics requests. This request will enter the CSS network at the company level.
**TPL Event #4a**: With the FO's assistance, the platoon leaders prepares a request for fire support to be forwarded to the company commander for approval.
**TPL Event #4b**: After deciding on a COA the platoon leader provides a brief back to the company commander for plan approval.
**TPL Event #4c**: Platoon leader provides administrative movement instructions for the platoon to the platoon SGT.
**TPL Event #4d**: Platoon SGT provides the platoon leader of the status of the administrative movement of the platoon.
**TPL Event #4e**: As OIC in charge of the movement the Platoon SGT receives the COP while conducting the movement.
**TPL Event #4f**: Platoon position is updated to the COP during administrative movement.
**TPL Event #4g**: Tentative Plan is provided to the platoon (4g, 4h, 4i, and 4j are conducted as one event)
**TPL Event #4h**: Tentative Plan is provided to the platoon (4g, 4h, 4i, and 4j are conducted as one event)
**TPL Event #4i**: Tentative Plan is provided to the platoon (4g, 4h, 4i, and 4j are conducted as one event)
**TPL Event #4j**: Tentative Plan is provided to the platoon (4g, 4h, 4i, and 4j are conducted as one event)
**TPL Event #5a:** Rehearsals are conducted and the platoon leader conducts after action review in which all platoon personnel participate (5a, 5b, 5c, and 5d occurs as one event.

**TPL Event #5b:** Rehearsals are conducted and the platoon leader conducts after action review in which all platoon personnel participate (5a, 5b, 5c, and 5d occurs as one event.
**TPL Event #5c:** Rehearsals are conducted and the platoon leader conducts after action review in which all platoon personnel participate (5a, 5b, 5c, and 5d occurs as one event.
**TPL Event #5d:** Rehearsals are conducted and the platoon leader conducts after action review in which all platoon personnel participate (5a, 5b, 5c, and 5d occurs as one event.
**TPL Event #6a:** The platoon leader conducts a reconnaissance of the target. He will often take key element leaders with him. Any changes in the situation are reported up the chain of command.
**TPL Event #6b:** The platoon leader informs the company commander of any proposed plan changes based on his recon of the target area.
**TPL Event #6c:** The platoon Leader issues the final order to the platoon. Often the platoon leader will only address changes to the original plan in the form of a FRAGO. The platoon members conduct final inspections of all mission critical equipment. Often the platoon leader will only address changes to the original plan. (6c-6h are completed as one event)
**TPL Event #6d:** The platoon Leader issues the final order to the platoon. Often the platoon leader will only address changes to the original plan in the form of a FRAGO. The platoon members conduct final inspections of all mission critical equipment. Often the platoon leader will only address changes to the original plan. (6c-6h are completed as one event)
**TPL Event #6e:** The platoon Leader issues the final order to the platoon. Often the platoon leader will only address changes to the original plan in the form of a FRAGO. The platoon members conduct final inspections of all mission critical equipment. Often the platoon leader will only address changes to the original plan. (6c-6h are completed as one event)
**TPL Event #6f:** The platoon Leader issues the final order to the platoon. Often the platoon leader will only address changes to the original plan in the form of a FRAGO. The platoon members conduct final inspections of all mission critical equipment. Often the platoon leader will only address changes to the original plan. (6c-6h are completed as one event)
**TPL Event #6g:** The platoon Leader issues the final order to the platoon. Often the platoon leader will only address changes to the original plan in the form of a FRAGO. The platoon members conduct final inspections of all mission critical equipment. Often the platoon leader will only address changes to the original plan. (6c-6h are completed as one event)
**TPL Event #6h:** The platoon Leader issues the final order to the platoon. Often the platoon leader will only address changes to the original plan in the form of a FRAGO. The platoon members conduct final inspections of all mission critical equipment. Often the platoon leader will only address changes to the original plan. (6c-6h are completed as one event)

## Operational Node Connectivity Description (OV-2) Ground Soldier System (GSS)

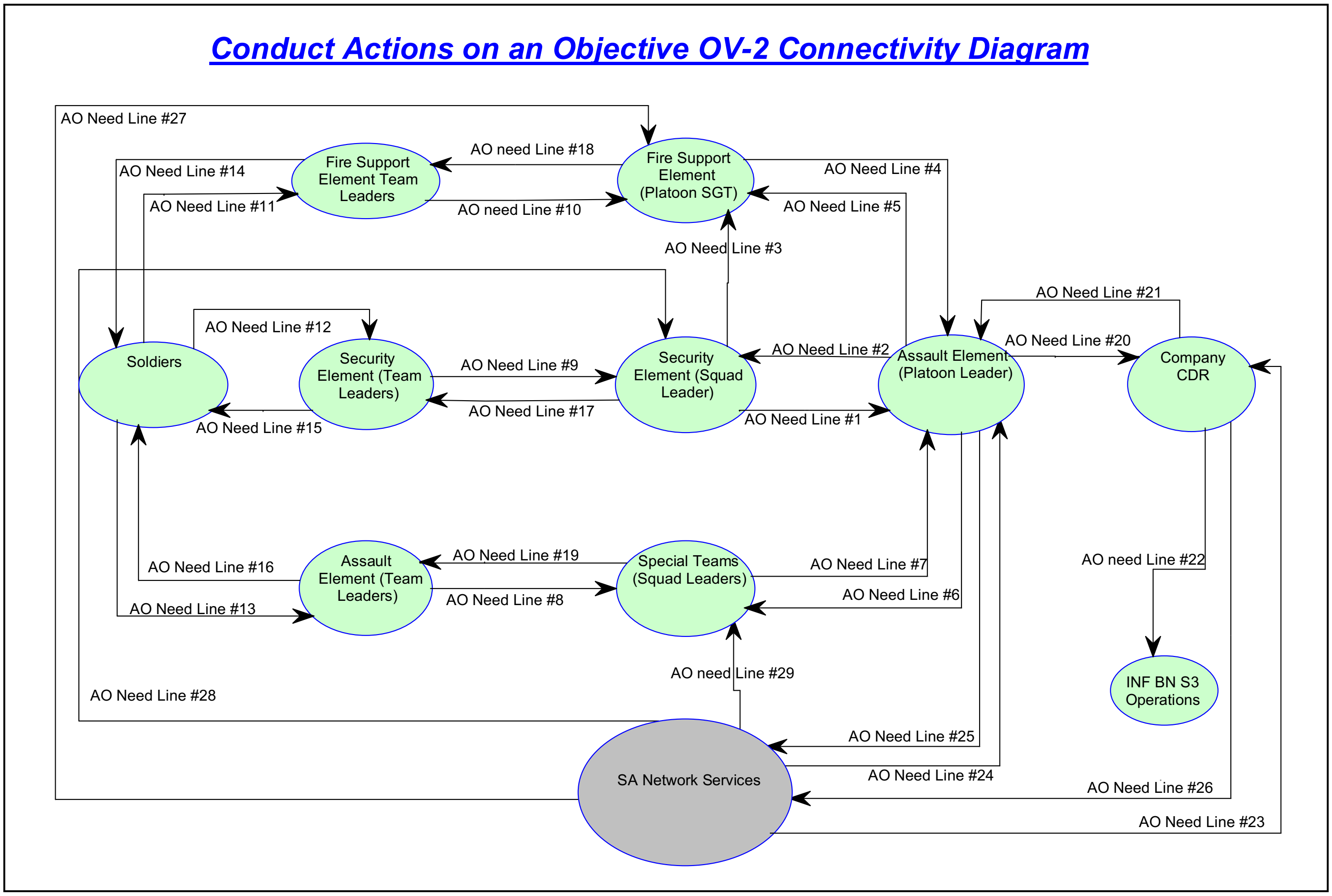



**Figure A-C-8: OV-2 – Operational Node Connectivity Description**

**Scope:** A GSS equipped Infantry platoon assaults an objective as part of a company offensive mission. The platoon is tasked organized according to mission requirements. This task organization is mission specific and does not represent all possible variations nor does it attempt to capture all communications above the platoon level. This diagram concentrates on the connectivity at the lowest levels of the assaulting element.

### Data Element Description: Need Lines

**AO Need Line #1** [Security Element (Squad Leaders) to Assault Element (Platoon Leader)]: Tactical Information: Mission Data
**AO Need Line #2** [Assault Element (Platoon Leader) to Security Element (Squad Leaders)]: Tactical Information: C2 Data
**AO Need Line #3** [Security Element (Squad Leaders) to Fire Support Element (Platoon SGT)]: Tactical Information: Mission Data
**AO Need Line #4** [Fire Support Element (Platoon SGT) to Assault Element (Platoon Leader)]: Tactical Information: Mission Data

**AO Need Line #5** [Assault Element (Platoon Leader) to Fire Support Element (Platoon SGT)]: Tactical Information: C2 Data
**AO Need Line #6** [Assault Element (Platoon Leader) to Special Teams (Squad Leaders)]: Tactical Information: C2 Data
**AO Need Line #7** [Special Teams (Squad Leaders) to Assault Element (Platoon Leader)]: Tactical Information: Mission Data
**AO Need Line #8** [Assault Element (Team Leader) to Special Teams (Squad Leaders)]: Tactical Information: Status Data
**AO Need Line #9** [Security Element (Team Leaders) to Security Element (Squad Leaders)]: Tactical Information: Status Data
**AO Need Line #10** [Fire Support Element (Team Leader) to Fire Support Element (Platoon SGT)]: Tactical Information: Status Data
**AO Need Line #11** [Soldiers to Fire Support Element (Team Leader)]: Tactical Information: Status Data
**AO Need Line #12** [Soldiers to Security Element (Team Leaders)]: Tactical Information: Status Data
**AO Need Line #13** [Soldiers to Assault Element (Team Leaders)]: Tactical Information: Status Data
**AO Need Line #14** [Fire Support Element (Team Leader) to Soldier]: Tactical Information: Status Data
**AO Need Line #15** [Security Element (Team Leaders) to Soldier]: Tactical Information: Status Data
**AO Need Line #16** [Assault Element (Team Leaders) to Soldier]: Tactical Information: Status Data
**AO Need Line #17** [Security Element (Squad Leaders) to Security Element (Team Leaders)]: Tactical Information: C2 Data
**AO Need Line #18** [Fire Support Element (Platoon SGT) to Fire Support Element (Team Leader)]: Tactical Information: C2 Data
**AO Need Line #19** [Special Teams (Squad Leaders) to Assault Element (Team Leader)]: Tactical Information: C2 Data
**AO Need Line #20** [Assault Element (Platoon Leader) to Company CDR)]: Tactical Information: Mission Data
**AO Need Line #21** [Company CDR to Assault Element (Platoon Leader)]: Tactical Information: C2 Data
**AO Need Line #22** [Company CDR to INF BN S3 Operations)]: Tactical Information: Mission Data
**AO Need Line #23** (SA Network Services to Company CDR): Tactical Information: COP Data
**AO Need Line #24** [SA Network Services to Assault Element (Platoon Leader)]: Tactical Information: COP Data
**AO Need Line #25** [Assault Element (Platoon Leader) to SA Network Services)]: Tactical Information: COP Data
**AO Need Line #26** (Company CDR to SA Network Services): Tactical Information: COP Data
**AO Need Line #27** [SA Network Services to Fire Support Element (Platoon SGT)]: Tactical Information: COP Data

**AO Need Line #28** [SA Network Services to Security Element (Squad Leaders)]: Tactical Information: COP Data
**AO Need Line #29** [SA Network Services to Special Teams (Squad Leaders)]: Tactical Information: COP Data

**Operational Activity Model (OV-5) Ground Soldier System (GSS)**

**Conduct Actions on the Objective**

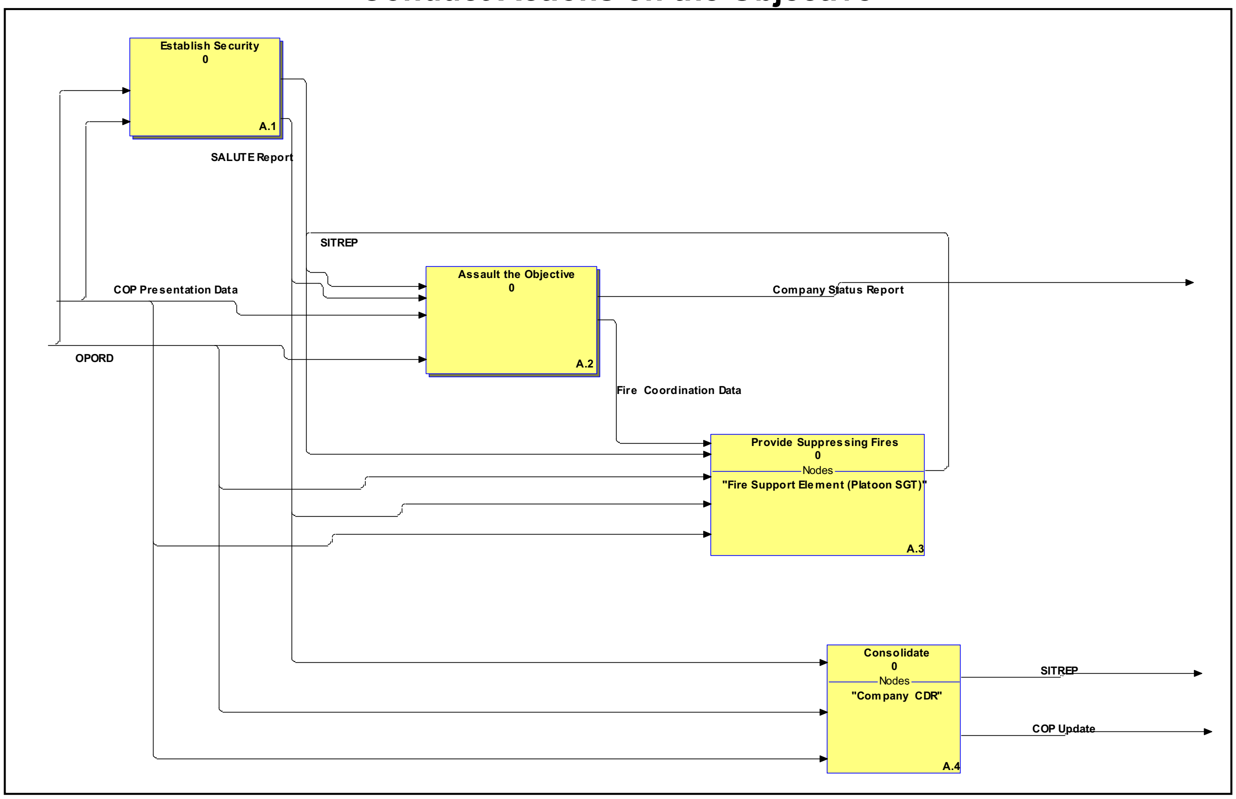
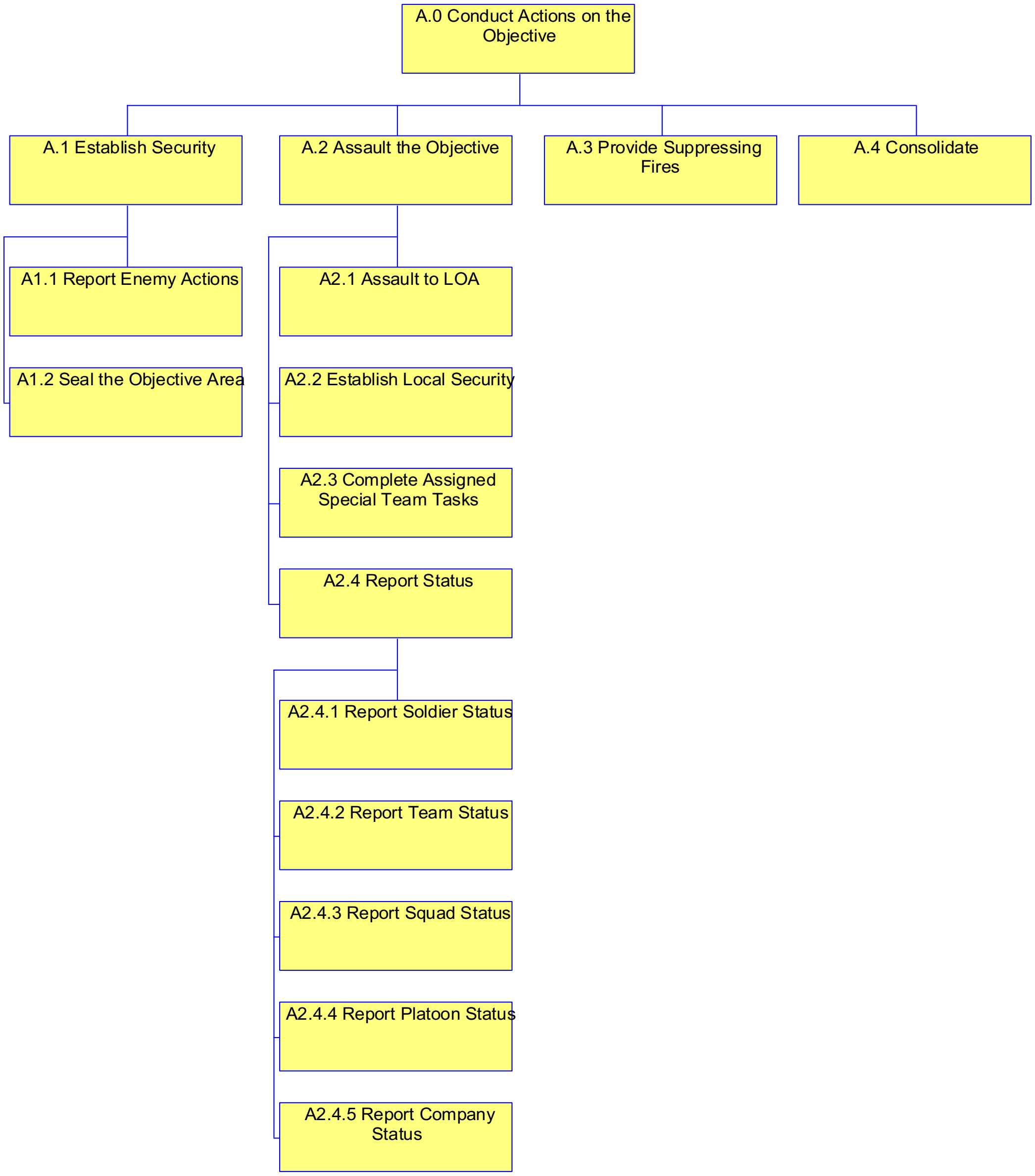



**Figure A-C-9: OV-5 – Operational Activity Model**

## GSS Conduct Actions on the Objective OV-5 Node Tree [OV-05 Node Tree]



**Operational Activity Model Operational View (OV-5)**
The following OV-5 activity modeling describes the operational activities executed by a GSS equipped infantry rifle platoon conducting actions on the objective. The platoon is the assault element for the company and therefore shows a representation of multiple activities (such as suppression and security).

**Scope:** This model depicts a GSS equipped Infantry platoon assaulting an objective. The platoon is tasked organized according to mission requirements. This model is mission specific and does not represent all possible variations nor does it attempt to capture activities above the platoon level.

**Diagram Properties for: ART 2.2.2 Conduct Actions on the Objective**
Audit ID: BENNNBT
IDEF0 Model:
Diagram Status: Working
Diagram Text: Develop the situation once contact is made concentrate the effects of combat power, and transition to a hasty attack or defense. Whether attacking or defending, commanders generate and sustain overwhelming combat power at the point combat forces collide to rapidly defeat the enemy**.**
Update Date: 7/17/2008

**Operational Activity: Establish Security**
Duration: Continuous during assault
Glossary Text: User Defined; Security teams secure the flanks of the objective site and provide early warning and seals off the objective from outside support or reinforcement
ICOM line: COP Presentation Data
    Input: coming from <off page>
Glossary Text: User requested information is presented in the tailored view of the user in the format decided by the user and is updated on a user defined schedule. Information determined by the Commander's information policy as essential to the user is also rendered as determined by that policy.
ICOM line: OPORD
    Input: coming from <off page>
Glossary Text:
ICOM line: SALUTE Report
    Output: going to <off page>
Glossary Text: Report of enemy activity which includes size, activity, location, unit, time, equipment.
ICOM line: SITREP
    Output: going to <off page>
Glossary Text: Situation report: Report submitted by a subordinate to a leader describing the current tactical situation

**Operational Activity: Assault the Objective**
Duration: METT-T Determined

Glossary Text: The assault element attacks and secures the objective
ICOM line: COP Presentation Data
Input: coming from <off page>
Glossary Text: User requested information is presented in the tailored view of the user in the format decided by the user and is updated on a user defined schedule. Information determined by the Commander's information policy as essential to the user is also rendered as determined by that policy.
ICOM line: Company Status Report
Output: going to <off page>
Glossary Text: Report from the company commander to his superior detailing the combat effectiveness of his unit and its ability to conduct combat operations.
ICOM line: Fire Coordination Data
Output: going to Provide Suppressing Fires as input
Glossary Text: Coordination between the assault element and the fire support element in order to avoid fratricide and fully suppress the enemy fire. Includes left and shift fires instructions.
ICOM line: OPORD
Input: coming from <off page>
Glossary Text:
ICOM line: SALUTE Report
Input: coming from <off page>
Glossary Text: Report of enemy activity which includes size, activity, location, unit, time, equipment.
ICOM line: SITREP
Input: coming from <off page>
Glossary Text: Situation report: Report submitted by a subordinate to a leader describing the current tactical situation

**Operational Activity: Provide Suppressing Fires**
Duration: METT-T Determined
Glossary Text: Supporting elements provide suppressing fires to gain fire superiority over enemy forces and provide cover for the assaulting element as it moves towards the objective.
ICOM line: COP Presentation Data
Input: coming from <off page>
Glossary Text: User requested information is presented in the tailored view of the user in the format decided by the user and is updated on a user defined schedule. Information determined by the Commander's information policy as essential to the user is also rendered as determined by that policy.
ICOM line: Fire Coordination Data
Input: coming from Assault the Objective as output
Glossary Text: Coordination between the assault element and the fire support element in order to avoid fratricide and fully suppress the enemy fire. Includes left and shift fires instructions.
ICOM line: OPORD
Input: coming from <off page>

Glossary Text:
ICOM line: SALUTE Report
    Input: coming from <off page>
Glossary Text: Report of enemy activity which includes size, activity, location, unit, time, equipment.
ICOM line: SITREP
    Output: going to <off page>
Glossary Text: Situation report: Report submitted by a subordinate to a leader describing the current tactical situation
ICOM line: SITREP
    Input: coming from <off page>
Glossary Text: Situation report: Report submitted by a subordinate to a leader describing the current tactical situation

**Operational Activity: Consolidate**
Duration: METT-T Determined
Glossary Text: Once enemy resistance on the objective has ceased, the attacking force must quickly take steps to consolidate and prepare to defend against counter attack.
ICOM line: COP Presentation Data
    Input: coming from <off page>
Glossary Text: User requested information is presented in the tailored view of the user in the format decided by the user and is updated on a user defined schedule. Information determined by the Commander's information policy as essential to the user is also rendered as determined by that policy.
ICOM line: COP Update
    Output: going to <off page>
Glossary Text: Update Common Operational Picture with current information.
ICOM line: OPORD
    Input: coming from <off page>
Glossary Text:
ICOM line: SALUTE Report
    Input: coming from <off page>
Glossary Text: Report of enemy activity which includes size, activity, location, unit, time, equipment.
ICOM line: SITREP
    Output: going to <off page>
Glossary Text: Situation report: Report submitted by a subordinate to a leader describing the current tactical situation

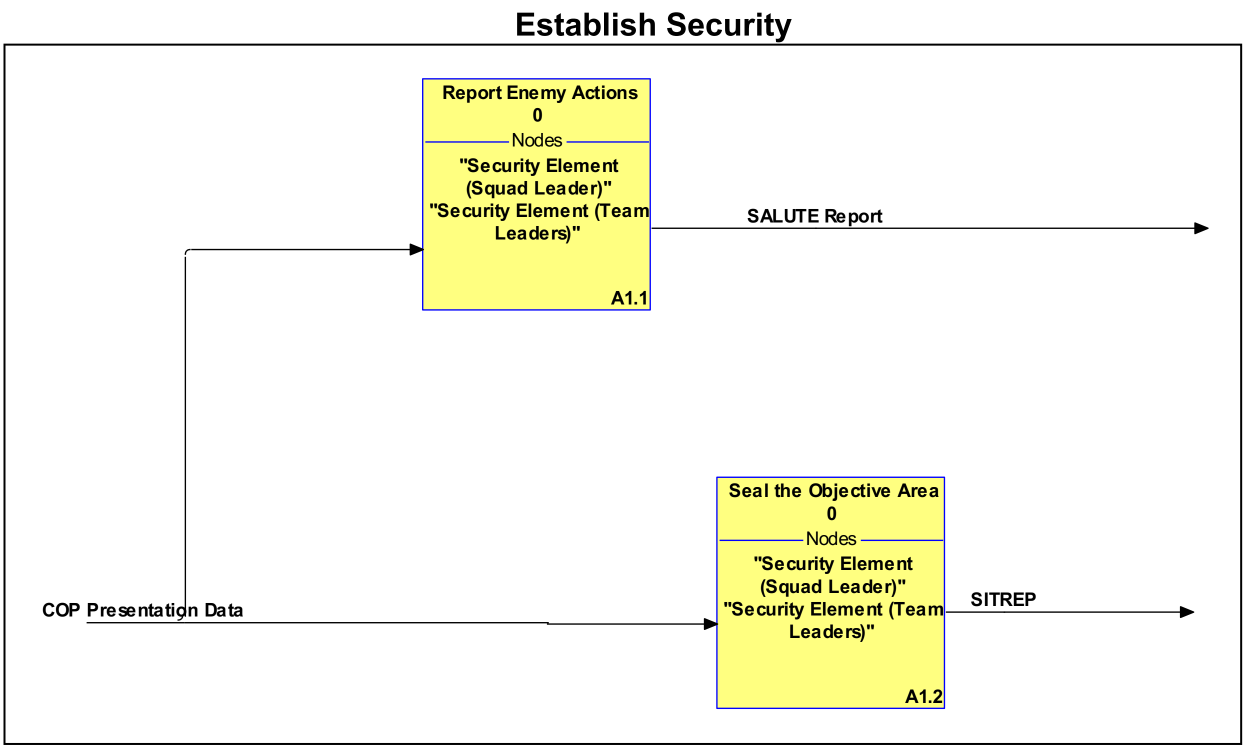



**Figure A-C-10: OV-5 – Operational Activity Model**

**Diagram Properties for: Establish Security**
Audit ID: BENNNBT
IDEF0 Model:
Diagram Status: Working
Diagram Text: Security teams secure the flanks of the objective site and provide early warning and seals off the objective from outside support or reinforcement.
Update Date: 7/17/2008

**Operational Activity: Report Enemy Actions**
Duration: 15 Seconds
Glossary Text: User defined: provide early warning of enemy movement towards or around the objective.
ICOM line: COP Presentation Data
Input: coming from <off page>
Glossary Text: User requested information is presented in the tailored view of the user in the format decided by the user and is updated on a user defined schedule. Information determined by the Commander's information policy as essential to the user is also rendered as determined by that policy.
ICOM line: SALUTE Report
Output: going to <off page>
Glossary Text: Report of enemy activity which includes size, activity, location, unit, time, equipment.

**Operational Activity: Seal the Objective Area**

Duration: Continuous
Glossary Text: Seals off the objective from outside support or reinforcement
ICOM line: COP Presentation Data
Input: coming from <off page>
Glossary Text: User requested information is presented in the tailored view of the user in the format decided by the user and is updated on a user defined schedule. Information determined by the Commander's information policy as essential to the user is also rendered as determined by that policy.
ICOM line: SITREP
Output: going to <off page>
Glossary Text: Situation report: Report submitted by a subordinate to a leader describing the current tactical situation.

## Assault the Objective

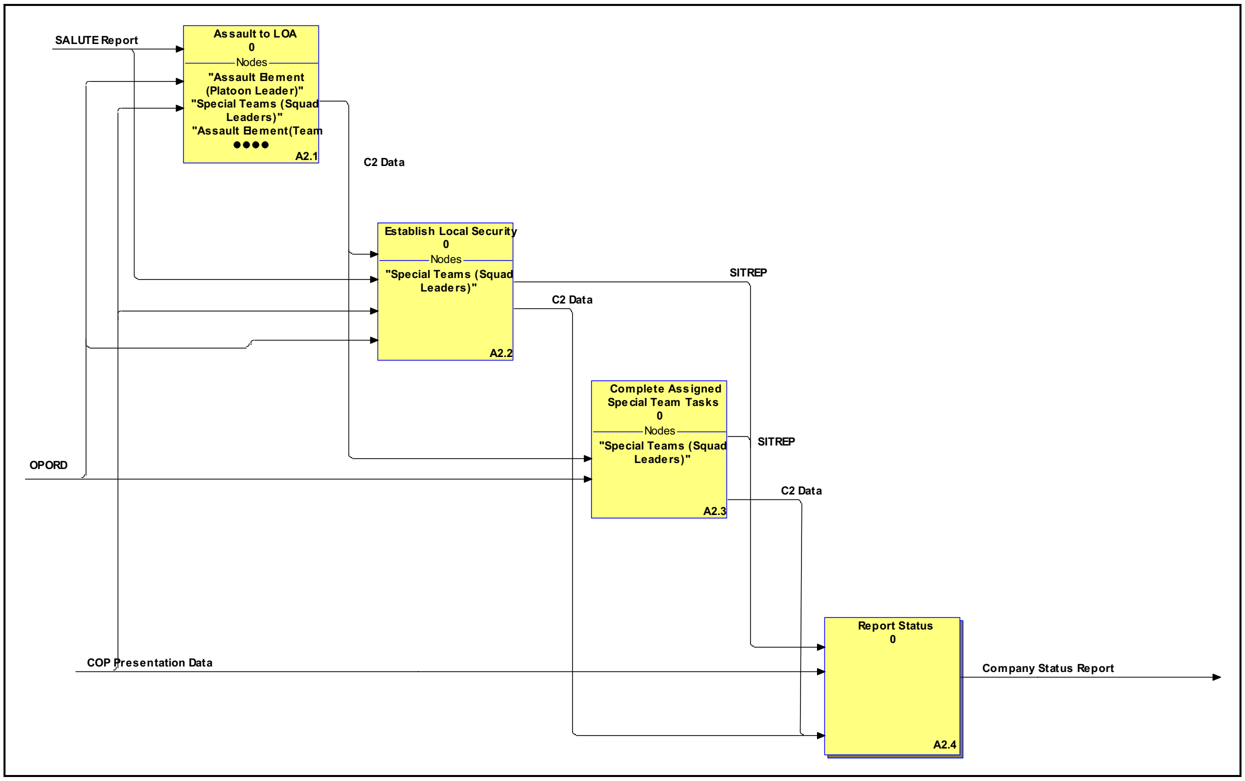



**Figure A-C-11: OV-5 – Operational Activity Model**

**Diagram Properties for: Assault the Objective**
Audit ID: BENNNBT
IDEF0 Model:
Diagram Status: Working
Diagram Text: The assault element attacks and secures the objective.
Update Date: 7/17/2008

**Operational Activity: Assault to LOA**
Duration: METT-T Determined
Glossary Text: Assault element assaults all the way through the objective to the designated limit of advance LOA.
ICOM line: C2 Data
Output: going to <off page>
Glossary Text: Leaders Instructions to a subordinate
ICOM line: COP Presentation Data
Input: coming from <off page>
Glossary Text: User requested information is presented in the tailored view of the user in the format decided by the user and is updated on a user defined schedule. Information determined by the Commander's information policy as essential to the user is also rendered as determined by that policy.
ICOM line: OPORD

Input: coming from <off page>
Glossary Text:
ICOM line: SALUTE Report
Input: coming from <off page>
Glossary Text: Report of enemy activity which includes size, activity, location, unit, time, equipment.

**Operational Activity: Establish Local Security**
Duration: METT-T Determined Time Measure: day
Glossary Text: Assault element leaders establish local security along the LOA.
ICOM line: C2 Data
Output: going to <off page>
Glossary Text: Leaders Instructions to a subordinate
ICOM line: C2 Data
Input: coming from <off page>
Glossary Text: Leaders Instructions to a subordinate
ICOM line: COP Presentation Data
Input: coming from <off page>
Glossary Text: User requested information is presented in the tailored view of the user in the format decided by the user and is updated on a user defined schedule. Information determined by the Commander's information policy as essential to the user is also rendered as determined by that policy.
ICOM line: OPORD
Input: coming from <off page>
Glossary Text:
ICOM line: SALUTE Report
Input: coming from <off page>
Glossary Text: Report of enemy activity which includes size, activity, location, unit, time, equipment.
ICOM line: SITREP
Output: going to <off page>
Glossary Text: Situation report:  Report submitted by a subordinate to a leader describing the current tactical situation

**Operational Activity: Complete Assigned Special Team Tasks**
Duration: METT-T Determined Time Measure: day
Glossary Text: On order, special teams accomplish all assigned tasks under the supervision of the Platoon Leader, who positions himself where required to maintain control.
ICOM line: C2 Data
Call: going to <off page>
Glossary Text: Leaders Instructions to a subordinate
ICOM line: C2 Data
Input: coming from <off page>
Glossary Text: Leaders Instructions to a subordinate
ICOM line: OPORD

Input: coming from <off page>
Glossary Text:
ICOM line: SITREP
Output: going to <off page>
Glossary Text: Situation report: Report submitted by a subordinate to a leader describing the current tactical situation

**Operational Activity: Report Status**
Duration: METT-T Determined
Glossary Text: Provide ammunition, casualty and equipment status (ACE) Report to higher HQ the combat capability of the unit.
ICOM line: C2 Data
Input: coming from <off page>
Glossary Text: Leaders Instructions to a subordinate
ICOM line: COP Presentation Data
Input: coming from <off page>
Glossary Text: User requested information is presented in the tailored view of the user in the format decided by the user and is updated on a user defined schedule. Information determined by the Commander's information policy as essential to the user is also rendered as determined by that policy.
ICOM line: Company Status Report
Output: going to <off page>
Glossary Text: Report from the company commander to his superior detailing the combat effectiveness of his unit and its ability to conduct combat operations.
ICOM line: SITREP
Input: coming from <off page>
Glossary Text: Situation report: Report submitted by a subordinate to a leader describing the current tactical situation

## Report Status

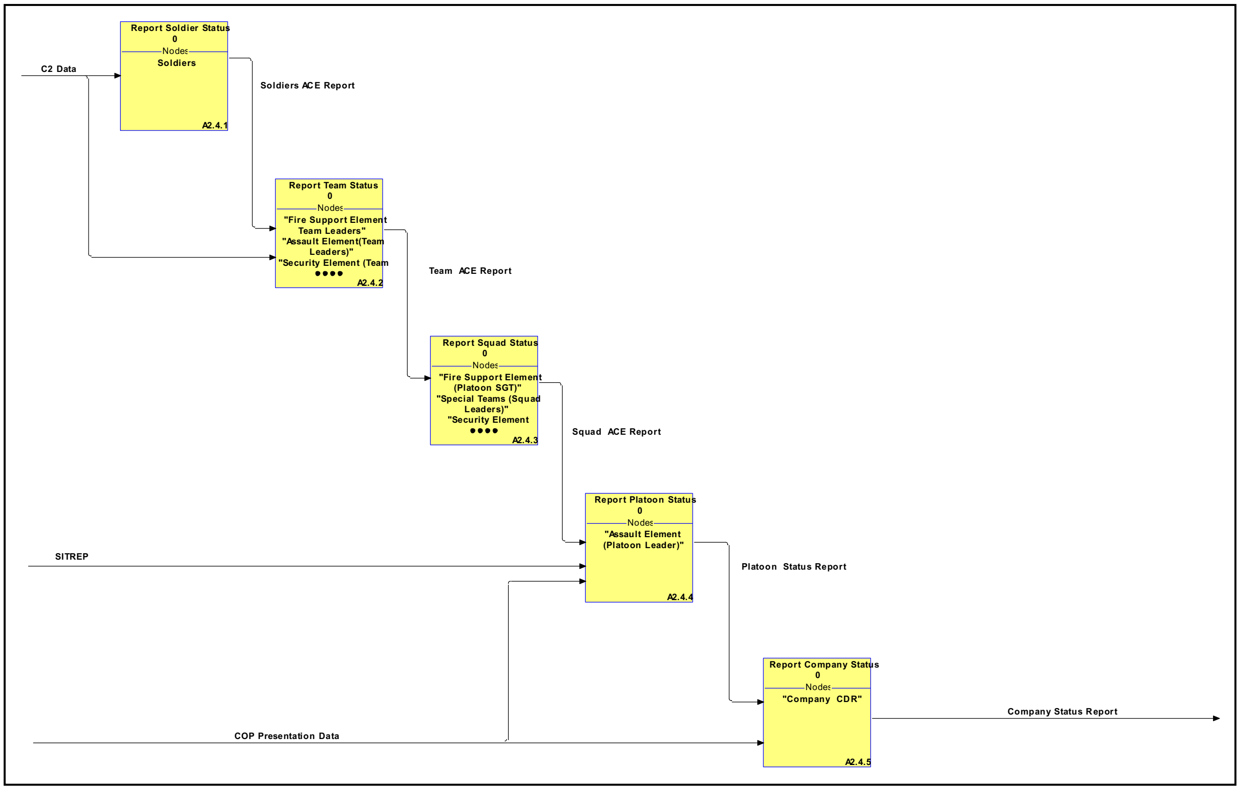



**Figure A-C-12: OV-5 – Operational Activity Model**

**Diagram Properties for: Report Status**
Audit ID: BENNNBT
IDEF0 Model:
Diagram Status: Working
Diagram Text: Provide ammunition, casualty and equipment status (ACE) Report to higher HQ the combat capability of the unit.
Update Date: 7/17/2008

**Operational Activity: Report Soldier Status**
Duration: 30 Seconds Time Measure: day
Glossary Text: Soldier reports physical injuries if any, on hand ammunition and the status of his equipment.
ICOM line: C2 Data
Input: coming from <off page>
Glossary Text: Leaders Instructions to a subordinate
ICOM line: Soldiers ACE Report
Output: going to Report Team Status as input
Glossary Text: Report of the status of a Soldier’s ammunition, medical condition, mission critical equipment

**Operational Activity: Report Team Status**
Duration: 60 Seconds
Glossary Text: Report team status by identifying ammunition status, causalities and mission essential equipment status.
ICOM line: C2 Data
Input: coming from <off page>
Glossary Text: Leaders Instructions to a subordinate
ICOM line: Soldiers ACE Report
Input: coming from Report Soldier Status as output
Glossary Text: Report of the status of a Soldier's ammunition, medical condition, mission critical equipment
ICOM line: Team ACE Report
Output: going to Report Squad Status as input
Glossary Text: Report of the status of a team ammunition, casualties, and mission critical equipment

**Operational Activity: Report Squad Status**
Duration: 90 seconds Time Measure: day
Glossary Text: Report squad status by identifying ammunition status, causalities and mission essential equipment status.
ICOM line: Squad ACE Report
Output: going to Report Platoon Status as input
Glossary Text: Report of the status of a squad ammunition, casualties, and mission critical equipment
ICOM line: Team ACE Report
Input: coming from Report Team Status as output
Glossary Text: Report of the status of a team ammunition, casualties, and mission critical equipment

**Operational Activity: Report Platoon Status**
Duration: 120 seconds Time Measure: day
Glossary Text: Report Platoon status by identifying ammunition status, causalities and mission essential equipment status.
ICOM line: COP Presentation Data
Input: coming from <off page>
Glossary Text: User requested information is presented in the tailored view of the user in the format decided by the user and is updated on a user defined schedule. Information determined by the Commander's information policy as essential to the user is also rendered as determined by that policy.
ICOM line: Platoon Status Report
Output: going to Report Company Status as input
Glossary Text: Report from a leader to his superior detailing the combat effectiveness of his unit and its ability to conduct combat operations.
ICOM line: SITREP
Input: coming from <off page>

Glossary Text: Situation report: Report submitted by a subordinate to a leader describing the current tactical situation
ICOM line: Squad ACE Report
Input: coming from Report Squad Status as output
Glossary Text: Report of the status of a squad ammunition, casualties, and mission critical equipment

**Operational Activity: Report Company Status**
Duration: 5 Minutes
Glossary Text: Commanders reports combat status of his unit to higher HQ.
ICOM line: COP Presentation Data
Input: coming from <off page>
Glossary Text: User requested information is presented in the tailored view of the user in the format decided by the user and is updated on a user defined schedule. Information determined by the Commander's information policy as essential to the user is also rendered as determined by that policy.
ICOM line: Company Status Report
Output: going to <off page>
Glossary Text: Report from the company commander to his superior detailing the combat effectiveness of his unit and its ability to conduct combat operations.
ICOM line: Platoon Status Report
Input: coming from Report Platoon Status as output
Glossary Text: Report from a leader to his superior detailing the combat effectiveness of his unit and its ability to conduct combat operations.

## Operational Information Exchange Matrix Operational View (OV-3).

The GSS OV-3 Operational Information Exchange Matrix is embedded as an excel spreadsheet below. The OV-3 details exchanges and identifies "who exchanges what information, with whom, why the information is necessary, and how the information exchange must occur.

Actions On the
Objective OV-3 .xls

# Operational Event-Trace Description (OV-6c) Ground Soldier System (GSS)

## *Conducts Actions on the Objective OV-6c Event Trace*

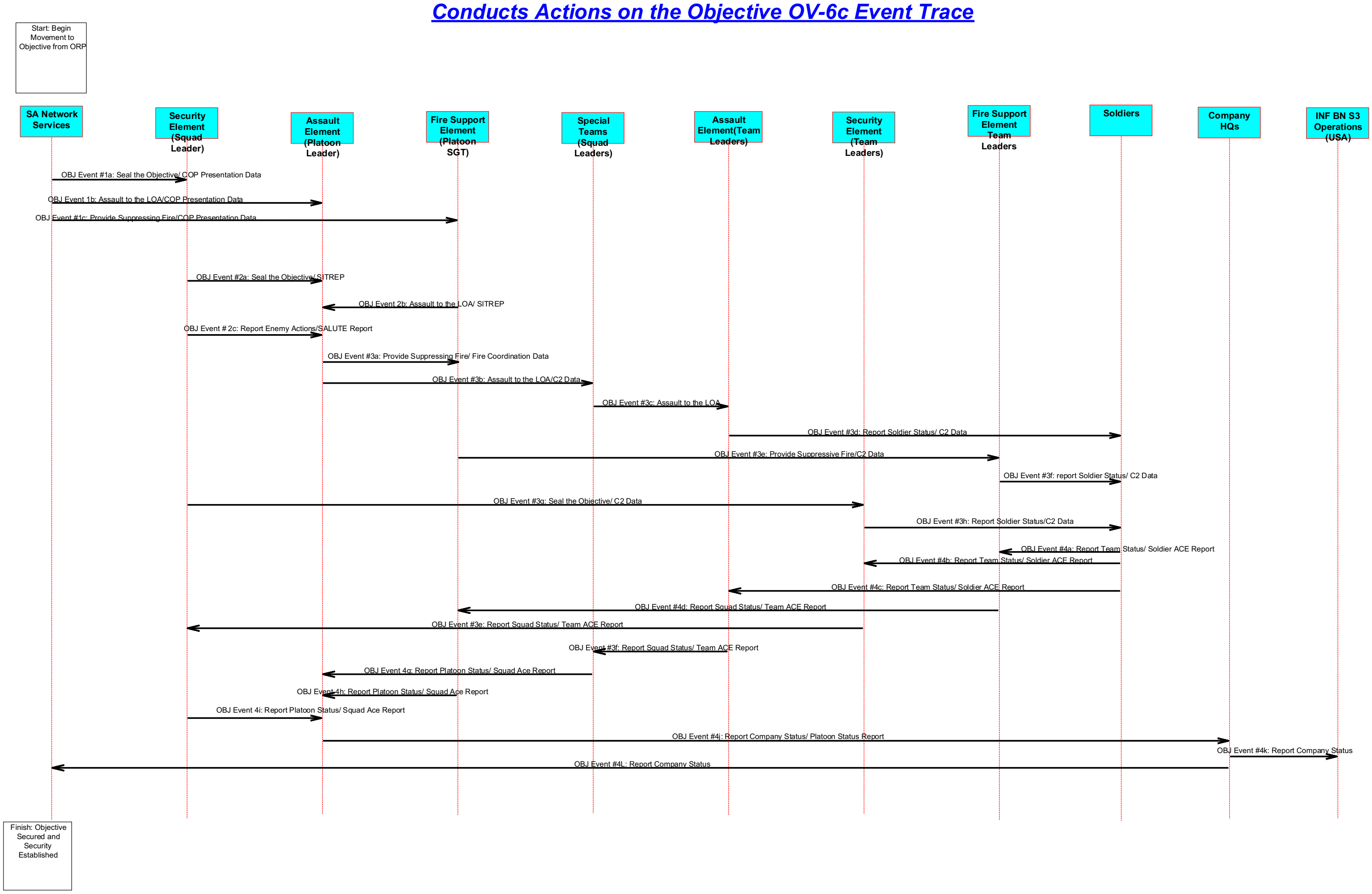



**Figure A-C-13: Operational Event Trace Description**

**Scope:** This thread depicts a GSS equipped Infantry platoon assaulting an objective. The platoon is tasked organized according to mission requirements. In this thread multiple activities are ongoing at the same time and there events that are constantly ongoing that may not be sequential.

**Data element Description**

**OBJ Event #1a:** COP Presentation Data is delivered to all assault elements and leaders during the entire operation.
**OBJ Event #1b:** COP Presentation Data is delivered to all assault elements and leaders during the entire operation.
**OBJ Event #1c:** COP Presentation Data is delivered to all assault elements and leaders during the entire operation

**OBJ Event #2a**: Upon the emplacement of the security element, the security element leaders provide a SITREP to the Platoon leader. This may be in the form of voice brevets code
**OBJ Event 2b**: The Fire Support Element provides Informs the Platoon Leader that it is in place.
**OBJ Event # 2c**: The Security Element reports any enemy movement to, from or around the objective to the Platoon Leader. This activity is conducted during the remainder of the operation.
**OBJ Event #3a**: Once the assault element is in position the platoon leader orders the fire support element to initiate the action by suppressing the objective. This coordination may take place as part of the planning process by the determination of a time initiated event or it may be initiated on command by voice. The Platoon leader will order the fire support element to lift and shift fires either by voice of by visual signal or both.
**OBJ Event #3b**: Platoon Leader provides Command and control of the platoon during the assault
**OBJ Event #3c**: Platoon Leader provides Command and control of the platoon during the assault
**OBJ Event #3d**: Team leaders lead their teams during the assault.
**OBJ Event #3e**: Platoon SGT provides Command and Control of the Fire support element during the assault.
**OBJ Event #3f**: Fire Support Team Leaders lead their teams during the assault
**OBJ Event #3g**: Security Element Leader provides Command and Control of the Security element during the assault.
**OBJ Event #3h**: Security team leaders lead their teams during the assault.
**OBJ Event #4a**: Once the objective is secure the Soldiers provide their team leaders with the status of their ammunition, injuries and equipment.
**OBJ Event #4b**: Once the objective is secure the Soldiers provide their team leaders with the status of their ammunition, injuries and equipment
**OBJ Event #4c**: Once the objective is secure the Soldiers provide their team leaders with the status of their ammunition, injuries and equipment
**OBJ Event #4d:** Team Leaders consolidate the Soldier’s ACE reports into a Team ACE report and provide it to the Squad leaders
**OBJ Event #4e**: Team Leaders consolidate the Soldier’s ACE reports into a Team ACE report and provide it to the Squad leaders
**OBJ Event #4f**: Team Leaders consolidate the Soldier’s ACE reports into a Team ACE report and provide it to the Squad leaders
**OBJ Event #4g**: Squad Leaders consolidate the Team ACE reports into a Squad ACE report and provide it to the platoon leaders
**OBJ Event #4h** Squad Leaders consolidate the Team ACE reports into a Squad ACE report and provide it to the platoon leaders
**OBJ Event #4i** Squad Leaders consolidate the Team ACE reports into a Squad ACE report and provide it to the platoon leaders
**OBJ Event #4j** Platoon Leader provides the company commander information on the combat readiness of his platoon based on the ACE reports given him.

**TAB D System View Architecture Products**
**Systems Communications Description (SV-2) Ground Soldier System (GSS)**

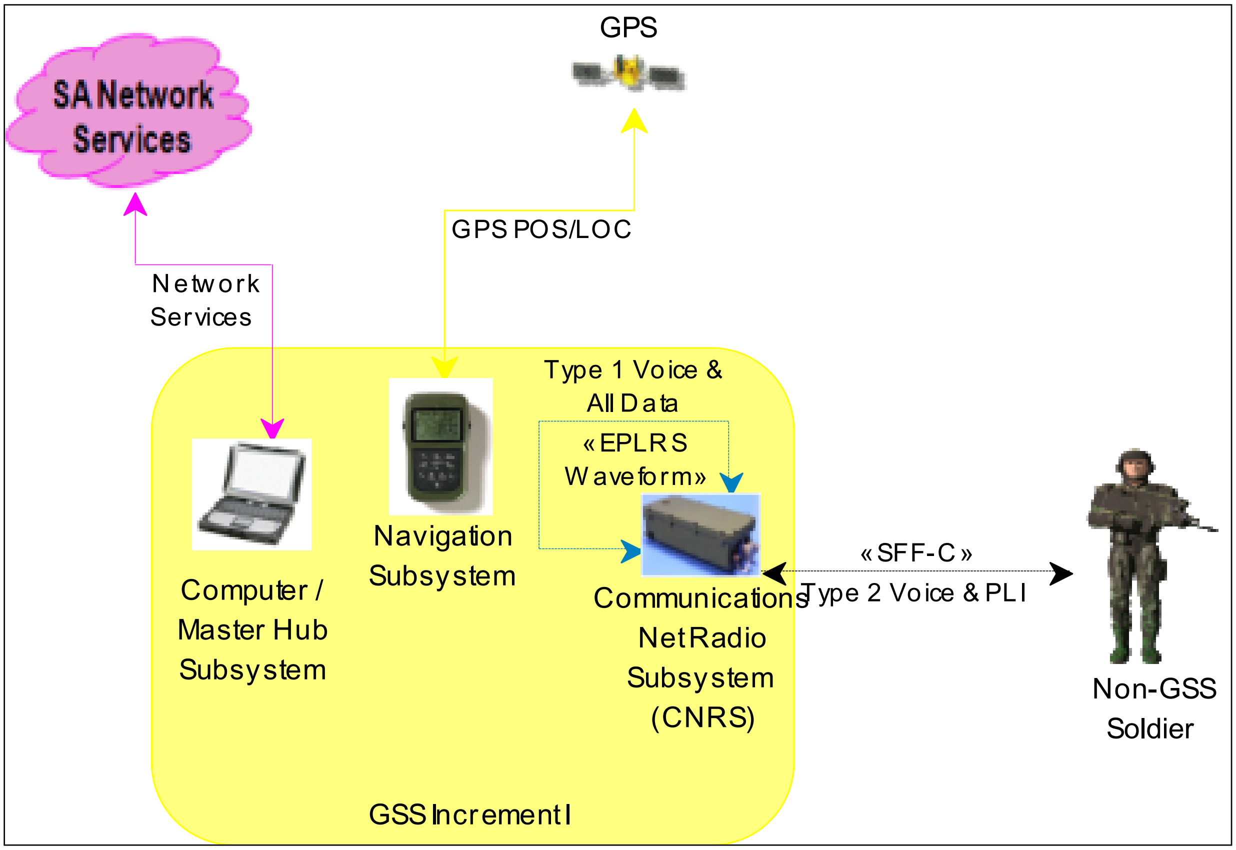



**Figure A-D-1: SV-2 – Systems Communications Description**

*Network Services*: The Network Services needline shown on the SV-2 (Figure A-D-1) are addressed by the following needlines on the GSS OV-2 (Conduct Troop Leading Procedures - Figure A-C-1 and Conduct Actions on the Objective - Figure A-C-8)

| GSS OV-2 (Figure A-C-1) | GSS OV-2 (Figure A-C-8) |
|---|---|
| TP18, TP19, TP20, TP21, TP22, TP23, TP24, TP25, TP26, TP27, TP28, TP29, TP30, TP31, TP32, TP33 | AO23, AO24, AO25, AO26, AO27, AO28, AO29 |

*Position Location Information*: The GPS POS/LOC needline shown on the SV-2 (Figure A-D-1) are addressed by the following needlines on the GSS OV-2 (Conduct Troop Leading Procedures - Figure A-C-1 and Conduct Actions on the Objective - Figure A-C-8)

| GSS OV-2 (Figure A-C-1) | GSS OV-2 (Figure A-C-8) |
|---|---|
| TP21, TP23, TP25, TP27, TP29, TP31, TP33 | AO1, AO3, AO11, AO12, AO13, AO25 |

*Leader to Leader Communication*: The Type 1 Voice & All Data needline shown on the SV-2 (Figure A-D-1) using the EPLRS waveform are addressed by the following needlines on the GSS OV-2 (Conduct Troop Leading Procedures - Figure A-C-1 and Conduct Actions on the Objective - Figure A-C-8)

| GSS OV-2 (Figure A-C-1) | GSS OV-2 (Figure A-C-8) |
|---|---|
| TP1, TP2, TP3, TP4, TP5, TP6, TP7, TP10, TP11, TP12, TP13, TP14, TP15, TP16 | AO1, AO2, AO3, AO4, AO5, AO6, AO7, AO8, AO9, AO10, AO11, AO12, AO13, AO14, AO15, AO16, AO17, AO18, AO19, AO20, AO21, AO22 |

*Leader to Soldier Communication*: The Type 2 Voice & PLI needline shown on the SV-2 (Figure A-D-1) using the SFF-C are addressed by the following needlines on the GSS OV-2 (Conduct Troop Leading Procedures - Figure A-C-1 and Conduct Actions on the Objective - Figure A-C-8)

| GSS OV-2 (Figure A-C-1) | GSS OV-2 (Figure A-C-8) |
|---|---|
| TP8, TP9, TP17, TP27, TP28 | AO11, AO12, AO13, AO14, AO15, AO16 |

The Ground Soldier System Increment I implementation of interfaces is accomplished via the following (See the SV-4 for a more complete description of system functions):

**Communications Net Radio Subsystem (CNRS)**
(a component of Integrated Body Subsystem Increment I)
Provides transmit/receive voice and data capability using EPLRS waveform.
Acronym: CNRS
It implements:
Activity: (F) Transmit Still-frame Video or Thermal Images
It utilizes:
Technology: EPLRS waveform
Enhance Position Location Reporting System (EPLRS) wave form
Communications Net Radio Subsystem (CNRS) has a:
Weight (in pounds) of '1.75'

**Computer / Master Hub Subsystem**
(a component of Integrated Body Subsystem Increment I and Integrated Body Subsystem Increment I P3I)
Computer/Master Hub Subsystem is the primary device for integrating data from the other sensors, components, and subsystems and displaying this information in the Leader's Helmet Mounted Display (HMD). The computer and software allows the Leader to access and process SA information received in mission operation orders and messages; and prepare mission reports to higher echelons, and process and send still frame video or thermal images from sensors and compress them for transmission via the radio subsystem.
Acronym: CSS

**Helmet / Display Subsystem**
(a component of GSS Increment I and GSS Increment I P3I)

Provides full-color display for the computer interface, and display of weapon-mounted sensor video; speaker and microphone.
Acronym : HSS
Helmet / Display Subsystem has a:
Weight (in pounds) of '1.39'

**Integrated Body Subsystem Increment I**
(a component of GSS Increment I)
Integrated Body Subsystem (IBS) consists of the Computer & Master Hub Subsystem, Soldier Control Unit Subsystem, Communications Net Radio Subsystem, Navigation Subsystem, Power Subsystem, and Personal Area Network.
Acronym : IBS

**Mission Data Support Equipment**
(a component of GSS Increment I, GSS Increment I P3I and GSS)
MDSE is the primary source of digital mission information used by system. It will generate, maintain, and transfer the Mission Data Packages (MDPs) that contain critical information required to enable conduct of missions. MDPs include digital maps, overlays, operations orders, Unit Task Organization (UTO) data, and images. The MDSE will prepare and edit SAM cards.
Acronym : MDSE

**Navigation Subsystem**
(a component of Integrated Body Subsystem Increment I and Integrated Body Subsystem Increment I P3I)
Navigation Subsystem provides position and navigation information. The Navigation Subsystem consists of the Global Positioning System (GPS) and antenna and the Dead Reckoning Device (DRD). GPS provides a position solution that is accurate to within 25 meters. The DRD augments GPS and maintains a position solution when the Soldier is operating in restrictive terrain (e.g., urban or thick tree canopy) and GPS may be degraded or unavailable. The DRD also provides a heading reference indication to assist in navigation. Algorithms process information from the GPS and DRD to provide a position solution, that is more accurate than either sensor alone.
Acronym : NSS

**Personal Area Network**
(a component of Integrated Body Subsystem Increment I and Integrated Body Subsystem Increment I P3I)
Personal Area Network and Power Subsystem is a distributed Universal Serial Bus (USB) network directing information between the subsystems and the Computer & Master Hub. The system is powered by Lithium Ion (LI 80 & LI 145) rechargeable batteries.
Acronym: PAN
Personal Area Network has a:
Weight (in pounds) of '1.99'

**Power Source Subsystem**
(a component of Integrated Body Subsystem Increment I and Integrated Body Subsystem Increment I P3I)
Provides centralized power from dual disposable or rechargeable batteries.
Acronym: PSS
Power Source Subsystem has a:
Weight (in pounds) of '4.5'

**Soldier Control Unit**
(a component of Integrated Body Subsystem Increment I and Integrated Body Subsystem Increment I P3I)
Soldier Control Unit (SCU) Subsystem allows the Leader to interface with the system, through a thumb stick for cursor mouse controls, mouse buttons, radio talk control buttons to control volume and video selection. The SCU is designed for right and/or left handed Leaders; and has a zeroize button allowing the Leader to zeroize sensitive data. The SCU contains a Soldier Access Module (SAM) card reader allowing Leaders to log-on. The SAM card contains the Unit Reference Number (URN) and Leader's Personal Identification Number (PIN); is unique for each Leader and made for the Leader on the Mission Data Support Equipment (MDSE).
Acronym: SCU
Soldier Control Unit has a:
Weight (in pounds) of '1.03'

The Ground Soldier System Increment I (P3I) functionality is provided via:

**Computer / Master Hub Subsystem**
(a component of Integrated Body Subsystem Increment I and Integrated Body Subsystem Increment I (P3I))
Computer/Master Hub Subsystem is the primary device for integrating data from the other sensors, components, and subsystems and displaying this information in the Leader's Helmet Mounted Display (HMD). The computer and software allows the Leader to access and process SA information received in mission operation orders and messages; and prepare mission reports to higher echelons, and process and send still frame video or thermal images from sensors and compress them for transmission via the radio subsystem.
Acronym: CSS
Computer / Master Hub Subsystem has a:
Weight (in pounds) of '1.4'

**GSS Increment I P3I**
It benefits from:
Plan: GSS Increment I P3I

As part of a P3I during increment I, GSS will transition to the JTRS Small Form Factor B utilizing the SRW. The decision to incorporate this new radio is dependent upon its technical maturity. The equipping focus of GSS Increment I will be to the IBCT with the ability to change to SBCT's or HBCT's. The focus of P3I is integrating the JTRS HMS SFF-B radio with the Soldier Radio Waveform as the transport layer for GSS. This radio replaces the EPLRS RT-1922B radio in Increment I.

It includes three components:

Helmet / Display Subsystem

Provides full-color display for the computer interface, and display of weapon-mounted sensor video; speaker and microphone.

Integrated Body Subsystem Increment I P3I

Mission Data Support Equipment

MDSE is the primary source of digital mission information used by system. It will generate, maintain, and transfer the Mission Data Packages (MDPs) that contain critical information required to enable conduct of missions. MDPs include digital maps, overlays, operations orders, Unit Task Organization (UTO) data, and images. The MDSE will prepare and edit SAM cards.

**Helmet / Display Subsystem**

(a component of GSS Increment I and GSS Increment I P3I)

Provides full-color display for the computer interface, and display of weapon-mounted sensor video; speaker and microphone.

Acronym: HSS

Helmet / Display Subsystem has a:

Weight (in pounds) of '1.39'

**Integrated Body Subsystem Increment I P3I**

(a component of GSS Increment I P3I)

It includes six components:

Computer / Master Hub Subsystem

Computer/Master Hub Subsystem is the primary device for integrating data from the other sensors, components, and subsystems and displaying this information in the Leader's Helmet Mounted Display (HMD). The computer and software allows the Leader to access and process SA information received in mission operation orders and messages; and prepare mission reports to higher echelons, and process and send still frame video or thermal images from sensors and compress them for transmission via the radio subsystem.

Joint Tactical Radio System (JTRS)

Navigation Subsystem

Navigation Subsystem provides position and navigation information. The Navigation Subsystem consists of the Global Positioning System (GPS) and antenna and the Dead Reckoning Device (DRD). GPS provides a position solution that is accurate to within 25 meters. The DRD augments GPS and maintains a position solution when the Soldier is operating in restrictive terrain (e.g., urban or thick tree canopy) and GPS may be degraded or unavailable. The DRD also provides a heading reference indication to assist in navigation. Algorithms process information from the GPS and DRD to provide a position solution, that is more accurate than either sensor alone.

Personal Area Network

Personal Area Network and Power Subsystem is a distributed Universal Serial Bus (USB) network directing information between the subsystems and the Computer &

Master Hub. The system is powered by Lithium Ion (LI 80 & LI 145) rechargeable batteries.

Power Source Subsystem

Provides centralized power from dual disposable or rechargeable batteries.

Soldier Control Unit

Soldier Control Unit (SCU) Subsystem allows the Leader to interface with the system, through a thumb stick for cursor mouse controls, mouse buttons, radio talk control buttons to control volume and video selection. The SCU is designed for right and/or left handed Leaders; and has a zeroize button allowing the Leader to zeroize sensitive data. The SCU contains a Soldier Access Module (SAM) card reader allowing Leaders to log-on. The SAM card contains the Unit Reference Number (URN) and Leader's Personal Identification Number (PIN); is unique for each Leader and made for the Leader on the Mission Data Support Equipment (MDSE).

**Joint Tactical Radio System (JTRS)**

(a component of Integrated Body Subsystem Increment I P3I) The Joint Tactical Radio System (JTRS) is the planned next-generation voice-and-data radio used jointly by the US Military. JTRS is a software-defined radio that will work with many existing military and civilian radios and associated waveforms. It includes integrated encryption and new Wideband Networking Software to create mobile ad hoc networks (MANETs). The JTRS is built on the Software Communications Architecture (SCA), an open-architecture framework that tells designers how hardware and software are to operate in harmony. It governs the structure and operation of the JTRS, enabling programmable radios to load waveforms, run applications, and be networked into an integrated system. A Core Framework, providing a standard operating environment, must be implemented on every hardware set. Interoperability among radio sets is increased because the same waveform software can be easily ported to all radios.

**Mission Data Support Equipment**

(a component of GSS Increment I, GSS Increment I P3I and GSS)

MDSE is the primary source of digital mission information used by system. It will generate, maintain, and transfer the Mission Data Packages (MDPs) that contain critical information required to enable conduct of missions. MDPs include digital maps, overlays, operations orders, Unit Task Organization (UTO) data, and images. The MDSE will prepare and edit SAM cards.

Acronym: MDSE

**Navigation Subsystem**

(a component of Integrated Body Subsystem Increment I and Integrated Body Subsystem Increment I P3I)

Navigation Subsystem provides position and navigation information. The Navigation Subsystem consists of the Global Positioning System (GPS) and antenna and the Dead Reckoning Device (DRD). GPS provides a position solution that is accurate to within 25 meters. The DRD augments GPS and maintains a position solution when the Soldier is operating in restrictive terrain (e.g., urban or thick tree canopy) and GPS may be degraded or unavailable. The DRD also provides a heading reference indication to

assist in navigation. Algorithms process information from the GPS and DRD to provide a position solution, that is more accurate than either sensor alone.
Acronym: NSS
It implements:
Activity: (F) Provide Position and Navigation Information
Navigation Subsystem has a:
Weight (in pounds) of '1.3'

**Personal Area Network**
(a component of Integrated Body Subsystem Increment I and Integrated Body Subsystem Increment I P3I)
Personal Area Network and Power Subsystem is a distributed Universal Serial Bus (USB) network directing information between the subsystems and the Computer & Master Hub. The system is powered by Lithium Ion (LI 80 & LI 145) rechargeable batteries.
Acronym: PAN
Personal Area Network has a:
Weight (in pounds) of '1.99'

**Power Source Subsystem**
(a component of Integrated Body Subsystem Increment I and Integrated Body Subsystem Increment I P3I)
Provides centralized power from dual disposable or rechargeable batteries.
Acronym: PSS
Power Source Subsystem has a:
Weight (in pounds) of '4.5'

**Soldier Control Unit**
(a component of Integrated Body Subsystem Increment I and Integrated Body Subsystem Increment I P3I)
Soldier Control Unit (SCU) Subsystem allows the Leader to interface with the system, through a thumb stick for cursor mouse controls, mouse buttons, radio talk control buttons to control volume and video selection. The SCU is designed for right and/or left handed Leaders; and has a zeroize button allowing the Leader to zeroize sensitive data. The SCU contains a Soldier Access Module (SAM) card reader allowing Leaders to log-on. The SAM card contains the Unit Reference Number (URN) and Leader’s Personal Identification Number (PIN); is unique for each Leader and made for the Leader on the Mission Data Support Equipment (MDSE).
Acronym: SCU
Soldier Control Unit has a:
Weight (in pounds) of '1.03'

**Systems Functionality Description (SV-4) Ground Soldier System (GSS)**

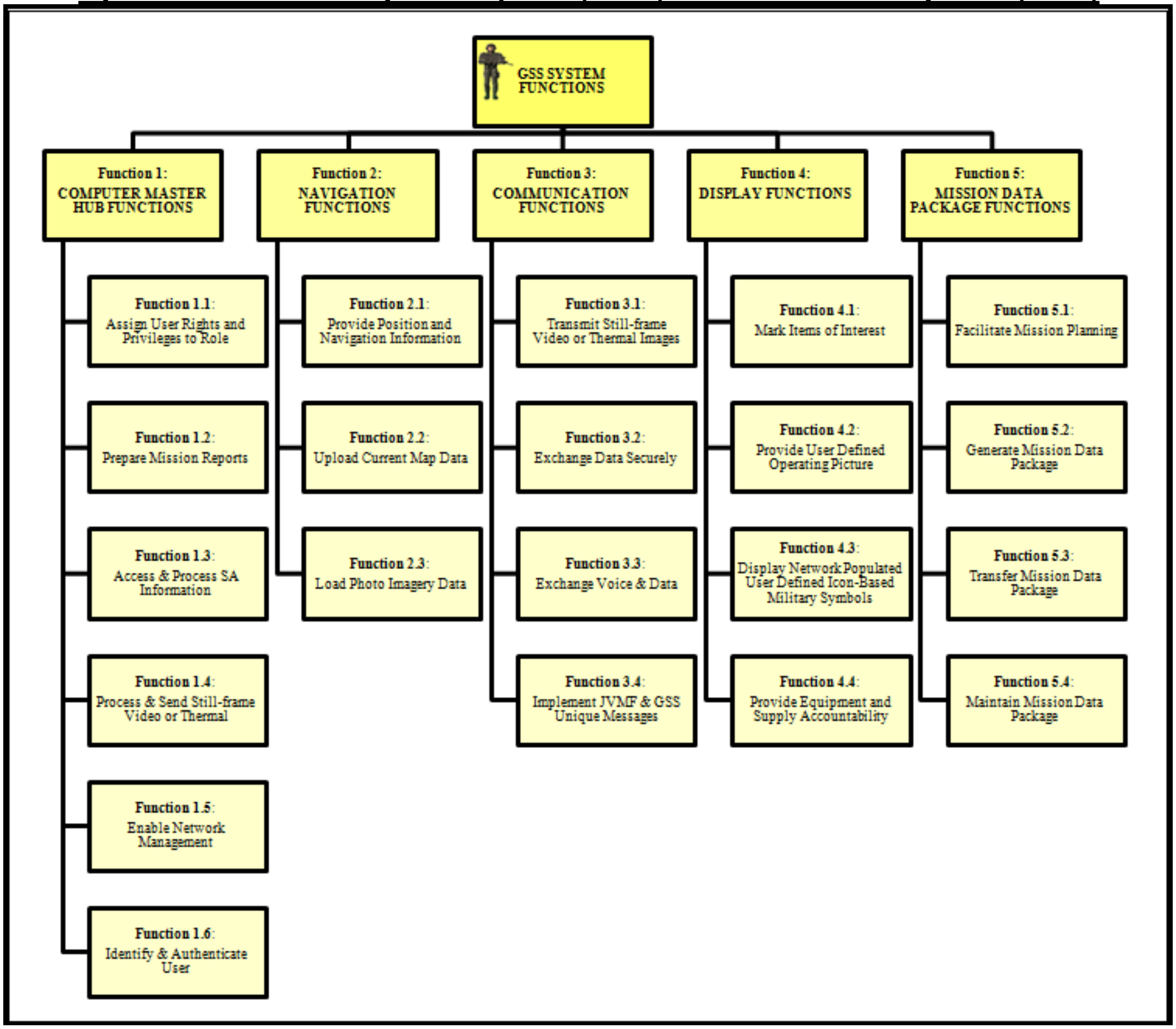



**Figure A-D-2: SV-4 – Systems Functionality Description**

Crosswalk of the GSS Inc I functions and the Joint Common System Function List (JCSFL).

2010.06.18_GSS
SV-4 TO JCSFL Mappi

The GSS System Functions include: Computer Master Hub Functions, Navigation Functions, Communication Functions, Display Functions, and Mission Data Package Functions

**Function 1: Computer Master Hub Functions**
(Component of GSS System Functions)

Computer Master Hub functions include, but are not limited to: Provide data entry interface, acquire/post/publish/register/store/retain data, perform Built-in-Test (BIT), receive and process signals, issue warings, alerts and orders, receive mission update, communicate status, communicate mission information, render useless, encrypt and decrypt data, provide tamper detection capability, manage and manipulate file and file systems.

**Function 1.1: Assign User Rights and Privileges to Role**
(Component of Computer/Master Hub Functions)

**Function 1.2: Prepare Mission Reports**
(Component of Computer/Master Hub Functions)

**Function 1.3: Access & Process SA Information**
(Component of Computer/Master Hub Functions)

**Function 1.4: Process & Send Still-frame Video or Thermal Images**
(Component of Computer/Master Hub Functions)

**Function 1.5: Enable Network Management**
(Component of Computer/Master Hub Functions)

**Function 1.6: Identify & Authenticate User**
(Component of Computer/Master Hub Functions)

**Function 2: Navigation Functions**
(Component of GSS System Functions)
Navigation functions include, but are not limited to: Detect/recognize/receive/process navigation signals, select navigational data source, distribute navigation and time data, invoke zeroization capability.

**Function 2.1: Provide Position and Navigation Information**
(Component of Navigation Functions)

**Function 2.2: Upload Current Map Data**
(Component of Navigation Functions)

**Function 2.3: Load Photo Imagery Data**
(Component of Navigation Functions)

**Function 3: Communication Functions**
(Component of GSS System Functions)
Communication functions include, but are not limited to: Establish communication networks, provide secure communications, transfer data, prioritize and provide message status, receive/transmit information, provide cryptographic services, assign roles and privileges, perform Built-in-Test (BIT).

**Function 3.1: Transmit Still-frame Video or Thermal Images**
(Component of Communication Functions)

**Function 3.2: Exchange Data Securely**
(Component of Communication Functions)

**Function 3.3: Exchange Voice & Data**
(Component of Communication Functions)

**Function 3.4: Implement JVMF & GSS Unique Messages**
(Component of Communication Functions)

**Function 4: Display Functions**
(Component of GSS System Functions)

Display functions include, but are not limited to: receive and process signals, display/post/publish data, provide Graphical User Interface (GUI) services, provide tactical status displays, overlays and reports, configure user defined displays, present user defined situational awareness information.

**Function 4.1: Mark Items of Interest**
(Component of Display Functions)

**Function 4.2: Provide User Defined Operating Picture**
(Component of Display Functions)

**Function 4.3: Display Network Populated User Defined Icon-Based Military Symbols**
(Component of Display Functions)

**Function 4.4: Provide Equipment and Supply Accountability**
(Component of Display Functions)

**Function 5: Mission Data Package Functions**
(Component of GSS System Functions)
Mission Data Package functions include, but are not limited to: Create/edit messages, generate/update Common Operational Picture (COP), create/update plans/orders, receive mission updates, communicate status/force information, manage accounts.

**Function 5.1: Facilitate Mission Planning**
(Component of Mission Data Package Functions)

**Function 5.2: Generate Mission Data Package**
(Component of Mission Data Package Functions)

**Function 5.3: Transfer Mission Data Package**
(Component of Mission Data Package Functions)

**Function 5.4: Maintain Mission Data Package**
(Component of Mission Data Package Functions)

## Operational Activity to Systems Function Traceability Matrix (SV-5)
## Ground Soldier System (GSS)

| Conduct Actions on the Objectives - OV-5 Node Tree | Conduct Troop Leading Procedures - OV-5 Node Tree | SV-5 Operational Activity to System Function | Function 1: COMPUTER MASTER HUB FUNCTIONS | Function 1.1: Assign User Rights and Privileges to Role | Function 1.2: Prepare Mission Reports | Function 1.3: Access & Process SA Information | Function 1.4: Process & Send Still-frame Video or Thermal | Function 1.5: Enable Network Management | Function 1.6: Identify & Authenticate User | Function 2: NAVIGATION FUNCTIONS | Function 2.1: Provide Position and Navigation Information | Function 2.2: Upload Current Map Data | Function 2.3: Load Photo Imagery Data | Function 3: COMMUNICATION FUNCTIONS | Function 3.1: Transmit Still-frame Video or Thermal Images | Function 3.2: Exchange Data Securely | Function 3.3: Exchange Voice & Data | Function 3.4: Implement JVMF & GS Unique Messages | Function 4: DISPLAY FUNCTIONS | Function 4.1: Mark Items of Interest | Function 4.2: Provide User Defined Operating Picture | Function 4.3: Display Network Populated User Defined Icon-Based Military Sy | Function 4.4: Provide Equipment and Supply Accountability | Function 5: MISSION DATA PACKAGE FUNCTIONS | Function 5.1: Facilitate Mission Planning | Function 5.2: Generate Mission Data Package | Function 5.3: Transfer Mission Data Package | Function 5.4: Maintain Mission Data Package |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| A.0 | | **CONDUCT ACTIONS ON THE OBJECTIVE** | | | | | | | | | | | | | | | | | | | | | | | | | | |
| A.1 | | **Establish Security** | X | X | | | | | X | X | | X | X | X | | | X | X | X | | X | X | | | | | | |
| A1.1 | | **Report Enemy Actions** | X | | X | | | X | | | | | | X | X | | | | X | | | X | | | | | | |
| A1.2 | | **Seal the Objective Area** | | | | | | | | X | | X | | | | | | | X | | X | | | | | | | |
| A.2 | | **Assault the Objective** | | | | | | | | | | | | | | | | | X | | X | | | | | | | |
| A2.1 | | **Assault to LOA** | | | | | | | | | | | | | | | | | X | | X | | | | | | | |
| A2.2 | | **Establish Local Security** | X | | | | | | X | X | | X | | | | | | | X | | | X | | | | | | |
| A2.3 | | **Complete Assigned Special Team Tasks** | | | | | | | | X | | X | X | | | | | | | | | | | | | | | |
| A2.4 | | **Report Status** | X | | X | | | X | | | | | | X | X | | | | X | | | X | X | | | | | |
| A2.4.1 | | **Report Soldier Status** | X | | X | | | X | | | | | | X | X | | | | X | | | X | X | | | | | |
| A2.4.2 | | **Report Team Status** | X | | X | | | X | | | | | | X | X | | | | X | | | X | X | | | | | |
| A2.4.3 | | **Report Squad Status** | X | | X | | | X | | | | | | X | X | | | | X | | | X | X | | | | | |
| A2.4.4 | | **Report Platoon Status** | X | | X | | | X | | | | | | X | X | | | | X | | | X | X | | | | | |
| A2.4.5 | | **Report Company Status** | X | | X | | | X | | | | | | X | X | | | | X | | | X | X | | | | | |
| A.3 | | **Provide Suppressing Fires** | X | | | X | | | X | X | X | | | X | | X | X | X | X | X | | | | X | X | X | X | |
| A.4 | | **Consolidate** | X | | | X | | | | X | X | X | X | | | | | | X | | X | | X | | | | | |
| | A.0 | **CONDUCT TROOP LEADING PROCEDURES** | | | | | | | | | | | | | | | | | | | | | | | | | | |
| | A.1 | **Receive the Mission** | X | | | | | X | X | | | | | X | | X | X | X | | | | | | X | X | X | X | X |
| | A1.1 | **Receive Orders** | X | | | | | X | X | | | | | X | | X | X | X | | | | X | | X | | X | X | X |
| | A1.2 | **Conduct Mission Analysis** | X | | | X | | | | X | X | X | X | | | | | | X | X | X | X | | X | X | | | |
| | A.2 | **Issue WARNORD** | | | | | | | | | | | | X | | X | X | X | | | | | | | | | | |
| | A.3 | **Make Tentative Plan** | | | | | | | | | | | | | | | | | X | X | | X | | | | | | |
| | A3.1 | **Conduct Detailed Mission Analysis** | X | | | X | | | | X | X | X | X | X | X | | | | X | X | | | | | | | | |
| | A3.2 | **Conduct Situation Analysis** | X | | | X | X | | | X | X | X | X | X | X | | | | X | | X | | | X | X | | | |
| | A3.3 | **Develop Courses of Action** | | | | | | | | X | | X | | X | X | | | | X | X | X | X | | X | X | | | |
| | A3.4 | **Decide Course of Action** | X | | | X | | | | X | | X | X | | | | | | X | | X | | | | | | | |
| | A.4 | **Start Necessary Movement** | | | | | | | | X | | X | | | | | | | X | | X | | | | | | | |
| | A.5 | **Reconnoiter** | | | | | | | | | | | | X | X | | | | | | | | | X | X | X | X | X |
| | A.6 | **Complete the Plan** | X | | X | | | X | | X | X | X | | X | X | X | X | X | X | X | | X | | X | | X | | |
| | A.7 | **Issue the Complete Order** | X | | | | | X | | | | | | X | | X | X | X | X | | | X | | | | | | |
| | A.8 | **Supervise** | | | | | | | | | | | | | | | | | | | | | | X | X | | | |
| | A.8.1 | **Conduct Inspections** | X | | | X | | X | | | | | | X | | X | X | X | X | | | | X | | | | | |
| | A8.1.1 | **Conduct Initial Inspections** | X | | | X | | X | | | | | | X | | X | X | X | X | | | | X | | | | | |
| | A8.1.2 | **Conduct Final Inspections** | X | | | X | | X | | | | | | X | | X | X | X | X | | | | X | | | | | |
| | A8.2 | **Conduct Rehearsals** | X | | | | | X | | X | X | X | X | X | X | | | | X | | | X | | X | X | | | |

**Figure A-D-5: SV-5 – Operational Activity to Systems Function**

The Ground Soldier System SV-5 Operational Activity to System Function matrix maps the relationship between the system functions to the operational activities. The matrix depicted above is embedded below for ease of use and readability.

2010.06.18 GSS
SV-5 Operational Acti

**Systems Data Exchange Matrix (SV-6) Ground Soldier System (GSS)**

The SV-6, GSS Increment I, System Data Exchange focuses on automated information exchanges between systems that are implemented in the Ground Soldier System. The SV-6 reports are embedded below for ease of use and readability.


SV- 6 Troop leading procedures.xlsx


SV-6 Actions on the OBJ.xls

**Systems Data Exchange Matrix (SV-6) Ground Soldier System (GSS)**

The SV-6, GSS Increment I P3I, System Data Exchange focuses on automated information exchanges between systems that are implemented in the Ground Soldier System. The report is embedded below for ease of use and readability.


SV-6 GSS Incr I P3I, Systems Data Exchar

## Technical Standards Profile (TV-1) and Technical Standards Forecast (TV-2) Ground Soldier System (GSS)

**Figure A-D-8: TV-1 – Technical Standards Profile and Technical Standards Forecast Ground Soldier System**

The Ground Soldier System TV-1, Technical Standards Profile along with the TV-2, Technical Standards Forecast generated from DISR online are embedded below for ease of use and readability.

2010.02.01_GSS Standards Profile TV-

Another detailed perspective on the structure of the Ground Soldier System Technical Standards Profile TV-1 is provided below with annotated data on the underlying standards. This is generated from DISR online (see http://disronline.disa.mil/). To view the TV-1 after it has been published, go to the SIPRNET https://jcpat.ncr.disa.smil.mil and select DISR online then the desired platform.

### Technical Standards Profile (TV-1) Ground Soldier System (GSS)

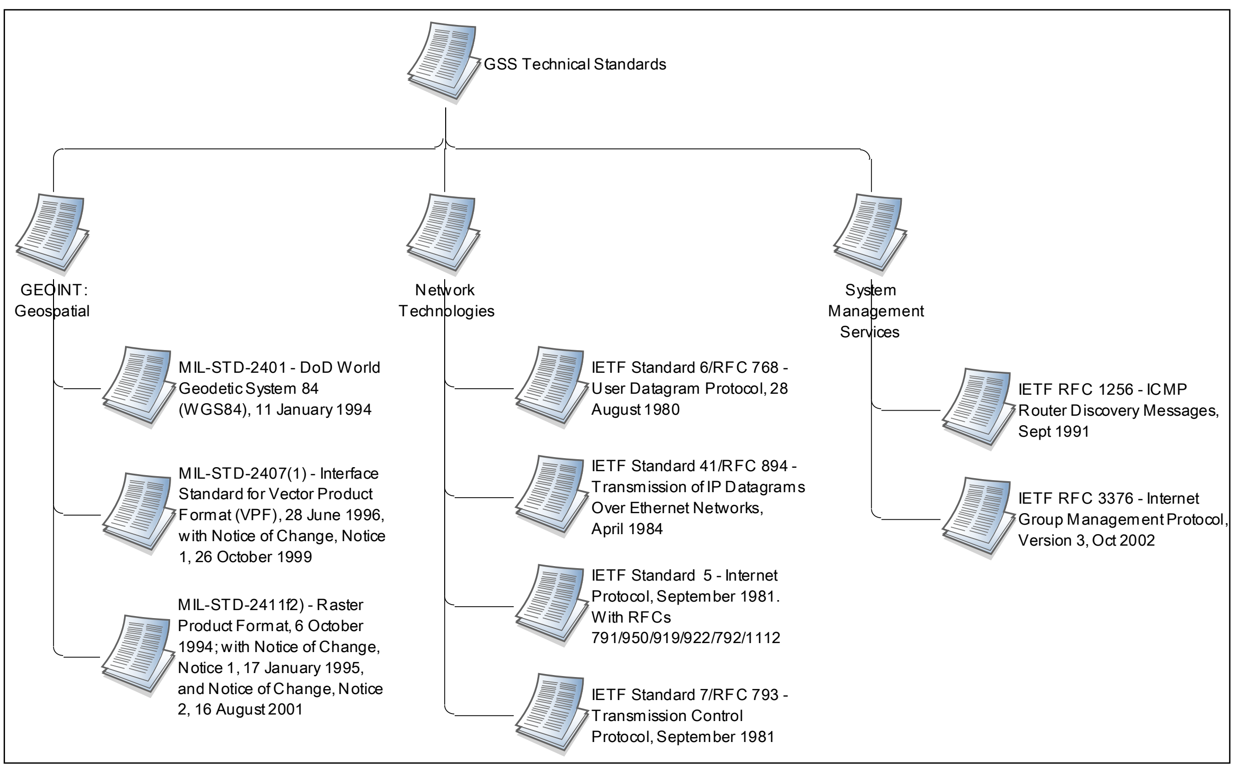



**Figure A-D-9: Ground Soldier System TV-1 – Technical Standards**

The Ground Soldier System Technical Standards Profile TV-1 includes three components:

**Geospatial**

MIL-STD-2401 - DoD World Geodetic System 84 (WGS84), 11 January 1994
MIL-STD-2407(1) - Interface Standard for Vector Product Format (VPF), 28 June 1996, with Notice of Change, Notice 1, 26 October 1999
MIL-STD-2411f2) - Raster Product Format, 6 October 1994; with Notice of Change, Notice 1, 17 January 1995, and Notice of Change, Notice 2, 16 August 2001

**Network Technologies**

ETF Standard 5 - Internet Protocol, September 1981. With RFCs 791/950/919/922/792/1112
IETF Standard 41/RFC 894 - Transmission of IP Datagrams Over Ethernet Networks, April 1984
IETF Standard 6/RFC 768 - User Datagram Protocol, 28 August 1980
IETF Standard 7/RFC 793 - Transmission Control Protocol, September 1981

**System Management Services**

IETF RFC 1256 - ICMP Router Discovery Messages, Sept 1991
IETF RFC 3376 - Internet Group Management Protocol, Version 3

**Appendix B – References**

AR 71-9, Materiel Requirements, 30 April 1997.

Army Operational Architecture Process and Procedures (AOAP2), Version 1.4, 6 August 2008.

Capability Production Document for Core Soldier System, June 2007.

Capability Production Document for Joint Tactical Radio System (JTRS).
Handheld, Manpack, and Small Form Fit (HMS) Rifleman Radio Increment I, 2008.

Capability Production Document for Land Warrior II, 20 July 2006.

Capstone Concepts for Joint Operations, Version 3.0, 15 January 2009.

Chairman Joint Chief of Staff Instruction (CJCSI) 3170.01G, Joint Capabilities Integration and Development System (JCIDS), 1 March 2009.

Combat Identification-Blue Force Tracking Joint Capability Document (CID-BFT JCD), 10 April 2009.

CJCSI 3312.01A, Joint Military Intelligence Requirements Certification, 23 February 2007.

CJCSI 6130.01D, 2007 CJCS Master Positioning, Navigation, and Timing Plan (MPNTP), 3 April 2007.

CJCSI 6212.01E, Interoperability and Supportability of Information Technology & National Security Systems, 15 December 2008.

Chairman Joint Chief of Staff Manual (CJCSM) 3500.04E, Universal Joint Task Manual, 25 August 2008.

Chemical, Biological, and Radiological Capstone Threat Assessment (S/NF), DI-1650-83, 06 June 2006

Cryptographic Modernization Mission Area Initial Capabilities Document (CM MA ICD), 14 August 2004

Concept Development Studies for Respiratory Protection System 21 dated September 1995.

DI-1577-37-07, DIA validated threat reference "Information Operations (IO) Capstone Threat Assessment, 6th edition, April 2007.

DOD Information Technology Standards Registry (DISR).

DOD Architecture Framework (DODAF), Version 1.5, 23 April 2007.

DODI 1322.26, Development, Management, and Delivery of Distributed Learning, 16 June 2006

DOD Test Method Standard for Environmental Engineering Considerations and Laboratory Tests MIL STD-810G

DODI 2010.06, Materiel Interoperability and Standardization with Allies and Coalition Partners, 29 July 2009.

DODD 4630.05, Interoperability and Supportability of Information Technology (IT) and National Security Systems (NSS), 5 May 2004.

DODD 4650.01, Policy and Procedures for Management and Use of the Electromagnetic Spectrum, 9 January 2009.

DODD 5000.01, The Defense Acquisition System, 12 May 2003.

DODI 5000.02, Operation of the Defense Acquisition System, 8 December 2008.

DODD 5250.01, Management of Signature Support Within the Department of Defense, 31 Jan 2008.

DODD 8320.02, Data Sharing in a Net-Centric Department of Defense, 23 April 2007.

DODD 8320.02-G, Guidance for Implementing Net-Centric Data Sharing, 12 April 2006.

DODD 8500.01E, Information Assurance (IA), 24 October 2002.

DODI 8500.2, Information Assurance (IA) Implementation, 6 February 2003.

DODI 8551.1, Ports, Protocols, and Services Management (PPSM), 13 August 2004.

DODD 8570.01, Information Assurance Training, Certification, and Workforce Management," 15 August 2004.

Department of the Army - approved LW System Mission Need Statement, 8 September 1993.

Department of the Army - approved Operational Requirements Document (ORD) for the LW (LW) System, 10 April 1994.

FM 7-15, The Army Universal Task List, Feb 09.

Field Manual 21-18 "Foot Marches", 1990.

General Accounting Office Report 04-678 Knowledge of Software Suppliers Needed to Manage Risks, May 2004.

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Global Positioning System Modernization Program, System Threat Assessment Report, (S/NF), NAIC-1574-0407-06, March 2006.

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Joint Service General Purpose Mask (JSGPM) System Threat Analysis Report (STAR), 2 June 1997.

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NATICK Study of Soldier Loads at the JRTC Training Center, 1998 TSM-Soldier Land Warrior KPP Analysis, 1999.

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NSTISSAM INFOSEC 1-99, The Insider Threat to US Government Information Systems, July 1999.

NSTISSAM number 11, National Information Assurance Acquisition Policy, 11 July 2003.

Physical Performance Benefits of Offloading the Soldiers Load, US Army Institute of Environmental Medicine (USARIEM) Soldier Load Study, 2003.

RESPO 21 Hazard Assessment, Phase 3 Summary Report, dated June 1997.

DIA validated SaaS System Threat Assessment Report (STAR), 21 January 2010.

STANAG 4579, Edition 1, – 10 December 1999 (NATO Restricted).

The Army Capstone Concept, Operational Adaptability: Operating under Conditions of Uncertainty and Complexity in an Era of Persistent Conflict 2016-2028, 21 Dec 2009

TRADOC - approved Operational Requirements Document (ORD) for Revised LW (LW), 3 August 1999.

TRADOC PAM 525-3-90, Objective Force Operational and Organizational Plan for the Maneuver Unit of Action, 30 June 2003.

TRADOC Joint Contingency Force Advanced Warfighting Experiment, 2000.

US Army NATICK Center, “Effects of Load on Weight, 2000.

Warfighter Information Network-Tactical Capability Production Document (WIN-T CPD) Increment 2, 5 December 2007.

**Appendix C – Acronym List**

| | |
|---|---|
| $A_m$ | Materiel Availability |
| $A_o$ | Operational Availability |
| AAE | Army Acquisition Executive |
| AAR | After-Action Review |
| ACH | Advanced Combat Helmet |
| ACOE | Army Communities of Excellence |
| AETF | Army Evaluation Task Force |
| AI | Area of Influence |
| ALDT | Administrative and Logistics Delay Time |
| AMC | Army Materiel Command, Air Mobility Command |
| AO | Area of Operations |
| AoA | Analysis of Alternatives |
| AOC | Area of Concentration |
| AP | Anti-Personnel |
| APG | Aberdeen Proving Ground |
| ARCIC | Army Capabilities Integration Center |
| ARFORGEN | Army Force Generation |
| AROC | Army Requirements Oversight Council |
| ASI | Additional Skill Identifier |
| ATD | Advanced Technology Demonstration |
| ATO | Army Technology Objective |
| ATO | Authorization To Operate |
| BC | Battle Command |
| BCS | Battle Command System |
| BCT | Brigade Combat Team |
| BIT | Built-In Test |
| BITE | Built-In Test Equipment |
| BOIP | Basis Of Issue Plan |
| BOIPFD | Basis Of Issue Plan Feeder Data |
| C4ISR | Command, Control, Communications, Computers, Intelligence, Surveillance, and Reconnaissance |
| CAS | Close Air Support |
| CADIE | Capabilities, Analysis, Development and Integration Environment |
| CB | Chemical and Biological |
| CBRNE | Chemical, Biological, Radiological, Nuclear and high yield Explosives |
| CBT | Combat Based Training |
| CC&D | Camouflage, Concealment & Deception |
| CCTT | Close Combat Tactical Trainer |
| CDD | Capability Development Document |
| CFE | Contractor Furnished Equipment |
| CJCSI | Chairman Joint Chiefs of Staff Instruction |
| CJCSM | Chairman Joint Chiefs of Staff Manual |
| CLS | Contractor Logistic Support |
| COMSEC | Communications Security |

| | |
|---|---|
| COMPUSEC | Computer Security |
| CNA | Computer Network Attack |
| CNE | Computer Network Exploitation |
| CNO | Computer Network Operations |
| CNRS | Communications Net Radio Subsystem |
| COA | Course of Action |
| CONOPS | Concept of Operations |
| COP | Common Operating Picture |
| COTS | Commercial-Off-The-Shelf |
| CPD | Capability Production Document |
| CSS | Core Soldier System |
| CTC | Combat Training Center |
| CUI | Controlled Unclassified Information |
| DAA | Designated Approval (Accrediting) Authority |
| DISR | Defense Information Standards Registry |
| DODAF | DOD Architecture Framework |
| DoD | Department of Defense |
| DODD | Department of Defense Directive |
| DODI | Department of Defense Instructions |
| DOD IEA | DOD Information Enterprise Architecture |
| DOS | Denial Of Service |
| DOTMLPF | Doctrine, Organization, Training, Materiel, Leadership and Education, Personnel, and Facilities |
| DRD | Dead Reckoning Device |
| EA | Electronic Attack |
| ECRG | Enhanced Compressed Raster Graphic |
| E3 | Electromagnetic Environmental Effects |
| EED | Electro Explosive Devices |
| EFF | Essential Function Failure |
| EME | Electromagnetic Environment |
| EMP | Electromagnetic Pulse |
| EOD | Explosive Ordinance Disposal |
| EPLRS | Enhanced Position Location Reporting System |
| ET | Embedded Training |
| EW | Electronic Warfare |
| FA | Force Application |
| FAA | Functional Area Analysis |
| FBCB2 | Force XXI Battle Command Brigade and Below |
| FCB | Functional Capabilities Board |
| FCS | Future Combat System |
| FDSC | Failure Definition and Scoring Criteria |
| FF | Future Force |
| FM | Field Manual |
| FNA | Functional Needs Analysis |
| FOC | Full Operational Capability |
| FOF | Force-on-Force |

| | |
|---|---|
| FOS | Family of Systems |
| FRAGO | Fragmentary Order |
| FBCT | Future Brigade Combat Team |
| FRP | Full Rate Production |
| FSA | Functional Solution Analysis |
| FUE | First Unit Equipped |
| GB-GRAM | Ground-Based GPS Receiver Application Module |
| GEOINT | Geospatial Intelligence |
| GESP | GIG Enterprise Service Profiles |
| GFE | Government-Furnished Equipment |
| GIG | Global Information Grid |
| GOTS | Government Off The Shelf |
| GI&S | Geospatial Information and Services |
| GPS | Global Positioning System |
| GSS | Ground Soldier System |
| GSS Inc I | Ground Soldier System Increment I |
| GTG | Global Information Grid (GIG) Technical Guidance |
| GUI | Graphic User Interface |
| HBCT | Heavy Brigade Combat Team |
| HEMP | High Altitude Electromagnetic Pulse |
| HERO | Hazards of Electromagnetic Radiation to Ordnance |
| HERP | Hazards of Electromagnetic Radiation to Personnel |
| HFE | Human Factors Engineering |
| HIA | Helmet Interface Assembly |
| HHA | Health Hazard Assessment |
| HIRF | High Intensity Radio Frequency |
| HLA | High-Level Architecture |
| HMS | Handheld, Manpack and Small Form Fit |
| HMD | Helmet Mounted Display |
| HNA | Host-Nation Approval |
| HSS | Helmet Subsystem |
| HSI | Human Systems Integration |
| HTML | HyperText Markup Language |
| HUMINT | Human Intelligence |
| IA | Information Assurance |
| IATO | Interim Approval To Operate |
| IBA | Interceptor Body Armor |
| IBCT | Infantry Brigade Combat Team |
| IBS | Integrated Body System |
| ICD | Initial Capabilities Document |
| ICS | Interim Contractor Support |
| IED | Improvised Explosive Device |
| ID | Identification |
| IDM | Information Dissemination Management |
| IER | Information Exchange Requirement |
| IET | Initial Entry Training |

| | |
|---|---|
| IETM | Interactive Electronic Technical Manual |
| IFTE | Integrated Family of Test Equipment |
| IKPT | Instructor and Key Personnel Training |
| ILS | Integrated Logistics Support |
| IMI | Interactive Multimedia Instruction |
| INWE | Initial Nuclear Weapons Effects |
| IO | Information Operations |
| IOC | Initial Operational Capability |
| IOT&E | Initial Operational Test and Evaluation |
| IP | Internet Protocol |
| ISP | Information Support Plan |
| ISR | Intelligence, Surveillance, and Reconnaissance |
| ISS | Individual Soldier Sensor |
| IT | Information Technology |
| ITS | Information Technology Systems |
| IW | Information Warfare |
| JBC-P | Joint Battle Command Platform |
| JCA | Joint Capability Area |
| JC2-A | Joint Command and Control - Army |
| JCIDS | Joint Capabilities Integration and Development System |
| JETS | Joint Effects Targeting System |
| JFC | Joint Future Concepts |
| JIM | Joint Interagency Multinational |
| JITC | Joint Interoperability Test Command |
| JNTC | Joint National Training Center |
| JOpsC | Joint Operational Concepts |
| JROC | Joint Requirements Oversight Council |
| JRTC | Joint Readiness Training Center |
| JSGPM | Joint Services General Purpose Mask |
| JSLIST | Joint Services Lightweight Integrated Suit Technology |
| JTAC | Joint Terminal Attack Controllers |
| JTRS | Joint Tactical Radio System |
| JVMF | Joint Variable Message Format |
| KIP | Key Interface Profile |
| KMDS | Knowledge Management Decision Support |
| KPP | Key Performance Parameter |
| KSA | Key System Attribute |
| LI | Lithium Ion |
| LIS | Logistics Information Systems |
| LMI | Logistics Management Information |
| LORA | Level of Repair Analysis |
| LOS | Line of Sight |
| LRM | Line Replaceable Module |
| LRU | Line Replaceable Unit |
| LSSP | Life-cycle Signatures Support Plan |
| LTT | Live Training Transformation |

| | |
|---|---|
| L-V-C | Live-Virtual-Constructive |
| LW | Land Warrior |
| LW-IC | Limited Initial Capability |
| LW-SI | Land Warrior Stryker Interoperable |
| LUT | Limited User Test |
| MA | Mission Area |
| MA ICD | Mission Area Initial Capabilities Document |
| MAC | Maintenance Allocation Charts |
| MAF | Mission Affecting Failure |
| MANPRINT | Manpower and Personnel Integration |
| MARC | Manpower Requirements Criteria |
| Max TTR | Maximum Time To Repair |
| MCA | Military Construction, Army |
| MCEB | Military Communications Electronics Board |
| MDA | Milestone Decision Authority |
| MDP | Mission Data Packages |
| MDR | Milestone Decision Review |
| MDSE | Mission Data Support Equipment |
| MERS | Marine Expeditionary Rifle Squad |
| METL | Mission Essential Task List |
| METT-TC | Mission, Equipment, Troops, Terrain - Time, Civil Considerations |
| MGRS | Military Grid Reference System |
| MHE | Military Load Handling Equipment |
| MILES | Multiple Integrated Laser Engagement System |
| MIL STD | Military Standard |
| NLT | Not Later Than |
| MNS | Mission Needs Statement |
| MOPP | Military Oriented Protective Posture |
| MOS | Military Occupational Specialty |
| MS | Milestone |
| MSS | Mounted Soldier System |
| MTBEFF | Mean Time Between Essential Function Failure |
| MTBMAF | Mean Time Between Mission Affecting Failures |
| MTC | Movement To Contact |
| MTP | Mission Training Plan |
| MTT | Mobile Training Teams |
| MTTR | Mean Time To Repair |
| MTW | Major Theater of War |
| MULE | Multi-function Utility/Logistics and Equipment Vehicle |
| MWSS | Mounted Warrior Soldier System |
| NBC | Nuclear, Biological, Chemical |
| NBCCS | Nuclear, Biological, and Chemical Contamination Survivability |
| NC/CSA | Netted Communications and Collaborative Situational Awareness |
| NCIC | National Crime Information Center |
| NCIE | Network Centric Information Environment |
| NCOW-RM | Net Centric Operations Warfare – Reference Model |

| | |
|---|---|
| NET | New Equipment Training |
| NGA | National Geospatial-Intelligence Agency |
| NIJ | National Institute of Justice |
| NLOS | Non-Line Of Sight |
| NM | Networked Management |
| NR-KPP | Net-Ready Key Performance Parameter |
| NRFTT | Networkable, Reconfigurable, Full Task Trainer |
| NSA | National Security Agency |
| NSN | National Stock Number |
| NSS | National Security System |
| NSTISSP | National Security Telecommunications and Information Systems Security Policy |
| NTISSAM | National Telecommunications and Information Systems Security Advisory Memorandum |
| NTC | National Training Center |
| O | Objective (requirement) |
| O&M | Operation and Maintenance |
| O&O | Operational and Organizational |
| OASD | Office of the Assistant Secretary of Defense |
| OE | Operational Environment |
| OHs | Operating Hours |
| OIF | Operation Iraqi Freedom |
| OMB | Office of Management and Budget |
| OMS/MP | Operational Mode Summary/Mission Profile |
| OneSAF | One Semi-Automated Forces |
| OPA | Other Procurement Army |
| OPORD | Operations Order |
| ORD | Operational Requirements Document |
| ORP | Objective Rally Point |
| OSUT | One Station Unit Training |
| OT | Operational Testing |
| OV | Operational View |
| PBA | Performance Based Agreements |
| PBL | Performance Based Logistics |
| PDSI | Project Development Skill Identifier |
| PFTEA | Post Fielding Training Effectiveness Analysis |
| PKI | Probability of Kill or Incapacitation |
| PL | Platoon Leader |
| PLI | Position Location Information |
| PM | Program Manager, Project Manager, Product Manager |
| PM-GS | Program Manager-Ground Soldier |
| PMCS | Preventive Maintenance Checks and Services |
| PME | Professional Military Education |
| POL | Petroleum, Oils, and Lubricants |
| POM | Program Objective Memorandum |
| PPS | Ports, Protocols and Services |
| PPS | Precise Positioning Service |

| | |
|---|---|
| R&M | Reliability and Maintainability |
| RAM | Reliability, Availability, and Maintainability |
| RDT&E | Research, Development, Test, & Evaluation Appropriation |
| RPMS | Remote Physiological Monitoring System |
| RF | Radio Frequency |
| RSTA | Reconnaissance, Surveillance, and Target Acquisition |
| RR | Rifleman Radio |
| SA | Situational Awareness |
| SAASM | Selective Availability Anti-Spoofing Module |
| SAALT | Under Secretary of the Army for Acquisition, Logistics, and Technology |
| SaaS | Soldier as a System |
| SASO | Stability and Support Operations |
| SBCT | Stryker Brigade Combat Team |
| SCI | Soldier Computer Interface |
| SCORM | Shareable Content Object Reference Model |
| SDD | System Development and Demonstration |
| SOF | Special Operations Forces |
| SOP | Standard Operating Procedures |
| SoS | System of Systems |
| SFF | Small Form Factor Radio |
| SRU | Shop Replaceable Unit |
| SRW | Soldier Radio Waveform |
| SS | Supportability Strategy |
| SSA | Safety System Assessment |
| SSC | Small Scale Contingency |
| SSI | Specialty Skill Identifier |
| SSP | Signatures Support Program |
| SSvA | Soldier Survivability Assessment |
| STAMIS | Standard Army Management Information Systems |
| STAR | System Threat Assessment Report |
| STRAP | System Training Plan |
| SU | Situational Understanding |
| SV | Systems View |
| T | Threshold (Requirement) |
| TA | Tactical |
| TACP | Tactical Air Control Party |
| TADSS | Training Aids, Devices, Simulations, and Simulators |
| TA/FCS | Target Acquisition/Fire Control System |
| TBD | To Be Determined |
| TDL | Tactical Data Link |
| TEMP | Test and Evaluation Master Plan |
| TESS | Tactical Engagement Simulation System |
| TEISS-D | The Enhanced Integrated Soldier System - Dismounted |
| TIC | Toxic Industrial Chemicals |
| TIM | Toxic Industrial Materials |
| TLE | Target Location Error |

| | |
|---|---|
| TLO | Terminal Learning Objective |
| TM | Technical Manual |
| TMDE | Test, Measurement, and Diagnostic Equipment |
| TMIP | Theater Medical Information Program |
| TOE | Table of Organization and Equipment |
| TRADOC | Training and Doctrine Command |
| TRAC-WSMR | TRADOC Analysis Center-White Sands Missile Range |
| TRL | Technology Readiness Level |
| TSP | Training Support Package |
| TTP | Tactics, Techniques, and Procedures |
| TTR | Time To Repair |
| TTSP | Test Training Support Package |
| UJTL | Universal Joint Task List |
| USAIS | US Army Infantry School |
| USARIEM | US Army Institute of Environmental Medicine |
| USF | Unit Set Fielding |
| USM | User Munitions Sensors |
| UTO | Unit Task Organization |
| WARSIM | War Simulation |
| WIN-T | Warfighter Information Network - Tactical |
| WMD | Weapons of Mass Destruction |

**Appendix D - GSS Increment II .**

**System Capabilities Required for GSS Increment II**. The primary focus of GSS Increment II is to strengthen digital communications connectivity among dismounted Leaders, while further reducing the burden on both these leaders and the sustainment tail. Increment II will strive to reduce the overall weight of GSS (system and associated batteries), while maintaining the needed mission profile established for Increment I. GSS Increment II will provide significant improvements in Soldier capabilities by providing increased lethal and non-lethal network fires, embedded training, multi-spectrum images, language translation, hands-free voice control, full color visor display, enhanced power for 48 hrs and Army BCT Modernized Battle Command interoperability. GSS Attributes for GSS Increment II will provide MOS specific Soldiers additional capabilities above what is provided by the Core Soldier System or GSS Increment I. Based on mission specific tasks, Soldiers will receive enhanced human sensing, area denial and non-lethal effects, enhanced micro-climate protection, auxiliary air supply, and enhanced fire protection.

(1) Statutory and Compliance Key Performance Parameters not appropriate for GSS.

(a) Survivability. The Survivability KPP does not apply because GSS is not a manned platform (as in tank, plane, ship, HMMWV, etc.). The GSS is a system of capabilities the Soldier will "wear." Just as body armor is not a "manned system."

(b) Force Protection. Force Protection is not applicable for the GSS CDD. The GSS is provided force protection through the CSS at no less than the current level of protection fielded at the time of GSS production. The Leader configuration will be physically compatible with the CSS force protection materiel and not require any modifications that will affect the integrity of the system.

(2) GSS Increment II KPPs**.** Table 6-5 contains the GSS Increment II KPPs with T and O values. All Soldiers receiving the GSS (either in the Leader or Soldier configuration) will have the following KPPs inherent in their individual system as described in Table 6-5 below. Additionally there are MOS specific Soldiers who will receive additional "attributes" based on mission specific tasks, i.e.; Engineers performing Explosive Ordinance Disposal (EOD) must have additional blast protection above what is provided by the Core or GSS.

| **JCA Tier 1/2** | KPP | Development Threshold (T) | Development Objective (O) |
|---|---|---|---|
| Net-Centric (Tier 1)<br>Enterprise Services (Tier 2)<br>Information Assurance (Tier 2)<br>Information Transport (Tier 2) | KPP 1: Net Ready: The system must support Net Centric military operations. The system must be able to enter and be managed in the network, and exchange data in a secure manner to enhance mission effectiveness. The system must continuously provide survivable, interoperable, | The capability, system, and/or service must fully support execution of joint critical operational activities and information exchanges identified in the DOD Enterprise Architecture and solution architectures based on integrated DODAF content, and must satisfy the technical requirements for transition to Net-Centric military operations to include:<br>1) Solution architecture products compliant with DOD Enterprise | The capability, system, and/or service must fully support execution of all operational activities and information exchanges identified in DOD Enterprise Architecture and solution architectures based on integrated DODAF content, and must satisfy the technical requirements for transition to Net-Centric military operations to include:<br>1) Solution architecture products compliant with DOD Enterprise |

| | | | |
|---|---|---|---|
| NET Management (Tier 2) | secure, and operationally effective information exchanges to enable a Net-Centric military capability. | Architecture based on integrated DODAF content, including specified operationally effective information exchanges<br>2) Compliant with Net -Centric Data Strategy and Net-Centric Services Strategy, and the principles and rules identified in the DOD Information Enterprise Architecture (DOD IEA), excepting tactical and non-IP communications<br>3) Compliant with GIG Technical Guidance to include IT Standards identified in the TV-1 and implementation guidance of GIG Enterprise Service Profiles (GESPs) necessary to meet all operational requirements specified in the DOD Enterprise Architecture and solution architecture views<br>4) Information assurance requirements including availability, integrity, authentication, confidentiality, and non-repudiation, and issuance of an Interim Authorization to Operate (IATO) or Authorization To Operate (ATO) by the Designated Accrediting Authority (DAA), and<br>5) Supportability requirements to include SAASM, Spectrum and JTRS requirements. | Architecture based on integrated DODAF content, including specified operationally effective information exchanges<br>2) Compliant with Net –Centric Data Strategy and Net-Centric Services Strategy, and the principles and rules identified in the DOD IEA, excepting tactical and non-IP communications<br>3) Compliant with GIG Technical Guidance to include IT Standards identified in the TV-1 and implementation guidance of GESPs, necessary to meet all operational requirements specified in the DOD Enterprise Architecture and solution architecture views<br>4) Information assurance requirements including availability, integrity, authentication, confidentiality,<br>and non-repudiation, and issuance of an ATO by the DAA, and<br>5) Supportability requirements to include SAASM, Spectrum and JTRS requirements. |
| Command & Control (Tier 1)<br>Understand (Tier 2)<br>Decide (Tier 2) | KPP 2. BC | Using standard communications devices and battle command applications the GSS must provide timely voice and data exchange between GSS Leaders as well as platform systems within the BCT and within Leaders area of operations (1km) (T). The GSS Leader configuration must implement JVMF and GSS unique messages (T).<br>The GSS must be interoperable with the RR in order to provide communications from Leader to Soldier. | Using standard communications devices and battle command applications the GSS must provide timely voice and data exchange between GSS Leaders as well as platform systems within the BCT and within Leaders area of influence (3.5km) (O) |
| Battlespace Awareness (Tier 1)<br>ISR (Tier2 | KPP 3 SA: <u>COP</u> | GSS Leader configuration will display at least 90% of all network populated user defined icon-based military symbols (friendly and enemy), received by the system within 5 seconds (T) within the Leader's area of operations (1km) (T) | GSS Leader configuration will display at least 90% of all network populated user defined icon-based military symbols (friendly and enemy), received by the system within 3 seconds (O) within the Leader's area of influence (3.5km) (O) |

| | | | |
|---|---|---|---|
| Logistics (Tier 1) Maintain (Tier 2) | KPP 4. Sustainability: Materiel Availability | The GSS shall have an Operational Availability (Ao) ≥ 90 % and Materiel Availability (Am) of ≥72% | The GSS shall have an Operational Availability (Ao) ≥ 95 % and Materiel Availability (Am) of ≥72 %. |
| Force Application (Tier 1) Maneuver (Tier 2) | KPP 5 Mobility: Weight | The GSS leader configuration will not weigh more than 10 lbs (T). | The GSS leader configuration will not weigh more than 8 lbs (O). |
| Force Application (Tier 1) Maneuver (Tier 2) Battlespace Awareness (Tier 1) ISR (Tier2 | KPP 6. Networked Lethality | Once the targeting information is populated on the COP, all systems within the platoon's area of operations and area of influence will receive this targeting information within 8 seconds (T) | T=O |

Table 6-5. GSS KPPs Increment II .

(3) KPP 1 Net-Ready. The capability, system, and/or service must fully support execution of joint (T) and all (O)critical operational activities and information exchanges identified in the DOD Enterprise Architecture and solution architectures based on integrated DODAF content, and must satisfy the technical requirements for transition to Net-Centric military operations to include: 1) Solution architecture products compliant with DOD Enterprise Architecture based on integrated DODAF content, including specified operationally effective information exchanges 2) Compliant with Net -Centric Data Strategy and Net-Centric Services Strategy, and the principles and rules identified in the DOD Information Enterprise Architecture (DOD IEA), excepting tactical and non-IP communications 3) Compliant with GIG Technical Guidance to include IT Standards identified in the TV-1 and implementation guidance of GIG Enterprise Service Profiles (GESPs) necessary to meet all operational requirements specified in the DOD Enterprise Architecture and solution architecture views 4) Information assurance requirements including availability, integrity, authentication, confidentiality, and non-repudiation, and issuance of an Interim Authorization to Operate (IATO) (T) or Authorization To Operate (ATO) (T=O) by the Designated Accrediting Authority (DAA), and 5) Supportability requirements to include SAASM, Spectrum and JTRS requirements.

*Rationale: The GSS will operate as part of a Joint Team across the full spectrum of military operations and, supporting operations vertically and horizontally, from alert through deployment and employment.*

(4) KPP 2 BC. Using standard communications devices and battle command applications, the GSS must provide timely voice and data exchange between GSS Leaders as well as platform systems within the BCT and within Leaders area of operation(1 km) (T) and area of influence (3.5 km) (O). The GSS Leader configuration must implement JVMF and GSS unique messages (T). The GSS must be interoperable with the RR in order to provide communications from Leader to Soldier. (T)

*Rationale: GSS requires a communications system to enable small units the ability to quickly and accurately communicate by both voice and data with leaders and subordinates in a rapidly* *changing operational environment with limited latency. The squad AO and AI drive the size of the Soldier and small unit battle space. For a quad the typical AO consists of the area that can be covered by the maximum effective range of the squad's weapons which is approximately 1200 meters. The AI includes the area in which operations are influenced by maneuver and fire support. The squad's area of influence (3.5 Km) provides the capability to engage threat systems outside of their maximum effective ranges. However, the AO and AI will expand or contract depending on the operational environment (urban, jungle) that the unit is in.*

(5) <u>KPP 3 SA: COP</u>. GSS Leader configuration will display at least 90% of all network populated user defined icon-based military symbols (friendly and enemy), received by the system within 5 seconds (T) and 3 seconds (O) within the Leader's area of operations (1km) (T) and area of influence (3.5km) (O).

*Rationale: Successful battle command demands timely and effective decisions based on applying judgment to available information. It requires commanders to not only know when and what to decide, but also requires them to continuously evaluate the quality of information and knowledge available to them. The networked Leader must have inherent confidence in the fidelity and timeliness of his system display as it updates his user defined operational picture. GSS enabled leaders will have the computer enhanced cognitive tools to more effectively exercise battle command in synchronizing forces and capabilities in time, space, and purpose to accomplish missions. GSS allows leaders to visualize the battlefield through a digital map or imagery display with terrain characteristics, while simultaneously displaying their location, their subordinates' locations, and enemy locations that are known or suspected. Continuous display of at least 90% of all network populated user defined icon-based military symbols (friendly and enemy) within the Leader's area of operations (1km) (T) and area of influence (3.5km) (O) provides appropriate fidelity to properly visualize the current situation as it relates to the battlefield geometry of disbursed friendly units, enemy disposition, and existing operational plans. GSS also allows leaders to influence the situation within their operational environment; it allows leaders to describe the desired endstate and direct the actions of subordinates through voice and digital communication of instructions, orders, tactical symbols (i.e., virtual chemlight), operational graphics, and overlays. Data received by the system from network architecture or created within GSS must be accurately displayed within 5 seconds (T) and 3 seconds (O) to enable appropriate and timely tactical actions, focus organic fires, or request joint supporting fires while minimizing the potential for fratricide and/or before the current situation represented by the digital operating picture changes significantly.*

(6) <u>KPP 4 Sustainment (Materiel Availability)</u>.

(<u>a</u>) Materiel availability, expressed as $A_o$ of the GSS in the field/operational environment, will be no less than 90 % (T) 95 % (O) during the 8-day wartime usage period (consisting of two aggregate 96-hour wartime scenarios) depicted in the OMS/MP, as well as on a steady state annualized basis. The assessment of $A_o$ includes both GSS CFE + GSS GFE combined and shall be consistent with equipment usage factors described in the OMS/MP, the repair cycle ALDT of 18.5 hours per failure projected for the field level support concept of fix-forward repair through replacement of GSS Line Replaceable Units (LRUs) at operator level, where EFFs as described by the GSS Reliability FDSC are the primary indicator of equipment not ready/available conditions.

*Rational: Requirement provides GSS-equipped leaders with the requisite high level of effectiveness when assessed on the basis of 148.7 GSS OHs for the 8-day wartime usage cycle, 4,462 OHs hours per year (wartime), the stated ALDT, and the GSS attribute requirement for the maximum time to repair. Objective requirement is consistent with (reflects) the $A_o$ required (band 2 requirement) for the future BCT modernized units. $A_o$ requirement for the high intensity pulse operating period (8-days) involving execution of the two aggregate 96-hour combat scenarios conducted back-to-back provides assurance that 90 % of the GSS will be in operational status (i.e., mission capable) throughout, wherein maintenance actions for all failures not resulting in an EFF are deferred until the operating pulse has been completed; $A_o$ requirement associated with steady state (e.g., annualized) GSS usage encompasses logistical impact of all malfunctions, to include those for which repair is deferrable.*

(b) Materiel Availability expressed as $A_m$ of the GSS in the field/operational environment shall be no less than .72% (T= O). The assessment of A(m) consists of the % of the total inventory of a system operationally capable (ready for tasking) of performing an assigned mission at a given time, based on materiel condition. This can be expressed mathematically as the number of operational end items divided by the total population and is measured over the total lifecycle of the entire fleet, including both peacetime and wartime operations. The A(m) addresses the total population of end items planned for operational use, including those temporarily in a non-operational status once placed into service (such as for depot-level maintenance). The total life-cycle timeframe, from placement into operational service through the planned end of service life, must be included.

***Rationale:*** *As a metric, Materiel Availability provides a measure of system performance, taking into account that all of the major sustainment elements are considered, planned for, and measured. The A(m) supports the management of the acquisition and materiel readiness processes supporting the required Operational Availability. For A(m) the entire population of the system must be accounted for, as does all time during the planned service life (using the ARFORGEN model this includes the Reset & Training Pool, Ready Pool, and Available Pool). Nonoperational units and time are included to provide a complete picture of the total investment and sustainment required across the entire program life cycle. For GSS the A(m) calculation, time starts when the system has been accepted by the unit (after NET and property transfer). A materially down system is not immediately available for tasking caused by failures, combat damage, maintenance delays, Reset, and packed for transportation time. There are no known improvements in any of these areas that will permit an increase of A(m) beyond the threshold value. Failure to achieve the A(m) value will decrease the number of operationally available systems causing undue risk for the combatant commander as he propagates the war fight. This increased risk will directly translate to an unacceptable increase in loss of life or equipment and may result in the inability of the Soldiers to achieve their assigned tasks or missions.*

(7) <u>KPP 5 Mobility (Weight).</u> The GSS leader configuration will not weigh more than 10.0 lbs (T) and 8 lbs (O). This weight is for the system (including GFE) and associated batteries.

*Rationale: The Soldier's load, both weight and cube, must not degrade the mobility of the dismounted Soldier or slow the momentum of small unit operations in close combat in complex terrain. The dismounted Soldier will maneuver through upper floor windows, underground sewers, holes in walls, over walls, and over rubble in all environments. He will have to roll left or right and then sprint three to five seconds, through varied terrain, to the next covered and concealed position, while under hostile enemy fire. The Soldier and small unit need the mobility to pursue and defeat a fleeting enemy in complex terrain. Reducing the collective weight of items carried by the Soldier improves his mobility and survivability in difficult terrain.*
*The combat load consists of mission essential equipment determined by the Commander to be required for the Soldier to fight and survive in ensuing combat operations. The combat load consists of two levels: approach march load and fighting load. Analytical studies and doctrine consistently show that fighting loads of 30% of Soldier body weight or less and approach march loads of 45% or less best support Soldier operational performance. FM 21-18 "Foot Marches", 1990: states that the fighting load should not exceed 30% of a Soldier's body weight and the approach march load should not exceed 45%. The 1998 Study of Soldier Loads at the Joint Readiness Training Center (JRTC) further validates this conclusion that the fighting load should not exceed 30% of a Soldier's body weight. Conclusions of the Physical Performance Benefits of Offloading the Soldier support a "30-33% of bodyweight recommend maximal load." US Army Institute of Environmental Medicine (USARIEM) Soldier Load Study, 2003.*
*CFE components for GSS are focused specifically within the fighting load, and are therefore additive to the fighting load of 57.950 lbs. The current estimate for the CFE components for GSS is 10 lbs. The table below illustrates the resultant fighting load when GFE and CFE are integrated or combined.*

| **Fighting Load** | **Weight (lbs)** |
|---|---|
| Fighting Load | 57.95 |
| GSS (w/power supply) | 10.00 |
| **Total** | **67.95** |

Table 6-5.1 Fighting load when GFE and CFE are integrated or combined

*The representative fighting load is common for all operating environments. Specific Soldier roles, specialized tasks and distribution of capabilities within the small unit can add to this load. Different extremes in operational environments can also modify this load.*

(8) KPP 6 Networked Lethality. The GSS allows leaders to populate the COP by providing a target description and location. The GSS must rapidly process, and exchange targeting information between systems. Once the targeting information is populated on the COP, all systems within the platoon's immediate operational environment will receive this targeting information within 8 seconds (T). The Platoon Leader will designate who will engage with what type weapon system. If the PL can not engage with his platoon organic weapon systems, he can pass off this targeting information to his higher headquarters. The PL passes off targeting information by populating the system within 8 seconds (T). All leaders who are equipped with a GSS will have the ability to click on a targeting icon to view this information. The icon will open within 5 seconds (T) with a target description, and location (T=O).

*Rationale**:** GSS capabilities must provide the Soldier a vision enhancement capability to accurately and effectively detect, classify, recognize, locate, and identify hard and soft targets during day, night, and periods of limited visibility beyond the range of the weapon being used. All Soldiers are sensors and vital intelligence gathering during operations increases Soldier survivability and situational awareness. Soldiers who identify sniper positions, enemy forces, potential IEDs, suspected persons of interest or any other intelligence can inform their leaders who can than populate the network for situational awareness or target hand off. These SPOT/SALUTE reports are vital to the success of any military operation and increases Soldier survivability and situational awareness. Employment of lethal and non-lethal means must not degrade the speed and momentum of small unit operations. Historical durations of small unit combat engagements (firefights) range from seconds to hours and days. Networked Lethality enables the FF Concept of cooperative engagement at the small unit level. The standard for immediate exchange and response enhances the leader's lethal overmatch through fire superiority.*

(9) KPP 7 Sustainability (Power). Provide a rechargeable power source and integrated power management solution to achieve an operational system runtime of 48 Mission Hours (T) and 72 Mission Hours (O). The system will have a recharging capability within tactical vehicles and a stand-alone recharging capability (T=O).

*Rationale: Sustainable and quickly restorable power supports the required momentum of small unit operations. "Dismounted forces must be self-sustaining during continuous operations for at least 24 Hours." BCT O&O, 6.12.10. The FF BCT must have the ability to operate for 72 hours before receiving replenishment of supplies. Units that are properly trained and ready for combat operations can sustain themselves until a unit re-supply is conducted. An enhanced power system will reduce the logistical burden of the GSS by reducing the weight, variety, and number of power sources carried by the Soldier and the logistical structure. Dismounted SCU, light infantry, airborne, air assault, Special Operations Soldiers require additional capabilities above CSS items to increase their ability to sustain combat operations and survive on today's battlefield.*

(10) Rationale for exclusion of "Survivability" as separate KPP*: The Survivability KPP does not apply because GSS is not a manned platform (as in tank, plane, ship, HMMWV, etc.). It's a system of capabilities that the Soldier will "wear." Just as body armor is not a "manned system."*

(11) Rationale for exclusion of "Force Protection" as separate KPP***:** Force Protection is not applicable for the GSS CDD. The GSS is provided force protection through the CSS at no less than the current level of protection fielded at the time of GSS production. The Leader configuration will be physically compatible with the CSS force protection materiel and not require any modifications that will affect the integrity of the system.*

g. GSS KSAs. The following table is a summary of the GSS Increment II KSAs.

| **JCA Tier ½** | **Key System Attribute** | **Development Threshold** | **Development Objective** |
|---|---|---|---|

| | | | |
|---|---|---|---|
| NET-Centric (Tier 1) Enterprise Services (Tier 2) | KSA 1. Geospatial Data Exchange (GEOINT) | Upon deployment units will be able to upload current map data for the theater of operations and have the ability to load photo imagery data for use at check points and within cities, villages and towns. T=O. | T=O |
| Logistics (Tier 1) Maintain (Tier 2) | KSA 2. Sustainability; Reliability | Operational reliability required of the GSS is a probability of at least .85 (T) of operating for 24 hours without incurring an EFF, as defined in the Reliability FDSC (T). The GSS CFE will not degrade reliability of GFE (T=O). | The GSS must have operational reliability of at least a .92 probability of operating for 24 hours without incurring an EFF (O). |
| Force Support (Tier 1) Force Preparation (Tier 2) | KSA 3. ET: Have embedded LVC training environments. | The GSS must support 90% of individual and collective task. | 100% of individual and collective task. |
| NET-Centric (Tier 1) Enterprise Services (Tier 2) Force Application (Tier 1) Engagement (Tier 2) | KSA 4. Far Target Locator | The GSS and its sensors must allow the Leader to detect, classify, recognize and identify a target under a variety of conditions. Probability of detection must be at least 70% at ranges up to 2500 meters. | The GSS and its sensors must allow the Leader to detect, classify, recognize and identify a target under a variety of conditions. Probability of detection must be at least 70% at ranges up to 4 km. |
| Force Application (Tier 1) Maneuver (Tier 2) | KSA 5. Leader (User) Interface: Hands Free Efficiency. | Keep hands free to execute critical combat tasks. Manual back-up | Execute all tasks during operations |
| Corporate Management and Support (Tier1) Acquisition (Tier 2) | KSA 6. Ownership Cost | The Average Procurement Unit Cost (APUC) for the GSS is (projected quantity of TBD). | T=O |

Table 6-6. GSS KSAs Increment II.

(1) KSA 1 Geospatial Data Exchange (GEOINT). The GSS, will build to an approved geospatial information architecture, to include the logical data models, formats, tools, applications, and standards; so as to enable the net-centric, interoperable, geospatial enterprise supporting battle command with a unified COP. Upon deployment units will be able to upload current map data for the theater of operations and have the ability to load photo imagery data for use at check points and within cities, villages and towns. (T=O).

*Rationale: Networking of Soldiers, platforms and sensors down to the individual and small unit with access to Army BCT modernized assets will facilitate real time or near real time information updates to GSS. The GSS receives data exchange (SA, orders, and overlay) information from unit Tactical Operations Centers and platforms. This data exchange facilitates intelligence supportability for all military operations.*

(2) KSA 2 Sustainability (Reliability). The GSS (GSS CFE + GSS GFE combined) must provide operational reliability of not less than a .85 probability (T) and .92 probability (O) of operating for 24 hours without incurring an EFF, as defined in the Reliability FDSC. The GSS CFE will not degrade reliability of GFE (T=O).

*Rationale: RAM analysis based on the OMS/MP, consideration of the LW-SI System support concept implemented in OIF and associated ALDT and repair time incurred per failure (factoring in GSS logistical considerations/implications relative to units other than the SBCT), as well as input from PM – Soldier Warrior representatives regarding the support concept currently envisioned for GSS (field level fix-forward repair through replacement of GSS LRUs at operator level, LRU repair at sustainment level (depot/vendor)) indicates the stated reliability requirements will enable the GSS to achieve both steady state and 8-day pulse operational availability of not less than the 90 % threshold and % objective (when assessed in conjunction with factors identified in the Materiel Availability KPP requirement and rationale paragraph). Requirements are defined operationally (as probability values) in accordance with current TRADOC guidance and have been expressed in relation to 24 hours of operation to maintain consistency (a common basis) with KSA 3 – Sustainability (Power); GSS threshold requirement correlates to 146 OHs MTBEFF and the objective requirement correlates to 291 OHs MTBEFF, based on expectation of the GSS having a constant failure rate).*

(3) KSA 3 Embedded Training (ET). The GSS must support 90% of individual and collective task (T) and 100% of individual and collective task (O). This system must provide a "reach back" capability to retrieve additional training products not hosted by the Soldier system (O). GSS must be compatible with current force-on-force tactical engagement simulation systems (T) and future embedded TESS (O). The system must facilitate collaborative course of action analysis and wargaming, mission rehearsal, and AARs (O).

*Rationale: The GSS ET will allow leaders the ability to access individual task such as Warrior Skill Levels 1-4 and Battle Drills. Embedded full task and TESS must provide leaders a full task training capability with a readily available system for planning, training and assessing tasks with subordinate units in any combination of LVC environments from distributed locations; provides feedback through the selected TSP and allow leaders to modify conditions within a TSP to adjust difficulty during training or build a TSP to support a mission rehearsal and provide the interface to Soldier System's ET capabilities.*

(4) KSA 4 Far Target Locator. The GSS and its sensors must allow the Leader to detect, classify, recognize and identify a target under a variety of conditions. Probability of detection must be at least 70% at ranges up to 2500 meters (T), 4 km (O), for man-sized targets under starlight, overcast, and obscured conditions. The GSS and its sensors will include a single Target Acquisition/Fire Control System (TA/FCS) enabling the Warfighter, during day and limited visibility conditions, to effectively detect, identify, determine range, and engage stationary and defilade personnel/vehicle targets within 5 seconds (T).

*Rationale: Leaders and small units in the BCT must influence a platoon radius of 8 Km. Leaders must be able to detect, classify, recognize, and identify targets in order to direct their attack, whether through systems that they carry or using external platforms and systems. Target* *detection is the initial step prior to determining whether the target is friend or foe. The probability of detection must enable overmatching capabilities against the threat. Leaders and their sensors must be able to detect targets through sound, movement, thermal energy, vibrations and other signatures. Once targets are identified, Leaders and sensors must then be able to distinguish between friend and foe. There will be a number of different vehicles and platform types on the joint and coalition battlefield for the Leader to distinguish. This capability will allow target detection, recognition and then cooperative target engagement by the small unit during close combat in close, complex terrain and tactical stand off engagements.*

(5) KSA 5 Leader (User) Interface: Hands Free Efficiency. Using a common interface, the system must streamline leader input processes for greater efficiency while allowing him control of all GSS Battle Command related functions and up to two unmanned systems. Leader data entry processes must keep dismounted leader's hands free to execute critical tasks while in combat (T); all tasks during operations under all environmental conditions (O). Each hands free capability must have a manual back up equivalent (T). The GSS must remotely control two or more systems concurrently (T), at least two or more platforms of any type using the integrated control (O).

*Rationale: The Leader must not lose focus on the close fight in complex terrain. User interfaces must not encumber - but enable. The Leader must maneuver tactically, operate his weapon, designate targets, engage targets with external assets and lead the small unit in combat without distractions from laborious, manual data entry tasks. The dismounted Leader using GSS must engage targets with his individual weapon and external assets simultaneously with the same level of operational efficiency as he would today from using only his individual weapon.*

(6) KSA 6 Ownership Cost. The Total Operating Cost for GSS associated with the Weapon Systems Review ACEIT model is $8.036B. The Average Procurement Unit Cost (APUC) for the GSS is (projected quantity of TBD). T=O

| | |
|---|---|
| RDTE | $0.367B |
| OPA | $5.807B |
| OMA | $1.836B |
| PA | $0.026B |
| Total | $8.036B |

Table 6-6.1Total Operating Cost

h. The following are additional attributes for GSS Increment II with threshold and objective values.

| Performance Parameter | Development Threshold (T) | Development Objective (O) |
|---|---|---|
| Army BCT Modernization Compatibility | Army BCT Modernized units | T=O |
| Family of Soldier Systems Compatibility | Reconfigurable to all Soldier variants | T=O |
| Combat Medic Integration | Combat Medic integrated with GSS | Combat Lifesaver bag integrated into load-carrying equipment |
| Remote Physiological Monitoring | Pulse and respiration reported when | Input data into TMIP. Provide remote |

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|---|---|---|
| System (RPMS) | queried electronically | triage |
| Immediate Casualty Aid | Facilitate immediate aid | Support remote triage |
| Recover Wounded and Dead | Locate, identify, and facilitate recovery of dead and wounded | Sustain wounded |
| Leader (User) Interface: Unmanned System Hand-Off | Hand off to other leaders | Hand off to geographically separated Soldiers |
| Leader (User) Interface: Light Discipline | GUIs must be viewable in daylight and at night without compromising light discipline | T=O |
| ET: Characteristics | System must have Instructions for operating and instructions for complex, perishable skills that the Soldier needs for combat | T=O |
| ET: Top Level Tasks | Top priority capabilities resident on GSS | Complex perishable skills |
| ET: Combined Arms and Joint Context | Facilitate training for mounted and dismounted individual and small unit combined arms and joint context | Collaborative training with other Soldiers and platforms |
| ET: Training Environment | LVC | T=O |
| ET: Skills | Tailored for specific units and skills | T=O |
| ET: Collaboration | Support collaboration by geographically dispersed Soldiers in field and garrison environments | T=O |
| ET: Conduct Religious Support | Conduct religious services-support across the full spectrum of operations | Extended religious support LVC spiritual training in real time |
| ET: Operational Integration | One integrated system for training and operations | T=O |
| ET: Cognitive Skills | Assist in task performance skills while Soldiers are in a degraded state | T=O |
| ET: After Action Review | Monitor, record, assess and playback training exercise and actual operations | Execute an operational pause to conduct an intermediate AAR |
| ET: Interface with Army BCT Modernized Battle Command | Interface with Networked Battle Command System collaborative, intuitive, full-range-of-mission planning and rehearsal functions | T=O |
| ET: Interface with the Army BCT Modernized AAR System | Interface with Army BCT Modernized AAR System to support Individual and Collective training and all data must be recorded in an embedded learning management system | T=O |
| ET: GSS Army BCT Modernization ET Interface | Interface with the Army BCT Modernization individual capability that supports multi-mode live-virtual-constructive training environments | Interface with the Army BCT Modernization individual and collective ET capability that supports multi-mode live-virtual-constructive training environments |
| ET: Interface with the Army BCT Modernization Networkable, Reconfigurable, Full Task Trainer (NRFTT) | Support individual, crew, and multi-echelon training without reconfiguration of the equipment | T=O |
| ET: GSS Interface with On Board BCT Modernization Repository | Interface to the Army BCT Modernization "on-board" repository of training products | T=O |
| ET: GSS interface with the Army BCT Modernization Multi-echelon Combined Arms Training (CAT). | Provide a multi-echelon CAT training capability including integration of current, SBCT and below | Provide a multi-echelon CAT training capability including integration of current, SBCT and JIM capabilities at |

| | | |
|---|---|---|
| | | BCT and above |
| ET: Interface with the Army BCT Modernization | Will generate, send, and receive engagement data that conforms to the Army Common Training Instrumentation Architecture (CTIA) and Joint Instrumentation Systems | Will be capable of transmitting tactical engagement data in accordance with the Army CTIA and Joint Instrumentation Systems |
| ET: One Tactical Engagement Simulation System (TESS) Interoperability with Physiological Status Monitoring | Provide simulated casualty data to the physiological status monitoring system during force-on-force collective training | T=O |
| ET: TADSS Training | System specific TADSS must be developed to facilitate training of Leader GSS tasks | Integrate with non-system specific TADSS |
| Area Lethality | Defeat threats within the small unit area of influence and area coverage. | T=O |
| Dismounted Target Hand-off and Designation | Geographically locate targets 0-2.5 Km with a 65m TLE and provide precision target location data to facilitate engagement by another Leader and/or another system. | Designation for precision-guided and loitering munitions 0-4 Km with a 10 m TLE. Capture and transmit target location data to GPS guided munitions in flight. |
| Target Recognition and Location | Must acquire, collate, process and transmit all necessary targeting information in formats required by all potential fires delivery systems. | Locate and laser range find targets to 20-meter accuracy. |
| NLOS/BLOS/LOS Target Engagement | Acquire, collate, process and transmit all necessary targeting information in formats required by all potential "shooter" systems. Populate COP and Networked Fires | Interactive dynamic re-tasking of multiple munitions as to target type and location |
| Optimized Attack Solutions. | Upon entry of targeting information (target designation), produce target engagement solution: < 1 second | T=O |
| Synchronization of Fires. | Facilitate the interactive synchronization of fires and effects for all small unit assets | All external assets |
| Report Battle Damage Assessment (BDA). | Provide BDA information to the Battle Command and Networked Fires systems following target engagements | T=O |
| Networked Lethality: Interoperability | Support Joint networked lethal and non-lethal effects | T=O |
| Detect/ Avoid Danger and Hazardous Areas | Sensors allow Soldier to detect, and/or bypass hazardous area | T=O |
| Precision Delivery of Troops and Equipment. | Transportable by the , Army BCT Modernization future tactical trucks (FTT), Rail, Army aviation, Army watercraft, US Air Force aircraft, and US Navy ships | T=O |
| Rapid Assembly of Dispersed Forces. | Enable quick and orderly assembly of dispersed forces | T=O |
| Personnel Visibility / Human Resources | On-demand enable Situational Awareness (SA) for friendly, enemy, and neutral locations personnel visibility | T=O |
| Equipment and Supply Accountability | Visibility via the COP. Forecast, request, allocate, store and distribute | Total Asset visibility data base. |

| | | |
|---|---|---|
| Embedded Mission Readiness | Monitor the status of mission-critical components and/or subsystems and consumables | Determine corrective actions to restore full mission capability |
| On-Line Fault Isolation. ID a single fault to an ambiguity group of no more than two operator replaceable (configurable) items | with at least 80 % accuracy without user action to modify the system configuration | with at least 95 % accuracy when used with guided (operator level) systems reconfiguration directions |
| Maximum Time to Repair (Max TTR) | Max TTR for 95 % of all GSS malfunctions correctable at field level of support in ≤ 20 minutes (dedicated maintenance personnel and operator repair actions assessed separately). | Max TTR for 95 % of all GSS malfunctions correctable at field level of support in ≤ 15 minutes (dedicated maintenance personnel and operator repair actions assessed separately). |
| Graceful Degradation | Loss of one system element must not terminate utility of the entire system | T=O |
| Protection from Work-Site Hazards | System and equipment must be protected from dust and contaminants generated by heavy machinery operations | T=O |
| Sensors: Individual Soldier Sensors | Light weight, reusable, operate 72 hours, placed by hand or thrown | Delivered by small team indirect fire, alert with Pd = 90% and Pcc = 85% human versus non-human; animal, clutter, noise within 5 meter radius of sensor |
| ISS: Mobility | Small, lightweight, re-usable, as disposable/consumable items which operate for 72 hours and able to be placed by hand (dropped or thrown) able to detect and alert to the presence of humans, animal, clutter, noise within 5 meter radius | Variants possible delivery by small team indirect fire systems providing greater stand-off |
| ISS: Interoperability | Alert with an audible/visual alert or via a wireless interface up to 500 meters away | Wireless up to 1000 meters away; capable of being used with an additional asset so that small networks of unattended sensors can be constructed |
| Body Protection: Sensor | Analyses will determine achievable threshold values | Detect projectiles of all types. Locate projectile source<br>Detect concealed explosives at standoff |
| COP: Enemy Location Tracking | Must track down enemy composition and disposition with a level of fidelity suitable for small unit operations | T=O |
| COP: Positive Biometric Identification (Facial Recognition) | Must uniquely and positively identify individuals within the AI or AO Friendly and Unknown | Friendly, Foe and Unknown |
| Intelligence: Sensor Control | From Vehicle: 5 Km | From Vehicle: 12 Km |
| Intelligence: Extended Operational Communications | Extended Distances | 150 km |
| Intelligence: Dissemination | Securely distribute and receive information within <4 sec | <1sec |
| Intelligence: Compatibility | Compatible with future BCS and DCGS-A | T=O |
| Intelligence: Air Warning | Air threat info within 6 seconds and updated every 2 seconds | Must disseminate received air threat information from Joint/National sensors |
| Intelligence: Language Translation | Must translate languages based on | 90% of two way speech and 98% of |

| | | |
|---|---|---|
| | voice recognition for all languages in the current AO. 80% of two way speech and 90% of text to speech and text to text in near real time | text to speech and text to text in near real time |
| Networked Communications: Interoperability | The Leader's communication system must allow full interoperability with the Army BCT Modernization C4ISR architecture (T). | The communications architecture must have the flexibility, scalability, and extensibility to allow every Leader access to the Army BCT Modernization network (O). The dismounted Leader must have communications capability for full BC/SA with or independent of a GCV vehicle (O). |

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Table 6-7. GSS Additional Performance Attributes for Increment II.

(1) Interoperability.

(a) Army BCT Modernization Compatibility. The GSS must be compatible with the Army BCT Modernized units (T=O).

*Rationale: GSS equipped Soldiers include all ground combatants in the FF.*

(b) Family of Soldier Systems Compatibility. The GSS must provide the capability to share components within and between other leader variants (T). Its open system architecture must be modular, scalable, interoperable and re-configurable between all leader systems, and must allow insertion of future technology (T=O).

*Rationale: The GSS architecture shall be fully interoperable to allow leaders the flexibility to interchange items within a family of subsystems and leader systems to support operations in diverse geographical regions and environmental conditions. The GSS equipped leader may find himself in a non-linear battlefield requiring the leader to enable scale to meet changing mission requirements. Geographic and the operational environments will include all environments that leaders perform missions, including arctic, desert, tropical, and mountainous regions, within a multi-dimensional urban or complex terrain. The GSS uses common architecture that is modular, interchangeable, interoperable, and scalable.*

(2) Medical Support.

(a) Combat Medic System (CM): Integration. The CMS is an integrated load management system for the combat medic; which addresses both Soldier load and medical load, and is designed to support the tenets of Tactical Combat Casualty Care (TCCC). The CMS component integrated with the GSS allows the combat medic to receive and interpret data from the RPMS. Additionally, the system will allow the Combat Medic to electronically document and transmit pertinent data from the point of wounding to the receiving level of care (T). The CLS load carrying solution must also be integrated into the GSS load-carrying equipment (O).

*Rationale: Currently there is no common load management solution for the combat medic that addresses both Soldier and medical load that intentionally supports the tenets of TCCC and* *minimizes load during the assault. Providing the leader and Combat Medic with RPMS data will enable them to determine where and when the medic's skill set is most needed. Providing the Combat Medic the ability to electronically transfer near-real-time relevant and actionable data to the receiving level of care, greatly increases the wounded individual's survivability by enabling outcome effecting decisions prior to arrival.*

(b) Remote Physiological Monitoring System (RPMS). The RPMS will integrate with the GSS to report pulse, respiration, body movement, geospatial location (T), anatomical orientation and other relevant, actionable physiological parameters (O). The RPMS will be a suite of sensors that rely on the GSS for power, geo-spatial location, data processing, interpretation, and transfer. The system will report monitored physiological parameters, the absence of physiological data, or a system sensor fault or failure. The RPMS shall provide for both active alerts and remote electronic query (T). The CME component integrated with the GSS allows the combat medic to receive and interpret data from the RPMS. Additionally, the system will allow the Combat Medic to electronically document and transmit pertinent data from the point of wounding to the receiving level of care (T). The GSS will integrate a physiological status monitoring capability that will provide remote triage including life status monitor, and other necessary physiological and environmental parameters (O). The GSS will allow for the collection and exchange of data input into the Theater Medical Information Program (TMIP) (O). The data from the sensor suite will produce actionable and relevant information for leaders and Unit medical personnel (O).

*Rationale: During recent conflicts, 20% of the medics killed in action were killed while trying to get to Soldiers that were already dead. Immediate detection of injuries with the initiation of first aid measures will improve the likelihood of Soldier survival after injury. The goal is to increase force effectiveness, accelerate the response to and treatment of casualties allowing the commander greater latitude and freedom of maneuver in the operational environment.*

(c) Immediate Casualty Aid. The GSS must facilitate immediate *First Responder* aid to the injured person and provide for life saving interventions until further treatment is available (T). The GSS must also support remote triage (O). This initially will be through communication from the medic, who has received information through the physiological monitoring system, to a first responder (buddy-aid or Combat Lifesaver (CLS)) and ultimately through medical devices and therapies incorporated into the GSS (remote treatment) (T).

*Rationale: Immediate casualty aid is critical in saving Soldier lives.*

(d) Recover Wounded and Dead. The GSS must provide a means to locate, identify, and facilitate recovery of wounded and dead (T), and sustain wounded (O), under all conditions while minimizing exposure to the effects of threat systems.

*Rationale: Recovery of wounded and dead is a critical reconstitution and morale support task. An enhanced performance system may include the capability for self casualty aid. The religious support team's data set further assists combatant commanders in assessing unit morale.*

(3) Leader (User) Interface.

(a) Leader (User) Interface: User Munitions Sensors (UMS) Hand-Off. The Leader must have the capability to hand-off control of an UMS to another Leader (T). This transfer of control must occur while both parties are geographically separated (O).

*Rationale: Exchanges of control for UMS's must not degrade the momentum of small unit operations. Leaders and small units must sustain continuous coverage of the same AI for SA/SU and lethal effects even during a relief in place. Rotation of Leaders and small units in the AO's must not result in gaps in coverage of their unmanned systems. If the UMS operator becomes a casualty, another Leader from any other location must be able to assume the operator role. UMS operators operating in shifts in a tactical situation may transfer control without unnecessary movement to meet in the same place.*

(b) Leader (User) Interface: Light Discipline. GUI's must be viewable in daylight, at night, and under other limited visibility conditions, without compromising light discipline (T=O).

*Rationale: The Leader will employ his system in all environments to include conditions of limited visibility. The system must not compromise light discipline.*

(4) GSS Training and Embedded Training.

(a) ET: Characteristics. The GSS must include ET that resides on the system or is interoperable with other Leader training systems (T=O).

*Rationale: ET will enhance training in all situations. The Leader must learn more skills and sustain higher proficiency levels in less time. ET must reduce both training time and the "learning curve." The ability to tailor training to unit types will ensure applicability to a wide range of needs. Soldier training accommodates many skills and takes many forms.*

(b) ET: Top Level ET Tasks. The top priority for training capabilities are those resident on (hosted by) the GSS (T). Top Level ET tasks may include instructions for operating and maintaining this system and instructions for complex, perishable skills the Leader needs for combat (O).

*Rationale: Preparation for combat can include training on mission specific skills. Mission tasks identified during the mission planning cycle can trigger execution of rehearsals and refresher training. The Leader needs a capability conveniently accessible on his own system.*

(c) ET: Combined Arms and Joint Context. The system must facilitate mounted and dismounted individual and small unit training in a combined arms and Joint context (T). The system must facilitate collaborative training with other Leaders and platforms (O).

*Rationale: Leaders fight as members of a combined arms team; therefore, they need the capability to train in a combined arms context.*

(d) ET: Training Environments. Supported training environments must include live, virtual and constructive training environments for all ground Soldiers (T=O).

*Rationale: Support for all training environments ensures maximization of training alternatives to sustain mission readiness.*

(e) ET: Skills. TSPs must be developed for GSS training tailored for specific units and skills (T). This capability must support all skill levels for individual Soldiers and leaders at the small unit level and collective training tasks at the small unit level (T=O).

*Rationale: Every unit has its own mission and training priorities. These priorities can change during operations. Unit must be able to customize their own list of training capabilities hosted by the GSS.*

(f) ET: Collaboration. The system must support collaborative participation by geographically dispersed Leaders in field and garrison environments (T=O).

*Rationale: Geographic dispersion of Leaders and small teams must not inconvenience and disrupt training. In certain tactical situations, bringing all unit members together in one location may be ill advised.*

(g) ET: Conduct Religious Support. Chaplains, Chaplain's Assistants, and the Combat Religious Support Team (CRST) must have the ability to conduct religious services/support across the full spectrum of operations (T). The delivery of comprehensive religious service/support requires the capability of two-way communication, net-centric technologies for spiritual fitness training products (T), extended religious support, the ability to interface with a predictive data base, and the ability to conduct LVC spiritual training in real time (O).

*Rationale: Chaplains, Chaplain Assistants, and the CRST provide a variety of religious services and support to Commanders and Soldiers at Home Station and when they are deployed. The database system provides the capability of predicting the morale and spiritual welfare of personnel as well as potential issues related to local religions and religious customs. The capability to manage the Religious Support COP is a vital command interest instrument to assess unit and individual morale and matters of religions which may affect the mission. Vestments and accoutrements are required for certain Chaplains to be able to provide religious support and must be integrated into and/ or compatible with the GSS. Chaplain's faith-specific items are designated by the individual Chaplain, which are then validated by the endorsing agency of the Chaplain. Chaplains also need to be able to conduct spiritual training for all phases of battlefield operations.*

(h) ET: Operational Integration. The Leader must use one integrated system for both training and actual operations (T=O).

*Rationale: Integrating operational and training capabilities enables the Leader to train like he fights and fight like he trains. Preparation for combat can include training on mission specific skills and enroute mission rehearsal.*

(i) ET: Cognitive Skills. The system must assist the Leader in task performance during operational periods when the Leader's cognitive skills are in a degraded state, for example, when the Leader is sleep deprived, in inclement weather, Soldier load, IED attacks and recent deaths of comrades (T=O).

*Rationale: Combat induced stress can degrade the Soldier's cognitive capabilities. This condition requires an enhanced capability to sustain important skills. With the use of the RPMS the Leader's vital signs and status can be monitored remotely by both medical personnel and leadership to determine their level of alertness or degradation.*

(j) ET: After Action Review (AAR). The GSS must contain an after AAR system to support individual and collective training (T). The AAR system will provide a means to monitor, record, assess and playback training exercise and actual operational events (T). The system must allow the trainer to flag events as they occur to facilitate locating specific events during playback (T). The system must be able to execute an operational pause during an exercise for the purpose of conducting an intermediate AAR (O). The system must be able to show effects of leadership decisions made during planning and execution (T). All AAR assessment and performance data must be recorded in an embedded learning management system (T). The same AAR system must capture lessons learned from actual combat operations (O).

*Rationale: The AAR is the heart of the Army's training strategy. Collection of training data is essential to support the analysis of performance and production AAR feedback. Results of battlefield actions must be collected so that effectiveness can be accurately measured and used for AAR and exercise feedback and TTP development. The AAR is not limited to the training environment. This tool is invaluable following reconnaissance missions, as well as direct action operations. Capturing lessons learned enables the Soldier and small unit to learn from their own and others' mistakes and adapt more quickly to changing battlefield conditions.*

(k) ET: Interface with Army BCT Modernization BCS. The GSS must interface with the Army BCT Modernized Networked BCS collaborative, intuitive, full-range-of-mission planning and rehearsal functions (T=O).

*Rationale: Mission planning and rehearsal is required during alert, deployment, employment, and redeployment. The Army BCT Modernization Networked BCS will have an enroute mission planning and rehearsal capability both during inter-theater and intra-theater movement of forces.*

(l) ET: Interface with the Army BCT Modernization AAR System. The GSS must interface with the Army BCT Modernization unit's AAR subsystem to support individual and collective training within each training domain as defined by FM 7-0 (T). The AAR subsystem will provide a means to assess, monitor, record, and play back exercise and operational events (T). Assessment uses diagnostic matrices that crosswalk performance standards to Mission Training Plan (MTP) with graduated degrees of scenario difficulty (T). The system must allow the trainer to flag events as they occur to facilitate locating specific events during playback (T). The system must be able to execute an operational pause during an exercise for the purpose of conducting an intermediate AAR. All AAR assessment and performance data must be recorded in an embedded learning management system (T=O).

*Rationale: The AAR is the heart of the Army's training strategy. Collection of training data is essential to support the analysis of performance and productive AAR feedback. Results of battlefield actions must be collected so that effectiveness can be accurately measured and used for AAR and exercise feedback and TTP development.*

(m) ET: GSS Army BCT Modernization ET Interface. The GSS must interface with the Army BCT Modernization individual (T) and collective ET capability that supports multi-mode LVC training environments (O).

*Rationale: "Embedded full task and TESS provide leaders with a readily available system for training and assessing Combat Critical Skills of Soldiers; and provide the capability to train and assess crew/sections (T), dismounted Soldiers, and multi-echelon combined arms skills (O)."*

(n) ET: Interface with Army BCT Modernization Networkable, Reconfigurable, Full Task Trainer (NRFTT). The GSS must interface with the Army BCT Modernization networked, embedded, virtual full task training capability which supports individual, crew, and multi-echelon training without reconfiguration of the equipment (T). The Army BCT Modernization ET systems interfaces that allow it to interoperate with TADSS, and with the current and future synthetic training environment that includes LVC simulators and simulations (e.g. range instrumentation, Combined Arms Tactical Trainer (CATT), War Simulation (WARSIM), and One Semi-Automated Forces (OneSAF) (T).

*Rationale: Soldiers and leaders of the BCT must train with the Army BCT Modernization NRFTT while equipped with their individual GSS.*

(o) ET: GSS Interface with On Board Ground Combat Vehicle Repository. The GSS must interface to the Ground Combat Vehicle "on-board" repository of training products including: TMs, MOS information, TTP, MTP, TSP, individual and unit morale and spiritual fitness sustainment training, interactive multimedia instruction courseware, and collective training task information that supports the sustainment of individual and collective skill levels at home station, the CTC, and when deployed (T). The Ground Combat Vehicle "on-board" repository will provide a capability to link via C4ISR to other AKE compliant repositories to obtain updates of stored data, access additional training products or exchange training products between center/school repositories. (T)

*Rationale: Soldiers and leaders within the BCT must be multi-functional. Immediate access to TTP associated with the myriad of BCT missions and tasks, particularly those low frequency tasks, will be vital to ensuring the readiness of BCT units to support full-spectrum operations.*

(p) ET: GSS interface with the Army BCT Modernization multi-echelon CAT. The GSS must interface with the Army BCT Modernization system which provides a multi-echelon CAT training capability for leaders, staffs, and units to build functional, CAT teams. ET will support multi-echelon training and rapid teaming including integration of current, SBCT and JIM capabilities at BCT and below (T) and above (O).

*Rationale: The Ground Combat Vehicle enroute mission planning and rehearsal system will require the same simulations capabilities as the training simulations but will require more rapid initialization and graphics file generation to meet operational timelines. Army BCT Modernization exercise development services must be extended to define projected AAR requirements.*

(q) ET: Interface with the Ground Combat Vehicle. The GSS must interface with the Ground Combat Vehicle SoS embedded capability which replicates friendly and enemy weapons effects and provides simulation/stimulation of other effects (NLOS, CBRNE, EW, ID, etc). The Ground Combat Vehicle embedded capability will generate, send, and receive engagement data that conforms to the Army Common Training Instrumentation Architecture (CTIA) and Joint Instrumentation Systems (T). The Ground Combat Vehicle must interoperate with current systems (Multiple Integrated Laser Engagement System (MILES) and Combat Training Center-Instrumentation System (CTC-IS) and future Army and Joint Engagement Simulation Systems via a High-Level Architecture (HLA) construct (T). Provide realistic simulation of the tactical environment during force-on-force and force-on-target tactical training exercises (T). Ground Combat Vehicle C4ISR system will be capable of transmitting tactical engagement data. A part of the Ground Combat Vehicle ET capability will include a tactical engagement system in accordance with the Army CTIA (T) and Joint Instrumentation Systems (O).

*Rationale: Embedded TESS capability provides the means to replicate the Operational Environment (OE). This capability must be able to interface with CTC/Joint National Training Center (JNTC), and home-station/deployed range and instrumentation systems. This capability allows training integration with current forces across all domains. This ability to have engagement data collection facilitates the development of AARs, which enhances individual and collective experiential learning and sustaining higher levels of task proficiency. Embedded TESS reduces the time required for live tactical engagement training.*

(r) ET: Interoperability with RPMS. The GSS training system must provide simulated casualty data to the RPMS during force-on-force collective training (T).

*Rationale: Individual and collective training must exercise medical systems. This capability enables the unit to assess readiness of maneuver sustainment systems. The intent is to incorporate training formerly enabled by MILES Casualty Cards.*

(s) ET: TADSS Training. The System specific TADSS must be developed (T) to facilitate training of leader GSS tasks. The GSS must be able to integrate with Non-system Specific TADSS (O).

*Rationale: It is anticipated that a large number of Leader tasks performed using the GSS will be more efficiently and effectively trained using TADSS. The use of TADSS will reduce power and other resources required by ET.*

(5) GSS Networked Lethality.

(a) Area Lethality. GSS weapon systems must provide Leaders the capability to quickly defeat threats within the small unit AI (T=O).

*Rationale: The small unit AO and AI drive Soldier engagement range capabilities. The FF BCT AI radius is 8 km for platoons and 16 km for the companies. Within the AI, Soldiers and small units must have a full range of capabilities that enables mutual supporting and massing of lethal and non-lethal effects.*
*While executing BCT missions, the Soldier and small unit will employ these capabilities to defeat threat targets in open and complex terrain that are line of sight and non-line of sight. Targets can include personnel, equipment and armored vehicles. These targets can be covered and concealed by vegetation, civilians and structural barriers such as earth and timber and masonry buildings.*
*Networking of Soldiers, sensors and leaders significantly improves Soldier and small unit lethality by enabling cooperative engagement. They can quickly mass effects while geographically dispersed to influence a larger area with greater precision and lethality. Collaborative SA/SU provided by the COP further enables this cooperative engagement.*

(b) Dismounted Target Hand-off and Designation. The GSS must enable the Leader to geographically locate targets at distances up to 2.5 Km with a Target Location Error (TLE) of 65m and provide precision target location data to facilitate engagement by another Leader and/or another system under all operational conditions (T). This capability must also provide designation by identified personnel for precision-guided and loitering munitions at distances up to 4 Km with a TLE of 10 m (T) delivered from LOS, NLOS, and BLOS weapons platforms and aircraft. Therefore, the system should be interoperable with current and future target designators (T). For GPS guided munitions, the GSS must capture and transmit target location data to these munitions in flight. The Leader should have the capability to designate threat aircraft for attack by appropriate weapon systems (O).

*Rationale: The Leader needs an organic target location and designation capability since platforms with such capabilities will not always be available. The Leader and small unit will need to designate targets during BCT air assault operations to operational depth. During close combat in close, complex terrain, the Leader and small unit will need to designate or hand-off targets in confined spaces where wheeled and tracked platforms cannot gain access. Distance requirements enable a standoff capability to dismounted Soldiers within the platoon AI and allow room for them to handoff targets to mounted platforms. The threshold TLE radius enables the Leader using his COP to distinguish between two reported targets. The TLE radius also prevents multiple sightings of the same target from resulting in a report of multiple targets. The objective TLE aligns the Leaders' capabilities with that of the supporting Ground Combat Vehicles.*

(c) Target Recognition and Location. Leaders tasked with performing as Fire Controllers / Forward Observers for Artillery fire must acquire, collate, process, and transmit all necessary targeting information in formats required by all potential fires delivery systems (T). The system will have optics allowing the fire support Leader to recognize targets during daylight at ranges up to 4.5km (T) and identify targets up to 8km (O). During periods of darkness, the system will have optics allowing the fire support Leader to recognize targets at ranges out to 350m (T) and identify targets out to 4km (O). The system will also locate and laser range find targets to a 20-meter accuracy standard at day and night ranges (O). The GSS will have the capability to mate with a lightweight laser designator (O). This will enable the observer to designate for the use of precision munitions.

*Rationale: Future engagements, enhanced by the small unit COP, will include employment of organic and external systems including precision-guided munitions. In complex terrain, this task will often require designation by a Fire Support Soldier. The Soldier needs to locate, receive, designate, and/or hand-off targets for combat systems, which include air assets and all indirect fire delivery systems. The Fire Support Soldier requires an organic target location and designation capability since platforms with such capabilities will not always be available. Efficient acquisition, collation, processing and transmission of targeting data will enable the Fire Support Soldier to act first and finish decisively with desired target effects. The Fire Support Soldier will acquire targeting data to facilitate employment of other systems.*

(d) <u>NLOS/BLOS/LOS Target Engagement</u>. The GSS must enable the Leader to engage targets using NLOS, BLOS, and LOS systems, at an individual through Company level and beyond, to achieve the desired effects with lethal and non-lethal systems (T). Immediately upon target recognition, the GSS must acquire, collate, process and transmit all necessary targeting information (see IERs) in formats required by all potential "shooter" systems into the COP and the Joint Networked Fires system (T). The system must accomplish this task with minimum interaction (manual data entry) by the user (T). The GSS must be capable of providing updates (target type, location, and velocity vector) by designated personnel to precision and loitering munitions in flight including interactive dynamic re-tasking of multiple munitions as to target type and location (O).

*Rationale: Networked Lethality and access to Networked Fires significantly improves the unit's lethality. BCT Modernization will include NLOS, BLOS, and LOS weapons systems for Soldiers and leaders to employ, whether mounted or dismounted, against a wide array of targets to include enemy personnel, light and heavy armored vehicles, aerial targets, and fortified positions. Efficient acquisition, collation, processing and transmission of targeting data will enable the small unit to act first and finish decisively with desired target effects. The Leader engages targets with his own system or acquires targeting data for other lethal systems.*

(e) <u>Optimized Attack Solutions</u>. Upon entry of targeting information (target designation), GSS must produce target engagement solutions in less than one second (T), same as threshold (O). GSS must balance the need for responsiveness against effective systems application.

*Rationale: Exploiting technology to calculate optimized attack solutions enables geographically dispersed Soldiers and small units to dynamically apply lethal and non-lethal effects. The time standard is a component of the Networked Lethality KPP timeline.*

(f) Synchronization of Fires. The GSS must accommodate synchronization of fires and effects for all small unit assets (T), all external assets (O). This capability includes planning, rehearsing and execution of such fires (T).

*Rationale: Efficient and timely synchronization of fires effects within the company's scheme of maneuver and fires is essential in order to decisively defeat enemy forces. Networking small unit leaders, sensors and Leaders must enable small unit members to collaboratively influence a larger area with greater precision, speed and a broader variety of lethal effects.*

(g) Report BDA. The GSS must provide BDA information to the BC and Networked Fires systems following target engagements regardless of which system engaged the target (T=O).

*Rationale: The COP requires a real time or near real time picture of the enemy and friendly situation. BDA closes the loop for target engagements. Friendly forces need the updates to the enemy situation that result from each employment of fires. Accurate and timely BDA helps leaders and staffs track current enemy disposition, composition and intent. Accurate and timely BDA supports the decision cycle and prevents wasting limited ammunition on re-engagement of targets that have already been killed.*

(h) Interoperability. The GSS must interface with the Army BCT Modernized network in order to support Joint networked lethal and non-lethal effects that achieve overmatch – out of contact and in contact, at tactical stand-off, and in close combat (T).

*Rationale: Attack from standoff positions using organic and joint assets is a key element of BCT operations. GSS provides the interface to this capability for the dismounted Soldier. Joint capabilities such as CAS must know the precise location of friendly forces while servicing targets.*

(i) Detect/Avoid Danger and Hazardous Areas. The GSS sensors must allow a Leader to detect, and/or bypass any danger and hazardous area (T).

*Rationale: Leaders will encounter a variety of danger and hazardous areas such as open areas, minefields and contaminated areas, which must be avoided or negotiated, while operating the system.*

(j) Precision Delivery of Troops and Equipment. The GSS equipped Leaders must be transportable by Army BCT Modernized units, Future Tactical Trucks, Rail, Army aviation, Army watercraft, US Air Force aircraft, and US Navy ships for strategic and tactical deployment to an area of operations or battlefield (T=O).

*Rationale: Leaders in the FF must be transportable by all strategic and tactical craft in order to deploy within the 96 hour required timeline and conduct operations once deployed. Leaders and small units will use their BC capability to collaboratively plan and rehearse missions while traveling.*

(k) Rapid Assembly of Dispersed Forces. The GSS must enable quick and orderly assembly of dispersed forces (T).

*Rationale: Recovery from geographical dispersions must not degrade the speed and momentum of small unit operations. Operational situations will disperse small unit members and their equipment for numerous reasons. During airborne operations, small unit members will not always land conveniently close to each other. They may become intermingled with other small units and separated from their equipment. Reacting to indirect fire can disperse and disorganize small unit members. Retrograding to the last designated rally point during dismounted maneuver can also disperse small unit members.*

(l) Personnel Visibility and Human Resources. The GSS must provide on-demand personnel visibility via the COP (T). The system must accommodate and streamline all personnel management information and processes performed by small unit leaders (T). The system must interface with human resource systems to provide hasty and accurate information and array it in a manner that enables all personnel services and personnel management activities and battlefield casualty information, which is provided by speech to text technology (O). The system must also enable the personnel accountability function by providing personnel visibility updates when required (T).

*Rationale: Commanders require real-time Soldier status as it applies to deployability, readiness, accountability and location whether at the embarkation point in the operational environment or replacement stream. Human resource information must be relevant, reliant, and readily available to commanders and staffs at all levels of command. Small unit leaders require one enterprise information system instead of multiple "stovepipe" systems to accomplish all required personnel management related tasks. Personnel management tasks for Small unit leaders include tracking individual training status, accountability of assigned individual equipment, and counseling. Personnel management tasks also include determining troops available during mission planning as part of mission analysis.*

(m) Equipment and Supply Accountability. The GSS must provide equipment and supply accountability via the COP (T). The system must provide the Leader and small unit with streamlined functionality to forecast, request, allocate, store and distribute supplies, especially supply classes I, III, V, batteries and water (T). The system and its repair parts must be visible in the Total Asset Visibility database using standard Automatic Identification Technology (T).

*Rationale: The small unit leader's COP capabilities include mission capability status of equipment. The Soldier will possess and must account for many equipment items. These items may be assigned permanently or temporarily based on METT-TC. The leader may cross-level equipment to manage the Soldiers' loads. The leader may reassign equipment from Soldiers who become casualties. A Soldier may replace his leader and assume responsibility for equipment accountability. Mission tailoring may include exchanging equipment and supplies between Soldiers and unmanned vehicles. The Soldier and leader must accomplish these tasks without losing accountability of equipment. The system must also give the leader visibility on the collective unit supply status.*

(n) Embedded Mission Readiness. The GSS must incorporate an embedded mission readiness system. This system must monitor the status of mission-critical components and/or subsystems and consumables (T). If a condition is detected that degrades mission capability, the embedded system will determine what action (repair, re-supply, crew rest period, etc.) is required to restore full mission capability (O).

*Rationale: Embedded Mission Readiness permits sustainment of forces with supplies they need, when they need them. It also facilitates cross leveling of supplies when required. Commanders at all levels must be provided simply displayed but comprehensive pictures of their sustainment status. This requirement is critical to the reduction of mechanics and the logistics footprint of the BCT as well as enabling the pulsed logistics concept.*

(o) On-Line Fault Isolation. The GSS shall allow the operator to identify a single fault to an ambiguity group of no more than two operator replaceable (configurable) items with at least 80 % accuracy without user action to modify the system configuration (T). On-Line fault isolation shall allow the operator to identify a single fault to an ambiguity group of no more than two operator replaceable (configurable) items with at least 95 % accuracy when used with guided (operator level) systems reconfiguration directions (O).

*Rationale: Ensure the GSS can execute assigned functions without the risk of mission failure and threat to life because of equipment malfunction. Retain operational tempo by identifying critical functional degradation, for which action can be taken to overcome if known, and minimizing maintenance and supply burdens caused by misdiagnosed failures.*

(p) Maximum Time to Repair (Max TTR). The Max TTR for 95 % of all malfunctions correctable at the field level of support will not exceed 20 minutes (T) 15 minutes (O) for dedicated maintenance personnel and for operators (each assessed separately).

*Rationale: Threshold requirement constrains the burden placed on dedicated field maintenance personnel and operators to the level achieved with the baseline LW-SI System in OIF. However, based on RAM and OMS/MP analysis, it will not preclude the need for one or more additional dedicated Table of Organization and Equipment (TOE) field level maintainers per BCT (dependent on the level of leaders to which the GSS is fielded) if the GSS support concept adopted were to involve traditional Army maintenance without the fix-forward benefits of the concept currently envisioned by PM-Soldier Warrior (field level Contractor Logistic Support (CLS) with fix-forward repair through replacement of GSS LRUs at operator level and back-up maintenance capability provided by dedicated Field Service Representatives (current concept employed for LW-SI System)). The fix-forward aspect of the GSS support concept currently envisioned by PM-Soldier Warrior, involving a designated Master Warrior operator with additional maintenance training who performs the operator level troubleshooting and repair within his respective platoon, results in a maintenance time burden of under one hour per day for that individual. Objective requirement reduces the burden placed on dedicated maintainers who provide back-up support to the platoon Master Warriors (reducing sustainment maintenance costs accordingly), as well as the Master Warrior's time spent performing corrective maintenance (which is an additional duty having the potential to adversely impact on his role as a leader); the need for one additional dedicated TOE maintainer (as applicable)* *would not be eliminated even at objective requirement level. The requirement for additional maintainer force structure within the BCTs will be verified through the Level of Repair Analysis (LORA) and Logistics Demonstration.*

(q) Graceful Degradation. The GSS must maintain some functionality if component systems fail. The loss of one element of the system must not terminate the utility of the entire suite of equipment (T=O).

*Rationale: Battlefield damage and extended use will result in a component of the system failing eventually. Graceful degradation will ensure the GSS equipped Leader maintains operational capabilities after such failures occur.*

(r) Protection from Work-Site Hazards. The GSS must be protected from the destructive effects of dust from construction tasks (e.g., rock dust, laterite), damages due to heavy machinery operation and maintenance, and must endure routine clean-up of the associated contaminants which are a by product of these tasks (T=O).

*Rationale: Some Soldiers routinely operate in environments, natural and man-made, and perform MOS-related missions where required chemicals, POL products, tars, detergents, water, and common battlefield contaminants (smoke, fuel spills, etc.) are prevalent. These degrade or eliminate the protective qualities of uniforms/fabrics to include CBRNE protections, and ruin or contaminate connectors of many types.*

*Some Soldiers are also subjected to more intense concentrations of dust, multi-frequency vibrations on all dimensional planes, abrasion and snagging due to use of tools such as heavy hand-tools, ropes and chains, chainsaws, drills, powered hammers and cutters, bulldozers and other heavy machinery, electrical wires, cables, barbed wire, electric arcs and cutting flames, etc..*

(6) GSS Sensors. Leaders deployed sensor capabilities must enhance SA/SU for areas of local or immediate interest and fill gaps left by Army BCT Modernized sensors (T). GSS Sensors at this level must directly support immediate Leader's SA/SU first and then, if desired, updates to the COP (O).

(a) Individual Soldier Sensors (ISS). ISS must enhance SA/SU for individual Leaders in their personal operational environment and squads in their AO (T). The ISS shall be multi-functional to support a wide range of missions in Complex (Jungle, Urban, and Forest) and open terrain (O). The primary use is for human intrusion detection in immediate area of concern, especially where Soldier worn imagers or weapon system sights do not have LOS to approaching targets. In addition, ISS should be used to detect threat personnel in caves, buildings, heavy foliage, underground spaces, etc (O). These sensors must have the capability to be used by individual Leaders in defensive (e.g. support at listening post/observation post) or offensive (e.g. throw sensor to detect potentially hidden threats) operations (O).

*Rationale: Leaders on the future battlefield will encounter natural and manmade obscuration and obstacles inhibiting vision. Leaders require the ability to defeat this obscuration in order to effectively and safely operate. This capability can be used for intrusion detection at any LP/OP,* *building clearing (during and after), detection of humans in caves/sewers (either in advance of entering or after clearing), etc.*

(b) ISS: Mobility. The ISS must be small and light enough so multiple sensors (T) may be easily carried by an individual Leader. The ISS must be re-usable but inexpensive enough that they can be considered as disposable or consumable items (T), operate for 72 hours (T) and have the capability to be turned on/off so they can be reused. Sensors must be able to be placed by hand (dropped or thrown) (T) and variants may be delivered by small team indirect fire (40 mm grenade launched) systems for greater stand-off (O). ISS must detect and alert the Leader to the presence of humans (Probability of detection = 90 %, Probability of classification = 85% (human vs. non-human; animal, clutter, noise) within 5 meter radius of sensor) (O).

*Rationale: This capability will give Leaders and small units the ability to have a reusable, but disposable human detection systems as an extension of and integrated with the GSS.*

(c) ISS: Interoperability. ISS must be able to be used individually or multiple at a time and must interface with the GSS (T). ISS must be capable of alerting a Leader with an Audible/Visual alert or via a wireless interface for the (wireless enabled) GSS (T). (1716) The ISS must interface with an (wireless enabled) GSS up to 500 meters away (T), 1000 meters away (O) and NLOS under all conditions; For wireless use, ISS must be capable of receiving geo-location information from the GSS (i.e. from Leaders known position) without requiring manual configuration by Leader. The ISS must use this information as approximate location when alerting Leaders. ISS must interface with any enabled GSS (O). Alerts received by the ISS must be received by the (enabled) GSS and displayed as appropriate (T). The GSS Leader will determine if further reports should be generated (T). ISS must be capable of being used with an additional asset (e.g. hub/gateway) so small networks of unattended sensors can be constructed (O).

*Rationale: This capability will allow the Leader to fill in local/immediate sensor gaps where LOS sensors (helmet mounted cameras or weapon sites) cannot see…Making this system reliable, easy to operate and reusable will promote its use much more than existing systems such as trip flares. By making this system multi-functional a wide range of missions can be supported, increasing the value of carrying such devices.*

(7) GSS Force Protection: Body Protection: Individual Soldier Sensor Kit. The GSS must warn the Soldier of threat materials. The system must detect projectiles of all types launched toward their location (O). An embedded system must locate the source of projectile launch in real time (O). The system must detect concealed explosives from standoff distance (O).

*Rationale: Primary threats to the GSS include fragments, bullets, blast, thermobaric, flame, and incendiary weapons. Delivery methods include projectiles, threat personnel on foot or using other transportation means, and concealed locations along friendly travel routes. Timely warning of threat materials enables the Leader to survive by avoiding the terminal effects. Significant threats to convoy operations include IEDs creatively concealed along travel routes. SASO operations require Soldiers to man checkpoints where approaching threats include explosives concealed on individuals and privately owned vehicles. Integration with Networked*

*Lethality enables the Soldier and small unit to convert the threat source location into actionable, retaliatory targeting information.*

(8) COP.

(a) COP: Enemy Location Tracking. The GSS must track known enemy composition and disposition with a level of fidelity suitable for small unit operations (T). The small unit leader must have a color capability to distinguish enemy from friendly forces in the AO. All icon-based symbols to identify enemy and friendly locations are in accordance with FM 101-5-1/MCRP 5-2A Operational Terms and Graphics. Leaders need the ability to template the enemy location and track the actual situation by assessing enemy composition and disposition down to the number of enemy personnel and their capabilities (T). Leaders must have the capability to identify enemy personnel through facial recognition technology for check point and clearing operations (T). The small unit leader must have the capability to zoom in and out on the COP display (T). Threat battle management will be automated when sensors are providing continuous feed; threat data will indicate time of last known composition and position and will grey out threat data that has exceeded threshold for composition or position update. The fidelity of entity presentation on the COP (e.g., individual entity, squad, company, battalion) is selectable (T=O).

*Rationale: The small unit leaders need this enemy tracking capability in order "to see first" and "understand first." All data provided to Small units conducting operations will be inputted using a computerized database. This database has been uploading using existing photographs and images provided by intelligence specialist and readily available to small unit leaders conducting missions. Leaders today, are using photographs and decks of cards to try to determine enemy and most wanted personnel. This technology is dated and leaves the Leaders vulnerable to surprise attacks. Leaders in the future will have the ability to photograph individuals while they are manning check points, moving prisoners and rendering medical attention to civilians and combatants. This technology will allow Leaders to determine enemy and most wanted suspects in real time. This facial recognition capability will be provided through the Leaders Computer/Master Hub Subsystem. This is the primary device for integrating data from other sensors, components, and subsystems and displaying this information in the Leader's HMD. The computer and software allows the Leader to access and process SA information received in mission OPORDS and intelligence summaries. This data will be uploaded by intelligence specialist using existing photographs and images of key enemy and most wanted fugitives. This facial query will come from a database for positive recognition provided through the COP and feed to the Leader through his HMD. The location of the enemy will be visible on the COP display in order to verify and designate as targets.*

(b) COP: Positive Biometric Identification (Facial Recognition). Leaders performing Military Police functions must uniquely and positively identify individual persons within the area of influence or operations; friendly and unknown (T), friendly, foe, and unknown (O).

*Rationale: The ability to positively identify friendly, foe and unknown persons mounted and dismounted to the level of who, what, when, where, how and why they are within the AI is* *paramount and has the ability to reduce threats to the MP Soldier while performing during missions.*

(9) Intelligence.

(a) Intelligence: Sensor Control. The GSS must interface with reconnaissance platforms in order to allow mounted or dismounted Leaders (Scouts) to use, in real time, platform-mounted sensors, recon, communication systems; to control assigned unmanned systems; to send sensor info to the platform; and provide a local and networked fires interface for the scout's SA/SU equipment while mounted or dismounted, to a distance of 5 km (T), 12 km (O) while the platform is stationary or moving.

*Rationale: Intelligence collection and remote sensor operation must provide the GSS equipped Soldier the means to enter the intelligence network in order to receive information supporting SA as well as the ability to feed information to that network. To enhance the survivability of reconnaissance elements and reduce their signature, they could be required to dismount and move away from the vehicle. They need the ability to conduct their reconnaissance and surveillance tasks in 24-hour operations "up to 12 km away from the vehicle." Tactical analytics demonstrate that a significant amount of reconnaissance and surveillance must be performed while dismounted. The Leader can use his own system as well as assets from his vehicle to conduct dismounted reconnaissance and surveillance.*

(b) Intelligence: Extended Operational Communications. The GSS must support purely dismounted human intelligence collection operations at extended distances (T), 150 Km (O).

*Rationale: BCT missions include air assault operations over extended distances (150 Km) in complex terrain; such operations may not include availability of manned platforms. The standard for dismounted units conducting reconnaissance operations is a minimum of 72 hours eyes on target. Human intelligence in accordance with these standards provides the best target intelligence.*

(c) Intelligence: Dissemination. The GSS must be able to securely distribute and receive relevant intelligence information in < 4 sec. (T) or, < 1 sec. (O), down to each individual Leader, and with a low probability of intercept.

*Rationale: Intelligence dissemination ensures that the SA/SU of any Leader is shared by all Leaders in the small unit. Leaders will share information through horizontal distribution and receive processed information through vertical distribution from higher. When the Leader identifies targets and items of interest, he must be able to quickly and conveniently designate them to the small unit COP. This capability enables the small unit to finish decisively using collaborative engagement and synchronized maneuver and fires. See IERs in Appendix A for prioritization of specific intelligence information.*

(d) Intelligence: Compatibility. The GSS must be compatible with the Future BCS and DCGS-A (T=O). The small unit leader must be able to access (pull and push data, information and intelligence from/to) Army (DCGS-A) and Joint intelligence repositories (T).

*Rationale: The Future BCS is the C2 system for the FF. DCGS-A is the Intelligence, Surveillance, and Reconnaissance (ISR) component of the Future BCS.*

(e) Intelligence: Air Warning. Air threat information must be provided within 6 seconds after track file establishment and updated within 2 seconds (T). The GSS network must receive and disseminate Air Threat Warning information from local sensors (such as Multi-Mission Radar), UE and Joint/National sensors and architectures such as the Joint Data Network, and Joint Composite Tracking Net (O).

*Rationale: In order to maintain survivability, each echelon in the BCT must have directed and tailored early warning in order to engage threat aircraft at standoff. The overall air threat will originate at the JIM level and the BCT must have interoperability in order "to see first." The Leader must receive this air warning with sufficient lead time to allow evasive action and target engagement. Platoons and companies will receive information to develop a complete air picture (friendly and hostile) within the platoon/company's vicinity (approx 16kms) to enable self-defense engagements and platoon/company reaction to air threat.*

(f) Intelligence: Language Translation. The GSS must translate languages based on voice recognition for all languages used in the current AO (T). This capability must translate from English to the foreign language of interest and vice versa (T). This capability must translate 80% of two-way speech and 90% of text to speech and text to text in near real-time (T), 90% of two-way speech and 98% of text to speech and text to text in near real time (O). This capability includes translation of both paper and electronic documents (O).

*Rationale: Leaders, especially military intelligence specialists, and civil affairs, will encounter situations in dealing directly with enemy and allied Soldiers and noncombatants on future battlefields and other environments. An embedded language and dialects translation capability will reduce misunderstandings and assist with coordinating coalition operations and interrogating Soldiers and noncombatants without the need for a translator. SASO operations often require significant interaction between friendly forces and local civilians. The amount of language content translated must be sufficient to allow the Soldier to "understand first" so he can "act first." Translation from English to the foreign language of interest and vice versa facilitates two-way communication.*

(g) NC: Interoperability. The Leader's communication system must allow full interoperability with the Army BCT Modernization C4ISR architecture (T). The communications architecture must have the flexibility, scalability, and extensibility to allow every Leader access to the Army BCT Modernization network (O). The dismounted Leader must have communications capability for full BC/SA with or independent of a Ground Combat vehicle (O).

***Rationale:*** *The Leader must be able to leverage the communications capability of their vehicle while mounted and while their vehicle is moving or stationary. The Leader must leverage available communication systems while traveling on any typical transportation means such as tactical trucks, aircraft and waterborne means. The Leader must also maintain SA without his* *vehicle because of terrain and mission considerations. The Leader will fight in complex terrain where ground platforms cannot gain access. BCT missions include Air Assault operations over extended distances (150 Km) in complex terrain, BCT O&O, Annex F. "The dismounted Soldier must operate up to 12km away from the vehicle."*

i. GSS Attributes for GSS Increment II. There are MOS specific Soldiers who will receive additional "attributes" based on mission specific tasks, i.e., Engineers performing Explosive Ordinance Disposal (EOD) that must have additional blast protection above what is provided by the Core or GSS. These Additional Performance Attributes for GSS Increment II are provided by programs that are GFE to GSS.

| Performance Parameter | Development Threshold (T) | Development Objective (O) |
|---|---|---|
| Enhanced Human Sensing | GSS will enhance the Soldier's natural senses at a distance of at least 300 meters in all levels of light, weather and battlefield conditions. | 500 meters |
| Personal Lethality: Speed | 0-50 M within 1 second | T=O |
| Personal Lethality: Personal Defense Weapon | Equal performance of current M-9 | Effective lethal fire at 200 M (O), conform to law enforcement capabilities |
| Area Denial and Non-Lethal Effects | The GSS weapon systems must provide the capability to deny access to, use of, or movement through a geographic area without debilitative injury to personnel. Delivery means of these effects may include Soldier emplaced, LOS, BLOS, NLOS, and unmanned systems. | T=O |
| Specialty Weapons | Specialized mission needs. Augment organic capabilities. | T=O |
| Sustainment Operations in Limited Visibility | Enable the Soldier to perform intricate actions at short distances under limited visibility conditions (i.e., eye focus at arms length) | T=O |
| Enhanced Micro-Climate Protection | Stand alone power | Powered off system source |
| Auxiliary Air Supply | GSS must be able to operate in areas devoid of oxygen or contain toxic gasses for 20 minutes | Must operate in poisonous underground air and ship holds for 60 minutes |
| Air Supply with Flotation Collar | The GSS will provide a small, compact, lightweight, breathing source with a regulated supply of oxygen. The amount of oxygen shall be sufficient for 50 seconds minimum breathing time at a 20-foot water depth and 55 º F water temperature, with a breathing rate of 40 liters per minute to facilitate escape from equipment and ascent to the surface (T). | The breathing device shall provide five minutes breathing time under the same conditions (O). |
| Crowd Control: Facial Protection | Lightweight protection from laser, blunt force. | Protect V-10 penetrations of projectiles 40 grains or less at 1500 feet/sec |
| Crowd Control: Body Protection Stab and Ballistic | Protect from sharp objects, ballistic protection against 9mm 124 gr @1400 ft per second | T=O |

| | | |
|---|---|---|
| Force Protection: Petroleum, Oil and Lubricants | Protection capability against common battlefield fuels | All petroleum based fuels |
| Enhanced Fire Protection | Fire Retardant | Fire and Fuel Resistant |
| Protection from Occupational Hazards | Protection enabling them to continue the mission after being contaminated, without a clothing change for a minimum of 12 hours | 24 hours, without the use of CBR-peculiar equipment. |
| Home Station Mission Protection | Provide specialized Soldier clothing and equipment for unique Home Station missions | T=O |
| Mobility: Create a Man-Sized Hole in a Wall | Enable Soldiers to create a man-sized hole through a wall, floor, or ceiling in a single stage of reduction (T). The hole must be through eight inches of double reinforced concrete (using #4 rebar on 12 inch centers) and have dimensions of thirty by fifty (30 x 50) inches (T) | fifty by seventy (50 x 70) inches (O). |
| Identify/ Reduce/ Defeat Obstacles | GSS sensors must be compatible with Ground Combat Vehicle capabilities for advance detection of obstacles and facilitate Soldiers' reduction and/or bypass of obstacles (T). | T=O |
| Soldier Sustainability: Improved Soldier Strength and Endurance | Soldiers must be enabled to improve their physical work performance by 50%. | Soldiers must be enabled to improve their physical work performance by 200% (O) Increase the endurance of and strength of the Soldier without the aid of drugs (O) |

Table 6.8. GSS Attributes for Increment II

(1) SA: Enhanced Human Sensing. The GSS will provide the Soldier the capability to determine the presence of humans on the far side of an opaque obstacle. Soldiers must be able to detect, locate, classify, recognize and identify threat personnel in caves, buildings, heavy foliage, and underground spaces (T) and when close to or attached to a structure (O). This sensing capability requires an operationally relevant level of area coverage, standoff distance and structural penetration. The GSS will enhance the Soldier's natural senses (such as hearing and vision to recognize personnel at a distance of at least 300 meters (T), 500 meters (O), in all levels of light, weather, and battlefield conditions (fog, haze, smoke), acquire targets, and navigate (O). This capability will improve the Soldier's ability to perform individual movement techniques during night and limited visibility conditions, including total darkness (T).

*Rationale: Enhanced human sensing requires units to use current SA/SU information to template enemy positions and potential hiding places. The small unit commander bears the responsibility based on current Rules of Engagement (ROE) to determine friendly, enemy and noncombatants. Soldiers on the future battlefield will encounter natural and manmade obscuration and obstacles that inhibit vision. Soldiers require the ability to defeat this obscuration in order to effectively and safely operate. This capability will increase accuracy in first time target engagements through obstacles, which significantly impacts reduction in ammunition expenditures and reduces the Soldier's load. Upon sensing a valid target through an obstacle the Soldier can designate it to the COP to facilitate cooperative engagement by the*

*small unit with organic and external assets. This capability improves the Soldier and small unit ability to "finish decisively."*
*Soldiers must be able to identify friend from foe at 300 meters range and negotiate the terrain over which they are moving during periods of limited visibility without the equipment hindering their movement. The distance requirement for recognition is intended to allow the Soldier to successfully engage targets at the maximum effective range of his individual weapon.*
*Soldiers must be able to identify friend from foe and other relevant audible noises at normal hearing ranges while using hearing protection devices. This capability must promote the Soldier's ability to perform his normal tasks given extremely high noise decibel environment but still distinguish relevant noise stimulus and still protect the Soldier's long-term hearing abilities.*

(2) GSS Lethality.

(a) Personal Lethality: Speed. The GSS must facilitate engagement of targets at distances of 0-50 meters within one second (T). This requirement applies to a Soldier with and without aided vision during the day and with enhanced vision capability during conditions of limited visibility (T=O).

*Rationale: Achieving lethal overmatch requires the Soldier to engage targets quickly and accurately. Recent historical experience consistently shows that threat personnel will seek to engage friendly forces in close combat in close, restrictive terrain in order to negate friendly overmatch in firepower and standoff. Threat personnel will use structural barriers such as masonry buildings to maneuver close to friendly forces in confined spaces where tracked and wheeled platforms cannot gain access. Targets may appear suddenly at distances of less than 50 meters and be fleeting in nature. The FFW ATD will determine the appropriate probability of kill or incapacitation (PKI) for a GSS equipped Soldier employing future weapons at short ranges.*

(b) Personal Lethality: Personal Defense Weapon. The GSS must provide designated Soldiers with an enhanced effectiveness pistol / side arm (T). The side arm will meet the performance of the current M-9 pistol in 9mm (T), and will use the service rifle cartridge (O), enabling Soldiers to engage enemy with effective lethal fire at 200 meters or more (O). The performance for garrison police duty side arms will conform to established law enforcement requirements (T=O).

*Rationale: Some GSS equipped Soldiers perform duties which make the carry of long arms impractical. This is especially true of Military Police and heavy equipment operators in selected MOSs. These Soldiers must have the capability to engage targets to 200 meters or more with effective fire.*

(c) Sub-Compact Weapon (SCW). The SCW provides the Warfighter with a small, compact, lethal weapon necessary for those individuals who normally require both hands to perform their primary military duties, e.g., selected leaders, vehicle operators, weapon platform crews, air crews, robotic system operators, Explosive Ordnance Disposal (EOD), law enforcement and military construction Military Occupational Specialties (MOS) etc. These Warfighters need a compact individual weapon with greater range and lethality than provided by a pistol. The SCW replaces selected rifles, carbines, and pistols across Army formations with a weapon that fires ammunition common to the rifle system, but is designed to provide a smaller, more compact package to allow hands free carry and to allow for more manageable access and employment in tight confines such as vehicle cabs or turrets, aircraft cockpits, etc. The SCW allows for accurate and lethal engagements from 0 to 150 meters (threshold) and 0 to 200 meters (objective) with degraded (compared to rifles/carbines) capability to 300 meters.

(d) Area Denial and Non-lethal Effects. The GSS weapon systems must provide the capability to deny access to, use of, or movement through a geographic area without debilitative injury to personnel (T). Delivery means of these effects may include Soldier emplaced, LOS, BLOS, NLOS, and unmanned systems (T=O).

*Rationale: The small unit AO and AI drive Soldier engagement range capabilities. The FF BCT AI radius is 8 km for platoons and 16 km for the companies. The Soldier and small unit will need to employ non-lethal effects in urban and complex terrain where the enemy may intermingle with non-combatants. Small units may have missions to control an area where strict rules of engagement may preclude injury to non-combatants. Non-lethal effects will aid the ground Soldier to see, understand and act first by providing the capability to dislocate enemy forces hiding in restricted and urban terrains or among the general population.*

(e) Specialty Weapons. GSS equipped Soldiers must employ lethal capabilities for specialized mission needs (T=O).

*Rationale: The Soldier and small unit will execute missions that require specialized lethal capabilities. They will employ these items as needed to augment their organic capabilities. Some lethal capabilities come in the form of a round of ammunition. They (e.g., AT-4 CS) may be assigned to anyone in the squad, but do not qualify as an individual or crew-served weapon. Targets include personnel, equipment and armored vehicles. These targets can be covered and concealed by vegetation, civilians and structural barriers such as earth and timber and masonry buildings. These "specialty" weapons must leverage the GSS capabilities.*

(3) Sustainment.

(a) Sustainment Operations in Limited Visibility. GSS must have enhanced vision devices to enable the Soldier to perform intricate actions at short distances under limited visibility conditions (i.e., eye focus within arms length) (T=O). Limited visibility conditions include darkness, smoke, and fog (T).

*Rationale: Soldiers must perform a broad variety of tasks during periods of limited visibility. Examples of such tasks include maintenance tasks, map reading, and combat lifesaving.*

(b) Enhanced Micro-Climate Protection. The GSS must provide a micro-climatic cooling/heating capability, and must incorporate modular design to facilitate varying levels of protection consistent with the potential for exposure to high thermal loads/cold cycles. Stand alone power (T), powered by system's power (O), and protect the Soldier from hot and extreme cold weather (T=O).

*Rationale: Some Soldiers will perform duties in extreme weather and environmental conditions, which includes thermal stresses of high temperature and cold temperature while engaged in manual labor, while wearing a variety of additional gear, as well as, Military Oriented Protective Posture (MOPP) equipment.*

(c) Auxiliary Air Supply. GSS must provide Soldiers a means to accept supplemental air for the purpose of descending into and working in areas which are devoid of oxygen, are low in oxygen, and / or contain toxic gasses which are not neutralized by the CBRNE filters (T). GSS must be able to operate in such conditions for 20 minutes (T) in poisonous underground air and ship holds, 60 minutes (O).

*Rationale: Some Soldiers are required to perform missions in areas such as in the holds of ships, in shipping containers, in chemical factories and warehouses, in underground facilities, etc. These locations have been known to cause the deaths of Soldiers and police due to the often un-detected hazards. Many of these spaces are devoid of oxygen or filled with poisonous gases, which are beyond the capabilities of CBRNE filters.*

(d) Air Supply with Flotation Collar. The GSS will provide a small, compact, lightweight, breathing source with a regulated supply of oxygen. The amount of oxygen shall be sufficient for 50 seconds minimum breathing time at a 20-foot water depth and 55 º F water temperature, with a breathing rate of 40 liters per minute to facilitate escape from equipment and ascent to the surface (T). The breathing device shall provide five minutes breathing time under the same conditions (O). GSS must provide sufficient buoyancy to Soldiers to surface and sustain flotation after ditching at a total weight of 300 lbs Soldier and gear combined (T). Inflation will be Soldier initiated (T). Oral re-inflation capability is required to maintain inflation for a minimum of 72 hours (T). The flotation device must automatically position the conscious or unconscious Soldier for unimpeded breathing while in the water (T).

*Rationale: Historically, equipment which falls into water has tended to rapidly invert and sink. Crew members may quickly find themselves disoriented under water in an inverted machine. Average breath holding time in a sudden immersion ranges from 40 seconds in 80 º F water to as low as ten seconds in 32 º F water. Time required to escape is a function of many variables, such as injuries sustained, extent of disorientation, location/condition of doors, canopies, and windows to provide escape areas, assistance needed to help other occupants, training, and depth at which exit is made. Coast Guard and Navy experiences have caused these services to specify a regulated air supply sufficient for two minutes at 20 feet depths and 55 º F water. Breathing rates vary widely between individuals depending on fear, physical condition, training, and other factors. Some persons may obtain up to five minutes from the same amount of air as others would use in two minutes.*

(4) GSS Force Protection.

(a) Crowd Control: Protection. Soldiers in urban operations often find themselves performing duties that require close contact with civilians. These duties may include crowd control/MP type missions. The GSS must protect these Soldiers from close-up hostile actions from individuals blending into the crowd (T).

1. Crowd Control: Facial Protection. The GSS must provide a lightweight and integrated headwear system providing facial protection with limited reflectivity to enemy observers, single or multi-layered, positively ventilated, suitable for daily wear in all light and weather conditions (T=O), must provide protection from laser exposure (T=O), must cover a minimum of two thirds of the exposed face (T), fully protect the face and neck (O), must be suitable for continuous wear with the combat helmet system (T=O), must provide automatic and manual adjusted light filters (T=O), must be effective against massive blunt trauma from sticks, bricks, rocks (T=O), and have a ballistic V-10 of 40 grains at 1,500 feet per second against both round ball and right cylinder projectiles (O).

*Rationale: Soldiers will be involved in a myriad of operations, such as hostile crowd and populace control, to include operating in urban terrain with the neutral local populace present. The facial system will protect the Soldier from facial injuries from thrown rocks and objects, which can cause serious injuries or fatalities, and from explosive launched projectiles and bullets. This attribute is special for MOS specific duties such as Military Police that may be required to control large crowds of non-combatants that have the potential to turn hostile.*

2. Crowd Control: Body Protection: Stab and Ballistic. The body armor shall consist of soft flexible textiles and/or soft fabric under garment armor that provides handgun ballistic protection against the 9mm (124 gr.) @ 1,400 ft/sec (T). It will protect the individual(s) from a variety of threats such as, handguns, small arms, knives, ice picks, and other sharpened metal objects fashioned as weapons (T).
*Rationale: It shall be ballistic resistance against handgun threats and will comply with National Institute of Justice (NIJ) Standard NIJ-STD-0101.3 for multi-hit, transient deformation and obliquity effects unless new military test standards documents contradictory requirements (T). If this does occur the military standards document will supersede the NIJ Ballistic standards and the California Ice Pick thrust test standard (O).*

(b) Force Protection: Petroleum, Oil and Lubricants (POL). Soldiers handling fuels must have protection capability against common battlefield fuels (T) and all petroleum-based fuels (O) while transporting fuel, performing fuel operations, pipeline operations, and repairs on fuel equipment in all environments under all operational environments. The GSS must provide hand, feet, body, and eye protection from contact with fuels during all types of fuel operations for a period of 120 days (T) and 180 days (O). The GSS must also provide protection from saturation for a period of at least two hours (T) and four hours (O). The protective material must also provide electrostatic protection and have a low thermal burden (T=O).

*Rationale: Fuels are considered both a health hazard and a safety hazard. Fuels also present a safety hazard due to their flammable nature. They pose a health hazard in that many of the additives found in certain fuels are rated as hazardous substances (i.e. are carcinogenic and/or toxic), as well as the caustic nature of petroleum-based fuels themselves. Direct contact with skin will result in burns and dermatitis.*

*Fuels are also a safety hazard due to the flammable nature of fuel. Soldiers who handle fuel or are at high risk from coming into contact with fuel are at risk from the fuel igniting (either the* *fuel source, or fuel-soaked clothing), which means Fuel Handlers and maintainers must also be provided electrostatic protection to preclude static electricity from causing fuel to ignite.*

1. Force Protection: Neutralize Electrostatic Discharge. Soldiers working in positions that have a high electrostatic threat must have the capability to neutralize electrostatic discharge prior to performing operations on electronic and explosive items (T). Soldiers require equal static dissipating potential as the equipment with which they operate or interface (T=O).

*Rationale: Static build-up and discharge from uniform fabrics, undergarment fabrics, materials, and the GSS can cause serious injury to the Soldier and damage to equipment. Static discharge can cause explosion, flash back, and fire, especially in fueling operations; and can cause damage to electrical circuits, computers, communication equipment, and other electrical systems.*

2. Enhanced Fire Protection. Soldiers performing fuel sustainment operations require additional protection from fire hazards (T). Fire protection in the threshold must be at least fire retardant (T), with a desired protection level of fire resistant and fuel resistant (O).

*Rationale: Soldiers driving bulk fuel tankers are at high risk from serious fire injuries in the event of a vehicle accident or from an IED. In a catastrophic event, these drivers can be instantly engulfed in fuel and flames. Without Fire Retardant (T), serious injury or death will occur. Both fire protection and fuel protection (O) are required to sufficiently protect the bulk fuel vehicle drivers. In addition, Fuel Handlers and maintainers providing support to aircraft require additional protection from the fuel they are handling, as well as from the risk of fire. As Soldiers become more modular and multifunctional, fuel handlers and maintainers must be prepared to perform all types of fuel operations, to include aviation support and bulk fuel drivers, and therefore require both fuel and fire protection (O).*

(c) Protection from Occupational Hazards. Some Soldiers are exposed to occupational contaminates while performing their MOS tasks. These troops must have protection enabling them to continue the mission after being contaminated without a clothing change for a minimum of 12 hours (T), 24 hours (O), without the use of CBR-peculiar equipment. This protection encompasses full body protection and may include special provisions for eye, facial, and skin in direct contact with hazards (T).

*Rationale: Mortuary Affairs Specialists, medical specialists, maintainers, water purification specialists, drillers, excavators, blasting personnel, stevedores, Military Load Handling Equipment (MHE) operators, transportation operators, laundry personnel, etc. routinely come into contact with chemicals, detergents, airborne pathogens, body fluids, solvents, powder and dust from stone and minerals, hydraulic oils of many types, etc. while performing MOS-related tasks. These contaminates can cause severe illness or injury if Soldiers are not sufficiently protected, and may prevent or impair mission performance. Operations in difficult environmental conditions and / or hostile scenarios, under heavy work loads and limited manpower availability, and under very limited re-supply / non-availability of clean-up facilities will frequently preclude the opportunity to change clothing in less than a full, extended work day.*

(d) Home Station Mission Protection. Provide specialized Soldier clothing and equipment for unique Home Station missions (T=O).

*Rationale: Some Soldiers have Home Station MOS missions requiring the wearing of personal clothing and equipment beyond which is covered by the CSS uniform. These stem from US law and regulatory guidance as well as DOD and HQDA policies and regulations. Each have a specific purpose, such as protecting the Soldier; presenting a clean, safe, and professional appearance; conforming to proper religious orders; and/or a less costly protective over-garment than an expensive field/tactical uniform. The following list is included as a guide; a comprehensive list is in development:*

| **Variant Specific Item** | **Unit of Issue** | **MOS** |
|---|---|---|
| Fuel Handler Gloves | 1 | 71 F,L,W, 92F |
| Fuel Handler Coveralls | 2 | 71 F,L,W, 92F |
| Safety Goggles | 2 | All MOS |
| Protective Footwear | 1 | 21,35,55,62,63,68,88 |
| Mechanics Coverall | 2 | All MOS |
| Cook whites | 6 | 92G,91M,91B |
| Safety Breathing Mask | | 92M |
| Safety Coveralls/Aprons | 2 | 92M |
| Hand protection (Mortuary Affairs) | | 92M |
| Protective Over garments (Mortuary Affairs) | | 92M |
| Eye Shield Protection | | 92M |
| Hearing Protection | 2 | All MOS |
| Chaplain garments | | 52 Series |
| Rigger Pocket shorts | | 92R |
| Band Uniforms (Blues) | | 02 Series |
| Band Uniforms (Greens) | | 02 Series |

Table 6-8.1. Specialized Soldier clothing and equipment.

(e) Urban Terrain Mobility: Create a Man-Sized Hole in a Wall. The GSS must enable Soldiers to create a man-sized hole through a wall, floor, or ceiling in a single stage of reduction (penetration and removal of all building materials) (T). The hole must be through eight inches of double reinforced concrete (using #4 rebar on 12 inch centers) and have dimensions of thirty by fifty (30 x 50) inches (T), fifty by seventy (50 x 70) inches (O).

*Rationale: Maneuver Support Soldiers are responsible to support combat and police actions to gain access to facilities for the purpose of enabling maneuver, defeating enemy, and performing rescue operations. A rapid, complete means of single stage reduction is often the difference between success and failure of the mission. The hole created must enable combat and reaction teams with their gear to pass through the wall wearing full tactical and mission gear (including stretchers) at a high rate of passage.*

(f) Identify/ Reduce/ Defeat Obstacles. GSS sensors must be compatible with Ground Combat Vehicle capabilities for advance detection of obstacles and facilitate Soldiers' reduction and/or bypass of obstacles (T).

*Rationale: Soldiers must be able to detect and reduce or bypass obstacles including concealed mines that are impeding or will impede their movement.*

(g) Soldier Sustainability: Improved Soldier Strength and Endurance. Increase the endurance of and strength of the Soldier without the aid of drugs (O). Physically demanding tasks must not drain energy or sap strength any more than would performing the requisite task motions with minimal load moved or work performed for a short duration. Soldiers must be enabled to improve their physical work performance by 50% (T) as compared to being unaided; performance improved by 200% (O).

***Rationale:*** *Engineer and Combat Sustainment Soldiers, and all Soldiers to a lesser degree, are required to routinely perform extremely taxing, difficult and long duration manual labor and repetitive tasks in the support of their primary tasks. Performance of these labor intensive activities leave Soldiers drained, and un-ready / unable to perform additional required tasks. Advances in nano-technology, synthetic muscles, weight-bearing structures, and robotics should be explored as a means to preserve the survivability of our troops, and enhance the effectiveness of the ever-shrinking military force structure.*